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CHAPTER 12

0

Mathematical Preliminaries

§ 0·1. Introduction. In this chapter we shall discuss some basic concepts and techniques of a number of mathematical topics that will facilitate the reader in the development and understanding of the theoretical and computational aspects of linear programming discussed in the succeeding chapters.

1. Matrices and Determinants.

§ 0·2. Definitions.

1. Matrix. *A matrix is a rectangular array of $m \times n$ numbers arranged in m rows and n columns as follows :*

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix} = [a_{ij}]_{m \times n}$$

a_{ij} is called the element of the matrix A in i-th row and j-th column.

2. Square Matrix. If $m = n$, matrix A is called a *square matrix* of order n.

3. Diagonal Matrix. A square matrix in which off-diagonal elements, a_{ij} for $i \neq j$, are all equal to zero is called a *diagonal matrix.*

4. Unit (or Identity) Matrix. A unit (or Identity) matrix is a diagonal matrix whose diagonal elements are equal to 1. In other words, a square matrix is called a *unit matrix* if $a_{ij} = 0$ for $i \neq j$ and $a_{ij} = 1$ for $i = j$. A unit matrix of order n is denoted by I_n or simply I.

Thus $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ etc.

5. Triangular Matrix. A square matrix A is called a *triangular matrix* if all $a_{ij} = 0$ for $i > j$ or if all $a_{ij} = 0$ for $i < j$.

6. Null Matrix. A matrix is called a *null matrix* if all its elements are equal to zero.

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7. Transposed Matrix. The *transpose* of matrix A is a matrix obtained by interchanging its rows and columns and is denoted by A^t or A'.

If the matrix A is of order $m \times n$ then the order of A' as $n \times m$.

8. Equality of Matrices. Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to be equal if

(i) they are of the same order,

and (ii) $a_{ij} = b_{ij}$, for all i and j.

§ 0·3. Operations of Matrix Addition and Multiplication.

1. Addition of Matrices.

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two matrices of the same order. Then the sum of these two matrices denoted by $A + B$ is given by

$$C = A + B = [a_{ij} + b_{ij}]$$

The order of $C = A + B$ is equal to the order of the matrices A and B.

It is important to note that the addition of matrices satisfy the following properties.

(*i*) $A + B = B + A$ (Commutative law)

(*ii*) $(A + B) + C = A + (B + C)$ (Associative law)

(*iii*) $(A + B)' = A' + B'$.

2. Multiplication of a matrix by a scalar.

Let $A = [a_{ij}]$ be a matrix of order $m \times n$ and c be a scalar, then the product of the matrix A by c, denoted by cA is given by

$$cA = [c\, a_{ij}].$$

The following properties hold for scalar multiplication of matrices

(*i*) $(a + b) A = aA + bA$. (Distributive law)

(*ii*) $a(A + B) = aA + aB$. (Distributive law)

3. Multiplication of Matrices.

Let $A = [a_{ij}]$ and $B = [b_{jk}]$ be two matrices of the orders $m \times n$ and $n \times p$ respectively. Then the product of the matrices A and B, denoted by AB is defined as the matrix $C = [c_{ik}]$ of order $m \times p$, such that

$$c_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk}$$

Note. The multiplication AB of matrices A and B is defined if and only if the number of columns of matrix A is equal to the number of the rows of matrix B.

The properties of matrix multiplication are as follows :

(*i*) $(AB)C = A(BC)$ (Associative law)

(*ii*) $(A + B)C = AC + BC$ (Distributive law)

(*iii*) $C(A + B) = CA + CB$

(*iv*) $a(AB) = (aA)B = A(aB)$

(*v*) $IA = A = AI$

(*vi*) $(AB)' = B'A'$.

It is important to note that the matrix multiplication is not commutative in general.

i.e. in general $AB \neq BA$.

§ 0·4. Sub-matrix. If we cross out all but r-rows and s-columns of an $m \times n$ matrix A, the resulting $r \times s$ matrix is called a *sub-matrix* of A.

§ 0·5. Minor of order k. For a $m \times n$ matrix A if R is the kth order sub-matrix obtained by deleting all but k-rows and k-columns of A. Then $|R|$ is called a *k-th order minor* of A.

§ 0·6. Determinant. There is a number associated with every square matrix and is called the determinant of the matrix.

Def. The *determinant* of square matrix $A = [a_{ij}]$ of order n, written as $|A|$ is defined to be the number computed from the following sum involving the n^2 elements in A

$$|A| = \Sigma (\pm)\, a_{1i} a_{2j} \dots a_{nr}$$

the sum being taken over all permutations of the second subscripts. A term is assigned a plus sign if $(i, j, \dots r)$ is an even permutation of $(1, 2, \dots n)$, and a minus sign if it is an odd permutation.

e.g.,

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{23}a_{31} + a_{12}a_{21}a_{33} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}$$

$$= a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

This is the expansion of the determinant with respect to the first row.

§ 0·7. Important properties of determinants. Now we give important properties of the determinants (without proof) that may be

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verified from definition of the determinant.

1. If every element of a row or column of a determinant is zero, then the value of the determinant is zero.
2. If $|B|$ is the determinant formed by interchanging any two rows or columns of a determinant $|A|$, then
$$|B| = -|A|$$
3. The value of a determinant is not changed if its rows and columns are interchanged.
4. If two rows or columns of a determinant are identical then the value of the determinant is zero.
5. If every element of a row or column of a determinant is multiplied by a fixed number k, then the value of the determinant is multiplied by k.
6. If to every element of a row or column of a determinant, we add k times the corresponding element of another row or column, then the value of the determinant is not changed.

§ 0·8. Minors. The *minor* of the element a_{ij} of determinant $|A|$ is the determinant obtained by deleting ith row and jth column of $|A|$ and is denoted by M_{ij}.

If
$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$
then M_{11} minor of element a_{11} is given by
$$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}.$$
It may be verified that
$$|A| = a_{11} M_{11} - a_{12} M_{12} + a_{13} M_{13}.$$

§ 0·9. Cofactors. The *cofactor* of the element a_{ij} of determinant $|A|$ is $(-1)^{i+j}$ times the determinant obtained by deleting ith row and jth column of $|A|$ and is denoted by C_{ij}.

i.e.
$$C_{ij} = (-1)^{i+j} M_{ij}.$$
If
$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$
then C_{11}, cofactor of a_{11}, is given by

$$C_{11} = (-1)^{1+1} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$
If may be verified that
$$|A| = a_{11} C_{11} + a_{12} C_{12} + a_{13} C_{13}.$$

§ 0·10. Rank of a Matrix. The *rank* of a matrix A is r if all the minors of order $r+1$ or more of the matrix A are zero but at least one minor of order r is not zero.

§ 0·11. Adjoint of a Matrix. Let A be a square matrix of order n. Then the *adjoint* of matrix A is the transpose of the matrix B, obtained from A by replacing each element a_{ij} of A by its cofactor C_{ij} and is denoted by adj. A.
$$\textit{i.e.}, \text{ adj. } A = \begin{bmatrix} C_{11} & C_{21} & \dots & C_{n1} \\ C_{12} & C_{22} & \dots & C_{n2} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ C_{1n} & C_{2n} & \dots & C_{nn} \end{bmatrix}$$

§ 0·12. Singular and Non-singular Matrices. A square matrix A is said to be a *non-singular matrix* if its determinant does not vanish and *singular matrix* if its determinant vanish.

i.e., A is *non-singular* if $|A| \neq 0$ and *singular* if $|A| = 0$.

§ 0·13. Inverse of a Matrix. Let A be a square matrix. If there exists a square matrix B which satisfies the relation
$$AB = I = BA$$
then B is called the *inverse* of A and is denoted by A^{-1}.

The inverse of the matrix A is given by
$$A^{-1} = \frac{1}{|A|} (\text{adj.} A).$$

Computation of the inverse by partioning.

A non-singular square matrix in partioned form is expressed as follows
$$M = \begin{bmatrix} I & Q \\ 0 & R \end{bmatrix}$$
where I (unit matrix) and R are square submatrices and Q, 0 (null matrix) are two matrices.

• If R^{-1} exists and is known, then the inverse of the matrix M is given by

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$$M^{-1} = \begin{bmatrix} I & -QR^{-1} \\ 0 & R^{-1} \end{bmatrix}$$

2. Vectors and Vector Spaces.

§ 0·14. Definitions.

1. Vector. Matrices having a single row or column are referred to as *vectors*. A matrix of a single row is called a *row vector*, and a matrix of a single column is called a *column vector*.

Vectors will be denoted by lower case bold face type. The n dimensional column vector $\begin{bmatrix} a_1 \\ a_2 \\ \cdots \\ a_n \end{bmatrix}$ is also written as $[a_1, a_2, ..., a_n]$ while the n-dimensional row vector is written as $(a_1, a_2, ..., a_n)$.

As vector $\mathbf{a} = (a_1, a_2, ..., a_n)$ is called an n-component vector, and $a_i, i = 1, 2, ..., n$ is said to be the ith component of vector $\mathbf{a}$.

As (a_1, a_2, a_3) is a point in three dimensional space, so the vector $\mathbf{a} = (a_1, a_2, ..., a_n)$ can be considered to be a point in the n-dimensional space. There is no geometrical distinction between row and column vectors.

2. Unit Vector. A *unit vector* denoted by e_i is a vector whose ith component is unity and all other components are zero.

For vectors with n components there are n unit vectors.

$$e_1 = (1, 0, ..., 0), e_2 = (0, 1, 0, ..., 0), ..., e_n = (0, 0, ..., 1).$$

3. Null vector. A vector whose all components are zero is called a *null vector* or zero vector and is denoted by 0, *i.e.*,

$$0 = (0, 0, ... 0).$$

4. Sum vector. A vector each of whose component is 1 is called a *sum vector* and is written as 1

$$1 = (1, 1, ..., 1).$$

5. Sum of vectors. The sum of two vectors

$$\mathbf{a} = (a_1, a_2, ..., a_n), \mathbf{b} = (b_1, b_2, ..., b_n)$$

denoted by $\mathbf{a} + \mathbf{b}$ is a vector defined by

$$\mathbf{a} + \mathbf{b} = (a_1 + b_1, a_2 + b_2, ..., a_n + b_n).$$

6. Subtraction of vectors. The difference of vectors a and b denoted by $\mathbf{a} - \mathbf{b}$ is a vector defined by

$$\mathbf{a} - \mathbf{b} = (a_1 - b_1, a_2 - b_2, ..., a_n - b_n).$$

7. Scalar Multiplication of a vector. The product of a vector a by a scalar k is defined by

$$k\mathbf{a} = (ka_1, ka_2, ..., ka_n).$$

8. Distance between two vectors. The distance between two n-component vectors a and b, denoted by $| \mathbf{a} - \mathbf{b} |$ is defined to be the positive number

$$| \mathbf{a} - \mathbf{b} | = \sqrt{\{(\mathbf{a} - \mathbf{b}) . (\mathbf{a} - \mathbf{b})\}} = \left\{ \sum_{i=1}^{n} (a_i - b_i)^2 \right\}^{1/2}$$

9. Scalar Product of two vectors. If $\mathbf{a} = (a_1, a_2, a_3)$ and $\mathbf{b} = (b_1, b_2, b_3)$ are two vectors then the scalar product of a and b denoted by a.b is defined by

$$\mathbf{a}.\mathbf{b} = | \mathbf{a} | \cdot | \mathbf{b} | \cos \theta$$

where θ is the angle between the two vectors.

It may be verified that

$$\mathbf{a}.\mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

Two vectors a and b are said to be orthogonal if $\mathbf{a}.\mathbf{b} = 0.$

§ 0·15. Euclidean space. An n-dimensional Euclidean space, denoted by E^n or R^n is the collection of all n component vectors (points) $\mathbf{a} = (a_1, a_2, ..., a_n)$. For these vectors, operations of addition, multiplication of a vector by a scalar are defined by the rules for matrix operations (as above). Furthermore, associated with any two vectors in the collection is a non-negative number called the distance between two vectors (defined in 8 of § 0·14).

Now in further discussion each vector will belong to some Euclidean space.

§ 0·16. Linear Dependence and Independence of vectors :

The set of vectors $\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_k$ from E^n is said to be *linearly dependent* (L.D.) if there exist scalars $\lambda_i, i = 1, 2, ... k$, with at least one $\lambda_i \neq 0$, such that

$$\sum_{i=1}^{k} \lambda_i \mathbf{a}_i = \lambda_1 \mathbf{a}_1 + \lambda_2 \mathbf{a}_2 + ... + \lambda_k \mathbf{a}_k = 0$$

where 0 is the null vector belonging to E^n.

If the only set of λ_i's which satisfies $\sum_{i=1}^{k} \lambda_i \mathbf{a}_i = 0$ is with $\lambda_1 = \lambda_2 = ... \lambda_k = 0$, then the set of vectors is said to be *linearly independent* (L.I.).

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§ 0·17. Linear Combination (L.C.) of vectors. Let $\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_k$ be the vectors from E^n. Then the vector $\mathbf{a}$ given by the expression

$$\mathbf{a} = \lambda_1 \mathbf{a}_1 + \lambda_2 \mathbf{a}_2 + ... + \lambda_k \mathbf{a}_k = \sum_{i=1}^{k} \lambda_i \mathbf{a}_i$$

where $\lambda_i, i = 1, 2, ..., k$ are scalars, is called the *linear combination* (L.C) of the given set of vectors and belong to E^n.

Important points.

1. The terms L.D. and L.I. are mutually exclusive *i.e.*, any set of vectors is either L.D. or L.I.
2. A set of vectors containing the 0 vector is L.D.
3. Any infinite set of vectors is said to be L.I. if its every subset is L.I., otherwise it is L.D.

§ 0·18. Spanning set. A set of vectors $\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_k$ from E^n is said to *span* or *generate* E^n if every vector in E^n can be written as a linear combination of vectors $\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_k$.

§ 0·19. Basis set. A spanning set whose vectors are L.I. is called a *basis set* or a *basis*. Thus a set $B \subset E^n$ form a basis for E^n if

(i) B is L.I.

and (ii) B span E^n *i.e.*, each vector in E^n can be expressed as a L.C of the vectors of B.

For example the set $B = \{e_1, e_2, ..., e_n\}$ form a basis of E^n.

§ 0·20. Some Useful Theorems of Linear Algebra. Now we shall state some theorems of linear algebra which are very useful in L.P. The proofs of these theorems have been omitted as their proofs may be seen in any standard book on Modern Algebra.

1. A set of vectors is L.D. iff some one of the vectors belonging to the set can be expressed as a L.C. of the others.
2. The representation of any vector in terms of a basis set of vectors in unique.
3. **Replacement Theorems.** Let $\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_n$ form a basis for a set A of vectors and $\mathbf{b} \in A$, $\mathbf{b} \neq \mathbf{a}_i$ $i = 1, 2, ..., n$. If $\mathbf{b}$ is non-null such that

$$\mathbf{b} = \sum_{i=1}^{n} \lambda_i \mathbf{a}_i \quad \text{and} \quad \lambda_i \neq 0,$$

then $\mathbf{b}$ can replace $\mathbf{a}_i$ representing the basis property of the new set.

4. Any two basis for the same set have the same number of vectors.
5. Any set of $(n + 1)$ or more vectors of E^n will be L.D.
6. Any set of n. L.I. vectors of E^n forms a basis of E^n.

3. Simultaneous Linear Equations.

§ 0·21. Simultaneous Linear Equations. A system of m simultaneous linear equations in $n\ (\geq m)$ unknowns $x_1, x_2, ..., x_n$ may be written as

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + ... + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + ... + a_{2n}x_n &= b_2 \\ ... \quad ... \quad ... \quad & ... \\ ... \quad ... \quad ... \quad & ... \\ a_{m1}x_1 + a_{m2}x_2 + ... + a_{mn}x_n &= b_m \end{aligned}$$

where a_{ij}, b_i are known constants, $i = 1, 2, ... m, j = 1, 2, ..., n$.

In matrix form the system can be written as follows :

$$A\mathbf{x} = \mathbf{b}$$

where $A = \begin{bmatrix} a_{11} & a_{12} & ... & a_{1n} \\ a_{21} & a_{22} & ... & a_{2n} \\ ... & ... & ... & ... \\ ... & ... & ... & ... \\ a_{m1} & a_{m2} & ... & a_{mn} \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ ... \\ x_m \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ ... \\ b_m \end{bmatrix}$

Matrix A is called the coefficient matrix.

The matrix $A\ \mathbf{b} = [A\ \mathbf{b}] = \begin{bmatrix} a_{11} & a_{12} & ... & a_{1n} & b_1 \\ a_{21} & a_{22} & ... & a_{2n} & b_2 \\ ... & ... & ... & ... & ... \\ ... & ... & ... & ... & ... \\ a_{m1} & a_{m2} & ... & a_{mn} & b_m \end{bmatrix}$

Any set of values of $x_1, x_2, ..., x_n$ which satisfy all the equations of the system is called the *solution* of the system. *A system of equations is said to be* **consistent** *if there exists a solution of the system, otherwise inconsistent.*

The necessary and sufficient condition for a system of equations to be consistent is that $r(A) = r[A\ \mathbf{b}]$.

Case 1. If $r(A) = r[A\ \mathbf{b}] = n$, the number of unknowns. This implies that $m \geq n$.

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When $m = n$, then $|A| \neq 0$. So we obtain a unique solution given by

$$x = A^{-1} \mathbf{b}.$$

Case II. *If* $r(A) = r[A\ \mathbf{b}] = K \leq n < m$ *(the number of equations).*

In this case k rows of A and $[A\ \mathbf{b}]$ are L.I. Hence the remaining $m - k$ rows (or say equations) can be written as a L.C. of the k independent rows. Thus the solution can be found by solving k independent equations which will also satisfy the remaining $(m - k)$ equations. In such cases we say that the $(m - k)$ equations are **redundant.**

Case III. If $r(A) = r[A\ \mathbf{b}] < n$: In this case the number of solutions is infinite.

1

LINEAR PROGRAMMING PROBLEMS

(Formulation and Graphical Solution)

§ 1·1. Introduction. In general, programming problems deal with determining optimal allocations of limited resources to meet given objectives. The resources may be in the form of men, materials and machines etc. and the objective may be to yield one or more products. There are certain restrictions on the total amount of each resource available, and on the quantity or quality of each resource available, and on the quantity or quality of each product made. Out of all permissible allocations of resources, one has to find the one which optimize (maximize or minimize) the total profit or cost.

Linear programming deals with that class of programming problems for which all relations among the variables are linear. The relations must be linear, both in the constraints and in the functions to be optimized. The term 'linear' means that all the relations in the particular problem are linear and the term 'programming' refers to the process of determining a particular programming or plan or action. Only a little over a decade has elapsed since George Dantzig formulated the general linear programming problem and developed the simplex method for the solution of this problem.

In this chapter, we shall discuss the properties of the linear programming problems and give graphical method for their soultion.

§ 1·2. General linear programming problems. (Definition).

[C. A. Int. Exam. May 85, May 86; Meerut 90, 94, 94(P) 94(R) 95, 95(BP), 98]

A linear programming problem includes a set of simultaneous linear equations which represent the conditions of the problem and a linear function which expresses the objective function of the problem.

The linear function which is to be optimized is called the **objective function** and the conditions of the problem expressed as simultaneous linear equations (or inequalities) are referred as **constraints.**

A general linear programming problem can be stated as follows :

Find $x_1, x_2, \ldots, x_n$, *which optimize the linear function*

$$Z = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n \qquad \ldots (1)$$

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subject to the constraints

$$\left.\begin{array}{l} a_{11}x_1 + a_{12}x_2 + \dots + a_{1j}x_j + \dots + a_{1n}x_n \ (\le = \ge)\ b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2j}x_j + \dots + a_{2n}x_n \ (\le = \ge)\ b_2 \\ \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ a_{i1}x_1 + a_{i2}x_2 + \dots + a_{ij}x_j + \dots + a_{in}x_n \ (\le = \ge)\ b_i \\ \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mj}x_j + \dots + a_{mn}x_n \ (\le = \ge)\ b_m \end{array}\right\} \quad \dots (2)$$

and non-negative restrictions

$$x_j \ge 0, \quad j = 1, 2, \dots n. \quad \dots (3)$$

where all a_{ij}'s, b_i's and c_j's are constants and x_j's are variables.

In the conditions given by (2) there may be any of the three signs $\le, =, \ge$.

The function z given by (1) is called the **objective function** and the conditions given by (2) are termed as the **constraints** of the linear programming problem.

We shall always assume that all $b_i \ge 0$. If any one is negative, we make it positive by multiplying both sides of the corresponding inequality by -1. By this multiplication the inequality is also reversed.

The above linear programming problem may also be stated in matrix form as follows :

Optimize $\quad Z = \mathbf{c}\,\mathbf{x}$

subject to

$$A\mathbf{x} \ (\le, =, \ge)\ \mathbf{b}$$

and $\quad \mathbf{x} \ge 0$

where $A = [a_{ij}]$, is the matrix of coefficient of order $m \times n$.

$\mathbf{c} = (c_1, c_2, \dots, c_n)$ is a row vector known as **price vector**

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} = [x_1, x_2 \dots\dots, x_n]^*$$

is column vector of variables.

*Throughout this book we shall write n-dimensional row vector as $(a_1, a_2, \dots, a_n)$ while m-dimensional column vector will be written as

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_m \end{bmatrix} = [b_1, b_2, \dots, b_m]$$

is column vector called the **requirement vector** and 0 is n dimensional null column vector.

The column vector formed by the coefficient of x_j in all the constraints is denoted by α_j

i.e., $$\alpha_j = \begin{bmatrix} a_{ij} \\ a_{2j} \\ \dots \\ a_{mj} \end{bmatrix} = [a_{ij}, a_{2j}, \dots, a_{mj}]$$

then $$A = (\alpha_1, \alpha_2, \dots, \alpha_n).$$

Note. In a general linear programming problem it is assumed that the number of rows of coefficient matrix A is less than equal to its number of columns.

§ 1·3. Mathematical Formulation of a L.P.P.

It is important to recognize a problem which can be handled by linear programming and then to formulate its mathematical model.

Several examples are given below where the mathematical models of given problem are formulated.

Illustrative Examples

Ex. 1. *A manufacturer of a line of patent medicines is preparing a production plant on medicines A and B. There are sufficient ingredients available to make 20,000 bottles of A and 40,000 bottles of B but there are only 45,000 bottles into which either of the medicines can be put. Further-more, it takes 3 hours to prepare enough material to fill 1,000 bottles of A, it takes one hour to prepare enough material to fill 1,000 bottles of B and there are 66 hours available for this operation. The profit is Rs. 8 /- per bottle for A and Rs. 7 /- per bottle of B.*

(a) *Formulate this problem as a linear programming problem ?*

(b) *How should the manufacture schedule production in order to maximize his profit ?*

Sol. (a) Let the manufacturer produce x_1 and x_2 bottles of medicines A and B respectively.

$\therefore$ Total profit (in Rs.) $\quad Z = 8x_1 + 7x_2. \quad \dots (1)$

The time required to prepare x_1 bottles of medicine A

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$= 3x_1/1000$ hours.

and the time required to prepare x_2 bottles of medicine B

$= x_2/1000$ hours.

$\therefore$ Total time required to prepare x_1 bottles of medicine A and x_2 bottles of medicine B is $\frac{3x_1}{1000} + \frac{x_2}{1000}$ hours.

Since total time available for this operation is 66 hours

$$\therefore \quad \frac{3x_1}{1000} + \frac{x_2}{1000} \le 66.$$

or $$3x_1 + x_2 \le 66{,}000.$$

Since there are only 45,000 bottles into which the medicines can be put

$$\therefore \quad x_1 + x_2 \le 45{,}000$$

Hence the linear programming problem of the given problem is as follows.

$$\text{Max. } Z = 8x_1 + 7x_2$$

subject to the constraints

$$3x_1 + x_2 \le 66{,}000,\ x_1 + x_2 \le 45{,}000$$
$$x_1 \le 20{,}000,\ x_2 \le 40{,}000,$$

and $$x_1 \ge 0,\ x_2 \ge 0.$$

(b) For the solution of the above problem see Ex. 8 on page 25.

Ex. 2. *A resourceful home decorator manufactures two types of lamps say A and B. Both lamps go through two technitions, first a cutter, second a finisher.*

Lamp A requires 2 hours of the cutter's time and 1 hour of the finisher's time. Lamp B requires 1 hour of cutter's and 2 hours of finisher's time. The cutter has 104 hours and finisher 76 hours of time available each month. Profit on one lamp A is Rs. 6·00 and on one lamp B is Rs. 11·00. Assuming that he can sale all that he produces, how many of each type of lamps should he manufacture to obtain the best return.

Sol. Formulation of the mathematical model.

Let the decorator manufacture x_1 and x_2 lamps of types A and B respectively.

$\therefore$ Total profit (in Rs.) $Z = 6x_1 + 11x_2$.

Total time of the cutter used in preparing x_1 lamps of type A and x_2 of type B is $2x_1 + x_2$.

Since cutter has 104 hours only for each month

$$\therefore \quad 2x_1 + x_2 \le 104.$$

Similarly the total time of the finisher used in preparing x_1 lamps of type A and x_2 of type B is

$$x_1 + 2x_2 \le 76$$

Hence the decorator problem is, to find x_1, x_2 which maximize

$$Z = 6x_1 + 11x_2$$

subject to the constraints

$$2x_1 + x_2 \le 104$$
$$x_1 + 2x_2 \le 76$$

and $$x_1 \ge 0,\ x_2 \ge 0.$$

Solution of the problem. For sol., see Ex. 9 on page 26.

Ex. 3. *A firm can produce three types of cloth say A, B and C. Three kinds of wool are required for it. say red wool, green wool and blue wool. One unit length of type A cloth needs 2 yards of red wool and 3 yards of blue wool; one unit length of type B cloth needs 3 yards of red wool, 2 yards of green wool and 2 yards of blue wool, and one unit length of type C cloth needs 5 yards of green wool and 4 yards of blue wool. The firm has only a stock of 8 yards of red wool, 10 yards of green wool and 15 yards of blue wool. It is assumed that the income obtained from one unit length of type A cloth is Rs. 3.00, of type B cloth is Rs. 5.00 and of type C cloth is Rs. 4.00. Determine how the firm should use the available material, so as to maximize the income from the finished cloth.*

Sol. Formulation of the mathematical model of the problem.

For clear understanding of the problem, first we construct a table 1·1 for the given datas. (See on page 16).

Let the firm produce x_1, x_2, x_3 yards of the three type of cloths A, B and C respectively.

Total profit in Rs. of the firm is given by

$$Z = 3x_1 + 5x_2 + 4x_3.$$

Total quantity of red wool required to prepare x_1, x_2, x_3 yards of three cloths of type A, B and C is

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Table 1.1

Kinds of wool	Types of cloth			Stock of wool available with the firm (in yards)
	$A(x_1)$	$B(x_2)$	$C(x_3)$	
Red	2	3	0	8
Green	0	2	5	10
Blue	3	2	4	15
Income from one unit length of cloth in Rs.	3.00	5.00	4.00	

$$2.x_1 + 3.x_2 + 0.x_3 \text{ yards.}$$

Since the stock of red wool available is only 8 yards,

$$\therefore \quad 2x_1 + 3x_2 + 0.x_3 \le 8$$

Similarly total quantity of green wool required is

$$0.x_1 + 2.x_2 + 5.x_3 \le 10$$

and total quantity of blue wool required is

$$3x_1 + 2x_2 + 4x_3 \le 15.$$

Hence the problem of the firm formulated as linear programming problem is as follows :

Find x_1, x_2, x_3 which maximize

$$Z = 3x_1 + 5x_2 + 4x_3$$

subject to the constraints

$$2x_1 + 3x_2 + 0.x_3 \le 8$$
$$0.x_1 + 2x_2 + 5x_3 \le 10$$
$$3x_1 + 2x_2 + 4x_3 \le 15$$

and $$x_1 \ge 0,\ x_2 \ge 0,\ x_3 \ge 0.$$

(Since firm cannot produce negative quantities)

Sol. of the problem.

For the solution of the problem See Ex. 3 of chapter 3.

Ex. 4. *The objective of a diet problem is to ascertain the quantities of certain foods that should be eaten to meet certain nutritional requirement at a minimum cost. The consideration is limited to milk : beef and eggs, and to vitamins A, B, C. The number of milligrams of each of these vitamins contained with a unit of each food is given below.*

Table 1.2.

Vitamin	Gallon of milk	Pound of beef	Dozen of eggs	Minimum daily requirements
A	1	1	10	1 mg.
B	100	10	10	50 mg.
C	10	100	10	10 mg.
Cost	Rs. 1.00	Rs. 1.10	Rs. 0.50	

What is the linear programming formulation for this problem ?

Sol. Let the daily diet consist of x_1 gallons of milk, x_2 pounds of beef and x_3 dozens of eggs.

$\therefore$ Total cost per day is Rs.

$$Z = x_1 + 1.10x_2 + 0.50x_3$$

Total amount of vitamin A in the daily diet is

$$x_1 + x_2 + 10x_3 \text{ mg}$$

which should be at least equal to 1 mg.

$$\therefore \quad x_1 + x_2 + 10x_3 \ge 1$$

Similarly total amount of vitamin B and C in the daily diet

$$100x_1 + 10x_2 + 10x_3 \ge 50$$

and $$10x_1 + 100x_2 + 10x_3 \ge 10$$

Hence the linear programming formulation to this diet problem is as follows :

Find x_1, x_2, x_3 which minimize $Z = x_1 + 1.1x_2 + 0.5x_3$

subject to the constraints

$$x_1 + x_2 + 10x_3 \ge 1$$
$$100x_1 + 10x_2 + 10x_3 \ge 50$$
$$10x_1 + 100x_2 + 10x_3 \ge 10, \text{ and } x_1, x_2, x_3 \ge 0.$$

§ 1·4. Basic Solution (B. S.)

Consider a system $Ax = \mathbf{b}$ of m equations in n unknowns (variables, $n > m$) and $r(A) = r(A\mathbf{b}) = m$. Then none of the equations is redundant.

If any $m \times n$ non-singular matrix (whose determinant is not zero) *is chosen from A and if all the $(n - m)$ variables not associated with the columns of this matrix are set equal to zero, the solution to the resulting system of equations is called* a **basic solution.** In other words, *a solution*

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obtained by setting any $(n-m)$ variables to zero is called a **basic solution** *provided the determinant of the coefficients of the remaining m variables is not zero.* Thus in a basic solution at least $(n-m)$ variables must vanish.

The m variables associated with the columns of the above non-singular matrix which may be different from zero are called **basic variables.**

It is important to note that the matrix formed by the coefficients of m basic variables is non-singular, hence the vectors associated to the basic variables are L.I. *Thus a solution in which the vectors associated to m variables are L.I. and the remaining $(n-m)$ variables are zero, is called a* **basic solution.**

If B is the matrix of m L.I. vectors of A and X_B is the vector of the corresponding variables (basic variables), then the basic solution is given by

$$X_B = B^{-1} b.$$

Since m vectors out of n (Number of columns of coefficient matrix A) can be selected in ${}^nC_m = \dfrac{n!}{m!\,(n-m)!}$ ways.

Hence maximum number of basic solutions is

$${}^nC_m = \frac{n!}{m!\,(n-m)!}.$$

Basic solutions are of two types,

1. Non-degenerate B. S. *A. B.S. is called non-degenerate B.S. if none of the basic variables is zero. In other words all the m basic variables are non-zero.*

2. Degenerate B.S. *A. B.S. is called degenerate B.S. if one or more of the basic variables are zero.*

§ 1·5. An Important Theorem.

A necessary and sufficient condition for the existence and non-degeneracy of all the basic solutions of $Ax = b$ is that every set of m columns of the augmented matrix $Ab = [A, b]$ is L.I.

Proof. Necessary Condition. First we consider that all the basic solution of $Ax = b$ exist and are non-degenerate. Therefore, every set of m-column vectors of A are L.I. Let $\alpha_1, \alpha_2, \ldots, \alpha_m$. be one set of m-column vectors of A, then from the given system, we have

$$\alpha_1 x_1 + \alpha_2 x_2 + \ldots + \alpha_m x_m = b.$$

But $x_1 \neq 0$, since each solution is non-degenerate. Therefore, the vector b can replace α_1 in the basis $\alpha_1, \alpha_2, \ldots, \alpha_m$. (By replacement

theorem of vectors). Thus the vectors $b, \alpha_2, \alpha_3, \ldots, \alpha_m$ also form a basis and hence are L.I. In the similar way $\alpha_1, b, \alpha_3, \ldots, \alpha_m$; $\alpha_1 \alpha_2, b, \alpha_4, \ldots \alpha_m$, etc. are L.I.

Thus the vector b with $(m-1)$ vectors of A form a L.I. set. Hence every set of m columns of the augmented matrix $Ab = [A, b]$ is L.I.

Sufficient Condition. Here we consider that every set of m columns of the augmented matrix $Ab = [A, b]$ is L.I. Obviously every set of m-column vectors of A are L.I. which implies that the basic solution is non-degenerate. Consider $\alpha_1, \alpha_2, \ldots \alpha_m$ be any m-column vectors of A which are L.I. and let $x_1, x_2, \ldots, x_m$ be the corresponding basic solutions.

$$\therefore \quad x_1 \alpha_1 + x_2 \alpha_2 + \ldots + x_m \alpha_m = b$$

$$\therefore \quad -1b + x_1 \alpha_1 + x_2 \alpha_2 + \ldots + x_m \alpha_m = 0$$

Now if $x_1 = 0$, then we have

$$-1.\,b + x_2 \alpha_2 + \ldots + x_m \alpha_m = 0$$

which implies that the vectors $b, \alpha_2, \ldots, \alpha_m$ are L.D. which is a contradiction since we have assumed that every set of m columns (vectors) of the augmented matrix $Ab = [A, b]$ is L.I.

Therefore $\quad x_1 \neq 0.$

Similarly we can prove that none of $x_2, x_3, \ldots, x_m$ are zero.

Hence the basic solution is non-degenerate.

In this way we can show that every basic solution is non-degenerate.

Cor. The necessary and sufficient condition for any given basic solution $X_B = B^{-1} b$ to be non-degenerate is the linear independence of b and every set of $(m-1)$ columns from B.

§ 1·6. Some Important Definitions.

1. A Feasible Solution (A.F.S.). **[Meerut 86, 94, 94(P)]**

(a) A feasible Solution *to a L.P. problem is the set of values of the variables which satisfies the set of constraints and the non-negative restrictions of the problem.*

(b) A feasible Solution *to a system of linear equations is the set of non-negative values of the variables which satisfies the given equations.*

2. Optimum (or optimal) solution. **[Meerut 94(P)]**

A feasible solution to a L.P. problem is said to be **optimum (or optimal) solution** *if it also optimizes the objective function Z of the problem.*

3. A Basic Feasible Solution. (A. B.F.S.)

[Meerut 86, 94, 94(P), 94(R), 95(P), 98]

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In a linear programming problem a feasible solution which is also basic is called a **basic feasible solution (B.F.S).** In other words, *a feasible solution to a L.P.P. in which the vectors associated to non-zero variables* (non zero variables are certainly positive in F.S.) *are L.I., is called a* **basic feasible solution.**

Since atmost m vectors of E^m (Euclidean space of m-dimensions) where m is the number of constraints may be L.I., hence a B.F.S. cannot have more than m non-zero (*i.e.*, positive) variables. Thus for a F.S. to be a B.F.S., at least $(n-m)$ variables must vanish.

B.F.S. are also finite in number and maximum number of them is ${}^nC_m$.

4. Optimal basic feasible solution. **[Meerut 90]**

A basic feasible solution to a L.P.P. is said to be **optimal B.F.S.** *if it also optimizes the objective function Z.*

5. Basic Variables. *The variables in a B.F.S. are called* **basic variables.**

6. Non-degenerate B.F.S. *A. B.F.S. of a L.P. problem is said to be* **non-degenerate B.F.S.** *if none of the basic variables is zero.*

7. Degenerate B.F.S. *A. B.F.S. of L.P. problem is said to be* **non-degenerate B.F.S.** *if at least one of the basic variables is zero.*

Illustrative Examples

Ex. 5. *Find all the basic solutions of the following system :*

$$x_1 + 2x_2 + x_3 = 4$$
$$2x_1 + x_2 + 5x_3 = 5$$

and prove that they are non-degenerate.

[Meerut L.P. 86, 90 (B.A.P. TDC)]

Sol. In matrix form the given system of equations can be written as

$$A\mathbf{x} = \mathbf{b}$$

where $A = (\alpha_1, \alpha_2, \alpha_3)$

$$\alpha_1 = \begin{bmatrix}1\\2\end{bmatrix}, \alpha_2 = \begin{bmatrix}2\\1\end{bmatrix}, \alpha_3 = \begin{bmatrix}1\\5\end{bmatrix}, \mathbf{x} = \begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}, \mathbf{b} = \begin{bmatrix}4\\5\end{bmatrix}$$

This problem can have at most ${}^3C_2 = 3$ basic solutions.

Now the three sets of two vectors, are

$$B_1 = [\alpha_1, \alpha_2] = \begin{bmatrix}1 & 2\\2 & 1\end{bmatrix}, B_2 = [\alpha_1, \alpha_3] = \begin{bmatrix}1 & 1\\2 & 5\end{bmatrix}$$

$$B_3 = [\alpha_2, \alpha_3] = \begin{bmatrix}2 & 1\\1 & 5\end{bmatrix}.$$

Here $\quad |B_1| = -3 \neq 0, \; |B_2| = 3 \neq 0, \; |B_3| = 9 \neq 0.$

Since none of these is zero, therefore, every set of two vectors of A are L.I. Hence all the three basic solutions exist.

If $X_{Bi}, i = 1, 2, 3$ are the vectors of the basic variables associated to the sets $B_i, i = 1, 2, 3$, respectively, then

$$\begin{bmatrix}x_1\\x_2\end{bmatrix} = X_{B_1} = B_1^{-1}\mathbf{b} = -\frac{1}{3}\begin{bmatrix}1 & -2\\-2 & 1\end{bmatrix}\begin{bmatrix}4\\5\end{bmatrix} = \begin{bmatrix}2\\1\end{bmatrix}$$

$$\begin{bmatrix}x_1\\x_3\end{bmatrix} = X_{B_2} = B_2^{-1}\mathbf{b} = \frac{1}{3}\begin{bmatrix}5 & -1\\-2 & 1\end{bmatrix}\begin{bmatrix}4\\5\end{bmatrix} = \begin{bmatrix}5\\-1\end{bmatrix}$$

$$\begin{bmatrix}x_2\\x_3\end{bmatrix} = X_{B_3} = B_3^{-1}\mathbf{b} = \frac{1}{9}\begin{bmatrix}5 & -1\\-1 & 2\end{bmatrix}\begin{bmatrix}4\\5\end{bmatrix}\begin{bmatrix}5/3\\2/3\end{bmatrix}$$

Hence the basic solutions are

$$x_1 = (2, 1, 0), x_2 = (5, 0, -1), x_3 = (0, 5/3, 2/3).$$

Now the determinants of matrices $[\alpha_1, \mathbf{b}]$ and $[\alpha_2, \mathbf{b}]$ are not zero, *i.e.*, $\mathbf{b}$ and every set of $m - 1 = 2 - 1 = 1$ column of B_1 are L.I. Hence by theorem 1·5 page 18 the basic solution x_1 is non-degenerate. Similarly, we can prove that the other basis solutions x_2 and x_3 are also non-degenerate.

Ex. 6. *If* $a_1 = \begin{bmatrix}3\\4\end{bmatrix}, a_2 = \begin{bmatrix}-1\\2\end{bmatrix}, a_3 = \begin{bmatrix}1\\4\end{bmatrix}$ *and* $\mathbf{b} = \begin{bmatrix}1\\4\end{bmatrix}$ *then determine whether all possible basic solutions exist for the following set of equations :*

$$[a_1, a_2, a_3]\,\mathbf{x} = \mathbf{b}.$$

Sol. Here the problem can have at most ${}^3C_2 = 3$ basic solutions.

Now three sets of two vectors are

$$B_1 = [a_1, a_2] = \begin{bmatrix}3 & -1\\4 & 2\end{bmatrix}, B_2 = [a_1, a_3] = \begin{bmatrix}3 & 1\\4 & 4\end{bmatrix}$$

and $$B_3 = [a_2, a_3] = \begin{bmatrix}-1 & 1\\2 & 4\end{bmatrix}.$$

Here $|B_1| = 10 \neq 0, \; |B_2| = 8 \neq 0$ and $|B_3| = -6 \neq 0.$

Since none of these is zero, therefore every set of two vectors of the matrix (a_1, a_2, a_3) are L.I. Hence all the three basic solutions exist.

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Ex. 7. *Find an optimal solution of the following linear programming problem without using the simplex algorithm.*

Max. $Z = 2x_1 + 3x_2 + 4x_3 + 7x_4$

s.t. $2x_1 + 3x_2 - x_3 + 4x_4 = 8$

$x_1 - 2x_2 + 6x_3 - 7x_4 = -3,$

$x_i \geq 0, i = 1, 2, 3, 4.$ **[Meerut 88, 90 (TDC), 94, 95(R)]**

Sol. We know that at least one extreme point will give the maximum value and the extreme point correspond to B.F. solution. Thus we proceed to find all the B.F. solutions.

In matrix form the given L.P.P. can be written as

$$Ax = b$$

where $A = (\alpha_1, \alpha_2, \alpha_3, \alpha_4)$

$$\alpha_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 3 \\ -2 \end{bmatrix}, \alpha_3 = \begin{bmatrix} -1 \\ 6 \end{bmatrix}, \alpha_4 = \begin{bmatrix} 4 \\ -7 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$b = \begin{bmatrix} 8 \\ -3 \end{bmatrix}.$$

This problem can have at most ${}^4C_2 = 6$ basic solutions. Now the sets of two vectors are

$$B_1 = [\alpha_1, \alpha_2] = \begin{bmatrix} 2 & 3 \\ 1 & -2 \end{bmatrix}, B_2 = [\alpha_1, \alpha_3] = \begin{bmatrix} 2 & -1 \\ 1 & 6 \end{bmatrix},$$

$$B_3 = [\alpha_1, \alpha_4] = \begin{bmatrix} 2 & 4 \\ 1 & -7 \end{bmatrix}, B_4 = [\alpha_2, \alpha_3] = \begin{bmatrix} 3 & -1 \\ -2 & 6 \end{bmatrix},$$

$$B_5 = [\alpha_2, \alpha_4] = \begin{bmatrix} 3 & 4 \\ -2 & -7 \end{bmatrix}, B_6 = [\alpha_3, \alpha_4] \begin{bmatrix} -1 & 4 \\ 6 & -7 \end{bmatrix}.$$

Here $|B_1| = -7, |B_2| = 13, |B_3| = -18, |B_4| = 16,$
$|B_5| = -13, |B_6| = -17.$

Since none of these is zero, therefore all these sets are L.I. Hence all the six basic solutions exist.

If $X_{B_i}, i = 1, 2, \ldots, 6$ are the vectors of the basic variables associated to the sets $B_i, i = 1, 2, \ldots, 6$ respectively, then

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = X_{B_1} = B_1^{-1} b = -\frac{1}{7} \begin{bmatrix} -2 & -3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 8 \\ -3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = X_{B_2} = B_2^{-1} b = \frac{1}{13} \begin{bmatrix} 6 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 8 \\ -3 \end{bmatrix} = \begin{bmatrix} 45/13 \\ -14/13 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_4 \end{bmatrix} = X_{B_3} = B_3^{-1} b = -\frac{1}{18} \begin{bmatrix} -7 & -4 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 8 \\ -3 \end{bmatrix} = \begin{bmatrix} 22/9 \\ 7/9 \end{bmatrix}$$

$$\begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = X_{B_4} = B_4^{-1} b = \frac{1}{16} \begin{bmatrix} 6 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 8 \\ -3 \end{bmatrix} = \begin{bmatrix} 45/16 \\ 7/16 \end{bmatrix}$$

$$\begin{bmatrix} x_2 \\ x_4 \end{bmatrix} = X_{B_5} = B_5^{-1} b = -\frac{1}{13} \begin{bmatrix} -7 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 8 \\ -3 \end{bmatrix} = \begin{bmatrix} 68/13 \\ -7/13 \end{bmatrix}$$

$$\begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = X_{B_6} = B_6^{-1} b = -\frac{1}{17} \begin{bmatrix} -7 & -4 \\ -6 & -1 \end{bmatrix} \begin{bmatrix} 8 \\ -3 \end{bmatrix} = \begin{bmatrix} 44/17 \\ 45/17 \end{bmatrix}$$

Since X_{B_2}, X_{B_5} do not satisfy the non-negative restriction of the variables and so these do not give the B.F. solutions.

Thus the B.F. solutions of the given L.P.P. are given by

$x_1 = (1, 2, 0, 0), x_2 = (22/9, 0, 0, 7/9)$ $x_3 = (0, 45/16, 7/16, 0),$
$x_4 = (0, 0, 44/17, 45/17).$

For these B.F. solutions the values of the objective function Z are

$$Z_1 = 2 \cdot 1 + 3 \cdot 1 + 4 \cdot 0 + 7 \cdot 0 = 8, Z_2 = 2 \cdot \frac{22}{9} + 3 \cdot 0 + 4 \cdot 0 + 7 \cdot \frac{7}{9} = \frac{31}{3},$$

$$Z_3 = 22 \cdot 0 + 3 \cdot 45/16 + 4 \cdot 7/16 + 7 \cdot 0 = 163/16,$$

and $Z_4 = 2 \cdot 0 + 3 \cdot 0 + 4 \cdot 44/17 + 7 \cdot 45/17 = 491/17.$

Since Max. $Z = Z_4 = 491/17.$

Hence the optimal solution to the given L.P.P. is given by
$x_1 = 0, x_2 = 0, x_3 = 44/17, x_4 = 45/17.$

§ 1·7. Solution of a linear programming problem.

In general we use the following two methods for the solution of a linear programming problem.

1. Geometrical (or graphical) method. If the objective function Z is a function of two variables only then the problem can be solved by graphical method. A problem of three variables, can be solved by this method, but it is complicated enough.

2. Simplex method. This is the most powerful tool of the linear programming as any problem can be solved by this method. This method is an algebraic procedure which progressively approaches the optimal

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Step three. The shaded region shown in fig. 1·1 is the permissible region for the values of the variables x_1 and x_2.

Step four. To find x_1, x_2 for which Z is maximum, we draw the line (dotted line through O) $Z = 8x_1 + 7x_2 = 0$ *i.e.* $\frac{x_1}{x_2} = \frac{-8}{7}$ and continue drawing parallel lines till we reach the point of the permissible region which is farthest away from the origin. As shown in fig. 1·1 this point is at point $P(10500, 34500)$ which is the point of intersection of lines

$$3x_1 + x_2 = 66{,}000$$

and
$$x_1 + x_2 = 45{,}000$$

$\therefore$ Z is maximum for $x_1 = 10,500$, $x_2 = 34{,}500$

and maximum $Z = 8 \times (10{,}500) + 7 \times (34{,}500) = 3{,}25{,}5{,}00.$

Ex. 9. *Solve the following linear programming problem ;*

$$\text{Max.} \quad Z = 6x_1 + 11x_2$$

$$\text{s.t.} \quad 2x_1 + x_2 \le 104$$

$$x_1 + 2x_2 \le 76$$

and
$$x_1 \ge 0,\ x_2 \ge 0.$$

Sol. Step one. First consider the constraints as equalities.

$$2x_1 + x_2 = 104$$

$$x_1 + 2x_2 = 76.$$

Step two. Here draw lines in two dim. plane.

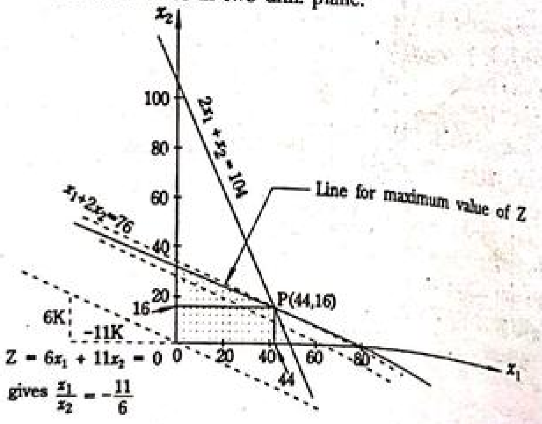

Fig. 1·2

Step three. The shaded region in the fig. 1·2 is the permissible region for the values of the variables x_1 and x_2.

Step four. As in Ex. 8, we see that max. Z is obtained at the point $P(44, 16)$ which is the point of intersection of lines

$$2x_1 + x_2 = 104$$

and
$$x_1 + 2x_2 = 6.$$

$\therefore$ for Max. Z, $x_1 = 44$, $x_2 = 16$

and Max. $Z = 6 \times 44 + 11 \times 16 = 440.$

Ex. 10. *Solve the following programming problem by graphical method ;*

$$\text{Max.} \quad Z = 5x_1 + 7x_2,$$

$$\text{s.t.} \quad x_1 + x_2 \le 4$$

$$3x_1 + 8x_2 \le 24$$

$$10x_1 + 7x_2 \le 35$$

and $x_1, x_2 \ge 0.$ **[Meerut 86, 87, 89, 90]**

Sol. Step one. First consider the constraints as equalities.

$$x_1 + x_2 = 4$$

$$3x_1 + 8x_2 = 24$$

$$10x_1 + 7x_2 = 35.$$

Step two. Here draw lines in two dim. plane.

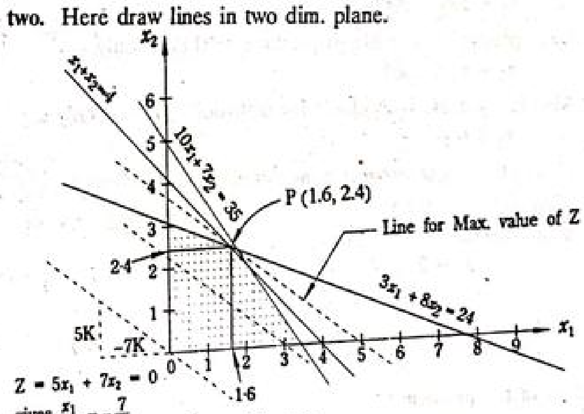

Fig 1·3

Step three. The shaded region in fig. 1·3 is the permissible region for values of the variables x_1 and x_2.

Step four. From the fig. 1·3, it can be seen that Z is Max. at the point (1.6, 2.4) which is the point of intersection of the lines,

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$$x_1 + x_2 = 4$$

and $\quad 3x_1 + 8x_2 = 24.$

$\therefore$ for Max. Z, $\quad x_1 = 1.6,\ x_2 = 2.4$

and $\quad$ Max. $Z = 5 \times (1.6) + 7 \times (2.4) = 24.8.$

Ex. 11. *A toy company manufactures two types of doll; a basic version-doll A and a deluxe version-doll B. Each doll of type B takes twice as long to produce as one of type A, and the company would have time to make a maximum 2,000 per day if it produces only the basic version. The supply of plastic is sufficient to produce 1,500 dolls per day (both A and B combined). The deluxe version requires a fancy dress of which there are only 600 per day available. If the company makes profit of Rs. 3.00 and 5.00 per doll respectively, on doll A and B ; how many of each should be produced per day in order to maximize profit ?*

[Meerut 90 (B.A.P. TDC)]

Sol. Formulation of the problem.

Let x_1 dolls of type A and x_2 dolls of type B be produced per day.

$\therefore$ Total profit $Z = 3x_1 + 5x_2$.

Total time per day consumed to prepare x_1 and x_2 dolls of type A and B is $x_1 + 2x_2$ which should be less than 2000.

$\therefore \quad x_1 + 2x_2 \le 2000.$

Since plastic is available to produce 1500 dolls only,

$\therefore \quad x_1 + x_2 \le 1500.$

Also fancy dress is available for 600 dolls per day only.

$\therefore \quad x_2 \le 600.$

Hence the linear programming problem is as follows :

Max. $Z = 3x_1 + 5x_2$

subject to the constraints

$$x_1 + 2x_2 \le 2000$$
$$x_1 + x_2 \le 1500$$
$$x_2 \le 600$$

and $\quad x_1 \ge 0, x_2 \ge 0.$

Solution of the problem :

Step one. First consider the constraints as equalities.

$$x_1 + 2x_2 = 2000$$
$$x_1 + x_2 = 1500$$
$$x_2 = 600.$$

Step two. Her we draw lines in two dim. plane.

Step three. The shaded region in the fig. 1·4 is the permissible region

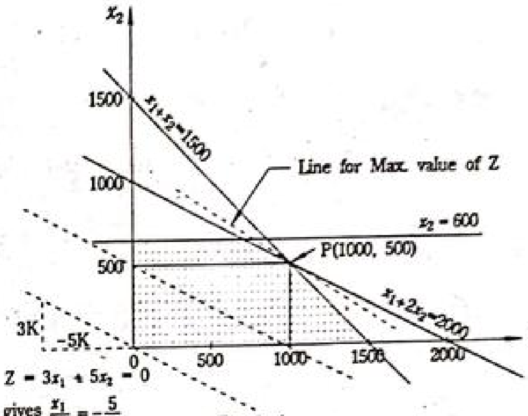

Fig 1·4

for the values of the variables x_1 and x_2.

Step four. From the fig. 1·4 it can be seen than Z is max. at the point P (1000, 500) which is the point of intersection of the lines

$$x_1 + 2x_2 = 2000$$

and $\quad x_1 + x_2 = 1500.$

$\therefore$ for Max. Z, $\quad x_1 = 1000,\ x_2 = 500.$

and $\quad$ Max. Z = Rs. $[3 \times (1000) + 5 \times (500)]$ = Rs. 5500.

Ex. 12. *Solve graphically the following linear programming problem.*

$$\text{Min. } Z = 3x_1 + 5x_2$$
$$\text{s.t.} \quad -3x_1 + 4x_2 \le 12$$
$$2x_1 - x_2 \ge -2$$
$$2x_1 + 3x_2 \ge 12$$
$$x_1 \le 4,\ x_2 \ge 2,$$
$$x_1, x_2 \ge 0.$$

[Meerut 88]

Sol. Step one. First consider the constraints as equalities.

$$-3x_1 + 4x_2 = 12$$
$$2x_1 - x_2 = -2$$
$$2x_1 + 3x_2 = 12$$
$$x_1 = 4,\ x_2 = 2.$$

Step two. Here draw lines in two dimensional plane.

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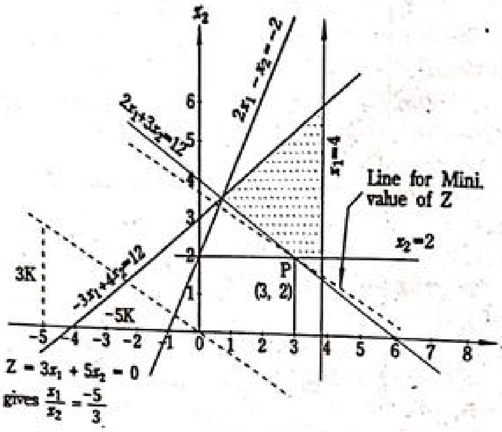

Fig. 1·5

Step three. The shaded region in fig. 1·5 is the permissible region for values of x_1 and x_2.

Step four. $Z = 3x_1 + 5x_2 = 0$ gives $\frac{x_1}{x_2} = \frac{-5}{3}$.

Draw this line through O (dotted line) and continue drawing lines parallel to it till we reach the point P of the permissible region which is nearest to the region.

$\therefore$ Z is min. at $P(3, 2)$ which is the point of intersection of lines $x_2 = 2$ and $2x_1 + 3x_2 = 12$.

$\therefore$ Z is Min. when $x_1 = 3, x_2 = 2$

and mini. $Z = 3 \times 3 + 5 \times 2 = 19$

Note. In this problem we need minimum value of Z, therefore, the point P of the permissible region which is nearest to O, give min. value of Z.

Ex. 13. *Solve the following linear programming problem*

Max. $Z = 0.75x_1 + x_2$

s.t. $x_1 - x_2 \geq 0$

$-0.5x_1 + x_2 \leq 1$

$x_1, x_2 \geq 0.$

Sol.

Step one. First consider the constraints as equalities.

$$x_1 - x_2 = 0$$
$$-0.5x_1 + x_2 = 1$$

Step two. Now draw the lines to each of the above equations in two dimensional plane.

Step three. The shaded region in the fig. 1.6 is the permissible region for the values of the variables x_1 and x_2.

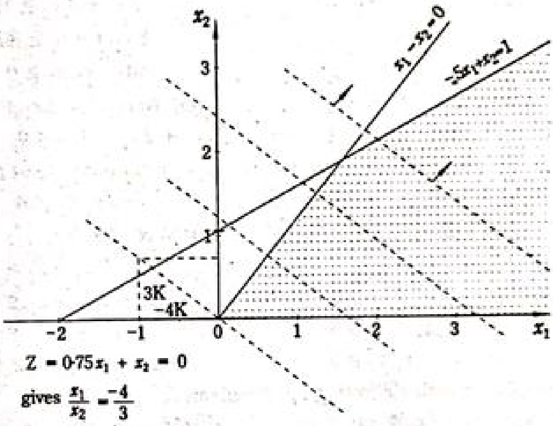

Fig. 1·6

Step four. Here the line of the objective function Z (dotted line) can be moved indefinitely, still containing a point of permissible region. Therefore, there is no finite minimum value of Z.

In such cases when the value of the objective function Z can be increased indefinitely, the problem is said to have an *unbounded solution.*

Note. Here it is not correct to say that the problem has no solution. The problem has a solution which is unbounded.

Exercise On Chapter I

1. Solve graphically the following L.P. Problems :

(i) Max. $Z = 2x_1 + 3x_2$
s.t. $x_1 + x_2 \leq 1$
$3x_1 + x_2 \leq 4$
and $x_1, x_2 \geq 0.$

(ii) Max. $Z = 6x_1 - 2x_2$
s.t. $2x_1 - x_2 \leq 2$
$x_1 \leq 3$ (x_1)
and $x_1, x_2 \geq 0$

(iii) Max. $Z = 3x_1 + 2x_2$
s.t. $2x_1 - x_2 \geq 2$

(iv) Max. $Z = 3x_1 + 2x_2$
s.t. $-2x_1 + x_2 \leq 1$

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$x_1 + 2x_2 \le 8$

and $x_1, x_2 \ge 0.$

(v) Max. $Z = 5x_1 + 3x_2$

s.t. $3x_1 + 5x_2 \le 15$

$5x_1 + 2x_2 \le 10$

and $x_1, x_2 \ge 0$

(vii) Max. $Z = 3x_1 + 4x_2$

s.t. $x_1 - x_2 \le -1$

$-x_1 + x_2 \le 0$

and $x_1, x_2 \ge 0$

(ix) Max. $Z = x_1 + x_2$

s.t. $-2x_1 + x_2 \le 1$

$x_1 \le 2$

$x_1 + x_2 \le 3$

and $x_1, x_2 \ge 0$

$x_1 \le 1$

$x_1 + x_2 \le 1$

and $x_1, x_2 \ge 0.$

(vi) Max. $Z = 3x_1 + 4x_2$

s.t. $5x_1 + 4x_2 \le 200$

$3x_1 + 5x_2 \le 150$

$5x_1 + 4x_2 \ge 100$

$8x_1 + 4x_2 \ge 80$

and $x_1, x_2 \ge 0.$

(viii) Max. $Z = 3x_1 + 2x_2$

s.t. $-2x_1 + 3x_2 \le 9$

$x_1 - 5x_2 \ge -20$

and $x_1, x_2 \ge 0$

(x) Max. $Z = 4x_1 + 5x_2$

s.t. $x_1 + x_2 \ge 1$

$-2x_1 + x_2 \le 1$

$4x_1 - 2x_2 \le 1$

and $x_1, x_2 \ge 0$

2. Solve graphically the following L.P. Problems.

(i) Min. $Z = 4x_1 + 2x_2$

s.t. $x_1 + 2x_2 \ge 2$

$3x_1 + x_2 \ge 3$

$4x_1 + 3x_2 \le 6$

and $x_1, x_2 \ge 0.$

(ii) Min. $Z = -x_1 + 2x_2$

s.t. $-x_1 + 3x_2 \le 10$

$x_1 + x_2 \le 6$

$x_1 - x_2 \le 2$

and $x_1, x_2 \ge 0$

(iii) Mini. $Z = 20x_1 + 10x_2$

s.t. $x_1 + 2x_2 \le 40$

$3x_1 + x_2 \ge 30$

$4x_1 + 3x_2 \ge 60$

and $x_1, x_2 \ge 0.$

(iv) Mini $Z = 2x_1 + 3x_2$

s.t. $x_1 + x_2 \le 4$

$6x_1 + 2x_2 \ge 8$

$x_1 + 5x_2 \ge 4$

$x_1 \le 3$

$x_2 \le 3$

and $x_1, x_2 \ge 0$

(v) Min. $Z = 2x_1 + 3x_2$

s.t. $-x_1 + 2x_2 \le 4$

$x_1 + x_2 \le 6$

$x_1 + 3x_2 \ge 9$

and $x_1, x_2 \ge 0$

(vi) Min. $Z = x_1 + x_2$

s.t. $5x_1 + 10x_2 \le 50$

$x_1 + x_2 \ge 1$

$x_2 \le 4$

and $x_1, x_2 \ge 0$

3. Determine $x \ge 0$ and $y \ge 0$, so as to maximize $Z = 2x + 3y$ subject to the constraints :

$x + y \le 30,\ y \le 12,\ x \le 20,\ y \ge 3,\ x - y \ge 0.$

4. Does the following L.P.P. has a feasible soultion ?

Maximize $Z = x_1 + x_2,$

Subject to $x_1 - x_2 \ge 0,\ 3x_1 - x_2 \le -3$

$x_1, x_2 \ge 0.$

Show with the help of a graph.

5. The postmaster of a local post office wishes to hire extra helpers during the Deepawali season, becuase of a large increase in the volume of mail handling and delivery. Because of the limited office space and the budgetary condition, the number of temporary helpers must not exceed 10. According to the past experience, men can handle 300 letters and 80 packages per day, on the average, and women can handle 400 letters and 50 packages per day. The post master believes that the daily volume of extra mail and packages will be no less than 3,400 and 680 respectively. A man receives Rs. 25 a day and a woman receives Rs. 22 a day. How many men and women helpers should be hired to keep the pay roll at a minimum ?

[Meerut 94]

6. A soft drink plant has two bottling machines A and B. It produces and sells 8 ounce and 16 ounce bottles. The following data is available :

Machine	8 Ounce	16 Ounce
A	100/minute	40/minute
B	60/minute	75/minute

The machines can be run 8 hrs. per day, 5 days per week. Weekly production of the drinks cannot exceed 3,00,000 ounces and the market can absorb 25,000 eight ounce bottles and 7,000 sixteen ounce bottles per week. Profit on these bottles is 15 paise and 25 paise per bottle respectively. The planner wishes to maximize his profit subject to all the production and marketing restrictions. Formulate it as a linear progrmming problem and solve graphically.

[Meerut 94(R), 98]

7. The ABC Electric Appliance Company produce two products; refrigerators and coolers. Production takes place in two separate departments. Refrigerators are produced in Department I and coolers produced in Department II. The company's two products are produced and sold on a weekly basis. The weekly production

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cannot exceed 25 refrigerators in Department I and 35 coolers in Department II, because of the limited available facilities in the two departments. The company regularly employs a total of workers in the two departments. A refrigerator requires 2 man-we of labour, while a cooler requires 1 man-week of labour. A refrigerator contributes a profit of Rs. 60 and a cooler contribu a profit of Rs. 40. How many units of refrigerators and cool should the company produce to realize maximum profit ?

8. A farmer has a 100 acre farm. He can sell the tomatoes, lettuce, radishes he can raise. The price he can obtain is Rs. 1 per kilogr for tomatoes, Rs. 0.75 per head for lettuce and Rs. 2 per kilogr for radishes. The average yield per acre is 2,000 kilogram tomatoes, 3,000 heads of lettuce and 1,000 kilograms of radish Fertilizer is available at Rs. 0.50 per kilogram and the amou required per acre is 100 kilograms each for tomatoes and lettu and 50 kilograms for radishes. Labour required for sowin cultivation and harvesting per acre is 5 man/day for tomatoes a radishes and 6 man/ day for lettuce. A total of 400 men/day of labo are available at Rs. 20 per man/day.

Formulate this problem as a L.P.P. to maximize the farme total profit. [Meerut 95 (BP

9. A firm manufactures two type of products A and B and sells the at a profit of Rs. 2.00 on type A and Rs. 3.00 on type B. Each produ is processed on two machines M_1 and M_2. Type A requires on minute of processing time on M_1 and two minutes on M_2; type requires one minute on M_1 and one minute on M_2. The machin M_1 is available for not more than 6 hours 40 minutes while machi M_2 is available for 10 hours during any working day.

Formulate the problem as L.P.P. and find how many produ of each type should the firm produce each day in order to g maximum profit.

10. Old hens can be bought for Rs. 2.00 each but young ones cost Rs 5.00 each. The old hens lay 3 eggs per week and the young ones eggs per week. Each egg being worth 30 paise. A hen cost Rs. 1.00 per week to feed. If I have only Rs. 80.00 to spend for hens how many of each kind should I buy to give a profit of more tha Rs. 6.00 per week assuming that I cannot house more than 20 hen Solve the L.P.P. graphically.

[Bhopal (State) 86; Meerut 90 (TDC), 94 (P), 9

11. An automobile manufacture makes automobiles and trucks in a factory that is divided into two shops. Shop A, which performs the basic assembly operation, must work 5 man-days on each truck but only 2 man-days on each automobile. Shop B, which performs finishing operation, must work 3 man-days for each automobile or truck that it produces. Because of men and machine limitations, shop A has 180 man-days per week available while shop B has 135 man-days per week. If the manufacturer makes a profit of Rs. 300 on each truck and Rs. 200 on each automobile, how many of each should he produce to maximize his profit ?

12. A manufacturer produces two types of models A and B. Each A model requires 4 hours of grinding and 2 hours of polishing whereas each B model requires 2 hours of grinding and 5 hours of polishing. The manufacturer has 2 grinders and 3 polishers. Each grinder works for 40 hours a week and each polisher works for 60 hours a week. Profit on an A model is Rs. 3.00 and on and B model is Rs. 4.00. Whatever is produced in a week is sold in the market. How should the manufacturer allocate his production capacity to the two types of models so that he may make the maximum profit in a week.

13. Solve Q. No. 12 when the profit on an A model is Rs. 30 and on a B model is Rs. 40.

14. A firm manufactures 3 products A, B and C. The profits are Rs. 3, Rs. 2 and Rs. 4 respectively. The firm has 2 machines and below is the required processing times in minutes for each machine on each product.

		Products		
		A	B	C
Machine	M_1	4	3	5
	M_2	2	2	4

Machines M_1 and M_2 have 2,000 and 2,500 machines minutes respectively. The firm must manufacture 100 A's, 200 B's and 50 C's but no more than 150 A's.

Set up an L.P.P. to maximize profit. Do not solve it.

15. A factory uses three different resources for the manufacture of two different products, 20 units of the resource A, 12 units of B and 16 units of C being available. 1 unit of the first product requires 2, 2

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and 4 units of the respective resources and 1 unit of the product requires 4, 2 and 0 units of the respective resou known that the first product gives a profit of 2 monetary unit and the second 3.

Formulate the linear programming problem. How many of each product should be manufactured for maximizing the Solve it graphically.

16. A pineapple firm produces two products - canned pineapp canned juice. The specific amounts of material, labou equipment required to produce each product and the availab each of these resources are shown in the table given below:

	Canned juice	Canned Pineapple	Availa Resour
Labour (man-hour)	3	2.0	12.0
Equipment (machine hours)	1	2.3	6.9
Material (unit)	1	1.4	4.9

Assuming one unit each of canned juice and canned pine has profit margins of Rs. 2 and Rs. 1 respectively. Determin product mix that will maximize the profit.

17. Diet Problem. Consider two different types of foodstuffs, s and F_2. Assume that these foodstuffs contain vitamins V_1, V_2 V_3 respectively. Minimum daily requirement of three vitamin 1 mg. of V_1, 50 mg. of V_2 and 10 mg. of V_3. Suppose that the stuff F_1 contains 1 mg. of V_1, 100 mg of V_2 and 10 mg. of V_3 wh the food stuff F_2 contains 1 mg. of V_1, 10 mg. of V_2 100 mg. of V_3. Cost of one unit of foodstuff F_1 is Rs. 1 that of F_2 is Rs. 1.5.

Find the minimum cost diet that would supply the body at the minimum requirements of each vitamin by graphical meth

[Meerut 9

18. A company sells two different product A and B. The comp makes a profit of Rs. 40 and Rs. 30 per unit on products A and respectively. The products are produced in a common prod process and are sold in two different markets. The prod process has a capacity of 30,000 man-hours. It takes 3 hour produce one unit of A and one hour to produce one unit of B.

market has been surveyed, and company officials feel that the maximum number of units of A that can be sold is 8,000 and the maximum of B is 12,000 units. Subject to these limitations, the products can be sold in any convex combination.

Formulate the above problem as a L.P.P. and solve it by graphical method.

19. Show that the feasible solution $x_1 = 1, x_2 = 0, x_3 = 1$ and $Z = 6$ to the system of equations

$$x_1 + x_2 + x_3 = 2$$
$$x_1 - x_2 + x_3 = 2,\ x_j \geq 0$$

which minimize $Z = 2x_1 + 3x_2 + 4x_3$ is not basic.

20. Find all the B.F.S. of following system

$$8x_1 + 6x_2 + 13x_3 + x_4 + x_5 = 6$$
$$9x_1 + x_2 + 2x_3 + 6x_4 + 10x_5 = 10.$$

21. Mark the feasible region represented by the constraint equations $x_1 + x_2 \leq 1,\quad 3x_1 + x_2 \geq 3,\quad x_1 \geq 0,\ x_2 \geq 0$ of a linear optimizing function $Z = x_1 + x_2$.

[C.A. Int. Exam May. 85]

22. Is $x_1 = 1, x_2 = \frac{1}{2}, x_3 = x_4 = x_5 = 0$ a basic solution to the following equations

$x_1 + 2x_2 + x_3 + x_4 = 2,\ x_1 + 2x_2 + \frac{1}{2}x_3 + x_5 = 2.$ **[Meerut 90]**

[**Hint :** $x_1 = 1, x_2 = \frac{1}{2}, x_3 = x_4 = x_5 = 0$ satisfy the equations and the values of $5 - 2 = 3$, variables are zero.

The determinant of column vectors corresponding to variables x_1, x_2 is

$$|\alpha_1 \alpha_2| = \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} = 2 - 2 = 0.$$

$\therefore$ the vectors α_1, α_2 are L.D.

Hence the solution is not basic solution].

Answers

1. (i) $x_1 = 0, x_2 = 1, Z = 3$ (ii) $x_1 = 3, x_2 = 4, Z = 10$

(iii) $x_1 = 8, x_2 = 0, Z = 24$ (iv) $x_1 = 2, x_2 = 1, Z = 8.$

(v) $x_1 = 1.053,\ x_2 = 2.368,\ Z = 12.37$

(vi) $x_1 = 30.7, x_2 = 11.5, Z = 138.1$

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(vii) No solution (viii) No solution

(ix) Infinite number of solutions (x) Unbounded solution

2. (i) $x_1 = 0.6, x_2 = 1.2, Z = 4.8$

(ii) $x_1 = 2, x_2 = 0, Z = -2$

(iii) $x_1 = 6, x_2 = 12, Z = 240$

(iv) $x_1 = 8/7, x_2 = 4/7, Z = 4$

(v) $x_1 = 1.2, x_2 = 2.6, Z = 10.2$

(vi) Infinite number of solutions.

3. $x = 18, y = 12, z = 72.4$. No. **5.** 6 Men and 4 women.

6. If x_1 and x_2 bottles of 8 and 16 ounces respectively, are produced then L.P.P. is Max. $Z = 0.15x_1 + 0.25x_2$

s.t. $8x_1 + 16x_2 \le 3{,}00{,}000$,

$2x_1 + 5x_2 \le 4{,}80{,}000$, $5x_1 + 4x_2 \le 7{,}20{,}000$,

$x_1 \le 25{,}000$, $x_2 \le 7{,}000$ and $x_1 \ge 0, x_2 \ge 0$,

$x_1 = 16{,}80{,}000/17$, $x_2 = 9{,}60{,}000/17$, Max. Z = Rs. 4,92,000/17

7. Refrigerators = 12.5, Coolers = 35, Max. profit = Rs. 2150.00.

8. If the farmer sow tomatoes, lettuce and radishes, in x_1, x_2, x_3 acres respectively, then L.P.P. is

Max. $Z = 1850x_1 + 2080x_2 + 1875x_3$

Note :- this is not a ~~the integer~~ Whole Numbers.

s.t. $x_1 + x_2 + x_3 \le 100, 5x_1 + 6x_2 + 5x_3 \le 400$

and $x_1, x_2, x_3 \ge 0$.

9. L.P.P. Max. $Z = 2A + 3B$; s.t. $A + B \le 400$, $2A + B \le 600$.

and $A, B \ge 0$,

$A = 0, B = 400, Z = 1200$.

10. Old hens = 0, Young hens = 16, Profit Z = Rs. 8.

11. Trucks = 30, Automobiles = 15 per week.

12. Model A = 2.5, Model B = 35, Profit Rs. 147.5.

13. Model A = 2.5, Model B = 35, Profit = Rs. 1475.

14. Max. $Z = 3x_1 + 2x_2 + 4x_3$.

Products A, B, C: x_1, x_2, x_3 respectively.

s.t. $4x_1 + 3x_2 + 5x_3 \le 2{,}000 : 2x_1 + 2x_2 + 4x_3 \le 2{,}500$;

$100 \le A \le 150$; $0 \le B \le 200$ and $0 \le C \le 50$.

15. First product = 2 units, second product = 4 units, profit = 16 monetary units.

16. Canned juice = 4, Pineapple juice = 0, Profit = Rs. 8.

17. $F_1 = 1$ unit, $F_2 = 0$ unit. Cost = Rs. 1.

18. If x_1 and x_2 are the units of products A and B respectively, L.P.P. is

Max. $Z = 40x_1 + 30x_2$; s.t. $3x_1 + x_2 \le 30{,}000$;

$x_1 \le 8{,}000$; $x_2 \le 12{,}000$; and $x_1, x_2 \ge 0$.

$x_1 = 6{,}000, x_2 = 12{,}000$.

20. (2/3, 0, 0, 2/3, 0), (50/71, 0, 0, 0, 26/71), (0, 26/35, 0, 54/35, 0), (0, 50/59, 0, 0, 54/59), (0, 0, 13/38, 59/38, 0), (0, 0, 25/64, 0, 59/64).

21. No. **22.** No.

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3

Simplex Method

§ 3·1 **Introduction.** It is either impossible or require great labour to search for optimal solution from amongst all the feasible solutions which may be infinite in number. The fundamental theorem of linear programming which states that "**if the given linear programming problem has an optimal solution, then at least one basic feasible solution must be optimal**" forms a firm base for the solution of L P. problem. Acccording to this theorem we can search the optimal solution among the basis feasible solutions only which are finite in number. Also it is easy to find an optimal among the basic feasibles then to find that among all the feasible solutions which may be infinite in number. In this way a L.P. problem can be solved be enumerating all the B.F. solutions. But it is also not an easy job to enumerate all the B.F. solutions even for small values of m (number of constraints) and n (no. of variables). To over come this difficulty a method known as **Simplex Method** (or **simplex Algorithm**) was developed by **George Dantizg** in 1947 which was made available in 1951. This method is an iterative (step by step) procedure in which we proceed in systematic steps from an intial B.F. solution to other B.F. solutions and finally in a finite number of steps, to an optimal B.F. solution, in such a way that the value of the objective function at each step (iteration) is better (or at least not worse) than at the preceding step. In other words the simplex algorithm consist of the following main steps,

(i) Finding a trial B.F.S. of the L.P.P.

(ii) Testing whether it is an optimal solution or not.

(iii) Improving the first trial B.F.S, (if it is not optimal) by a set of rules.

(iv) Repeating the steps (ii) and (iii) till we get an optimal solution.

This method also indicates whether there is bounded solution of the given L.P.P.

§ 3·2 **Slack and Surplus Variables.**

[Meerut 90 TDC, 90 (B.A. P TDC) 94 (R), 98 (OLD)]

1. **Slack Variables.**

If a constraint has a sign $\leq$, then in order to make it an equality we have to add some thing positive to the left hand side.

The positive variables which are added to L.H. sides to the constraints to convert them into equalities are called the slack variables.

For example, consider the liner programming problem

Max. $Z = 2x_1 + 3x_2 + 4x_3$

s.t. $\left.\begin{array}{l} x_1 + x_2 + x_3 \le b_1 \\ 2x_1 + 4x_2 - x_3 \le b_2 \end{array}\right\}$ ...(1)

$x_1, x_2, x_3 \ge 0.$

In order to convert constraints (1) into equalities (equations) we add x_4 and x_5 on L.H.S. of (1), then we have

$$x_1 + x_2 + x_3 + x_4 = b_1.$$

and $2x_1 + 4x_2 - x_3 + x_5 = b_2.$

Here x_4 and x_4 are called the slack variables.

2. Surplus Varibles. If a constraint has a sign $\ge$, then in order to make it an equality we have to subtract something positive from its L.H.S. **The Positive variables which are subtracted from the L.H. sides of the constraints to convert them into equalities are called the surplus variables.**

For example, consider the linear programming problem.

Max. $Z = 2x_2 + 4x_2 + 4x_3$

s.t. $\left.\begin{array}{l} x_1 + x_2 + 2x_3 \le b_1 \\ 2x_1 + 4x_2 + 6x_3 \ge b_2 \\ x_1 - 2x_2 + 4x_3 \ge b_3 \end{array}\right\}$ ... (2)

The constraints (2) are inequalities. In order to convert them into equalities we have to add some thing in the L.H.S. of first and subtract some thing from the L.H.S. sides of the second and the third inequalities of (2), *i.e.* we have following equalities

$$x_1 + x_2 + 2x_3 + x_4 = b_1$$
$$2x_1 + 4x_2 + 6x_3 - x_5 = b_2$$
$$x_1 - 2x_2 + 4x_3 - x_6 = b_3$$

Here, x_4 is the slack variables while x_5 and x_6 are the surplus variables.

§ 3·3. Some Definitions and Notations.

In this section we shall introduce some notations and definitions which are used in further discussion.

Let the L.P. problem after introducing the slack or surplus variables be as follows :

Max. $Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n + 0.x_{n+1} + 0.x_{n+2} + \ldots + 0.x_{n+m}$

subject to ,

$$\begin{array}{llll} a_{11}x_1 + a_{12}x_2 + \ldots + a_{ij}x_j + \ldots + a_{in}x_n + x_{n+1} & & & = b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2j}x_j + \ldots + a_{2n}x_n & + x_{n+2} & & = b_2 \\ \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mj}x_j + \ldots + a_{mn}x_n & & + x_{n+m} & = b_m \end{array}$$

$x_i \ge 0$ forall $i = 1, 2, \ldots, (n+m)$

$b_1, b_2, \ldots b_m$ are positive.

The above L.P. problem may also be stated as

Max. $Z = \mathbf{c}\,\mathbf{x}$

subject to $A\mathbf{x} = \mathbf{b}$

and $\mathbf{x} \ge 0$

where $\mathbf{c} = \{(c_1, c_2, \ldots c_n, 0, 0, \ldots 0)_{1 \times (m+n)}$ arrow vector of order $1 \times m + n$

$A = [a_{ij}]_{m \times (m+n)}$ the matrix of coefficient of order $m \times (m+n)$.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \\ x_{n+m} \end{bmatrix}_{(n+m)\times 1} = [x_1, x_2, \ldots x_n, x_{n+1}, \ldots, x_{n+m}]$$

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1} = [b_1, b_2, \ldots, b_m]$$

If α_j denotes the jth column of the matrix A then

$$A = (\alpha_1, \alpha_2, \ldots, \alpha_{n+m}).$$

(i) First we shall denote by B a $m \times m$ non-singular matrix whose column vectors are m number of linearly independent columns of the matrix A. If these columns are denoted by $\beta_1, \beta_2 \ldots \beta_m$ then

$$B = (\beta_1, \beta_2, \ldots, \beta_m).$$

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Matrix B is called the **basis matrix.**

(ii) The variables corresponding to $\beta_1, \beta_2, \ldots, \beta_m$, are called the **basic variables** and are denoted by $x_{B1}, x_{B2}, \ldots, x_{Bm}$, respectively.

The column vector of these m basic variables is denoted by $\mathbf{x}_B$

i.e. $\mathbf{x}_B = [x_{B1}, x_{B2}, \ldots, x_{Bm}]$

where $\mathbf{x}_B = B^{-1}\mathbf{b}$.

and is called the B.F.S. of the L.P.P.

(iii) The coefficients of basic varibles $x_{B1}, x_{B2} \ldots, x_{Bm}$ in the objective function Z will be denoted by $c_{B1}, c_{B2}, \ldots, c_{Bm}$, so that

$$C_B = (c_{B1}, c_{B2}, \ldots, c_{Bm})$$

(iv) Since $B = (\beta_1, \beta_2 \ldots, \beta_m)$ is a non-singular matrix of order $m \times m$, and the vectors $\beta_1, \beta_2 \ldots, \beta_m$ are linearly independent, they form the basis of E^m. Therefore each vector in E^m can be expressed as a linear combination of vector in B.

Let $\alpha_j = \beta_1 y_{1j} + \beta_2 y_{2j} + \ldots + \beta_m y_{mj}$

$$= (\beta_1, \beta_2, \ldots, \beta_m) \begin{bmatrix} y_{ij} \\ y_{2j} \\ \vdots \\ y_{mj} \end{bmatrix} = BY_j, \quad Y_j = \begin{bmatrix} y_{ij} \\ y_{2j} \\ \vdots \\ y_{mj} \end{bmatrix}_{m \times 1}$$

$y_{1j}, y_{2j}, \ldots, y_{mj}$, are scalars required to express α_j, $(j = 1, 2, 3, \ldots, n+m)$ as L.C. of $\beta_1, \beta_2 \ldots, \beta_m$.

$$\therefore \quad Y_j = B^{-1}\alpha_j.$$

(v) In the end we write

$$Z_j = C_B Y_j = c_{B1} y_{1j} + c_{B2} y_{2j} + \ldots + c_{Bm} y_{mj}.$$

§ 3·4. Fundamental Theorem of Linear Programming.

If a L.P. problem

Max. $Z = \mathbf{c}\mathbf{x}$, *s.t.* $A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0$

where A is a $m \times (m+n)$ matrix of coefficients given by $A = (\alpha_1, \alpha_2, \ldots, \alpha_{m+n})$, has an optimal Solution, then at least one basic feasible solution must be optimal.

[Meerut 87, 90 (TDC), 90 (B.A. P TDC), 95 ; Raj. 85, 86 ; B.H.U. B.Sc. Hons. 85]

Proof. Let $\mathbf{x}^* = [x_1, x_2, \ldots, x_{m+n}]$, be an optimal feasible

solution of the given L.P. problem and $Z^* = \sum_{i=1}^{m+n} c_i x_i$ be the maximum value of the objective function corresponding to this solution $\mathbf{x}^*$.

Suppose that k components (variables) in $\mathbf{x}^*$ are non-zero and remaining $(m+n-k)$ components are zero. We can assume these non-zero components as the first k components of $\mathbf{x}^*$.

$$\textit{i.e.,} \quad \mathbf{x}^* = [x_1, x_2, \ldots, x_k, \overbrace{0, 0, \ldots, 0}^{(m+n-k)}]$$

$$\therefore \quad \sum_{i=1}^{k} x_i \alpha_i = \mathbf{b} \qquad \ldots(1)$$

and
$$Z^* = \sum_{i=1}^{k} c_i x_i \qquad \ldots(2)$$

Now there are two possibilities.

1. If $\alpha_1, \alpha_2 \ldots, \alpha_k$ are L.I. If the vectors $\alpha_1, \alpha_2, \ldots, \alpha_k$ are L. I. then by defination $\mathbf{x}^*$ is a B. F. S. which is also optimal. Hence the theorem is true in this case. This solution is degenerate if $k < m$ and non-degenerate if $k = m$.

2. If $\alpha_1, \alpha_2 \ldots, \alpha_k$ are L. D. This is the case when $k > m$. In this case we can reduce the number of non-zero variables step by step as follows.

Since $\alpha_1, \alpha_2 \ldots, \alpha_k$, are L. D., therefore there exist scalars $\lambda_1, \lambda_2 \ldots, \lambda_k$, with at least one $\lambda_i \neq 0$, s. t.

$$\sum_{i=1}^{k} \lambda_i \alpha_i = 0 \qquad \ldots(3)$$

Now suppose that at least one λ_i is positive because if none is positive then we can multiply the equation by -1 and get positive λ_i.

Let
$$v = \underset{1 \leq i \leq k}{\text{Max.}} \left(\frac{\lambda_i}{x_i} \right). \qquad \ldots(4)$$

Clearly $v > 0$, since $x_i \geq 0$, for $i = 1, 2, \ldots, k$ and at least one λ_i is positive.

Multiplying (3) by $\frac{1}{v}$ and subtracting from (1), we have

$$\sum_{i=1}^{k} \left(x_i - \frac{\lambda_i}{v} \right) \alpha_i = \mathbf{b}. \qquad \ldots(5)$$

From (5), it follows that

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$$\mathbf{x}' = \left[x_1 - \frac{\lambda_1}{v}, x_2 - \frac{\lambda_2}{v}, \ldots, x_k - \frac{\lambda_k}{v}, \overbrace{0, 0, \ldots, 0}^{(m+n-k)}\right] \quad \ldots (6)$$

is also a solution of the system $A\mathbf{x} = \mathbf{b}$.

Also from (4), we have

$$v \geq \frac{\lambda_i}{x_i} \quad \forall\, i = 1, 2, \ldots, k,$$

or $$x_i \geq \frac{\lambda_i}{v}$$

or $$x_i - \frac{\lambda_i}{v} \geq 0, \forall\, i = 1, 2, \ldots, k,$$

i.e. all the components of the solution $\mathbf{x}'$ given by (6) are non-negative and thus $\mathbf{x}'$ is a F.S. to the given L. P. P.

Also (4) implies that, at least for one value of i

$$v = \frac{\lambda_i}{x_i}$$

or $$x_i = \frac{\lambda_i}{v} \quad \text{or} \quad x_i - \frac{\lambda_i}{v} = 0$$

i.e., $x_i - \frac{\lambda_i}{v}$ vanishes at least for one value of i. Therefore the B. S. $\mathbf{x}'$ given by (6) cannot have more than $(k-1)$ non-zero components (variables).

Thus we have derived a new F. S. from the given optimal solution, containing less number of non-zero variables.

Now we shall prove that $\mathbf{x}'$ *is also an optimal solution.*

Let Z' be the value of the objective function corresponding to this solution.

$$\therefore Z' = \sum_{i=1}^{k} c_i \left(x_i - \frac{\lambda_i}{v}\right)$$

$$= \sum_{i=1}^{k} c_i x_i - \frac{1}{v}\sum_{i=1}^{k} c_i \lambda_i = Z^* - \frac{1}{v}\sum_{i=1}^{k} c_i \lambda_i \quad \ldots (7)$$

Now if Z' is optimal then $Z' = Z^*$, which is possible only if $\sum_{i=1}^{k} c_i \lambda_i = 0$.

To prove $\sum_{i=1}^{k} c_i \lambda_i = 0$, we shall use contradiction.

If possible, suppose that $\sum_{i=1}^{k} c_i \lambda_i \neq 0$.

$\therefore$ either (i) $\sum_{i=1}^{k} c_i \lambda_i < 0$ or (ii) $\sum_{i=1}^{k} c_i \lambda_i > 0$.

$\therefore$ We can find a real number r (negative in case (i) and positive in case (ii) such that

$$r. \sum_{i=1}^{k} c_i \lambda_i > 0$$

or $$\sum_{i=1}^{k} c_i (r\lambda_i) > 0$$

or $$\sum_{i=1}^{k} c_i \left(r\lambda_i\right) + \sum_{i=1}^{k} c_i x_i > \sum_{i=1}^{k} c_i x_i$$

or $$\sum_{i=1}^{k} c_i \left(x_i + r\lambda_i\right) > Z^* \quad \ldots (8)$$

Now multiplying (3) by r and adding to (1), we have

$$\sum_{i=1}^{k} \left(x_i + r\lambda_i\right) \alpha_i = \mathbf{b}$$

Which implies that

$\left[x_1 + r\lambda_1, x_2 + r\lambda_2, \ldots, x_k + r\lambda_k, \overbrace{0, 0, \ldots, 0}^{(m+n-k)}\right]$, for all values of r is also a solution of the system $A\mathbf{x} = \mathbf{b}$.

If we choose r, s.t.

$$x_i + r\lambda_i \geq 0 \quad \forall\, i = 1, 2, \ldots, k$$

or $$r\lambda_i \geq -x_i$$

$$\therefore \quad r \geq -\frac{x_i}{\lambda_i} \quad \text{if } \lambda_i > 0$$

$$r \leq -\frac{x_i}{\lambda_i} \quad \text{if } \lambda_i < 0,$$

and r unrestricted, when $\lambda_i = 0$.

Thus $x_i + r\lambda_i \geq 0$ for $i = 1, 2, \ldots, k$, if we choose r such that

$$\underset{\substack{i \\ (\lambda_i > 0)}}{\text{Max.}} \left(-\frac{x_i}{\lambda_i}\right) \leq r \leq \underset{\substack{i \\ (\lambda_i < 0)}}{\text{Min.}} \left(-\frac{x_i}{\lambda_i}\right) \quad \ldots (9)$$

Also we have

$$\underset{\substack{i \\ (\lambda_i > 0)}}{\text{Max.}} \left(-\frac{x_i}{\lambda_i}\right) < 0 \text{ and } \underset{\substack{i \\ (\lambda_i < 0)}}{\text{Min}} \left(-\frac{x_i}{\lambda_i}\right) > 0,$$

which implies that interval given by (9) is non-empty interval.

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Hence when r lies in the non-empty interval given by (9) then the solution given by

$$[x_i + r\lambda_1, x_2 + r\lambda_2, \ldots, x_k + r\lambda_k, \underbrace{0, 0, \ldots\ldots, 0}_{(m+n-k)}]$$

is also a F.S. of the system $A\mathbf{x} = \mathbf{b}$.

and from (8), this solution gives a value Z' of Z which is greater than Z^* (optimal value or greatest value). Which is a contradiction.

$$\therefore \quad \sum_{i=1}^{k} c_i\lambda_i = 0 \text{ or } Z' = Z^*$$

i.e. $\mathbf{x}'$ given by (6) is also an optimal solution.

In this way from a given optimal solution we have constructed (derived) a new optimal solution containing less number of non-zero variables. This solution is optimal B.F.S. if the column vectors associated to non-zero variables in this new solution are L.I. Hence the theorem is true. If these associated vactors are not L.I. (*i.e.* solution not B.F.S.) then by repeating the above process we can get another optimal B.F.S. of the given system containing not more than $(k-2)$ non zero variables. Continueing in this way for finite number of times we will certainly get an optimal B.F.S. of the system. Hence the theorem is true.

Note. In the F.S. $\mathbf{x}'$ the variable x_r will become zero if

$$\text{subject } \; v = \underset{1 \le i \le k}{\text{Max.}} \left\{\frac{\lambda_i}{x_i}\right\} = \frac{\lambda_r}{x_r}.$$

§ 3·5. To obtain B.F.S. from F.S.

Theorem. *If a L.P.P.*

$$Max.\ Z = \mathbf{cx}, s.t.\ A\mathbf{x} = \mathbf{b}, \mathbf{x} \ge 0,$$

where A is $m \times (m+n)$ matrix of coefficients given by $A = (\alpha_1, \alpha_2, \ldots, \alpha_{m+n})$, has at least one feasible solution. then it has at least one basic feasible solution also.*

[Meerut 95 (BP), 97 (BP), 98 (OLD); Raj. 87]

Proof. Let $\mathbf{x} = [x_1, x_2, \ldots, x_{m+n}]$ be a feasible solution of the given L.P.P.

Suppose that k components (variables) in $\mathbf{x}$ are non-zero and remaining $(m+n-k)$ components are zero. We can assume these non-zero components as the first k components of $\mathbf{x}$

i.e. $$\mathbf{x} = [x_1, x_2, \ldots, x_k, \underbrace{0, 0, \ldots, 0}_{(m+n-k)}]$$

* In L.P.P. any solution which satisfies $x \ge 0$ will be termed as feasible solution.

$$\therefore \quad \sum_{i=1}^{k} x_i\alpha_i = \mathbf{b}. \quad \ldots (1)$$

Now there are two possibilities.

1. If the vectors $\alpha_1, \alpha_2, \ldots, \alpha_k$ are L.I. If the vectors $\alpha_1, \alpha_2, \ldots, \alpha_k$ are L.I., then by defination $\mathbf{x}$ is a B.F.S. Hence the theorem is true in this case. This solution is degenerate if $k < m$ and non-degenerate if $k = m$.

2. If the vectors $\alpha_1, \alpha_2, \ldots, \alpha_k$ are L.D. This is the case when $k > m$. In this case we can reduce the number of non-zero variables step by step, so that the feasible solution become B.F.S.

Proceeding as in § 3·4, can reduce the F.S.

$$\mathbf{x}' = [x_1, x_2, \ldots, x_k, \underbrace{0, 0, \ldots, 0}_{(m+n-K)}]$$

to a new basic feasible solution.

$$\mathbf{x}' = \left[x_1 - \frac{\lambda_1}{y}, x_2 - \frac{\lambda_2}{v}, \ldots, x_k - \frac{\lambda_k}{v}, \underbrace{0, 0, \ldots, 0}_{(m+n-k)}\right]$$

which cannot contain more than $(k-1)$ non-zero components. Thus we have derived a new F.S. from the given F.S. containing less number of non-zero (positive) variables.

If the column vectors associated to non-zero variable in this new sol^n are L.I., then this sol^n is B.F.S. and hence the theorem is true. But if these associated vectors are not L.I. then this solution is not B.S.F. In this case we repeat the above process again and get another B.F.S. of the given system containing not more than $(k-2)$ non-zero variables. Continueing in this way for finite number of times we will certainly get an optimal B.F.S. of the system. Hence the theorem is true.

Note. The two theorems 3·4 and 3·5 in combined form may be stated *as follows.*

If the L.P.P. Max. $Z = \mathbf{cx}$, s.t. $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \ge 0$ has feasible solution, then at least one of the B.F. solution will be optimal.

Illustrative Examples

Ex. 1. *If $x_1 = 2, x_2 = 3, x_3 = 1$, be a feasible solution of the L.P.P.*

$$\begin{aligned} \textit{Max.} \quad & Z = x_1 + 2x_2 + 4x_3 \\ \text{s.t.} \quad & 2x_1 + x_2 + 4x_3 = 11 \\ & 3x_1 + x_2 + 5x_3 = 14 \\ & x_1, x_2, x_3 \ge 0 \end{aligned}$$

then find a B.F.S. [Meerut 89, 94 (P), 95]

Sol. The given L.P.P. can be written as

Max. $Z = x_1 + 2x_2 + 4x_3$

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 = \text{b} \quad \dots (1)$$

$$x_1, x_2, x_3 \geq 0$$

where $\alpha_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}, \text{b} = \begin{bmatrix} 11 \\ 14 \end{bmatrix}$

Here m (number of constraints) $= 2$, so the B.F.S. of this L.P.P. can not have more that 2 non-zero variables.

$\therefore$ The given feasible solution $x_1 = 2, x_2 = 3, x_3 = 1$ is not B.F.S. In order to reduce the F.S. to a B.S.F. we have to make at least one variable zero. For this we proceed as follows.

Since the vectors $\alpha_1, \alpha_2, \alpha_3$, associated with the non-zero variables are L.D., therefore one of these vectors may be expressed as a L.C. of the remaining two.

Let $\quad \alpha_1 = \lambda_2 \alpha_2 + \lambda_3 \alpha_3$

or $\quad \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \lambda_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \lambda_3 \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} \lambda_2 + 4\lambda_3 \\ \lambda_2 + 5\lambda_3 \end{bmatrix}$

$\therefore \quad \lambda_2 + 4\lambda_3 = 2$ and $\lambda_2 + 5\lambda_3 = 3$.

Solving, we have $\lambda_2 = -2$ and $\lambda_3 = 1$

$$\therefore \alpha_1 = -2\alpha_2 + 1\alpha_3$$

or $\quad \alpha_1 + 2\alpha_2 - \alpha_3 = 0 \quad \dots (2)$

or $\quad \sum_{i=1}^{3} \lambda_i x_i = 0$, where $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = -1$

Now $\quad v = \underset{1 \leq i \leq 3}{Max} \left(\frac{\lambda_i}{x_i}\right)$

$$= \text{Max.} \left(\frac{\lambda_1}{x_1}, \frac{\lambda_2}{x_2}, \frac{\lambda_3}{x_3}\right)$$

$$= \text{Max.} \left(\frac{1}{2}, \frac{2}{3}, \frac{-1}{1}\right) = \frac{2}{3} = \frac{\lambda_2}{x_2}.$$

$\because$ The variables x_2 should be zero or the vector α_2 should be eliminated.

Substituting the given feasible solution $x_1 = 2, x_2 = 3, x_3 = 1$ in (1), we have

$$2\alpha_1 + 3\alpha_2 + \alpha_3 = \text{b} \quad \dots (3)$$

Substituting the value of α_2 from (2) in (3), we have

$$2\alpha_1 + 3\left(\frac{\alpha_3 - \alpha_1}{2}\right) + \alpha_3 = \text{b}$$

or $\quad \frac{1}{2}\alpha_1 + \frac{5}{2}\alpha_3 = \text{b}$

or $\quad \frac{1}{2}\alpha_1 + 0.\alpha_2 + \frac{5}{2}\alpha_3 = \text{b}.$

$\therefore$ The new feasible solution is $x_1 = 1/2, x_2 = 0, x_3 = 5/2$.

Here the vector $\alpha_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$, associated to non-zero varibles are L.I. Hence the F.S. $x_1 = 1/2, x_2 = 0, x_3 = 5/2$ is B.F.S.

Note 1. Another B.F.S. of the L.P.P. is $x_1 = 3, x_2 = 5, x_3 = 0$.

2. The new values of the variables can also be found by using the formula.

$$x_i' = x_i - \frac{\lambda_i}{v}.$$

Ex. 2. *If $x_1 = 1, x_2 = 2, x_3 = 1, x_4 = 3$ be a F.S. to the set of equations*

$$5x_1 - 4x_2 + 3x_3 + x_4 = 3$$

$$2x_1 + x_2 + 5x_3 - 3x_4 = 0$$

$$x_1 + 6x_2 - 4x_3 + 2x_4 = 15$$

then find a B. F. S. [Meerut 87]

Sol. The given set of equation can be written as

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \alpha_4 x_4 = \text{b} \quad \dots (1)$$

where $\alpha_1 = \begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} -4 \\ 1 \\ 6 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 3 \\ 5 \\ -4 \end{bmatrix}, \alpha_4 = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$,

$$\text{b} = \begin{bmatrix} 3 \\ 0 \\ 15 \end{bmatrix}.$$

Here $m = 3$, so the B.F.S. of the set of equations can not have more than 3 non-zero variables.

$\therefore$ The given F.S. $x_1 = 1, x_2 = 2, x_3 = 1, x_4 = 3$ is not B.F.S. In order to reduce this F.S. to B.F.S. we have to make at least one variable zero. For this we proceed follows.

Let $\quad \alpha_1 = \lambda_2 \alpha_2 + \lambda_3 \alpha_3 + \lambda_4 \alpha_4$

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or $\begin{bmatrix}5\\2\\1\end{bmatrix} = \lambda_2 \begin{bmatrix}-4\\1\\6\end{bmatrix} + \lambda_3 \begin{bmatrix}3\\5\\-4\end{bmatrix} + \lambda_4 \begin{bmatrix}1\\-3\\2\end{bmatrix}$

$$= \begin{bmatrix}-4\lambda_2 + 3\lambda_3 + \lambda_4\\ \lambda_2 + 5\lambda_3 - 3\lambda_4\\ 6\lambda_2 - 4\lambda_3 + 2\lambda_4\end{bmatrix}$$

$\therefore$ $-4\lambda_2 + 3\lambda_3 + \lambda_4 = 5$

$\lambda_2 + 5\lambda_3 - 3\lambda_4 = 2$

$6\lambda_2 - 4\lambda_3 + 2\lambda_4 = 1.$

Solving, we have $\lambda_2 = \frac{22}{43}, \lambda_3 = \frac{139}{86}, \lambda_4 = \frac{189}{86}.$

$\therefore$ $\alpha_1 = \frac{22}{43}\alpha_2 + \frac{139}{86}\alpha_3 + \frac{189}{86}\alpha_4$

or $86\alpha_1 - 44\alpha_2 - 139\alpha_3 - 189\alpha_4 = 0$

or $\sum_{i=1}^{4} \lambda_i \alpha_i = 0$, where $\lambda_1 = 86$, $\lambda_2 = -44$, $\lambda_3 = -139$, $\lambda_4 = -$

Now $v = \underset{1 \le i \le 4}{Max} \left(\frac{\lambda_i}{x_i}\right)$

$= \text{Max.} \left(\frac{\lambda_1}{x_1}, \frac{\lambda_2}{x_2}, \frac{\lambda_3}{x_3}, \frac{\lambda_4}{x_4}\right)$

$= \text{Max.} \left(\frac{86}{1}, \frac{-44}{2}, \frac{-139}{1}, \frac{-189}{3}\right) = \frac{86}{1} = \frac{\lambda_1}{x_1}$

$\therefore$ The variable x_1 should be zero or the vector α_1 shoul eliminated.

Substituting he given F.S. (1), we have

$\alpha_1 + 2\alpha_2 + \alpha_3 + 3\alpha_4 = \mathbf{b}$

substituting the value of α_1 from (2) in (3), we have

$\frac{44\alpha_2 + 139\alpha_3 + 189\alpha_4}{86} + 2\alpha_2 + \alpha_3 + 3\alpha_4 = \mathbf{b}$

or $0.\alpha_1 + \frac{216}{86}\alpha_2 + \frac{225}{86}\alpha_3 + \frac{447}{86}\alpha_4 = \mathbf{b}$

$\therefore$ The new F.S. is $x_1 = 0, x_2 = \frac{216}{86}, x_3 = \frac{225}{86}, x_4 = \frac{447}{86}$. Here

the vectors $\alpha_2 = \begin{bmatrix}-4\\1\\6\end{bmatrix}, \alpha_3 = \begin{bmatrix}3\\5\\-4\end{bmatrix}, \alpha_4 = \begin{bmatrix}1\\-3\\2\end{bmatrix}$, associated to n-zero variables are L.I. Hence this F.S. is B.S.F.

te. Another basic feasible solution of the set of equations is

$$x_1 = \frac{25}{139}, \; x_2 = \frac{234}{139}, \; x_3 = 0, \; x_4 = \frac{228}{139}.$$

§ 3·6. To Determine Improved B.F.S.

Now we shall prove the following important theorem which lps us to determine an improved B.F.S. from a given B.F.S. of a P.P.

Theorem. *Let $\mathbf{x}_B = B^{-1}\mathbf{b}$ be a B. F. S. of a L. P. P. with $= c_B \, x_B$ as the value of the objective function. If for any column in A, but not in B, the condition $c_j - Z_j > 0$ or $Z_j - c_j < 0$ holds, and at least one $y_{ij} > 0, i = 1, 2, \ldots, m$, then it is possible to obtain a new F. S. by replacing one of the columns in* **B** *by α_j and if the new value the objective function is Z' then $Z' \ge Z$. If the given B. F. S. is n-generate then $Z' > Z$.*

Proof. Let $\mathbf{x_B} = [x_{B1}, x_{B2}, \ldots, x_{Bm}]$, be a B. F. S. of the given P.P.

$$\text{Max. } z = \mathbf{c}\,\mathbf{x}, A\mathbf{x} = \mathbf{b}, \mathbf{x} \ge 0$$

where $A = (\alpha_1, \alpha_2, \ldots, \alpha_{m+n})$ and the basis matrix

$$B = (\beta_1, \beta_2, \ldots, \beta_m)$$

$$B \, \mathbf{x}_B = \mathbf{b} \quad \ldots (1)$$

et $\alpha_j \in A$, s.t. $\alpha_j \notin B$.

Since $\beta_1, \beta_2, \ldots, \beta_m$ form the basis of A, therefore α_j can be xpressed as the L.C. of these vectors.

$$\alpha_j = \sum_{i=1}^{m} \beta_i y_{ij} = \beta_1 y_{1j} + \beta_2 y_{2j} + \ldots + \beta_m y_{mj} \quad \ldots (2)$$

If $y_{rj} \ne 0$, then we can replace β_r in B by α_j and still B is a basis atrix.

$\therefore$ assuming that $y_{rj} \ne 0$, from (1), we have

$$\beta_r = \frac{1}{y_{rj}}\alpha_j - \frac{y_{1j}}{y_{rj}}\beta_1 - \ldots - \frac{y_{r-1,j}}{y_{rj}}\beta_{r-1} - \frac{y_{r+1,j}}{y_{rj}}\beta_{r+1} - \ldots - \frac{y_{mj}}{y_{rj}}\beta_m$$

$$\beta_r = \frac{1}{y_{rj}}\alpha_j - \sum_{\substack{i=1\\ i \ne r}}^{m} \frac{y_{ij}}{y_{rj}}\beta_i \quad \ldots (3)$$

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From (1), we have

$$\beta_1 x_{B1} + \beta_2 x_{B2} + \ldots + \beta_r x_{Br} + \ldots + \beta_m x_{Bm} = \mathbf{b}$$

or

$$\sum_{\substack{i=1 \\ i \neq r}}^{m} \beta_i x_{Bi} + \beta_r x_{Br} = \mathbf{b} \quad \ldots (4)$$

Substituting the value of β_r from (3) in (4), we have

$$\sum_{\substack{i=1 \\ i \neq r}}^{m} \beta_i \left[x_{Bj} - \frac{y_{ij}}{y_{rj}} x_{Br}\right] + \frac{x_{BR}}{y_{rj}} \alpha_j = \mathbf{b} \ldots (5)$$

Comparing with (4), the new basic solution to $A\mathbf{x} = \mathbf{b}$ is given by

$$x'_B = [x'_{B1}, x'_{B2}, \ldots, x'_{Bm}]$$

$$\left.\begin{aligned} \text{where} \quad x'_{Bi} &= x_{Bi} - \frac{y_{ij}}{y_{rj}} x_{Br}, \; i = 1, 2, \ldots, m; i \neq r \\ \text{and} \quad x'_{Br} &= \frac{x_{Br}}{y_{rj}} \end{aligned}\right\} \quad \ldots (6)$$

The basic solution will be feasible if

$$x_{Bi} - \frac{y_{ij}}{y_{rj}} x_{Br} \geq 0, \qquad (i = 1, 2, \ldots, m, i \neq r) \quad \ldots (7)$$

and

$$\frac{x_{Br}}{y_{rj}} \geq 0 \quad \ldots (8)$$

Since x_B is a B. F. S. of the L.P. P. $\therefore x_{Br} \geq 0$
$i = 1, 2, \ldots, m$

Thus we see that (7) and (8) are satisfied only if $y_{rj} > 0$ and $y_{ij} \leq 0$ $(i = 1, 2, \ldots, m; i \neq r)$.

If $y_{rj} > 0$ and $y_{ij} > 0$, then (7) and (8) are satisfied only if

$$\frac{x_{Bi}}{y_{ij}} - \frac{x_{Br}}{y_{rj}} \geq 0$$

or

$$\frac{x_{Br}}{y_{rj}} \geq \frac{x_{Br}}{y_{rj}}$$

$$\text{where} \quad \frac{x_{Br}}{y_{rj}} = \underset{i}{Mini} \left\{\frac{x_{Bi}}{y_{ij}}, y_{ij} > 0\right\}.$$

Thus we can obtain a new B. F. S. from the initial B. F. S. by removing the column vector β_r of the basis matrix B, by α_j, if $y_{rj} > 0$ where r is to be selected such that

$$\frac{x_{Br}}{y_{rj}} = \underset{i}{Mini} \left(\frac{x_{Bi}}{y_{ij}}, y_{ij} > 0\right) = v \quad \ldots (9)$$

This relation (9) gives the criterion, termed as *simplex criterion* for selecting the vectors β_r which is to be deleted (removed) from the basis matrix *B*.

It is important to note that in the new B.F.S. the variable at the rth position is different from that in the previous B.F.S. while at all other positions there are the same variables, usually with different values. The variables at the rth position in the previous B.F.S. is driven to zero.

From (9) if $v = 0$ which is possible only if $x_{Br} = 0$, then we conclude that the initial (original) B.F.S. is degenerate B.F.S.

To prove that $Z' \geq Z$. We have

$$Z = \sum_{i=1}^{m} c_{Bi} x_{Bi} \text{ and } Z' = \sum_{i=1}^{m} c'_{Bi} x'_{Bi}$$

where c'_{Bi} are the coefficients of the basic varaibles x'_{Bi} $(i = 1, 2, \ldots, m)$ in the objective function *Z*.

It is obvious that

$$c'_{Bi} = c_{Bi} \qquad (i = 1, 2, \ldots, m; i \neq r)$$

and

$$c'_{Br} = c_j$$

Now using (6), we have

$$Z' = \sum_{\substack{i=1 \\ i \neq r}}^{m} c'_{Bi} \left(x_{Bi} - \frac{y_{ij}}{y_{ri}} x_{Br}\right) + c'_{Br} \cdot \frac{y_{Br}}{y_{rj}}$$

$$= \sum_{\substack{i=1 \\ i \neq r}}^{m} c_{Bi} \left(x_{Bi} - \frac{y_{ij}}{y_{ri}} x_{Br}\right) + c_j \cdot \frac{y_{Br}}{y_{rj}}$$

$$= \sum_{i=1}^{m} c_{Bi} \left(x_{Bi} - \frac{y_{ij}}{y_{ri}} x_{Br}\right) + c_j \cdot \frac{y_{Br}}{y_{rj}}$$

$$\text{Since} \quad c_{Bi} \left(x_i - \frac{y_{ij}}{x_{rj}} x_{Br}\right) = 0 \quad \text{for } i = r,$$

$$= \sum_{i=1}^{m} c_{Bi} x_{Bi} + \frac{x_{Br}}{y_{rj}} \left(c_j - \sum_{i=1}^{m} c_{Bi} y_{ij}\right)$$

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$$= Z + \frac{x_{Br}}{y_{rj}}(c_j - Z_j) \text{ where } Z_j = \sum_{i=1}^{m} c_{Bi} y_{ij} = C_B Y_j$$

[Note. Z_j correspond to the original B.F.S.

or $$Z' = Z + v(c_j - Z_j) \qquad \dots (1$$

Thus the new value of the objective function is equal to the original value plus the quantity $v(c_j - Z_j)$.

Now $Z' \geq Z$ if $v(c_j - Z_j) \geq 0$

Since $v \geq 0$

$\therefore$ $Z' \geq Z$ only if $c_j - Z_j \geq 0$

or $Z_j - c_j < 0$.

Thus by choosing the vector α_j for which $c_j - Z_j > 0$ and at least one $y_{ij} > 0$, we obtain new basic solution with $Z' \geq Z$.

If the original B.F.S. was non-degenerate then $v > 0$ and in this case we have

$Z' > Z$. Hence the theorem is proved.

Note 1. If $v = 0$, then x_{Br} *i.e.* the original B.F.S. is degenerate B.F.S. In this case from (6), we have $x'_{Bi} = x_{Bi}, i \neq r$ and $x'_{Br} = 0 = x_{Br}$. Thus we see that values of the variables common to the two solutions (old and new) do not change, and the new variable is at zero level (at r th position). Hence the new B.F.S. is also degenerate B.F.S.

2. If $y_{ij} < 0$, for every x_{Bi} which is zero in the original B.F.S. then none of these variables enter in the calculation of $\frac{x_{Br}}{y_{rj}}$ in (9). $\therefore v > 0$ in this case and the new B.F.S. will be non-degenerate B.F.S.

3. The vector β_r to be deleted from the basis B is called the departing (outgoing) vector while the vector α_j to be introduced into the basis is called the entering (in coming) vector and the element y_{ij} (in matrix A) is called the key element.

§ 3·7. Unbounded Solutions. In theorem 3·6 we assumed that if we insert α_j into the basis B and remove some column to form a new basis then at least one $y_{ij} > 0, i = 1, 2, \dots, m$. Now we shall discuss the case if we insert an α_j for which all $y_{ij} \leq 0, i = 1, 2, \dots, m$. For this we prove the following theorem.

Theorem. *If for any basic feasible solution of a L.P.P., there is some column α_j in A but not B for which $c_j - Z_j > 0$ and $y_{ij} \leq 0$ ($i = 1, 2, \dots, m$), then the problem has an *unbounded solution if the objective function is to be maximized.*

[Meerut 94, 96, 97 (BP), 97]

Proof. Let $\mathbf{x_B} = [x_{B1}, x_{B2}, \dots, x_{Bm}]$, be a B.F.S. of the given L.P.P.,

Max. $Z = \mathbf{cx}$ s.t. $A\mathbf{x} = \mathbf{b}$ $\mathbf{x} \geq 0$

Where $A = (\alpha_1, \alpha_2, \dots, \alpha_{m+n})$ and with the basis matrix $B = (\beta_1, \beta_2, \dots, \beta_m)$

$$\therefore \quad B\,\mathbf{x_B} = \mathbf{c} \quad \text{or} \quad \sum_{i=1}^{m} x_{Bi}\,\beta_i = \mathbf{b} \qquad \dots (1)$$

The value of the objective function for this B.F.S. is given by

$$Z = C_B\,\mathbf{x_B} = \sum_{i=1}^{m} c_{Bi} x_{Bi} \qquad \dots (2)$$

Let us insert the column α_j in B, where α_j is in A but not in B and $c_j - Z_j > 0$ and $y_{ij} \leq 0$ $i = 1, 2, \dots, m$.

If λ is any scalar then adding and subtracting $\lambda\alpha_j$ in (1), we have

$$\sum_{i=1}^{m} x_{Bi}\beta_i - \lambda\alpha_j + \lambda\alpha_j = \mathbf{b} \qquad \dots (3)$$

Since $\alpha_j \in A$ s.t. $\alpha_j \notin B$.

Since $\beta_1, \beta_2, \dots, \beta_m$ form the basis of A, therefore α_j can be expressed as the L.C. of these vectors.

i.e. $$\alpha_j = \sum_{i=1}^{m} \beta_i y_{ij}$$

or $$-\lambda\alpha_j = -\lambda\sum_{i=1}^{m} \beta_i y_{ij}$$

Substituting in (3), we have

$$\sum_{i=1}^{m} x_{Bi}\beta_i - \lambda\sum_{i=1}^{m} \beta_i \lambda_{ij} + \lambda\alpha_j = \mathbf{b}$$

or $$\sum_{i=1}^{m} \left(x_{Bi}\beta_i - \lambda y_{ij}\right)\beta_i + \lambda\alpha_j = \mathbf{b} \qquad \dots (4)$$

When $\lambda > 0$ then $x_{Bi} - \lambda y_{ij} \geq 0$

*A problem is said to have an unbounded solution if the objective function can be increased or decreased arbitrarity i.e., if there is no finite optimum of the objective function.

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Since $y_{ij} \leq 0, i = 1, 2, \ldots, m$

$\therefore$ from (4) we obtain a new feasible solution

$$x'_B = [x'_{B1}, x'_{B2}, \ldots, x'_{Bm}, \lambda]$$

where $\quad x'_{Bi} = x_{Bi} - \lambda y_{ij} \quad i = 1, 2, \ldots, m$

in which $(m+1)$ varaibles can be different from zero.

In the feasible solution x'_B the number of positive varaibles is less than or equal to $(m+1)$. This may be less than $(m+1)$ since $x_{Bi} - \lambda y_{ij}$ may be zero for some value of i. If the number of positive variable in this solution x'_B is equal to $(m+1)$ then this solution is non-basic feasible solution.

Now if Z' is the value of the objective function for this new feasible solution then

$$Z' = \sum_{i=1}^{m} c_{Bi}(x_{Bi} - \lambda y_{ij}) + c_j . \lambda$$

or
$$Z' = \sum_{i=1}^{m} c_{Bi} x_{Bi} + \lambda (c_j - c_{Bi} y_{ij})$$

or
$$Z' = Z + \lambda (c_j - Z_j) \quad \ldots (5)$$

Since $c_j - Z_j > 0$, we can make Z' as large as we please by taking λ sufficiently large. Hence the problem has an unbounded solution if the objective function is to be maximized.

Note. Similarly if for some $\alpha_j, Z_j - c_j > 0$ and $y_{ij} \leq 0$, $i = 1, 2, \ldots, m$, then the L.P.P. has an unbounced solution if the objective function is to be minimized.

§ 3·8. Optimality conditions.

Theorem. *Let $x_B = B^{-1}b$ be the basic feasible solution to the L.P.P. Max. $Z = cx$, s.t. $Ax = b, x \geq 0$ and $Z^* = c_B x_B$ the value of the objective function for this B.F.S. If $c_j - Z_j \leq 0$ for every column a_j in A but not in B, then Z^* is the optimum (maximum) value of the objective function Z and this B.F.S. x_B is an optimal B.F.S.*

[Meerut 88, 89]

Proof. Let $x_B = [x_{B1}, x_{B2}, \ldots, x_{Bm}]$ be the B.F.S. of the given L.P.P. Max. $Z = cx$, s.t. $Ax = b$, $x \geq 0$

where $A = (\alpha_1, \alpha_2, \ldots, \alpha_{m+n})$ and with the basis matrix

$$B = (\beta_1, \beta_2, \ldots, \beta_m).$$

$$\therefore Bx_B = b \quad \text{or} \quad x_B = B^{-1}b.$$

The value of the objective function for this B.F.S. is given by

$$Z^* = c_B . x_B = \sum_{i=1}^{m} c_{Bi} . x_{Bi} \quad \ldots (1)$$

Also it is given that $c_j - Z_j \leq 0$ for every column α_j in A but not in B.

If $x' = [x_1', x_2', \ldots, x'_{n+m}]$ is any F.S. of the given problem then

$$B x_B = b = A x' \quad \ldots (2)$$

If Z' is the value of the objective function for this solution, then

$$Z' = cx' = \sum_{i=1}^{n+m} c_j x'_j \quad \ldots (3)$$

Now we proceed to prove that $\quad Z^* \geq Z'$.

From (2), we have

$x_B = B^{-1}b = B^{-1}(Ax')$

$= (B^{-1}A)x'$

$= Yx'$ where $Y = [Y_1, Y_2, \ldots, Y_{m+n}]$

$= (Y_1, Y_2, \ldots, Y_{m+n})[x_1', x_2', \ldots, x'_{m+n}]$

$$= \begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1j} & \cdots & y_{1,m+n} \\ y_{21} & y^{22} & \cdots & y_{2j} & \cdots & y_{2,m+n} \\ \cdots & \cdots & & \cdots & & \cdots \\ \cdots & \cdots & & \cdots & & \cdots \\ y_{m1} & y_{m2} & \cdots & y_{mj} & \cdots & y_{m,m+n} \end{bmatrix} \begin{bmatrix} x_1' \\ x_2' \\ \vdots \\ \cdots \\ x'_{m+n} \end{bmatrix}$$

or $\quad [x_{B1}, x_{B2}, \ldots, x_{Bi}, \ldots, x_{Bm}]$

$$= \left[\sum_{j=1}^{m+n} y_{1j} . x_j', \sum_{j=i}^{m+n} y_{2j} . x_j', \ldots, \sum_{i=1}^{m+n} y_{ij} . x'_j, \ldots, \sum_{j=1}^{m+n} y_{mj} x_j' \right]$$

$$\therefore \quad x_{Bi} = \sum_{j=1}^{m+n} y_{ij} x_j'. \quad \ldots (4)$$

Now it is given that $c_j - Z_j \leq 0$, for all j for which $\alpha_j \notin B$.

Now if $\alpha_j \notin B$ say $\alpha_j = \beta_i$

then $\alpha_j = \beta_i = 0.\beta_1 + 0.\beta_2 + \ldots + 0.\beta_{i-1} + 1.\beta_i + 0.\beta_{i+1} + \ldots + 0.\beta_m$

which shows that $y_j = e_i$ (a vector whose ith component is unity).

Since $\quad \alpha_j = \beta i \quad \therefore \; c_j = c_{Bi}$

i.e. $\quad c_j - Z_j = c_j - c_B Y_j = c_j - c_B e_i = c_j - c_{Bi} = 0$

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have $r(A) = r(A\mathbf{b})$ *i.e.* the constraint equations (after introducing the slack and artificial variables) should be consistent. Since in simplex method we always have $r(A) = r(A\mathbf{b}) = m$.

If the system $A\mathbf{x} = \mathbf{b}$ involves artificial variables. Then we can not say whether this system is consistent of there is any redundancy. Below we give the cases (without proof) to decide about the consistency and redundancy is such systems.

Case I. *If the basis B contains no artificial vector and the optimality condition is satisfied (at any iteration) then the current solution is a* B.F.S. *of the problem.*

Case II. *If one or more artificial vector appears in the basis B at a zero level i.e. the value of the artificial variables corresponding to artificial vectors in B are zero and the optimality condition is satisfied (at any iteration) then the system is consistent. Further more if $y_{ij} = 0$. $\forall\ j$ and $x_{Br} = 0$ and r corresponds to the row containing an artificial vector, then the r th constraint equation is* **redundant.**

Case III. *If at least one artificial vector appears in the basis B at a positive level i.e. the value of at least one artificial variable corresponding to artificial vector in B is non-zero and the optimality condition is satisfied (at any iteration), then there exists* **no feasible** *solution of the problem.*

§ 3·11. To determine starting B.F.S.

In the constraints of a general linear programming problem there may be any of the the three signs $\leq, =, \geq$. Assuming that the requirement vector $\mathbf{b} \geq 0$ (if any of the b_i's is negative, multiply the corresponding constraint by -1) we shall discuss the following three cases.

Case I. To find intial B.F.S. when all the original constraints have $\leq$ sign.

In this case all the constraints are converted into equations by inserting slack varibles only. Inserting slack variables the equations obtained are as follows.

$$\begin{array}{llllll}
a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n & + 1.x_{n+1} & & & & = b_1 \\
a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n & & + 1x_{n+2} & & & = b_2 \\
\dots\ \dots\ \dots\ \dots\ \dots & \dots & \dots & \dots & & \dots \\
\dots\ \dots\ \dots\ \dots\ \dots & \dots & \dots & \dots & & \dots \\
a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n + & & & & + 1.x_{n+m} & = b_m
\end{array}$$

Here $x_{n+1}, x_{n+2}, \dots, x_{n+m}$ are slack variables.

In matrix form these, equations can be written as

slack vectors

$$\begin{bmatrix} a_{11} & a_{12} \cdots a_{1n} & 1 & 0 \dots 0 \\ a_{21} & a_{22} \cdots a_{2n} & 0 & 1 \dots 0 \\ \dots & \dots\ \dots & \dots & \dots \\ \dots & \dots\ \dots & \dots & \dots \\ a_{m1} & a_{m2}\ \ a_{mn} & 0 & 0 \dots 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \\ x_{n+m} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

Here we take the initial basis matrix

$$B = I_m. \text{ (unit matrix of order } m \times m).$$

$\therefore$ *the initial basic solution* is given by

$$\mathbf{x}_B = B^{-1}\mathbf{b} = I_m\,\mathbf{b} = \mathbf{b} \geq o$$

Thus the initial B.F.S.is

$$x_{n+1} = x_{B1} = b_1, x_{n+2} = x_{B2} = b_2, \dots x_{n+m} = x_{Bm} = b_m.$$

Which can be *obtained by writing all the non-basic variables (i.e given variables) $x_1, x_2, \dots, x_n$ equal to zero and solving the equations for the remaining variables (i.e. slack variables)* $x_{n+1}, x_{n+2}, \dots, x_{n+m}$.

Case II. To find Initial B.F.S. when all the original constraints have $\geq$ sign.

In this case all the constraints are converted into equations by inserting surplus variables. Inserting surplus variables the equations obtained are as follows.

$$\begin{array}{lllll}
a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n & - x_{n+1} & & & = b_1 \\
a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n & & - x_{n+2} & & = b_2 \\
\dots\ \dots\ \dots\ \dots\ \dots & \dots & \dots & \dots & \dots \\
\dots\ \dots\ \dots\ \dots\ \dots & \dots & \dots & \dots & \dots \\
a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n & & & - x_{n+m} & = b_m
\end{array}$$

Here $x_{n+1}, x_{n+2}, \dots, x_{n+m}$ are surplus variables.

In matrix form, these equations can be written as

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surplus vectors

$$\begin{bmatrix} a_{11} & a_{12} \cdots a_{1n} & -1 & 0\ldots 0 \\ a_{21} & a_{22}\cdots a_{2n} & 0 & -1\ldots 0 \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} \; a_{mn} & 0 & 0\ldots -1 \end{bmatrix} \begin{bmatrix} x_1\\ x_2\\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \\ x_{n+m}\end{bmatrix} = \begin{bmatrix} b_1\\ b_2\\ \vdots\\ b_m\end{bmatrix}$$

Here taking the initial basis matrix $B = -I_m$.

we have $x_B = B^{-1}b = -I_m b = -b \leq 0$

This basic solution is not B.F.S.

In this case to avoid this difficulty we add one more variable to each constraint. These variables are called "artificial variables".

Adding surplus and artificial variables are called, the constraints of a given L.P.P. are converted to the following equations.

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n - x_{n+1} \quad & + x_{n+m+1} &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n \quad - x_{n+2} \quad & + x_{n+m+2} &= b_2 \\ \cdots \quad \cdots \quad \cdots \quad & \cdots &\cdots \\ \cdots \quad \cdots \quad \cdots \quad & \cdots &\cdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n \quad - x_{n+m} \quad & + x_{n+m+m} &= b_m \end{aligned}$$

Here $x_{n+m+1}, x_{n+m+2}, \ldots, x_{n+m+m}$ are the artificial variables.

In matrix form, these equations can be written as

Artificial vectors

$$\begin{bmatrix} a_{21} & a_{12} & \cdots & a_{1n} & -1 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ a_{21} & 22 & \cdots & a_{2n} & 0 & -1 & \cdots & 0 & 0 & 1 & \cdots & 0 \\ \cdots & & \cdots & & & & \cdots & & & & \cdots & \\ \cdots & & \cdots & & & & \cdots & & & & \cdots & \\ a_{m1} & a_{m2} & \cdots & a_{mn} & 0 & 0 & \cdots & -1 & 0 & 0 & \cdots & 1 \end{bmatrix} \times$$

$$\begin{bmatrix} x_1\\ x_2\\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \\ x_{n+m} \\ x_{n+m+1} \\ \vdots \\ \vdots \\ x_{n+m+m}\end{bmatrix} = \begin{bmatrix} b_1\\ b_2\\ \vdots\\ b_m\end{bmatrix}$$

Here we can take the basis matrix $B = I_m$.

$\therefore \quad x_B = B^{-1}b = I_m b = b \geq 0$

which is a B.F.S.

Thus in this case the B.F.S. is

$x_{n+m+1} = x_{B1}, = b_1, x_{n+m+2} = x_{B2} = b_2 \ldots, x_{n+m+m} = x_{B3} = b_m$.

Which can be *obtained by writting all the non-basic variables (i.e. given variables and surplus variables)* $x_1, x_2, \ldots, x_n; x_{n+1}, x_{n+2}, \ldots, x_{n+m}$ *equal to zero and solving the equations for the remaining basic variables (i.e. artificial variables)* $x_{n+m+1}, x_{n+m+2}, \ldots, x_{n+m+m}$.

Case III. To find initial B.F.S when the constraints have '$\leq$', '$\geq$' and '$=$' signs.

In this case the constraints are converted into equations by inserting slack, surplus and artificial variables. Here the basis matrix $B = I_m$ is obtained by taking the slack and aritfical vectors.

In ths case also *the initial BS.F. is obtained by writing all the non-basic variables equal to zero and solving for remaining basic variables. Here also the initial B.F.S.* $x_B = b \geq 0$.

Note 1. It is important to note that in all the three cases the initial B.F.S. consist of the R.H.S. constants $b_1, b_2, \ldots, b_m$, (note that $b_1, b_2, \ldots, b_m$ are all positive).

2. Some times it happens that an identity is present in A without introducing the artificial variables. So we do not introduce the artificial variables in such case and get a B.F.S. with its basis matrix $B = I_m$. (See Ex. 8).

§ 3·12. Computational procedure of the simplex method for the solution of a maximization L.P.P. [Meerut 87, 95 (P)]

It consists of the following steps systematically.

Step 1. If the problem is of minimization, convert it into the maximization problem.

The values of $x_1, x_2, \ldots, x_n$ that minimize the function

$$Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n.$$

maximize the function

$$Z' = -Z = -C_1x_1 - c_2x_2 - \ldots - c_nx_n.$$

Thus in the problems of minimization, we maximize the function $Z' = -Z$. If $Z' = v$ is the maximum value of Z' then $-v$ will be the minimum value of Z.

Step 2. Make all the bi's positive. If any of the bi's is negative, multiply the corresponding constraint by -1.

Step 3. (a) Convert the constraints into equations by introducing the non negative slack or surplus variables defined in § 3·2.

Also introduce **arificial variables** in the constrains where surplus variables are inserted and which do not form the column of unit matrix.

Step 4. To find initial B.F.S. Find the initial B.F.S. by the method of § 3·11. If in a **L.P.P.** artificial variables are also introduced in step 3 then we follow two-phase method. In phase 1 we proceed to get the starting B.F.S. in terms of non-artificial variables (See § 3·13). Big M-Method can also be used in such cases.

5. Construction of starting simplex table.

Now we construct the starting simplex table as follows : (Given on page 81)

Here the columns corresponding to the coefficients of $x_1, x_2, \ldots$, are shown by $Y_1, Y_2, \ldots$, and c_j is the row of coefficients of the variables in the objective function.

Step 6. Test of starting B.F.S. for optimality.

This is done by computing an evaluation Δ_j for each non-basic variable (zero variables) x_j by the formula $\Delta_j = c_j - C_B Y_j$.

Note that Δ_j for basic variables are zero.

Insert the value of Δ_j's for zero variables as a row in the starting simplex table as shown in table 3·1 (on page 81).

(i) If $\Delta_j \le 0$ for each j, the solution under test is **optimal.**

(a) If none of Δ_j is positive, but any zero, then other optimal solution exist with the same value of Z.

Starting Simplex Table

Table 3·1

B	C_B	$\frac{c_j}{X_B}$	$\frac{c_1}{Y_1(=\alpha_1)}$	$\frac{c_2}{Y_2(=\alpha_2)}$	...	$\frac{c_k}{Y_k(=\alpha_k)}$	...	$\frac{c_n}{Y_n(=\alpha_n)}$	$\frac{c_{n+1}}{Y_{n+1}(\beta_1)}$	$\frac{c_{n+2}}{Y_{n+2}(\beta_2)}$	...	$\frac{c_{n+m}}{Y_{n+m}(\beta_m)}$	Mini-Ratio
y_{n+1}	$c_{B1}=0$	$x_{B1}=b_1$	$y_{11}=a_{11}$	$y_{12}=a_{12}$	...	$y_{1k}=a_{1k}$	...	$y_{1n}=a_{1n}$	1	0	...	0	
y_{n+2}	$c_{B2}=0$	$x_{B2}=b_2$	$y_{21}=a_{21}$	$y_{22}=a_{22}$	...	$y_{2k}=a_{2k}$	...	$y_{2n}=a_{2n}$	0	1	...	0	→
⋮	⋮	⋮	⋮	⋮	...	⋮	...	⋮	⋮	⋮	...	⋮	
y_{n+r}	$c_{Br}=0$	$x_{Br}=b_r$	$y_{r1}=a_{r1}$	$y_{r2}=a_{r2}$	...	$\boxed{y_{rk}=a_{rk}}$	...	$y_{rn}=a_{rn}$	⋮	⋮	...	⋮	
⋮	⋮	⋮	⋮	⋮	...	⋮	...	⋮	⋮	⋮	...	⋮	
y_{n+m}	$c_{Bm}=0$	$x_{Bm}=b_m$	$y_{m1}=a_{m1}$	$y_{m2}=a_{m2}$	...	$y_{mn}=a_{mn}$	...	$y_{mn}=a_{mn}$	0	0	...	1	
$Z = C_B X_B = 0$		Δ_j	Δ_1	Δ_2		Δ_k		Δ_n	Δ_{n+1}	Δ_{n+2}		Δ_{n+m}	

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(ii) If all of Δ_j are negative (less than zero), the solution under test is **unique optimal solution.**

(iii) If $\Delta_j > 0$ for and j *i.e.*, if one or more of Δ_j are positive, the solution under test is not optimal.

If the solution under test is not optimal, we must proceed to the next step.

(c) If corresponding to maximum positive Δ_j all the elements in the column Y_j are negative or zero, then the solution under test will be **unbounded.**

(d) If the value of at least one artificial variable appearing in the basis is non-zero and the optimality condition is satisfied, then we shall say that the problem has **no feasible solution.**

Step 7. To find in-coming (or entering) and out going vectors.

To improve the above solution (which is not optimal) we find the vector entering the basis matrix (called **in-coming vector**) and the vector to be removed from the basis matrix (called **out going vector**) by the following rules.

To find in-coming vector. The incoming vector will be taken as α_k if $\Delta_k = \text{Max.}\, \Delta_j$.

If max, value of Δ_j occurs for more than one α_j, then any one of these may be taken as an incoming vector.

To find out going vector. The out going vector β_r is taken corresponding to that value of r for which.

$$\frac{x_{Br}}{y_{rk}} = \underset{i}{Mini}\left\{\frac{x_{Bi}}{y_{ik}}, y_{ik} > 0\right\},$$

when α_k is the in-coming vector.

If minimum is not unique, *i.e.*, the minimum occurs at more than one value of i, then more than one variable will vanish in the next solution. Therefore the next solution will become a degenerate B.F.S. for which the out going vector is selected in a different way discussed in chapter 4.

Step 8. When α_k is the incoming vector and β_r the out going vector then the element $y_{rk} (= a_{rk})$ is called the **key element** or **pivot element** which is at the intersection of minimum ratio arrow ($\leftarrow$) and in coming vector arrow ($\uparrow$). We mark this element in $\square$ as shown in table 3·1.

In order to bring β_r in place α_k (Y_k) there shhould be unity at the position $\square$ *i.e.*, the key element $y_{rk} (= a_{rk})$ should be equal to unity (= 1). If it is not 1 then divide all the element of this row by this key element a_{rk} : Then subtract appropriate multipliers of this row (containing key element) from all the other rows and obtain zero (= 0) at all other positions of this column $\alpha_k (= Y_k)$. Now bring β_r in place of $\alpha_k (= Y_k)$ and construct new (revised) simplex table.

In this way we get improved basic feasiable solution.

Step 9. Now test the above improved B.F.S. for optimality as in step 6.

If this solution is not optimal then repeat step (7) and (8) until an optimum solution is finally obtained.

For the clear understanding of the procedure, few illustrative examples are given here.

Illustrative Examples

Ex.3. *Solve the L.P. problem.*

Maximize $Z = 3x_1 + 5x_2 + 4x_3$

subject to
$$2x_1 + 3x_2 \le 8$$
$$2x_2 + 5x_3 \le 10$$
$$3x_1 + 2x_2 + 4x_3 \le 15$$

and $x_1, x_2, x_3 \ge 0.$ **[Meerut 86, 90 (T.D.C.), 95]**

Sol. We shall solve this example step wise so that the students may understand the procedure of the simplex method.

Step 1. The problem is a problem of maximization.

Step 2. All the bi's are already positive.

Step 3. Now the inequalities are converted to equations by the introduction of slack variables x_4, x_5 and x_6 as follows.

$$2x_1 + 3x_2 + 0.x_3 + x_4 = 8$$
$$0.x_1 + 2x_2 + 5x_3 + x_5 = 10$$
$$3x_1 + 2x_2 + 4x_3 + x_6 = 15$$

Step 4. Taking $x_1 = 0, x_2 = 0, x_3 = 0$, we get $x_4 = 8, x_5 = 10$ and $x_6 = 15$ which is the starting B.F.S.

Step 5. Now we construct the starting simplex table 3·2.

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Starting Simplex Table.

Table 3·2

		c_j	3	5	4	0	0	0	Mini Ra[illegible]
B	C_B	X_B	Y_1 (α_1)	Y_2 (α_2)	Y_3 (α_3)	Y_4 (β_1)	Y_5 (β_2)	Y_6 (β_3)	$\frac{X_B}{Y_2}$
Y_4	0	8	2	$\boxed{3}$	0	1	0	0	8/3 (Mi[illegible] $\longrightarrow$
Y_5	0	10	0	2	5	0	1	0	10/2=[illegible]
Y_6	0	15	3	2	4	0	0	1	15/2
$Z = C_B X_B = 0.$		Δ_j	3	5	4	0	0	0	

$\uparrow$ incoming vector $\downarrow$ outgoing vector

Step 6. Here we compute Δ_j for all zero variab[illegible] (non-basic) $x_j, i = 1, 2, 3$ by the formula.

$$\Delta_j = c_j - C_B Y_j$$

$$\Delta_1 = c_1 - C_B Y_1 = 3 - (0, 0, 0), (2, 0, 3) = 3$$

$$\Delta_2 = c_2 - C_B Y_2 = 5 - (0, 0, 0), (3, 2, 2) = 5$$

$$\Delta_3 = c_3 - C_B Y_3 = 4 - (0, 0, 0), (0, 5, 4) = 4.$$

Since all Δ_j are not less than or equal to zero, therefore [illegible] solution is not optimal.

So we proceed to the next step.

Step 7. To find incoming vector.

Since $\Delta_2 = 5$ is max. of $\Delta_1, \Delta_2, \Delta_3$, $\therefore \alpha_2 (= Y_2)$ is inco[illegible] vector.

To find outgoing vector.

Since $Y_2 (\alpha_2)$ is incoming vector therefore we consider [illegible] ratio.

$$\frac{X_B}{Y_2} = \left(\frac{x_{B1}}{y_{12}}, \frac{x_{B2}}{y_{22}}, \frac{x_{B3}}{y_{33}}\right) = \left(\frac{8}{3}, \frac{10}{2}, \frac{15}{2}\right).$$

$$\frac{x_{Br}}{y_{r2}} = \underset{i}{Mini}\left[\frac{x_{Bi}}{y_2}, y_2 > 0\right] = \underset{i}{Mini}\left(\frac{x_{B1}}{y_{12}}, \frac{x_{B2}}{y_{22}}, \frac{x_{B3}}{y_{22}}\right) = \frac{8}{3} = [illegible]$$

$\therefore r = 1$ *i.e.*, $\beta_1 (= Y_4)$ is the outgoing vector.

Step 8. Since $Y_2 (\alpha_2)$ is incoming vector and $\beta_1 (= Y_4)$ is outgoing vector.

$\therefore$ the key element is $y_{12} (= a_{12})$ as shown in the table 3·2 which is equal to 3 (not equal to 1).

$\because$ In order to bring Y_2 in place of Y_4 in the basis we make the following operations.

Divide the first row containing key element $a_{12} = 3$ by 3, to get unity at this position and then subtract 2 times of the first row (obtained after dividing by 3) from the second and the third rows and subtract 5 times of this row from the second and the third rows and subtract 5 times of this row from the row Δ_j's to get the new values of Δ_j's. *

Thus we construct the second simplex table in which $\beta_i (= Y_4)$ is replaced by $\alpha_2 (= Y_2)$ as follows.

Second Simplex Table.

Table 3·3

		c_j	3	5	4	0	0	0	
B	C_B	X_B	Y_1 (α_1)	Y_2 (β_1)	Y_3 (α_3)	Y_4 (α_4)	Y_5 (β_2)	Y_6 (β_3)	Mini Ratio X_B/Y_3
Y_2	5	8/3	2/3	1	0	1/3	0	0	–
Y_5	0	14/3	–4/3	0	$\boxed{5}$	–2/3	1	0	14/15(Mini) $\longrightarrow$
Y_6	0	29/3	5/3	0	4	–2/3	0	1	29/12
$Z = C_B X_B = \frac{40}{3}$		Δ_j	–1/3	0	4	–5/3	0	0	

$\uparrow$ Incoming vector $\downarrow$ outgoing vector

To check the values of Δ_j, we also compute Δ_j's by using the formula $\Delta_j = c_j - C_B Y_j$, for x_1, x_3, and x_4.

$$\Delta_1 = c_1 - C_B Y_1 = 3 - (5, 0, 0).\left(\tfrac{2}{3}, -\tfrac{4}{3}, \tfrac{5}{3}\right) = 3 - \tfrac{10}{3} = -1/3$$

$$\Delta_3 = c_3 - C_B Y_3 = 4 - (5, 0, 0).(0, 5, 4) = 4$$

$$\Delta_4 = c_4 - C_B Y_4 = 0 - (5, 0, 0).\left(\tfrac{1}{3}, -\tfrac{2}{3}, -\tfrac{2}{3}\right) = -5/3.$$

* The values of Δ_j's for the new simplex table are also calculated by row operations. Thus a lot of time is saved. Even then the students are advised to check them by computing form the formula $\Delta_j = c_j - C_B Y_j$.

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The values of $\Delta_2, \Delta_5, \Delta_6$ will be zero as they correspond to unit column vectors.

Since in table 3·3 all Δ_j are not less than or equal to zero, therefore this solution is also not optimal.

Since $\Delta_3 = 4$ is max of the Δ_j's.

$\therefore \quad \alpha_3 = (= Y_3)$ is the incoming vector.

$$\text{Also } \frac{x_{Br}}{y_{r3}} = \underset{i}{\text{Mini}} \left[\frac{x_{Bi}}{y_{i3}}, y_{i3} > 0 \right]$$

$$= \text{Mini} \left[\frac{x_{B2}}{y_{23}}, \frac{x_{B3}}{y_{33}} \right], \frac{x_{B1}}{y_{13}} \text{ not considered } \because y_{13} = 0$$

$$= \text{Mini} \left[\frac{14}{15}, \frac{29}{12} \right] = \frac{14}{15} = \frac{x_{B2}}{y_{23}} \therefore r = 2.$$

i.e., $\beta_2 (= Y_5)$ is the outgoing vector and $y_{23} = a_{23} = 5$ is the key element.

In order to bring Y_3 in place of $\beta_2 = (Y_5)$, we divide the 2nd row containing the key element by 5 to get 1 at this position, then subtract 4 times of the second row thus obtained from third row and also from the row of Δ_j's.

$\therefore$ The third simplex table 3·4 in which $\beta_2 (= Y_5)$ is replaced by Y_3 is as follows.

Table 3·4

B	C_B	c_j / X_B	3 / Y_1 (α_1)	5 / Y_2 $(=\beta_1)$	4 / Y_3 $(=\beta_2)$	0 / Y_4 (α_4)	0 / Y_5 (α_5)	0 / Y_6 $(=\beta_3)$	Mini Ratio X_B / Y_1
Y_2	5	8/3	2/3	1	0	1/3	0	0	4
Y_3	4	14/15	-4/15	0	1	-2/15	1/5	0	$-\frac{7}{2}$ (Neg).
Y_6	0	89/15	$\boxed{\frac{41}{15}}$	0	0	-2/15	-4/15	1	$\frac{89}{41}$ (Mini) $\longrightarrow$
$Z = C_B X_B = 256/15$		Δ_j	11/15	0	0	$-\frac{17}{15}$	-4/5	0	

$\uparrow$ Incoming vector (under Y_1) $\qquad \uparrow$ Outgoing vector

The check the values of Δ_j, we again compute Δ_j by using the formula $\Delta_j = c_i - C_B Y_j$.

$\Delta_1 = c_1 - C_B Y_1 = 3 - (5, 4, 0)(2/3, -4/15, 41/15)$
$= 3 - \{10/3 - 16/15\} = 11/15$

$\Delta_4 = c_4 - C_B Y_4 = 0 - (5, 4, 0).(1/3, -2/15, -2/15) = -17/15$

$\Delta_5 = c_5 - C_B Y_5 = 0 - (5, 4, 0).(0, 1/5, -4/5) = -4/5.$

Also $\Delta_2 = 0 = \Delta_3 = \Delta_6$.

Since all Δ_j are not less than or equal to zero therefore the solution is not optimal.

Since Δ_1 is max. of all the Δ_j's.

$\therefore \quad \alpha_1 (= Y_1)$ is the incoming vector.

$$\text{Also} \quad \frac{x_{Bi}}{y_{r1}} = \underset{i}{\text{Mini}} \left[\frac{x_{Bi}}{y_{i1}}, y_{i1} > 0 \right]$$

$$= \text{Mini} \left[\frac{x_{B1}}{y_{11}}, \frac{x_{B3}}{y_{31}} \right], \frac{x_{B2}}{y_{21}} \text{ not considered}$$

as y_{21} is negative as

$$= \text{Mini} \left[4, \frac{89}{41} \right] = \frac{89}{41} = \frac{x_{B3}}{y_{31}}$$

$\therefore \quad r = 3.$

i.e. $\beta_3 (= Y_6)$ is the outgoing vector and $y_{31} = a_{31} = \frac{41}{15}$ is the key element.

Again in order to bring y_1 in place of $\beta_3 (Y_6)$ we divide the 3rd row containing the key element by 41/15 to get 1 at this position, then subtract 2/3 times of the third row (thus obtained) from the second row and from first row, add 4/15 times of third row to the row of Δ_j's. subtract 11/15 times of third row from the row of Δ_j's.

Thus the fourth simplex table in which $\beta_3 (= Y_6)$ is replaced by Y_1 is as follows. (See table 3·5 on page 88)

To check the value of Δ_j's we also compute Δ_j by using the formula $\Delta_j = c_j - C_B Y_j$.

$$\Delta_4 = c_4 - C_B Y_4 = 0 - (5, 4, 3) \left(\frac{15}{41}, \frac{-6}{41}, \frac{-2}{41} \right) = -\frac{45}{41}.$$

Similarly $\Delta_5 = -24/41, \Delta_6 = -11/41, \Delta_1 = 0 = \Delta_2 = \Delta_3$.

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Table 3·5

B	C_B	c_j / X_B	3 / Y_1 (β_3)	5 / Y_2 (β_1)	4 / Y_3 (β_2)	0 / Y_4 (α_4)	0 / Y_5 (α_5)	0 / Y_6 (α_6)	Mini Ratio
Y_2	5	$\frac{50}{41}$	0	1	0	$\frac{15}{41}$	$\frac{8}{41}$	$-\frac{10}{41}$	
Y_3	4	$\frac{62}{41}$	0	0	1	$-\frac{6}{41}$	$\frac{5}{41}$	$\frac{4}{41}$	
Y_1	3	$\frac{89}{41}$	1	0	0	$-\frac{2}{41}$	$-\frac{12}{41}$	$\frac{15}{41}$	
$Z = C_B X_B = \frac{765}{41}$		Δ_j	0	0	0	$-\frac{45}{41}$	$-\frac{24}{41}$	$-\frac{11}{41}$	

Since all the Δ_j 's for non-basic variables are negative so this solution is optimal.

Optimal solution is $x_1 = 89/41, x_2 = 50/41, x_3 = 62/41$

and Max. $Z = Rs. 765/41$.

All tables 3·2, 3·4 and 3·5 can be computed in a single table as in table 3·6.

Table 3·6

B	C_B	c_j / X_B	3 / Y_1	5 / Y_2	4 / Y_3	0 / Y_4	0 / Y_5	0 / Y_6	Mini Ratio X_B/Y_2
Y_4	0	8	2	$\boxed{3}$	0	1	0	0	8/3 (Mini) →
Y_5	0	10	0	2	5	0	1	0	10/2 = 5
Z_6	0	15	3	2	4	0	0	1	15/2
$Z = C_B X B = 0$		Δ_j	3	5 ↑	4	0 ↓	0	0	X_B/Y_3
Y_2	5	8/3	2/3	1	0	1/3	0	0	–
Y_5	0	14/3	−4/3	0	$\boxed{5}$	−2/3	1	0	14/15 (Mini) →
Y_6	0	29/3	5/3	0	4	−2/3	0	1	29/15
$Z = C_B X_B = 40/3$		Δ_j	−1/3	0	4 ↑	−5/3	0 ↓	0	X_B/Y_1

B	C_B	c_j / X_B	3 / Y_1	5 / Y_2	4 / Y_3	0 / Y_4	0 / Y_5	0 / Y_6	Mini Ratio X_B/Y_2
Y_2	5	8/3	2/3	1	0	1/3	0	0	4
Y_3	4	14/15	− 4/15	0	1	−2/15	1/5	0	–
Y_6	0	89/15	$\boxed{41/15}$	0	0	−2/15	−4/5	1	89/41 (Mini) →
$Z = C_B X_B = 256/45$		Δ_j	11/15 ↑	0	0	−17/15	−4/5	0 ↓	
Y_2	5	50/41	0	1	0	15/41	8/41	−10/41	
Y_2	4	62/41	0	0	1	− 6/41	5/41	4/41	
Y_1	3	89/41	1	0	0	− 2/41	−12/41	15/41	
$Z = C_B X_B = 765/41$		Δ_j	0	0	0	− 45/41	−24/41	11/41	

Ex. 4. *Solve by simplex method the following L. P. problem.*

Minimize $Z = x_1 - 3x_2 + 2x_3$

Subject to
$$3x_1 - x_2 + 2x_3 \le 7$$
$$-2x_1 + 4x_2 \le 12$$
$$-4x_1 + 3x_2 + 8x_3 \le 10$$
$$x_1, x_2, x_3 \ge 0.$$

[Meerut O. R. 87, L.P. 89, 90]

Sol. First we convert the problem of minimization to maximization problem by taking the objective function as
$$Z' = -Z$$
Max. $Z' = -Z = -x_1 + 3x_2 - 2x_3$.

Now the equations obtained by introducing slack variables x_4, x_5, x_6 are as follows:
$$3x_1 - x_2 + 2x_3 + x_4 = 7$$
$$-2x_1 + 4x_2 + 0x_3 + x_5 = 12$$
$$-4x_1 + 3x_2 + 8x_3 + x_6 = 10$$

Taking $x_1 = 0, x_2 = 0, x_3 = 0$ we get $x_4 = 7, x_5 = 12, x_6 = 10$ which is the starting B.F.S.

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Starting Simplex Table. Table 3·7

B	C_B	c_j	-1	3	-2	0	0	0	Mini Ratio (X_B/Y_2)
		X_B	Y_1 $(=\alpha_1)$	Y_2 $(=\alpha_2)$	Y_3 $(=\alpha_3)$	Y_4 $(=\beta_1)$	Y_5 $(=\beta_2)$	Y_6 $(=\beta_3)$	
Y_4	0	7	3	-1	2	1	0	0	-7 (Neg.)
Y_5	0	12	-2	$\boxed{4}$	0	0	1	0	3(Mini)→
Y_6	0	10	-4	3	8	0	0	1	10/3
$Z = C_B X_B = 0$		Δ_j	-1	3 ↑	-2	0	0 ↓	0	

Incoming vector   Outgoing vector

$$\Delta_1 = c_1 - C_B Y_1 = -1 - (0,0,0).(3,-2,-4) = -1$$

$$\Delta_2 = c_2 - C_B Y_2 = 3 - (0,0,0).(-1,4,3) = 3$$

$$\Delta_3 = c_3 - C_B Y_3 = -2 - (0,0,0).(2,0,8) = -2.$$

Since all Δ_j are not less than or equal to zero, therefore the solution is not optimal.

Here incoming vector is $\alpha_2 (= Y_2)$

and by mini. ratio rule, outgoing vector is $\beta_2 (Y_5)$

$\therefore$ key element $= y_{22} = a_{22} = 4.$

In order to bring $\alpha_2 (= Y_2)$ in plane of $\beta_2 (= Y_5)$, we divide the elements of 2nd row containing key element $a_{22} = 4$ by 4 and add 2nd row thus obtained to 1st row and subtract its 3 times from 3rd row and also from the row of Δ_j 's the second simplex table is as follows.

(See table 3·8 on page 90)

Since all Δ_j are not less than or equal to zero, therefore the solution is not optimal.

Second Simplex Table. Table 3·8

B	C_B	c_j	-1	3	-2	0	0	0	Mini Ratio X_B/Y_1
		X_B	Y_1 (α_1)	Y_2 $(=\beta_2)$	Y_3 (α_3)	Y_4 $(=\beta_1)$	Y_5 (α_5)	Y_6 $(=\beta_3)$	
Y_4	0	10	$\boxed{\frac{5}{2}}$	0	2	1	$\frac{1}{4}$	0	4 (Mini)→
Y_2	3	3	$-\frac{1}{2}$	1	0	0	$\frac{1}{4}$	0	-6 (neg.)
Y_6	0	1	$-\frac{5}{2}$	0	8	0	$-\frac{3}{4}$	1	$-\frac{2}{5}$ (neg.)
$Z' = C_B X_B = 9$		Δ_j	$\frac{1}{2}$ ↑	0	-2	0 ↓	$-\frac{3}{4}$	0	

Here $\Delta_1 = 1/2$ is max.

$\therefore$ Y_1 is the incoming vector and by the minimum ratio rule we find $\beta_1 (= Y_4)$ as the outgoing vector.

$\therefore$ key element $= y_{11} = 5/2.$

In order to bring Y_1 in place of $\beta_1 (= Y_4)$, we divide 1st row containing the key element $y_{11} = 5/2$ by $5/2$ and add $1/2$, $5/2$ and $-1/2$ times the 1st row thus obtained in the 2nd, 3rd and the row of Δ_j's of respectively.

Thus we get the third simplex table as follows :

Third Simplex Table. Table 3·9

B	C_B	c_j	-1	3	-2	0	0	0	Mini-Ratio
		X_B	$Y_1 = \beta_1$	$Y_2 = \beta_2$	Y_3 α_3	Y_4 α_4	Y_5 α_5	Y_6 $= \beta_3$	
Y_1	-1	4	1	0	4/5	2/5	1/10	0	
Y_2	3	5	0	1	2/5	1/5	3/10	0	
Y_6	0	11	0	0	10	1	$-1/2$	1	
$Z' = C_B X_B = 11$		Δ_j	0	0	$-12/5$	$-1/5$	$-4/5$	0	

All tables 3·7, 3·8, 3·9 can be computed in a single table as in table 3·10.

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Table 3·10

		c_j	−1	3	−2	0	0	0	Min. Ratio
B	C_B	X_B	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	X_B/Y_2
Y_4	0	7	3	−1	2	1	0	0	Neg.
Y_5	0	12	−2	$\boxed{4}$	0	0	1	0	12/4 = 3(Mini) →
Y_6	0	10	−4	3	8	0	0	1	10/3
$Z' = C_B X_B = 0$		Δ_j	−1	3	−2	0	0 ↓	0	X_B/Y_1
Y_4	0	10	$\boxed{5/2}$	0	2	1	1/4	0	4 (Mini) →
Y_2	3	3	−1/2	1	0	0	1/4	0	Neg.
Y_6	0	1	−5/2	0	8	0	−3/4	1	Neg.
$Z' = C_B X_B = 9$		Δ_j	1/2 ↑	0	−2	0 ↓	−3/4	0	
Y_1	−1	4	1	0	4/5	2/5	1/10	0	
Y_2	3	5	0	1	2/5	1/5	3/10	0	
Y_6	0	11	0	0	10	1	−1/2	1	
$Z' = C_B X_B = 11$		Δ_j	0	0	−12/5	−1/5	−4/5	0	

Since all Δ_j for all non-basic variables are negative, so this solution is optimal.

Optimal solution is $x_1 = 4, x_2 = 5, x_3 = 0$

and Mini. $Z = -Z' = -11$.

Ex. 5. *Using simplex Algorithm solve the problem*

Max. $Z = 2x_1 + 5x_2 + 7x_3$

subject to
$$3x_1 + 2x_2 + 4x_3 \le 100$$
$$x_1 + 4x_2 + 2x_3 \le 100$$
$$x_1 + x_2 + 3x_3 \le 100$$
$$x_1, x_2, x_3 \ge 0$$

[Agra 86]

Sol. The equations obtained by introducing slack variables x_4, x_5, x_6 are as follows :

$$x_1 + 4x_2 + 2x_3 + x_5 = 100$$
$$x_1 + x_2 + 3x_3 + x_6 = 100$$

Taking $x_1 = 0, x_2 = 0, x_3 = 0$ we get $x_4 = 100, x_5 = 100, x_6 = 100$, which is starting B.F.S.

All computation work is done in the following table 3·11.

Table 3·11

		c_j	2	5	7	0	0	0	Mini. Ratio
B	C_B	X_B	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	X_B/Y_3
Y_4	0	100	3	2	$\boxed{4}$	1	0	0	25(Mini-) →
Y_5	0	100	1	4	2	0	1	0	50
Y_6	0	100	1	1	3	0	0	1	100/3
$Z' = C_B X_B = 0$		Δ_j	2	5	7 ↑	0 ↓	0	0	X_3/Y_2
Y_3	0	25	3/4	1/2	1	1/4	0	0	50
Y_5	0	50	−1/2	$\boxed{3}$	0	−1/2	1	0	50/3 (Min.) →
Y_6	0	25	−5/4	−1/2	0	−3/4	0	1	Neg.
$Z = C_B X_B = 175$		Δ_j	−13/4	3/2 ↑	0	−7/4	0 ↓	0	
Y_3	7	50/3	5/6	0	1	1/3	−1/6	0	
Y_2	5	50/3	−1/6	1	0	−1/6	1/3	0	
Y_6	0	100/3	−4/3	0	0	−5/6	1/6	1	
$Z = C_B X_B = 200$		Δ_j	−3	0	0	−3/2	−1/2	0	

Since all Δ_j's are zero or negative, so the solution is optimal.

$\therefore$ Optimal solution is $x_1 = 0, x_2 = 50/3, x_3 = 50/3$ and Max. $Z = 200$.

§ 3·13. Artificial Variables Technique.

In L.P.P. some constraints may have the sings $\ge$ or $=$ with all b_i's positive. In such problems we introduce surplus variables in the constraints with sign $\ge$. In these problems we can not get the

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starting basic matrix $B = I_m$. So to avoid this difficulty we add one more variable to each of such constraints. These variables are called **'artificial variables'**. As the name implies these variables are fictitious and represent no physical entities. The artificial variable technique is merely a device to get the starting B.F.S. so that we may proceed with simplex method to get the optimal solution. Such problems are solved by two methods.

Method I. Two Phase Method.

Phase I. We first remove the artificial variables from the basis matrix and introduce other variables (non-artificial variables). This part of the solution in which we remove the artificial variables is called phase I.

Phase II . After the removal of artificial variables we proceed for the simplex routine called the phase II.

For the clear understanding of the two pahse method for the solution of a L.P.P. see Ex. 6.

Method II Big M-Method (Carners M-Method). **[Meerut 87]**

We have already discussed in § 3·7, that if the basis B (basis matrix) contains no artificial vector and the optimality condition is satisfied (at any iteration) then the current solution is a B.F.S. of the L.P. problem. On the other hand if the problem does not have a solution then one or more artificial variables will appear in the basis B at a positive level *i.e.* at least one artificial variable will appear in the final solution with positive value. Keeping this in mind a method known as **big M-Method or Carners M-Method** for the solution of L.P.P. involving artificial variables is gives here. In this method a very large negative price say $-M$ is assigned to each artificial variables in the objective function. Due to these negative prices, the objective function cannot be improved in the presence of the artificial variable. So we first remove these artificial vectors from the basis matrix. Once an artificial vectors leaves the basis we forget about it for ever and never consider it as a vector to enter into the basis at any iteration.

For the clear understanding of the procedure see Ex. 7.

IllustrativeExamples

Example Solved by Two Phase Method.

Ex. 6. *Solve the following L. P. problem.*

Min $\quad Z = x_1 + x_2$

subject to $\quad 2x_1 + x_2 \geq 4$

$\quad x_1 + 7x_2 \geq 7$

$x_1 x_2 \geq 0.$ **[Meerut 95; Raj. 87]**

Sol. First we convert the problem of minimization to the maximization problem by taking the objective function as $Z' = -Z$

Max $\quad Z' = -Z = -x_1 - x_2.$

Introducing the surplus variables x_3 and x_4 (positive) we get the following equations

$$\left.\begin{aligned} 2x_1 + x_2 - x_3 \qquad &= 4 \\ x_1 + 7x_2 \qquad - x_4 &= 7 \end{aligned}\right\}$$

Here we cannot get the starting B.F.S. so we introduce the artificial variables (positive variables) x_5 and x_6. Thus the above equations may be written as

$$\left.\begin{aligned} 2x_1 + x_2 - x_3 \qquad + x_5 \qquad &= 4 \\ x_1 + 7x_2 \qquad - x_4 \qquad + x_6 &= 7 \end{aligned}\right\}$$

Clearly $x_5 < x_3, x_6 < x_4$.

The problem will be solved in two phases.

Phase I. *This phase consists of the removal of arificial variables.*

Taking $x_1 = 0, x_2 = 0, x_3 = 0, x_4 = 0$, we get $x_5 = 4, x_6 = 7$.

Now we construct the first table as follows.

Table 3·12

	X_B	Y_1	Y_2	Y_3	Y_4	$A_1(\beta_1)$	$A_2(\beta_2)$	
A_1	4	2	1	−1	0	1	0	
A_2	7	1	$\boxed{7}$	0	−1	0	1	→
	x_j	0	0	0	0	4	7	
			↑				↓	

First we shall remove the artificial variable vectors (Columns). A_1 and A_2 from the basis matrix. In place of artificial variable vectors the entering vector sholud he so chosen that the revised solution is non-negative (B.F.) solution.

We can remove A_2 and introduce Y_2 in its place in the basis matrix. For this we divide the second row by 7 and then subtract this new row from the first row. Thus we get the following table.

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Table 3·13

	X_B	Y_1	$Y_2\,(\beta_2)$	Y_3	Y_4	$A_1\,(\beta_1)$	$A_2^{\bullet}$	
A_1	3	$\boxed{13/7}$	0	−1	1/7	1	−1/7	→
Y_2	1	1/7	1	0	−1/7	0	1/7	
	x_j	0	1	0	0	3	0	
		↑				↓		

It may be seen that if any of Y_1, Y_3, Y_4, is entered in place of A_2 then revised solution is not non-negative. So we can not enter either of them, in place of A_2. Since artificial variable x_6 become zero, so we forget about $A_2^{\bullet}$ for ever and will not consider it in any other table.

Now we proceed to remove A_1 and introduce Y_1 in its place in the basis matrix. For this we multiply first row by 7/13 and subtract 1/7 times of this new row from the second row. Thus we get the following table.

Table 3·14

	X_B	$Y_1\,(\beta_1)$	$Y_2\,(\beta_2)$	Y_3	Y_4	$A_1^{\bullet}$
Y_1	21/13	1	0	−7/13	1/13	7/13
Y_2	10/13	0	1	1/13	−2/13	−1/13
	x_j	21/13	10/13	0	0	0

Since artificial variable x_5 become zero, so we forget about A_1 and will not consider it again.

Thus we get the following solution in phase I.

$$X_1 = \frac{21}{13},\ x_2 = \frac{10}{13},\ x_3 = 0,\ x_4 = 0.$$

Which is the B.F.S. with which we can proceed to get the optimal solution by simplex method.

Phase II. The starting simplex table is as follows.

Table 3·15

		c_j	−1	−1	0	0	
B	C_B	X_B	$Y_1\,(\beta_1)$	$Y_2\,(\beta_2)$	Y_3	Y_4	Mini Ratio
Y_1	−1	21/13	1	0	7/13	1/13	
Y_2	−1	10/13	0	1	1/13	−2/13	
$Z' = C_B X_B = -31/13$		Δ_j	0	0	−6/13	−1/13	

$$\Delta_1 = 0 = \Delta_2$$

$$\Delta_3 = c_3 - C_B Y_3 = 0 - (-1, -1) \cdot (-7/13, 1/13)$$
$$= -(7/13 - 1/13) = -6/13$$

$$\Delta_4 = c_4 - C_B Y_4 = 0 - (-1, -1) \cdot (1/13, -2/13)$$
$$= -(-1/13 + 2/13) = -1/13$$

Since Δ_j's for zero variables are negative, so the solution is optimal.

$\therefore$ the optimal solution is $x_1 = 21/13, x_2 = 10/13$

and Min Z= − Max. Z' = 31/13.

Example Solved by Big M-Method

Ex. 7. *Solve the following L.P. problem*

$$\text{Max. } Z = x_1 + 2x_2 + 3x_3 - x_4$$

subject to
$$x_1 + 2x_2 + 3x_3 = 15$$
$$2x_1 + x_2 + 5x_3 = 20$$
$$x_1 + 2x_2 + x_3 + x_4 = 10$$
$$x_1, x_2, x_3, x_4 \geq 0.$$

[Raj. 86, 88; Meerut 87, 88, 90, 94, 96]

Sol. Examining the constraints carefully we note that in order to get an identity matrix of order 3 × 3 we need two more unit vectors (columns of the unit matrix), as one unit vector (column of unit matrix, formed by coefficients of x_4) is already present in the constraints. Thus we need only two artificial variables in the first two constraints. Introducing the artificial variables x_5 and x_6 in the first two constraints, and assiging large negative price $-M$ to the artificial variables, the problem reduces to the following form.

$$\text{Max. } Z = x_1 + 2x_2 + 3x_3 - x_4 - Mx_5 - Mx_6$$

s.t.
$$x_1 + 2x_2 + 3x_3 + 0.x_4 + x_5 = 15$$
$$2x_1 + x_2 + 5x_3 + 0.x_4 + x_6 = 20$$
$$x_1 + 2x_2 + x_3 + x_4 = 10$$
$$x_i \geq 0,\ \forall\, i = 1, 2, \ldots, 6.$$

Taking $x_1 = 0, x_2 = 0, x_3 = 0$, we get $x_4 = 10, x_5 = 15, x_6 = 20$, which is the starting B.F.S.

All computation work is done in the following table 3·16.

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Table 3·16

B	C_B	c_j / X_B	1 / Y_1	2 / Y_2	3 / Y_3	−1 / Y_4	$-M$ / A_1	$-M$ / A_2	Mini Ratio X_B/Y_3
A_1	$-M$	15	1	2	3	0	1	0	5
A_2	$-M$	20	2	1	$\boxed{5}$	0	0	1	4 (Mini) →
Y_3	−1	10	1	2	1	1	0	0	10
$Z = C_B X_B$ $= -35M - 10$		Δ_j	$3M + 2$	$3M + 4$	$8M + 4$ ↑	0	0	0 ↓	X_B/Y_2
A_1	$-M$	3	−1/5	$\boxed{7/5}$	0	0	1		15/7(Mini) →
Y_3	3	4	2/5	1/5	1	0	0		4
Y_4	−1	6	3/5	9/5	0	1	.0		10/3
$Z = C_B X_B$ $= -3M + 6$		Δ_j	$\frac{(-M+2)}{5}$	$\frac{7M+16}{5}$ ↑	0	0	0 ↓		X_B/Y_1
Y_2	2	15/7	−1/7	1	0	0			Neg.
Y_3	3	25/7	3/7	0	1	0			25/3
Y_4	−1	15/7	$\boxed{6/7}$	0	0	1			5/2 (Mini) →
$Z = C_B X_B$ $= 90/7$		Δ_j	6/7 ↑	0	0	0			
Y_2	2	5/2	0	1	0	1/6			
Y_3	3	5/2	0	0	1	−1/2			
Y_1	1	5/2	1	0	0	7/6			
$Z = C_B X_B$ $= 15$		Δ_j	0	0	0	−1			

Here no $\Delta_j > 0$, so this solution is optimal.

$\therefore$ Optimal solution is,

$$x_1 = 5/2, x_2 = 5/2, x_3 = 5/2, x_4 = 0 \text{ and Max. } Z = 15.$$

Ex. 8. *Using simplex algorithm solve the L.P. problem.*

$$\text{Mini} \quad Z = 4x_1 + 8x_2 + 3x_3$$

subject to

$$x_1 + x_2 \geq 2$$
$$2x_1 + x_3 \geq 5,$$
$$x_1, x_2, x_3 \geq 0.$$

Sol. First we convert the problem of minimization to maximization problem by taking the objective function as Z'.

$\therefore$ Max. $Z' = -Z = -4x_1 - 8x_2 - 3x_3$.

Introducing the surplus variables x_4, x_5 the equations obtained are

$$x_1 + x_2 + 0.x_3 - x_4 = 2$$
$$2x_1 + 0.x_2 + x_3 - x_5 = 5.$$

Examining the equations carefully we note that the columns of the coefficient of x_2 and x_3 form a unit matrix. Therefore there is no need to introduce the artificial variables.

Taking $x_1 = 0, x_4 = 0, x_5 = 0$, we have $x_2 = 2, x_3 = 5$ which is the starting B.F.S.

All computation work is done in the following table 3·17.

Table 3·17

B	C_B	c_j / X_B	−4 / Y_1	−8 / Y_2	−3 / Y_3	0 / Y_4	0 / Y_5	Mini-Ratio X_B/Y_1
Y_2	−8	2	$\boxed{1}$	1	0	−1	0	2 (Mini.) →
Y_3	−3	5	2	0	1	0	−1	5/2
$Z' = C_B X_B$ $= -31$		Δ_j	10 ↑	0 ↓	0	−8	−3	X_B/Y_4
Y_1	−4	2	1	1	0	−1	0	Neg.
Y_3	−3	1	0	−2	1	$\boxed{2}$	1	1/2 (Mini) →
$Z' = C_B X_B$ $= -11$		Δ_j	0	−10	0 ↓	2 ↑	−3	
Y_1	−4	5/2	1	0	1/2	0	−1/2	
Y_4	0	1/2	0	−1	1/2	1	−1/2	
$Z' = C_B X_B$ $= -10$		Δ_j	0	−8	−1	0	−2	

Here no $\Delta_j > 0$, so this solution is optimal.

$\therefore$ Optimal solution is

$$x_1 = 5/2, x_2 = 0, x_3 = 0 \text{ and Mini } Z = -Z' = 10.$$

Problem having no Feasible Solution

Ex. 9. *Solve the L.P.P.*

$$\textit{Max.} \quad Z = -x_1 - x_2$$

$$\textit{s.t.} \quad 3x_1 + 2x_2 \geq 30$$

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$$-2x_1 + 3x_2 \le -30$$
$$x_1 + x_2 \le 5.$$
$$x_1, x_2 \ge 0.$$

Sol. Multiplying second constraint by -1 (to make the right hand side positive), and adding slack and surplus variables, the given problem reduces to the form.

Max. $Z = -x_1 - x_2 + 0.x_3 + 0.x_4 + 0.x_5$

s.t.
$$3x_1 + 2x_2 - x_3 = 30$$
$$2x_1 - 3x_2 - x_4 = 30$$
$$x_1 + x_2 + x_5 = 5$$

In order to get a unit matrix I_3 we have to add two artificial variables x_6 and x_7 in the first two constraints. Thus assigning a large negative price vector $-M$ to the artificial variables the given problem reduces to the following form

Max. $Z = -x_1 - x_2 + 0.x_3 + 0.x_4 + 0.x_5 - Mx_6 - Mx_7$
$$3x_1 + 2x_2 - x_3 + 0.x_4 + 0.x_5 + x_6 + 0.x_7 = 30$$
$$2x_1 - 3x_2 + 0.x_3 - x_4 + 0.x_5 + 0.x_6 + x_7 = 30$$
$$x_1 + x_2 + 0.x_3 + 0.x_4 + x_5 + 0.x_6 + 0.x_7 = 5$$

Taking $x_1 = 0, x_2 = 0, x_3 = 0, x_4 = 0$ we get $x_5 = 5, x_6 = 30, x_7 = 30$, which is the starting B.F.S.

All computation work is done in the following table 3·18.

Table 3·18

B	C_B	c_j / X_B	-1 / Y_1	-1 / Y_2	0 / Y_3	0 / Y_4	0 / Y_5	$-M$ / A_1	$-M$ / A_2	Mini Ratio X_B/Y_1
A_1	$-M$	30	3	2	-1	0	0	1	0	10
A_2	$-M$	30	2	-3	0	-1	0	0	1	15
A_6	0	5	$\boxed{1}$	1	0	0	1	0	0	5 (Mini)
$Z = C_B X_B = -60M$		Δ_j	$5M-1$ ↑	$-M$	$-M$	$-M$	0 ↓	0	0	
A_1	$-M$	15	0	-1	-1	0	-3	1	0	
A_2	$-M$	20	0	-5	0	-1	-2	0	1	
Y_1	-1	5	1	1	0	0	1	0	0	
$Z = 35M - 5$		Δ_j	0	$-6M$	$-M$	$-M$	$-5M$	0	0	

Here no $\Delta_j > 0$. Hence the optimality condition is satisfied and therefore this solution is optimal. $\therefore$ Optimal solution is

$x_1 = 5, x_2 = 0, x_3 = 0, x_4 = 0, x_5 = 0, x_6 = 15, x_7 = 20.$

Here the artificial vectors A_1, A_2 appear in the basis at positive level, which immediately indicates that the given problem has no feasible solution.

Problem having an Unbounded Feasible solution but bounded optimal solution.

Ex. 10. *Solve the L.P.P.*

Max. $Z = 6x_1 - 2x_2$

s.t.
$$2x_1 - x_2 \le 2$$
$$x_1 \le 4$$
$$x_1 + x_2 \ge 0.$$

Sol. Adding slack variables x_3 and x_4, the inequalities reduces to the following equations
$$2x_1 - x_2 + x_3 = 2$$
$$x_1 + 0.x_2 + x_4 = 4$$

Taking $x_1 = 0$, $x_2 = 0$, we get $x_3 = 2, x_4 = 4$, which is the starting B.F.S.

All computation work is done in the following table 3·19.

Table 3·19

B	C_B	c_j / X_B	6 / Y_1	-2 / Y_2	0 / Y_3	0 / Y_4	MiniRatio X_B/Y_1
Y_3	0	2	$\boxed{2}$	-1	1	0	1 (Mini) →
Y_4	0	4	1	0	0	1	4
$Z = C_B X_B = 0$		Δ_j	6 ↑	-2	0 ↓	0	X_B/Y_2
Y_1	6	1	1	$-\frac{1}{2}$	$\frac{1}{2}$	0	Neg.
Y_4	0	3	0	$\boxed{\frac{1}{2}}$	$-\frac{1}{2}$	1	6 (Mini)
$Z = C_B X_B = 6$		Δ_j	0	1 ↑	-3	0 ↓	
Y_1	6	4	1	0	0	1	
Y_2	-2	6	0	1	-1	2	
$Z = C_B X_B = 12$		Δ_j	0	0	-2	-2	

$\therefore$ The optimal solution is $x_1 = 4, x_2 = 6$ and Max. $Z = 12$.

From the starting simplex table we note that the elements of

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column vector Y_2 are negative or zero which indicates that the feasible region is not bounded.

Hence we conclude that a L.P.P. having unbounded feasible solution may have a bounded optimal solution.

Problem having unbounded solution.

Ex 11. *Solve the following L.P.P.*

$$\text{Max.} \quad Z = 3x_1 + 2x_2 + x_3$$

$$\text{s.t.} \quad -3x_1 + 2x_2 + 2x_3 = 8$$

$$-3x_1 + 4x_2 + x_3 = 7,$$

$$\text{and} \quad x_1, x_2, x_3 \geq 0$$

[Meerut 95]

Sol. Introducing the artificial variables, the given L.P.P. reduces to the following form

$$\text{Max.} \quad Z = 3x_1 + 2x_2 + x_3 - Mx_4 - Mx_5$$

$$-3x_1 + 2x_2 + 2x_3 + x_4 = 8$$

$$-3x_1 + 4x_2 + x_3 + x_5 = 7$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

Taking $x_1 = 0, x_2 = 0, x_3 = 0$, we get $x_4 = 8, x_5 = 7$ which is the starting B.F.S.

All computation work is done in the following table 3·20

Table 3·20.

B	C_B	c_j / X_B	3 / Y_1	2 / Y_2	1 / Y_3	$-M$ / A_1	$-M$ / A_2	Mini Ratio X_B/Y_2
A_1	$-M$	8	-3	2	2	1	0	4
A_2	$-M$	7	-3	$\boxed{4}$	1	0	1	$\frac{7}{4}$ (Mini) →
$Z = C_B X_B = -15M$		Δ_j	$-6M+3$	$6M+2$ ↑	$3M+1$	0	0 ↓	X_B/Y_3
A_1	$-M$	9/2	$-3/2$	0	$\boxed{\frac{3}{2}}$	1		3 (Mini) →
Y_2	2	7/4	$-3/4$	1	1/4	0		7
$Z = C_B X_B = 7/2 - 9M/2$		Δ_j	$(-3M+9)/2$	0 ↑	$(3M+1)/2$	0 ↓		X_B/Y_1
Y_3	1	3	-1	0	1			
Y_2	2	1	$-1/2$	1	0			
		Δ_j	5 ↑	0	0			

Here Y_1 is the incoming vector. All the elements of this vector are negative. Thus we cannot select the outgoing vector.

Hence in this case the solution is unbounded.

Ex. 12. *Solve the following L.P.P.*

$$\text{Max.} \quad Z = 2x_1 + x_2$$

$$\text{s.t.} \quad x_1 - x_2 \leq 10$$

$$2x_1 - x_2 \leq 40$$

$$\text{and} \quad x_1 \geq 0, x_2 \geq 0.$$

Sol. Entering the slack variables, the inequalities reduces to the following form.

$$x_1 - x_2 + x_3 = 10$$

$$2x_1 - x_2 + x_4 = 40.$$

Taking $x_1 = 0, x_2 = 0$, we get $x_3 = 10, x_4 = 40$ which is the starting B.F.S.

The starting simplex table is as follows. Table 3·21.

Table 3·21

B	C_B	c_j / X_B	2 / $Y_1(\alpha_1)$	1 / $Y_2(\alpha_2)$	0 / $Y_3(\beta_1)$	0 / $Y_4(\beta_2)$	Mini Ratio
Y_3	0	10	1	-1	1	0	
Y_4	0	40	2	-1	0	1	
$Z = C_B X_B = 0$		Δ_j	2	1	0	0	

Here the column $\alpha_2 \in A$ but $\alpha_2 \notin B$ and for this column $c_j - Z_j = c_2 - C_B Y_2 = 1 > 0$ and $y_{i2} \leq 0, i = 1, 2$. Hence the solution of this problem is unbounded, (See § 3·7).

§ 3·14. L.P.P. With Unrestricted Variable.

If in a L.P.P. a variable say x_i is unrestricted in sign then x_i is replaced by $x_i' - x_i''$ where $x_i', x_i'' \geq 0$, and the problem is solved as usual. In case more than one variable is unrestricted in sign then we replace all these variable by the difference of two non-negative variables and slove as usual L.P.P. method. For clear understanding of the method see Ex. 13.

Ex. 13. Max. $Z = 2x_1 + 3x_2$

$$\text{s.t.} \quad -x_1 + 2x_2 \leq 4$$

$$x_1 + x_2 \leq 6$$

$$x_1 + 3x_2 \leq 9, \quad x_1, x_2 \text{ unrestricted.}$$

Sol. Taking $x_1 = x_1' - x_1''$, $x_2 = x_2' - x_2''$ s.t.

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$x_1', x_1'', x_2', x_2'' \geq 0$ and introducing the slack variables x_3, x_4 and x_5 the equations can be written as

$$\text{Max.} \quad Z = 2x_1' - 2x_1'' + 3x_2' - 3x_2''$$

$$\begin{aligned} -x_1' + x_1'' + 2x_2' - 2x_2'' + x_3 \qquad\qquad &= 4 \\ x_1' - x_1'' + x_2' - x_2'' \qquad + x_4 \qquad &= 6 \\ x_1' - x_1'' + 3x_2' - 3x_2'' \qquad\qquad + x_5 &= 9 \end{aligned}$$

Taking $x_1' = 0 = x_1'' = x_2' = x_2''$, we get $x_3 = 4, x_4 = 6, x_5 = 9$ which is the starting B.F.S.

All computation work is done is the following table.

Table 3·22

B	C_B	C_j / X_B	2 Y_1'	−2 Y_1''	3 Y_2'	−3 Y_2''	0 Y_3	0 Y_4	0 Y_5	Mini Ratio X_B/Y_3'
Y_3	0	4	−1	1	$\boxed{2}$	−2	1	0	0	2 (Mini) →
Y_4	0	6	1	−1	1	−1	0	1	0	6
Y_5	0	9	1	−1	3	−3	0	0	1	3
$Z = C_B X_B = 0$		Δ_j	2	−2	3 ↑	−3	0 ↓	0	0	X_B/Y_1'
Y_2'	3	2	−1/2	1/2	1	−1	1/2	0	0	Neg.
Y_2	3	4	3/2	−3/2	0	0	−1/2	1	0	8/3
Y_3	3	3	$\boxed{5/2}$	−5/2	0	0	−3/2	0	1	6/5 (Mini) →
$Z = C_B X_B = 6$		Δ_j	7/2 ↑	−7/2	0	0	−3/2	0	0 ↓	X_B/Y_3
Y_2'	3	13/5	0	0	1	−1	1/5	0	1/5	13
Y_4	0	11\5	0	G	0	0	$\boxed{2/5}$	1	−3/5	11/2(Mini) →
Y_1'	2	6/5	1	−1	0	0	$\boxed{-3/5}$	0	2/5	Neg.
$Z = C_B X_B = 51/5$		Δ_j	0	0	0	0	3/5 ↑	0 ↓	−7/5	
Y_2'	3	3/2	0	0	1	−1	0	−1/2	1/2	
Y_3	0	11/2	0	0	0	0	1	5/2	−3/2	
Y_1'	2	9/2	1	−1	0	0	0	3/2	−1/2	
$Z = C_B X_B = 27/2$		Δ_j	0	0	0	0	0	−3/2	−1/2	

∴ Since no $\Delta_j > 0$, the optimal solution to the given L.P.P. is

$x_1' = 9/2, x_1'' = 0, x_2' = 3/2, x_2'' = 0$ and Max. $Z = 27/2$.

i.e. $x_1 = x_1' - x_1'' = 9/2, x_2 = x_2' - x_2'' = 3/2$

Max. $Z = 27/2$.

Problem having more than are optimal solution.

Ex. 14. *Solve the L.P. Problem*

$$\begin{aligned} \text{Max.} \quad & Z = 6x_1 + 4x_2 \\ \text{s.t.} \quad & 2x_1 + 3x_2 \leq 30 \\ & 3x_1 + 2x_2 \leq 24 \\ & x_1 + x_2 \geq 3 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Is your answer unique? If not, give 3 different solutions.

Sol. Introducing the slack, surplus and artificial variables, the given L.P.P. reduces to the following form.

$$\text{Max.} \quad Z = 6x_1 + 4x_2 + 0.x_3 + 0.x_4 + 0.x_5 - Mx_6.$$

$$\begin{aligned} \text{s.t.} \quad & 2x_1 + 3x_2 + x_3 \qquad\qquad\qquad = 30 \\ & 3x_1 + 2x_2 \qquad\qquad + x_4 = 24 \\ & x_1 + x_2 \qquad\qquad - x_5 + x_6 = 3 \\ \text{and} \quad & x_1, x_2, \ldots, x_6 \geq 0. \end{aligned}$$

Here x_6 is the artificial variable.

Taking $x_1 = 0$, $x_2 = 0$, $x_5 = 0$, $x_3 = 30$, $x_4 = 24$, $x_6 = 3$, which is the starting B.F.S.

All computational work is done is the following table.

Table 3·23

B	C_B	c_j / X_B	6 Y_1	4 Y_2	0 Y_3	0 Y_4	0 Y_5	−M A_1	Min. Ratio X_B/Y_i
Y_3	0	30	2	3	1	0	0	0	15
Y_4	0	24	3	2	0	1	0	0	8
A_1	−M	3	$\boxed{1}$	1	0	0	−1	1	3 (Min.) →
$Z = C_B X_B = -3M$		Δ_j	6 + M ↑	4 + M	0	0	−M	0	X_B/Y_5
Y_3	0	24	0	1	1	0	2		12
Y_4	0	15	0	−1	0	1	$\boxed{3}$		5 (Min.) →
Y_1	6	3	1	1	0	0	−1		
$Z = C_B X_B = 18$		Δ_j	0	−2	0	0	6 ↑		

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Y_3	0	14	0	$\boxed{5/3}$	1	$-2/3$	0	
Y_5	0	5	0	$-1/3$	0	$1/3$	1	
Y_1	6	8	1	$2/3$	0	$1/3$	0	
$Z = C_B X_B = 48$		Δ_j	0	0	0	-2	0	

Since here all $\Delta_j \le 0$, so this is an optimal solution.

$\therefore$ One optimal solution is $x_1 = 8$, $x_2 = 0$ and Max $Z = 48$.

Since $\Delta_2 = 0$ and $\alpha_2 (= Y_2)$ is not is the basis B

$\therefore$ We enter Y_2 and delete Y_3 from the basis. Thus we get the following simplex table.

Table 3·24.

B	C_B	c_j / X_B	6 / Y_1	4 / Y_2	0 / Y_3	0 / Y_4	0 / Y_5	Min. Ratio
Y_2	4	42/5	0	1	3/5	$-2/5$	0	
Y_5	0	39/5	0	0	1/5	1/5	0	
Y_1	6	12/5	1	0	$-2/5$	3/5	1	
$Z = C_B X_B = 48$		Δ_j	0	0	0	-2	0	

Since all $\Delta_j \le 0$, solution is also an optimal solution, having same max. value of Z.

$\therefore$ Second optimal solution is

$x_1 = 12/5$, $x_2 = 42/5$ and Max. $Z = 48$.

To find third optimal solution.

We know that the convex combination of B.F. solutions is also the optimal solution of the problem.

From the last iteration of table 3·23 and table 3·24 the two B.F. solutions are

$$x_{B1} = \begin{bmatrix} 8 \\ 0 \\ 14 \\ 0 \\ 5 \end{bmatrix}, \qquad x_{B2} = \begin{bmatrix} 12/5 \\ 42/5 \\ 0 \\ 0 \\ 39/5 \end{bmatrix}$$

Let $x^* = \lambda x_{B1} + (1-\lambda) x_{B2} =$ $\begin{bmatrix} 8\lambda + (1-\lambda)\,12/5 \\ 0 + (1-\lambda)\,42/5 \\ 14\lambda \\ 0 \\ 5\lambda + (1-\lambda).39/5 \end{bmatrix}$

$0 \le \lambda \le 1$.

x^* give infinite number of non-basic optimal solution of the given L.P.P., for different values of λ, s.t. $0 \le \lambda \le 1$

Taking a partcular value of λ, say $\lambda = 1/2$, we obtain

$$x^* = \begin{bmatrix} 26/5 \\ 21/5 \\ 7 \\ 0 \\ 32/5 \end{bmatrix}$$

$\therefore$ the third optimal solution is

$x_1 = 26/5$, $x_2 = 21/5$ and Max. $Z = 6 \cdot 26/5 + 4 \cdot 21/5 = 48$.

Here the required three optimal solutions of the given L.P.P. are

$x_1 = 8, x_2 = 0$; $x_1 = 12/5, x_2 = 12/5$; $x_1 = 26/5, x_2 = 21/5$. and Max. $Z = 48$.

§ 3·15. Solution of system of simultaneous linear equations by Simplex Method.

To solve a system of simultaneous linear equations by simplex method, we introduce a dummy objective function Z with cost zero to all given variables and cost -1 to each artificial variable and then solve by simplex method us usual. For clear understanding of the method see Ex. 15.

Ex. 15. *Solve the following system of simultaneous linear equations by usual simplex method.*

$$\begin{aligned} x_1 - x_3 + 4x_4 &= 3 \\ 2x_1 - x_2 &= 3 \\ 3x_1 - 2x_2 - x_4 &= 1 \\ x_1, x_2, x_3, x_4 &\ge 0. \end{aligned}$$

[Meerut 95 (B.P.)]

Sol. Introducing the artificial variables x_5, x_6, x_7, and the dummy objective function Z with costs zero to each given variable x_1, x_2, x_3, x_4 and cost -1 to each artificial variable x_5, x_6, x_7, the given problem, can be written as a L.P.P. in the following form.

Max. $Z = 0.x_1 + 0.x_2 + 0.x_3 + 0.x_4 - 1.x_5 - 1.x_6 - 1.x_7$,

s.t.

$$\begin{aligned} x_1 + 0.x_2 - x_3 + 4x_4 + x_5 &= 3 \\ 2x_1 - x_2 + 0.x_3 + 0.x_4 + x_6 &= 3 \\ 3x_1 - 2x_2 + 0.x_3 - x_4 + x_7 &= 1 \end{aligned}$$

and $x_1, x_2, x_3, x_4, x_5, x_6, x_7, \ge 0$.

Taking $x_1 = 0 = x_2 = x_3 = x_4$, we get $x_5 = 3, x_6 = 3$ and $x_7 = 1$ which is the starting B.F.S.

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All computation work is shown in the following table.

Table 3·25

B	C_B	c_j / X_B	0 / Y_1	0 / Y_2	0 / Y_3	0 / Y_4	−1 / A_1	−1 / A_2	−1 / A_3	Min. Ratio X_B/Y_1
A_1	−1	3	1	0	−1	4	1	0	0	3
A_2	−1	3	2	−1	0	0	0	1	0	3/2
A_3	−1	1	$\boxed{3}$	−2	0	−1	0	0	1	1/3 (Min.) →
$Z = C_B X_B = -7$		Δ_j	6 ↑	−3	−1	3	0	0	0 ↓	X_B/Y_4
A_1	−1	8/3	0	2/3	−1	$\boxed{13/3}$	1	0		8/3 (Min.) →
A_2	−1	7/3	0	1/3	0	2/3	0	1		7/2
Y_1	0	1/3	1	−2/3	0	−1/3	0	0		Neg.
$Z = C_B X_B = -5$		Δ_j	0	1	−1	5 ↑	0 ↓	0		X_B/Y_2
Y_4	0	8/3	0	$\boxed{2/13}$	−3/13	1		0		4 (Min.) →
A_2	−1	25/13	0	3/13	2/13	0		1		25/3
Y_1	0	7/13	1	−8/13	−1/13	0		0		Neg.
$Z = -25/13$		Δ_j	1	3/13 ↑	2/13	0 ↓		0		X_B/Y_3
Y_3	0	4	0	1	−3/2	13/2		0		Neg.
A_2	−1	1	0	0	$\boxed{1/2}$	−3/2		1		2 (Mini) →
Y_1	0	3	1	0	−1	4		0		Neg.
$Z = -1$		Δ_j	0	0	1/2 ↑	−3/2		0 ↓		
Y_2	0	7	0	1	0	2				
Y_3	0	2	0	0	1	−3				
Y_1	0	5	1	0	0	1				
$Z = 0$		Δ_j	0	0	0	0				

Since no $\Delta_j > 0$, ∴ optiomal solution to the given problem is $x_1 = 5, x_2 = 7, x_3 = 2$ and $x_4 = 0$.

§ 3·16. To compute the inverse of a matrix for which one column is different from that of a matrix whose inverse is known.

Let $B = (\beta_1, \beta_2, \ldots, \beta_n)$ be a non-singular matrix whose inverse *i.e.*, B^{-1} is known.

To find the inverse of the matrix

$B_\alpha = (\beta_1, \beta_2, \ldots, \beta_{r-1}, \alpha, \beta_r, \ldots, \beta_n)$.

B_α is obtain from B by replacing its r-th column by the column vector α.

If B and B_α of are considered as bases in E^n, then obviously β_α is obtained from B by changing a single column vector.

Now, if $\alpha = \sum_{i=1}^{n} y_i \beta_i$.

then B_α will be non-singular iff $y_r \neq 0$.

If $y_r \neq 0$, then

$\beta_r = B_\alpha . A_r$

where $A_r = \left[-\dfrac{y_1}{y_r}, \ldots, -\dfrac{y_{r-1}}{y_r}, \dfrac{1}{y_r}, \dfrac{-y_{r+1}}{y_r}, \ldots, \dfrac{-y_n}{y_r}\right]$

obciously and $y = [y_1, y_2, \ldots, y_n] = B^{-1}.\alpha$

$$B = B_\alpha E.$$

where $E = (e_1, e_2, \ldots, e_{r-1}, A_r, e_{r+1}, \ldots, e_n)$

$\Rightarrow \quad B^{-1} = (B_\alpha E)^{-1} = E^{-1} B_\alpha^{-1}$

$\Rightarrow \quad B_\alpha^{-1} = EB^{-1}$

Ex. 15. *Compute the inverse of the matrix*

$$B_\alpha = (\beta_1, \beta_2, \alpha),$$

where $B = (\beta_1, \beta_2, \beta_3)$,

$$\alpha = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \text{ and } B^{-1} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

[Meerut 94 (P), 94 (R) 95, 98]

Sol. We have

$$y = [y_1, y_2, y_3] = B^{-1}.\alpha = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$\Rightarrow \quad y_1 = 2, y_2 = -1, y_3 = 1$

$\therefore \quad A_3 = \left[-\dfrac{y_1}{y_3}, -\dfrac{y_2}{y_3}, \dfrac{1}{y_3}\right] = [-2, 1, 1]$

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$$\therefore \quad E = (e_1, e_2, A_3) = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$\text{Hence } B_a^{-1} = EB^{-1} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & -3 & -5 \\ -2 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

§ 3·17. Inverse of a matrix by Simplex Method

Let A be any $n \times n$ non-singular real matrix.

Then by introducing a dummy $n \times 1$ real matrix **b**, consider the following system of equations.

$$A\mathbf{x} = \mathbf{b}, \qquad \mathbf{x} \geq 0.$$

Introducing the artificial variables $\mathbf{x}_a \geq 0$, and a dummy objective function Z, with cost zero to variables in **x** and cost 1 to each artificial variable find the solution of the following L.P. problem

Max. $\quad Z = 0\mathbf{x} - 1.\mathbf{x}_a$.

s.t. $\quad A\mathbf{x} = \mathbf{b}, \quad \mathbf{x}_a \geq 0$ and $\mathbf{x} \geq 0$.

Then, in the last iteration of the simplex method, when the columns of A becomes the columns of I, the inverse of matrix A is given by those column vectors in the last iteration of the simplex method, which were the columns of the initial basis in the starting of the simplex method.

Ex. 16. *Use simplex method to find the inverse of the following matrix.* $\begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$

[Meerut 97]

Sol. Introducing the dummy real column vector $\mathbf{b} = [4\,6]$, Consider the following system of equations.

$$\begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix} \quad x_1, x_2 \geq 0.$$

Introducing the artificial variables x_3, x_4 and the dummy objective function Z, we obtain the following L.P. problem.

Max. $\quad Z = 0.x_1 + 0.x_2 - 1.x_3 - 1.x_4$

s.t.
$$4x_1 + 0x_2 + x_3 = 4$$
$$3x_1 + 2x_2 + x_4 = 6$$

and $\quad x_1, x_2, x_3, x_4 \geq 0.$

Taking $x_1 = 0, x_2 = 0, x_3 = 4, x_4 = 6$, which is the starting basic feasible solution.

Table 3·26

B	C_B	c_j / X_B	0 / Y_1	0 / Y_2	−1 / A_1	−1 / A_2	Min. Ratio X_B/Y_1
A_1	−1	4	$\boxed{4}$	3	1	0	4/4 = 1 (min.) ⟶
A_2	−1	6	3	2	0	1	6/3 = 2
$Z = C_B X_B$ = − 10		Δ_j	7 ↑	5	0	0	
Y_1	0	1	1	3/4	1/4	0	
A_2	−1	3	0	$\boxed{-1/4}$	−3/4	1	⟶
$Z = C_B X_B$ = − 3		Δ_j	0	− 1/4 ↑	−7/4	0	

Since all Δ_j are less than or equal to zero, so an optimal solution has been obtained. But the given matrix A is still not converted into the unit matrix. In order to do this we introduce Y_2 into the basis and drop A_2 from the basis. Thus we get the following table.

Table 3·27

B	C_B	c_j / X_B	0 / Y_1	0 / Y_2	−1 / A_1	−1 / A_2
Y_1	0	10	1	0	−2	3
Y_2	0	−12	0	1	3	−4
$Z = C_B X_B$ = 0		Δ_j	0	0	−1	−1

Now we arrive to an optimal infeasible solution. But since the columns of given matrix A are converted to the columns of unit matrix, so the column vectors A_1, A_2 of initial basis in the last table, give the inverse of the matrix. A.

$$\text{Hence} \qquad A^{-1} = \begin{bmatrix} -2 & 3 \\ 3 & -4 \end{bmatrix}$$

Exercise on Chapter III

1. Write short note on the simplex method.

or

Give computational procedure for simplex method in linear programming.

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2. What is meant by L.P.P. ? Give a brief description of the problem with illustration. How the same can be solved graphically.
3. What do you understand by optimization ? State clearly a L.P.P. and give an acconut of the simplex method of solving this problem. Make a list of the type of errors which can be made during simplex calculations.
4. (a) Let $x_1 = 2, x_2 = 4$ and $x_3 = 1$ be a F.S. to the system of equations

$$2x_1 - x_2 + 2x_3 = 2, \quad x_1 + 4x_2 = 18,$$

Reduce the F.S. to a B.F.S.

(b) (1, 2, 3,) is a feasible solution to the system of equations

$$x_1 + x_2 + 2x_3 = 4, \quad 2x_1 - x_2 + x_3 = 2$$

Reduce the given F.S. to a B.F.S.

5. $x_1 = 1, x_2 = 1, x_3 = 1, x_4 = 0$ is a F.S. to the system of equations

$$x_1 + 2x_2 + 4x_3 + x_4 = 7, \quad 2x_1 - x_2 + 3x_3 - 2x_4 = 4$$

Reduce the F.S. to two different B.F. solutions.

[Meerut 94 (R), 98 (Old)]

6. Solve the following problems by simplex method.

(i) Max. $Z = 2x_1 + 4x_2 + x_3 + x_4$
s.t. $x_1 + 3x_2 + x_4 \le 4$
$2x_1 + x_2 \le 3$
$x_2 + 4x_3 + x_4 \le 3$
$x_1, x_2, x_3, x_4 \ge 0$
[Meerut 94 (R),98 (Old)]

(ii) Max. $Z = 2x_1 + 4x_2 + x_3$
s.t. $x_1 + 2x_2 \le 4$
$2x_1 + x_2 \le 3$
$x_2 + 4x_3 \le 3$
$x_1, x_2, x_3 \ge 0.$

(iii) Max. $Z = x_1 + x_2 + 3$
s.t. $x_1 + 3x_2 \le 9$
$2x_1 + x_2 \le 8$
$3x_1 + 4x_2 \ge 12$
$x_1, x_2, \ge 0.$ **[Meerut 97]**

(iv) Max. $Z = 2x_1 + 4x_2$
s.t. $2x_1 + 3x_2 \le 48$
$x_1 + 3x_2 \le 42$
$x_1 + x_2 \le 21$
$x_1, x_2 \ge 0.$

(v) Max. $Z = 5x_1 + 3x_2$
s.t. $3x_1 + 5x_2 \le 15$
$5x_1 + 2x_2 \le 10$
$x_1, x_2 \ge 0.$
[Meerut 87]

(vi) Max. $Z = 3x_1 - x_2$
s.t. $2x_1 + x_2 \ge 2$
$x_1 + 3x_2 \le 3$
$x_2 \le 4$
$x_1, x_2 \ge 0.$

(vii) Max. $Z = 3x_1 + 4x_2$
s.t. $x_1 + 3x_2 \le 9$

(viii) Max. $Z = 7x_1 + 5x_2$
s.t. $x_1 + 2x_2 \le 6$

(vii) (contd.)
$2x_1 - x_2 \le 8$
$x_1 + x_2 \le 5$
$x_1, x_2 \ge 0.$ **[Meerut 97 (B.P.)]**

(viii) (contd.)
$4x_1 + 3x_2 \le 12$
$x_1, x_2 \ge 0.$

I (ix) Max. $Z = 4x_1 + 10x_2$
s.t. $2x_1 + x_2 \le 50$
$2x_1 + 5x_2 \le 100$
$2x_1 + 3x_2 \le 90$
$x_1, x_2 \ge 0.$

(x) Max. $Z = 4x_1 + 5x_2 + 9x_3 + 11x_4$
s.t. $x_1 + x_2 + x_3 + x_4 \le 15$
$7x_1 + 5x_2 + 3x_3 - 2x_4 \le 120$
$3x_1 + 5x_2 + 10x_3 + 15x_4 \le 100$
$x_1, x_2, x_3, x_4 \ge 0.$

(xi) Max. $Z = 2x_1 - x_2 + x_3$
s.t. $x_1 + x_2 - 3x_3 \le 8$
$4x_1 - x_2 + x_3 \ge 2$
$2x_1 + 3x_2 - x_3 \ge 4$
and $x_1, x_2, x_3 \ge 0.$

(xii) Max. $Z = 3x_1 + 2x_2$
VII s.t. $2x_1 + x_2 \le 2$
$3x_1 + 4x_2 \ge 12$
$x_1, x_2 \ge 0.$

7. Solve the following problems by simplex method.

(i) Mini $Z = x_1 + x_2 + 3x_3$
VIII s.t. $3x_1 + 2x_2 + x_3 \le 3$
$2x_1 + x_2 + 2x_3 \le 2$
$x, x_2, x_3 \ge 0.$

(ii) Mini $Z = 2x_1 + 9x_2 + x_3$
VI. s.t. $x_1 + 4x_2 + 2x_3 \ge 5$
$3x_1 + x_2 + 2x_3 \ge 4$
$x, x_2, x_3 \ge 0.$

(iii) Mini $Z = x_1 + x_2 + x_3$
s.t. $x_1 - x_4 - 2x_6 = 5$
$x_2 + 2x_4 - 3x_5 + x_6 = 3$
$x_3 + 2x_4 - 5x_5 + 6x_6 = 5$
$x_i \ge 0, (i = 1, 2, \ldots, 6).$

(iv) Mini $Z = 2x_1 + x_2$
s.t. $3x_1 + x_2 = 3$
$4x_1 + 3x_2 \ge 6$
$x_1 + 2x_2 \le 3$
$x_1, x_2 \ge 0.$

(v) Mini $Z = 4x_1 + 2x_2$
s.t. $3x_1 + x_2 \ge 27,$
$x_1 + 2x_2 \ge 30,$
$-x_1 - x_2 \le -21$
and $x_1, x_2 \ge 0.$ **[Meerut 94 (P)]**

8. Use two phase simplex method to solve the following L.P.P.

(i) Max. $Z = 5x_1 + 8x_2$
s.t. $3x_1 + 2x_2 \ge 3$
$x_1 + 4x_2 \ge 4$
$x_1 + x_2 \le 5$
$x_1 \ge 0, x_2 \ge 0.$
[Meerut 95]

(ii) Max. $Z = 3x_1 - x_2$
s.t. $2x_1 + x_2 \ge 2$
$x_1 + x_2 \le 2$
$x_2 \le 4$
$x_1, x_2 \ge 0.$
[Meerut 94 (R), 98 (Old)]

(iii) Max. $Z = 3x_1 + 2x_2 + x_3 + 4x_4$
s.t. $4x_1 + 5x_2 + x_3 - 3x_4 = 5$

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$$2x_1 - 3x_2 - 4x_3 + 5x_4 = 7$$
$$x_1 + 4x_2 + 2.5x_3 - 4x_4 = 7$$

9. Solve the following problems by 'Big M' method,

II (I) Max. $Z = -2x_1 - x_2$

s.t. $3x_1 + x_2 = 3$

$4x_1 + 3x_2 \geq 6$

$x_1 + 2x_2 \leq 4$

$x_1, x_2 \geq 0$ **[Meerut 95]**

III (II) Min. $Z = 5x_1 + 6x_2$

s.t. $2x_1 + 5x_2 \geq 1500$

$3x_1 + x_2 \geq 1200$

$x_1, x_2 \geq 0$

10. Solve the following L.P.P. by simplex method ·

IV Max. $Z = 4x_1 + 3x_2$

s.t. $x_1 + x_2 \leq 50$

$x_1 + 2x_2 \geq 80$

$3x_1 + 2x_2 \geq 140$

and $x_1, x_2 \geq 0$ **[Meerut 94]**

11. A company produces two types of leather belts, say type A and B. Belt A is a superior quality and belt B is of a lower quality. Profits on the two type of belts are 40 and 30 paise per belt respectively. Each belt of type A requires twice as much as required by a belt of type B. If all belts were of type B, the company could produce 1,000 belts per day. But the supply of leather is sufficient only for 800 belts per day. But A requires a fancy buckle and only 400 fancy buckles are available for this, per day. For belt of type B, only 700 buckles are available per day.

How should the company manufacture the two types of belts in order to have maximum over all profit ? **[Rohilkhand 88]**

12. A manufacture of leather belts makes three types of belts A, B and C. Which are processed on three machines M_1, M_2, and M_3. Belt A requires 2 hours on machine M_1 and 3 hours on machine M_3. Belt B requires 3 hours on machine M_1, 2 hours on machine M_2 and 2 hours on machine M_3 and belt C requires 5 hours on machine M_2 and 4 hours on machine M_3. There are 8 hours of time per day available on machine M_1, 10 hours of time per day available on machine M_2 and 15 hours of time per day available on machine M_3. The profit gained from belt A is Rs. 3.00 per unit, from belt B is Rs. 5.00 per unit from belt C is Rs. 4.00 per unit. What should be the daily production of each type of belts so that the profit is maximum.

13. Two products A and B are processed on three machines M_1, M_2, and M_3. The processed times per unit, machine availability and profit per unit are as follows :

Machine	Processing time (hours) A	Processing time (hours) B	Availabilty (hours)
M_1	2	3	1500
M_2	3	2	1500
M_3	1	1	1000
profit per unit	10	12	

Formulate the methematical model, solve it by using the simplex method and also find the number of hours machine M_3 remains unutilised.

14. A manufacturing firm has discontinued production of a certain unprofitable product line. This created considerable excess production capacity. Management is considering to devote this excess capacity to one or more of three products, call them products 1, 2 and 3. The available capacity on the machine which might limit output is summarized in the following table.

	Productivity (hour per unit)			Available time (hours per week)
Product → Machine type ↓	A	B	C	
Milling Machine	8	2	3	250
Lathe	4	3	0	150
Grinder	2	–	1	50

The unit profit would be Rs. 20, 6 and Rs. 8 respectively for products A, B and C. Find how much of each product the firm should produce in order to maximize profit.

15. Solve the following system of simultaneous linear equations by using simplex method.

(i) $x_1 + x_2 = 1,\ 2x_1 + x_2 = 3.$ **[Meerut 90]**

(ii) $3x_1 + 2x_2 = 4,\ 4x_1 - x_2 = 6$ **[Meerut 94]**

(iii)
$$x_1 + 2x_2 + x_3 + x_4 = 10$$
$$x_1 + 2x_2 + 3x_3 = 15$$
$$2x_1 + x_2 + 5x_3 = 20$$

16. Use simplex method to find the inverse of the following matrix

(i) $\begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} 3 & 2 \\ 4 & -1 \end{bmatrix}$

[Meerut 96]

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17. Find all the optimal B.F. solutions of the following L.P.P.

VI.

$$\begin{aligned} \text{Max.}\quad & 6x_1 + 2x_2 + 9 \\ \text{s.t.}\quad & 3x_1 + x_2 \le 6 \\ & x_1 + 3x_2 \le 6 \\ & x_1, x_2 \ge 0. \end{aligned}$$

[Meerut 96 (BP)]

18. Find all optimal basis feasible solutions of the following L.P.P.

$$\begin{aligned} \text{Min.}\quad & 3x_1 + 6x_2 + 7 \\ \text{s.t.}\quad & 2x_1 + x_2 \ge 3 \\ & x_1 + 2x_2 \ge 3 \\ & x_1, x_2 \ge 0. \end{aligned}$$

[Meerut 96]

19. Determine two different B.F. solution of the L.P.P.

$$\begin{aligned} \text{Max.}\quad & Z = 2x_1 + 3x_2 + x_3 + 0.x_4 - M.x_5 - M.x_6 \\ \text{s.t.}\quad & 2x_1 + 3x_2 + 4x_3 + x_4 = 6 \\ & x_1 + 2x_2 + 2x_3 - x_5 + x_6 = 3 \\ \text{and}\quad & x_1, x_2, ..., x_6 \ge 0. \end{aligned}$$

[Meerut 95 (BP)]

20. The constraints of a L.P.P. are as follows :

V.

$$\begin{aligned} & 3x_1 + x_2 \ge 8 \\ & 2x_1 + 5x_2 \le 14 \\ & 3x_1 - x_2 \le 36 \\ & 6x_1 + x_2 \le 99 \\ & 0 \le x_2 \le 15, \quad x_1 \ge 0. \end{aligned}$$

Its objective function is $f(x_1, x_2) = 6x_1 + x_2$.

Find graphically, the maximum and minimum values of the function and the corresponding solutions. Also mark on the graph, the set of feasible solutions at which the value of the objective function is 30. [Meerut 95 (BP)]

21. Consider a matrix $B = (b_1, b_2, b_3)$ whose inverse is

$$B^{-1} = \begin{bmatrix} 2 & 0 & 4 \\ 5 & 1 & 6 \\ 7 & 9 & 3 \end{bmatrix}$$

Replace the second column of B with $\alpha = [1, 8, 6]$, to yield the matrix $B\alpha = (b_1, \alpha, b_3)$.

Compute B_α^{-1}.

ANSWERS

4. (a) $x_1 = 26/9, x_2 = 34/9, x_3 = 0.$ (b) $x_1 = 2, x_2 = 2, x_3 = 0.$

5. $x_1 = 0, x_2 = 1/2, x_3 = 3/2, x_4 = 0$ and $x_1 = 3, x_2 = 2, x_3 = 0 = x_4$.

6. (i) $x_1 = 1, x_2 = 1, x_3 = 1/2, x_4 = 0, Z = 13/2.$

(ii) $x_1 = 2/3, x_2 = 5/3, x_3 = 1/3, Z = 25/3.$

(iii) $x_1 = 3,\ x_2 = 2,\ Z = 8.$

(iv) $x_1 = 6, x_2 = 12, Z = 60.$

(v) $x_1 = 20/19, x_2 = 45/19, Z = 235/19.$

(vi) $x_1 = 3, x_2 = 0,\ Z = 9.$ (vii) $x_1 = 3,\ x_2 = 2,\ Z = 17.$

(viii) $x_1 = 3, x_2 = 0, Z = 21.$

(ix) $x_1 = 0, x_2 = 20$ or $x_1 = 75/4, x_2 = 25/2, Z = 200.$

(x) $x_1 = 50/7, x_2 = 0, x_3 = 55/7, x_4 = 0, Z = 695/7.$

(xi) Unbounded Solution. (xii) No. F.S.

7. (i) $x_1 = 0 = x_2 = x_3, Z = 0.$ (ii) $x_1 = 0 = x_2, x_3 = 5/2, Z = 5/2.$

(iii) $x_1 = 213/30, x_2 = 0 = x_3 = x_5, x_4 = 13/10, x_6 = 2/5, Z = 213/30.$

(iv) $x_1 = 3/5, x_2 = 6/5, Z = 12/5.$

(v) $x_1 = 3,\ x_2 = 18, Z = 48.$

8. (i) $x_1 = 0, x_2 = 5, Z = 40.$ (ii) $x_1 = 2, x_2 = 0, Z = 6.$

(iii) Unbounded.

9. (i) $x_1 = 3/5, x_2 = 6/5, Z = -12/5.$

(ii) $x_1 = 4500/13, x_2 = 2100/13, Z = 2700$

10. No feasible solution.

11. 200 of type A, 600 of type B, Max. Profit Rs. 260.

12. $A = 89/41, b = 50/41, C = 62/41,$ Max. Profit Rs. 765/41.

13. $A = 300,\ B = 300$ units, 400 hours.

14. $A = 0$ unit, $B = 50$ units, $C = 50$ units per week, Profit Rs. 700.

15. (i) $x_1 = 2, x_2 = -1.$ (ii) $16/11, \ -2/11$

(iii) $x_1 = 5/2,\ x_2 = 5/2,\ x_3 = 5/2,\ x_4 = 0.$

16. (i) $\frac{1}{4}\begin{bmatrix} 2 & -2 \\ -1 & 3 \end{bmatrix}$ (ii) $\frac{1}{11}\begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$

17. $x_1 = 2,\ x_2 = 0,\ Z = 21$

18. $x_1 = 1,\ x_2 = 1,\ Z = 16.$

21. $\frac{1}{49}\begin{bmatrix} -32 & -26 & 40 \\ 5 & 1 & 6 \\ -142 & 344 & -435 \end{bmatrix}$

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6

Duality

§ 6·1. Introduction. Every linear programming problem is associated with another linear programming problem called the dual of the problem. The original problem is called **'primal'** while the other is called its **'dual'**. The optimal solution of either problem reveals information concerning the optimal solution of the other. If the optimal solution of either problem (primal or its dual) is known then the optimal solution of the other is also available.

§ 6·2. Symmetric Dual Problems.

[Meerut 94, 95 (BP), 98 (old)]

A symmetric relation between a primal and its dual problem exists.

Consider a L.P. problem

Find $x_1, x_2, \ldots, x_n$, which

$$\left.\begin{array}{ll} \text{Max. } & Z_p = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n \\ \text{s.t.} & a_{11} x_1 + a_{12} x_2 + \ldots + a_{1n} x_n \le b_1 \\ & a_{21} x_2 + a_{22} x_2 + \ldots + a_{2n} x_n \le b_2 \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \\ & a_{m1} x_1 + a_{m2} x_2 + \ldots + a_{mn} x_n \le b_m \end{array}\right] \quad \ldots (1)$$

and $\quad x_1, x_2, \ldots\ldots, x_n \ge 0$

where the signs of all parameters a, b and c's are arbitrary.

The dual problem of the above L.P. problem is obtained by

(i) Transposing the coefficient matrix.

(ii) Interchanging the role of constant terms and the coefficients of the objective function.

(iii) Reverting the inequalities and

(iv) Minimizing the objective function instead of maximizing it.

The **dual problem** is as follows :

Find $w_1, w_2, \ldots, w_m$, for which

$$\begin{aligned} \text{Mini} \quad & Z_D = b_1 w_1 + b_2 w_2 + \ldots + b_m w_m \\ \text{s.t.} \quad & a_{11} w_1 + a_{21} w_2 + \ldots + a_{m1} w_m \ge c_1 \\ & a_{12} w_1 + a_{22} w_2 + \ldots + a_{m2} w_m \ge c_2 \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ & a_{1n} w_1 + a_{2n} w_2 + \ldots + a_{mn} w_m \ge c_n \end{aligned} \quad \ldots (2)$$

and $\quad w_1, w_2, \ldots, w_m \ge 0.$

In matrix notation the primal and dual problems can be written as follows :

Primal problem. Find a column vector $\mathbf{x}$, which

Max. $\quad Z_p = \mathbf{cx}$

s.t. $\quad A\mathbf{x} \le \mathbf{b}$

and $\quad \mathbf{x} \ge 0.$

Dual Problem. Find a column vector $\mathbf{w}$, which

Mini $\quad Z_D = \mathbf{b'w}$

s.t. $\quad A'\mathbf{w} \ge \mathbf{c'}$

and $\quad \mathbf{w} \ge 0$

where $\quad \mathbf{w} = [w_1, w_2, \ldots, w_m]$

and $A', \mathbf{b}', \mathbf{c}'$ are the transposes of A, $\mathbf{b}$ and $\mathbf{c}$ respectively.

§ 6·3. Unsymmetrical Dual Problem.

Primal Problem. Find a column vector $\mathbf{x}$ which

Max. $\quad Z_p = \mathbf{cx}$

s.t. $\quad A\mathbf{x} = \mathbf{b}$

and $\quad \mathbf{x} \ge 0.$

Dual Problem. Find a column vector $\mathbf{w}$, which

Mini $\quad Z_D = \mathbf{b'w}$

s.t. $\quad A'\mathbf{w} \ge \mathbf{c'}$.

It is important to note that the dual variables are unrestricted in sign.

§ 6·4. The Dual of a mixed system.

If a system consists of a mixture of equations, inequalities (in either direction), non-negative variables or unrestricted variables then the dual of the problem can be obtained by reducing to the form (1), § 6·2. For this the following procedure is adopted.

(i) If a constraint has a sign $\ge$, then multiply both sides by -1 and make the sign $\le$.

(ii) If a constraint has a sign $=$, *i.e.* an equation, then it is replaced by two constraints involving the inequalities going in opposite directions.

for Example. an equation $\sum_{j=1}^{n} a_{ij} x_j = b_i$

is replaced by two constraints (inequalities)

$$\sum_{j=1}^{n} a_{ij} x_j \le b_i$$

and

$$\sum_{j=1}^{n} a_{ij} x_j \ge b_i$$

The second inequality can be written as

$$-\sum_{j=1}^{n} a_{ij} x_j \le -b_i$$

(iii) Every unrestricted variable is replaced by the difference of two positive variables.

Note. The dual variables which correspond to primal equality constraints, must be unrestricted in sign and those associated with the primal inequalities must be non-negative.

§ 6·5. Standard form of the primal.

A L.P. problem is said to be in standard primal form if

(i) All the constraints involve the sign $\le$ if it is a problem of maximization.

or (ii) All the constraints involve the sign $\ge$ if it is a problem of minimization.

§ 6·6. Theorem. Dual of the dual of a given primal, is the primal itself. [Meerut 95 (BP), 98 (Old); Raj. 85]

Proof. Consider the L.P. problem

$$\begin{aligned} \text{Primal. Max.} \quad & Z_p = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n \\ \text{s.t.} \quad & a_{11} x_1 + a_{12} x_2 + \ldots + a_{1n} x_n \le b_1 \\ & a_{21} x_1 + a_{22} x_2 + \ldots + a_{2n} x_n \le b_2 \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ & a_{m1} x_1 + a_{m2} x_2 + \ldots + a_{mn} x_n \le b_m \end{aligned} \quad \ldots (1)$$

and $\quad x_1, x_2, \ldots, x_n \ge 0.$

Dual. The dual of the above primal (1) is given by

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$$\left.\begin{array}{ll} \text{Mini} & Z_D = b_1 w_1 + b_2 w_2 + \ldots + b_m w_m \\ \text{s.t.} & a_{11} w_1 + a_{21} w_2 + \ldots + a_{m1} w_m \geq c_1 \\ & a_{12} w_1 + a_{22} w_2 + \ldots + a_{m2} w_m \geq c_2 \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ & a_{1n} w_1 + a_{2n} w_2 + \ldots + a_{mn} w_m \geq c_n \\ \text{and} & w_1, w_2, \ldots, w_m \geq 0. \end{array}\right] \quad \ldots (2)$$

Now to write the dual of the dual (2), we first write (2) in the standard primal form (1). In standard primal form, the dual (2) can be written as

$$\left.\begin{array}{ll} \text{Max.} & (-Z_D) = -b_1 w_1 - b_2 w_2 - \ldots - b_m w_m \\ \text{s.t.} & -a_{11} w_1 - a_{21} w_2 - \ldots - a_{m1} w_m \leq -c_1 \\ & -a_{12} w_1 - a_{22} w_2 - \ldots - a_{m2} w_m \leq -c_2 \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ & -a_{1n} w_1 - a_{2n} w_2 - \ldots - a_{mn} w_m \leq -c_n \\ \text{and} & w_1, w_2, \ldots, w_m \geq 0 \end{array}\right] \quad \ldots (3)$$

Dual of the dual. Now (3) is of the form (1)

$\therefore$ Considering (3) as a primal its dual is given by

$$\left.\begin{array}{ll} \text{Min.} & Z_{D_1} = -c_1 v_1 - c_2 v_2 - \ldots - c_n v_n \\ \text{s.t.} & -a_{11} v_1 - a_{12} v_2 - \ldots - a_{1n} v_n \geq -b_1 \\ & -a_{21} v_1 - a_{22} v_2 - \ldots - a_{2n} v_n \geq -b_2 \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ & -a_{m1} v_1 - a_{m2} v_2 - \ldots - a_{mn} v_n \geq -b_m \\ \text{and} & v_1, v_2, \ldots, v_n \geq 0. \end{array}\right] \quad \ldots (4)$$

The dual (4) of the dual (2) can be written as

$$\left.\begin{array}{ll} \text{Max.} & Z_{D_2} = c_1 v_1 + c_2 v_2 + \ldots + c_n v_n \\ \text{s.t.} & a_{11} v_1 + a_{12} v_2 + \ldots + a_{1n} v_n \leq b_1 \\ & a_{21} v_1 + a_{22} v_2 + \ldots + a_{2n} v_n \leq b_2 \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ & a_{m1} v_1 + a_{m2} v_2 + \ldots + a_{mn} v_n \leq b_m \\ \text{and} & v_1, v_2, \ldots, v_n \geq 0 \end{array}\right] \quad \ldots (5)$$

which is identical to (1).

Hence the dual of the dual is the primal itself.

Note :- Instead of writing the primal (dual (2)) in the standard primal form (1), the dual of (2) can directly be written as (1).

Thus we conclude that in the sets (1) and (2) if either problem is considered to be primal, the other is its dual.

Aliter. Consider the following L.P.P. in matrix form.

$$\textbf{Primal.} \quad \left.\begin{array}{ll} \text{Max.} & Z_P = \mathbf{cx} \\ \text{s.t.} & A\mathbf{x} \leq \mathbf{b} \\ \text{and} & \mathbf{x} \geq 0 \end{array}\right] \quad \ldots (6)$$

Dual. The dual of the primal (6) is

$$\left.\begin{array}{ll} \text{Min.} & Z_D = \mathbf{b}'\mathbf{w} \\ \text{s.t.} & A'\mathbf{w} \geq \mathbf{c}' \\ \text{and} & \mathbf{w} \geq 0 \end{array}\right] \quad \ldots (7)$$

The dual (7) can be written as

$$\left.\begin{array}{ll} \text{Max.} & (-Z_D) = -\mathbf{b}'\mathbf{w} \\ \text{s.t.} & -A'\mathbf{w} \leq -\mathbf{c}' \\ \text{and} & \mathbf{w} \geq 0 \end{array}\right] \quad \ldots (8)$$

Dual of the dual. Now (8) is of the form (6).

$\therefore$ Considering (8) as a primal, its dual is given by

$$\left.\begin{array}{ll} \text{Min.} & Z_{D_1} = (-\mathbf{c}')'.\mathbf{v} = -\mathbf{c}.\mathbf{v} \\ \text{s.t.} & (-A')'\mathbf{v} \geq (-\mathbf{b}')' \\ \text{and} & \mathbf{v} \geq 0 \end{array}\right] \quad \ldots (9)$$

(9) can be written as

$$\left.\begin{array}{ll} \text{Max.} & Z_{D_2} = \mathbf{c}.\mathbf{v} \\ \text{s.t.} & A\mathbf{v} \leq \mathbf{b} \\ \text{and} & \mathbf{v} \geq 0 \end{array}\right] \quad \ldots (10)$$

which is identical to (6).

Hence the dual of the dual is the primal itself.

Illustrative Examples.

Ex. 1. *Write the dual of the problem.*

$$\begin{array}{ll} \textit{Min.} & Z = 3x_1 + x_2 \\ \textit{s.t.} & 2x_1 + 3x_2 \geq 2 \\ & x_1 + x_2 \geq 1 \\ \textit{and} & x_1, x_2 \geq 0. \end{array}$$

Sol. The given L.P.P. is in the standard primal form. In matrix form the given problem can be written as

Min. $Z = (3, 1)[x_1, x_2] = \mathbf{cx}$

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$$\text{s.t.} \quad \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \geq \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

or $\qquad A\mathbf{x} \geq \mathbf{b}.$

$\therefore$ The dual of the given problem is

Max. $\quad Z_D = \mathbf{b}'\mathbf{w} = (2, 1)\,[w_1, w_2] = 2w_1 + w_2$

s.t. $\quad A'\mathbf{w} \leq \mathbf{c}'$

or
$$\begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \leq \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

or $\qquad 2w_1 + w_2 \leq 3$

$\qquad\quad 3w_1 + w_2 \leq 1$

and $\qquad w_1, w_2 \geq 0.$

Ex. 2. *Write the dual of the problem.*

Min $\quad Z = 2x_2 + 5x_3$

s.t. $\quad x_1 + x_2 \geq 2$

$\qquad 2x_1 + x_2 + 6x_3 \leq 6$

$\qquad x_1 - x_2 + 3x_3 = 4.$

and $\qquad x_1, x_2, x_3 \geq 0.$

Sol. First we shall write the given problem in the standard primal form as follows.

(i) Since it is a minimization problem, all the constraints must involve the sign $\geq$.

$\therefore$ We multiply second constraint by -1, and get

$$-2x_1 - x_2 + 6x_3 \geq -6.$$

(ii) The third constraints is an equality, so we replace it by two constraints

$\qquad x_1 - x_2 + 3x_3 \leq 4$

and $\qquad x_1 - x_2 + 3x_3 \geq 4.$

The first on multiplying by -1, reduce to

$$-x_1 + x_2 - 3x_3 \geq -4.$$

Thus the given problem in the standard primal form is as follows:

Min. $\quad Z = 0.x_1 + 2x_2 + 5x_3 = (0, 2, 5)\,.\,[x_1, x_2, x_3] = \mathbf{c}\,\mathbf{x}$

s.t. $\quad x_1 + x_2 + 0.x_3 \geq 2$

$\qquad -2x_1 - x_2 - 6x_3 \geq -6$

$\qquad -x_1 + x_2 - 3x_3 \geq -4$

$$x_1 - x_2 + 3x_3 \geq 4$$

or
$$\begin{bmatrix} 1 & 1 & 0 \\ -2 & -1 & -6 \\ -1 & 1 & -3 \\ 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \geq \begin{bmatrix} 2 \\ -6 \\ -4 \\ 4 \end{bmatrix}$$

or $\qquad A\mathbf{x} \geq \mathbf{b}.$

$\therefore$ The dual of given problem is

Max. $Z_D = \mathbf{b}'\mathbf{w} = (2, -6, -4, 4)\,[w_1, w_2, w_3', w''_3]$

$\qquad = 2w_1 - 6w_2 - 4w'_3 + 4w''_3$

s.t. $\quad A'\mathbf{w} \leq \mathbf{c}'$

or
$$\begin{bmatrix} 1 & -2 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ 0 & -6 & -3 & 3 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3' \\ w_3'' \end{bmatrix} \leq \begin{bmatrix} 0 \\ 2 \\ 5 \end{bmatrix}$$

or $\qquad w_1 - 2w_2 - (w_3' - w_3'') \leq 0$

$\qquad w_1 - w_2 + (w_3' - w_3'') \leq 2$

$\qquad 0\,.\,w_1 - 6w_2 - 3(w_3' - w_3'') \leq 5$

and $\qquad w_1, w_2, w_3', w_4'' \geq 0$

writting $w_3' - w_3'' = w_3$, the dual problem is

Max. $\quad Z_D = 2w_1 - 6w_2 - 4w_3$

s.t. $\quad w_1 - 2w_2 - w_3 \leq 0$

$\qquad w_1 - w_2 + w_3 \leq 2$

$\qquad -6w_2 - 3w_3 \leq 5$

and $\qquad w_1, w_2 \geq 0$, w_3 unrestricted in sign.

Ex. 3. *Give the dual of the following L.P.P.*

Max. $\quad Z = 2x_1 + 3x_2 + x_3$

s.t. $\quad 4x_1 + 3x_2 + x_3 = 6$

$\qquad x_1 + 2x_2 + 5x_3 = 4$

and $\qquad x_1, x_2, x_3 \geq 0.$ **[Meerut 90 (B.A. P TDC)]**

Sol. First we shall write the given problem in the standard primal form as follows.

(i) Since it is a maximization problem, all the constraints must involve the sign $\leq$.

(ii) The first equation (constraint) is equivalent to

$$4x_1 + 3x_2 + x_3 \leq 6$$

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and $\qquad 4x_1 + 3x_2 + x_3 \geq 6$

the second can be written as

$$-4x_1 - 3x_2 - x_3 \leq -6.$$

(iii) The second equation (constraint) is equivalent to

$$x_1 + 2x_2 + 5x_3 \leq 4$$

and $\qquad x_1 + 2x_2 + 5x_3 \geq 4$

Thus second can be written as

$$-x_1 - 2x_2 - 5x_3 \leq -4.$$

Thus the given L.P.P. in the standard primal form is

Max. $\quad Z = 2x_1 + 3x_2 + x_3 = (2, 3, 1) . [x_1, x_2, x_3] = \mathbf{cx}$

s.t.
$$4x_1 + 3x_2 + x_3 \leq 6$$
$$-4x_1 - 3x_2 - x_3 \leq -6$$
$$x_1 + 2x_2 + 5x_3 \leq 4$$
$$-x_1 - 2x_2 - 5x_3 \leq -4$$

or
$$\begin{bmatrix} 4 & 3 & 1 \\ -4 & -3 & -1 \\ 1 & 2 & 5 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \leq \begin{bmatrix} 6 \\ -6 \\ 4 \\ -4 \end{bmatrix}$$

or $\qquad A\mathbf{x} \leq \mathbf{b}$

and $\qquad x_1, x_2, x_3 \geq 0.$

$\therefore$ The dual of the given problem is

Min. $\quad Z_D = \mathbf{b}' \mathbf{w} = (6, -6, 4, -4) [w_1', w_1'', w_2', w_2'']$

$$= 6w_1' - 6w_1'' + 4w_2' - 4w_2''$$

s.t. $\qquad A' \mathbf{w} \geq \mathbf{c}'$

or
$$\begin{bmatrix} 4 & -4 & 1 & -1 \\ 3 & -3 & 2 & -2 \\ 1 & -1 & 5 & -5 \end{bmatrix} \begin{bmatrix} w_1' \\ w_1'' \\ w_2' \\ w_2'' \end{bmatrix} \geq \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

or
$$4w_1' - 4w_1'' + w_2' - w_2'' \geq 2$$
$$3w_1' - 3w_1'' + 2w_2' - 2w_2'' \geq 3$$
$$w_1' - w_1'' + 5w_2' - 5w_2'' \geq 1$$

and $\qquad w_1', w_1'', w_2', w_2'' \geq 0$

writing $\quad w_1' - w_2'' = w_1, w_2' - w_2'' = w_2$ the dual problem is

Mini $\quad Z_D = 6w_1 + 4w_2$

s.t $\quad 4w_1 + w_2 \geq 2$

$$3w_1 + 2w_2 \geq 3$$
$$w_1 + 5w_2 \geq 1$$

and $\qquad w_1, w_2$ are both unrestricted in sign.

Ex. 4. *Find the dual of the following L.P.P.*

Min $\qquad Z = x_1 + x_2 + x_3$

s.t.
$$x_1 - 3x_2 + 4x_3 = 5$$
$$x_1 - 2x_2 \leq 3$$
$$2x_2 - x_3 \geq 4$$

$x_1, x_2 \geq 0, x_3$ *is unrestricted in sign.* **[Meerut 87, 90 (TDC), 94]**

Sol. First we shall write the given problem in the standard primal form as follows.

(i) It is minimization problem, so all the constraints must contain the sign $\geq$..

(ii) The variable x_3 is unrestricted in sign.

$\therefore$ We write $x_3 = x_3' - x_3''$ where $x_3', x_3'' \geq 0$

(iii) The first constraint (equation) is equivalent to

$$x_1 - 3x_2 + 4(x_3' - x_3'') \geq 5$$

and $\qquad x_1 - 3x_2 + 4(x_3' - x_3'') \leq 5$

The second can be written as

$$-x_1 + 3x_2 - 4x_3' + 4x_3'' \geq -5.$$

(iv) Multiplying second constraint by -1, we have

$$-x_1 + 2x_2 + 0.(x_3' - x_3'') \geq -3$$

Thus the given problem in the standard primal form is

Min. $\quad Z = x_1 + x_2 + x_3' - x_3'' = (1, 1, 1, -1) [x_1, x_2, x_3', x_3'']$

$$= \mathbf{cx}$$

s.t.
$$x_1 - 3x_2 + 4x_3' - 4x_3'' \geq 5$$
$$-x_1 + 3x_2 - 4x_3' + 4x_3'' \geq -5$$
$$-x_1 + 2x_2 + 0.x_3' - 0.x_3'' \geq -3$$
$$0.x_1 + 2x_2 - x_3' + x_3'' \geq 4$$

Or
$$\begin{bmatrix} 1 & -3 & 4 & -4 \\ -1 & 3 & -4 & 4 \\ -1 & 2 & 0 & 0 \\ 0 & 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3' \\ x_3'' \end{bmatrix} \geq \begin{bmatrix} 5 \\ -5 \\ -3 \\ 4 \end{bmatrix}$$

Or $\qquad A\mathbf{x} \geq \mathbf{b}$

and $\qquad x_1, x_2, x_3', x_3'' \geq 0.$

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$\therefore$ The dual of the given problem is

Max. $Z_D = \mathbf{b}'\mathbf{w} = (5, -5, -3, 4)\,[w_1', w_1'', w_2, w_3]$

$= 5w_1' - 5w_1'' - 3w_2 + 4w_3$

s.t. $A'\mathbf{w} \le \mathbf{c}'$

or
$$\begin{bmatrix} 1 & -1 & -1 & 0 \\ -3 & 3 & 2 & 2 \\ 4 & -4 & 0 & -1 \\ -4 & 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} w_1' \\ w_1'' \\ w_2 \\ w_3 \end{bmatrix} \le \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

or
$$w_1' - w_1'' - w_2 + 0.w_3 \le 1$$
$$-3w_1' + 3w_1'' + 2w_2 + 2w_3 \le 1$$
$$4w_1' - 4w_1'' + 0.w_2 - w_3 \le 1$$
$$-4w_1' + 4w_1'' + 0.w_2 + w_3 \le -1$$

and $w_1', w_1'', w_2, w_3 \ge 0.$

Writing $w_1' - w_1'' = w_1$, the dual problem is

Max. $Z_D = 5w_1 - 3w_2 + 4w_3$

s.t.
$$w_1 - w_2 \le 1$$
$$-3w_1 + 2w_2 + 2w_3 \le 1$$
$$4w_1 - w_3 = 1$$

$w_2, w_3 \ge 0$, w_1 unrestricted in sign.

EXERCISE (6·1)

Write the dual of the following L.P. problems :

1. Max. $Z = x_1 + 2x_2$
   s.t. $2x_1 - 3x_2 \le 3$
   $4x_1 + x_2 \le -4$
   and $x_1, x_2 \ge 0$

2. Min. $Z = 4x_1 + 6x_2 + 18x_3$
   s.t. $x_1 + 3x_3 \ge 3$
   $x_2 + 2x_3 \ge 5$
   and $x_1, x_2, x_3 \ge 0$ **[Raj. 87]**

3. Max. $Z = x_1 + 2x_2 - x_3$
   s.t. $2x_1 - 3x_2 + 4x_3 \le 5$
   $2x_1 - 2x_2 \le 5$
   $3x_1 - x_3 \ge 4$
   and $x_1, x_2, x_3 \ge 0$

4. Max. $Z = 3x_1 + 5x_2 + 4x_3$
   s.t. $2x_1 + 3x_2 \le 8$
   $2x_2 + 5x_3 \le 10$
   $3x_1 + 2x_2 + 4x_3 \le 15$
   and $x_1, x_2, x_3 \ge 0$

5. Min. $Z = 2x_1 + 2x_2 + 4x_3$
   s.t. $2x_1 + 3x_2 + 5x_3 \ge 2$
   $3x_1 + x_2 + 7x_3 \le 3$
   $x_1 + 4x_2 + 6x_3 \le 5$
   and $x_1, x_2, x_3 \ge 0$

6. Max. $Z = x_1 + 3x_2$
   s.t. $3x_1 + 2x_2 \le 6$
   $3x_1 + x_2 = 4$
   and $x_1, x_2 \ge 0$

7. Max. $Z = 6x_1 + 4x_2 + x_3 + 7x_4 + 5x_5$
   s.t. $3x_1 + 7x_2 + 8x_3 + 5x_4 + x_5 = 2$
   $2x_1 + x_2 + 3x_3 + 2x_4 + 9x_5 = 6$
   and $x_1, x_2, x_3, x_4 \ge 0$, x_5 unresricted, in sign.

8. Min. $Z = x_1 - 3x_2 - 2x_3$
   s.t. $3x_1 - x_2 + 2x_3 \le 7$
   $2x_1 - 4x_2 \ge 12$
   $-4x_1 + 3x_2 + 8x_3 = 10$
   $x_1, x_2 \ge 0$ and x_3 is unrestricted. **[Meerut 95 (P)]**

ANSWERS

1. Min. $Z_D = 3w_1 - 4w_2$
   s.t. $2w_1 + 4w_2 \ge 1$
   $-3w_1 + w_2 \ge 2$
   and $w_1, w_2 \ge 0$

2. Max. $Z_D = 3w_1 + 5w_2$
   s.t. $w_1 \le 4$
   $w_2 \le 6$
   $3w_1 + 2w_2 \le 18$
   and $w_1, w_2, w_3 \ge 0$

3. Min. $Z_D = 5w_1 + 5w_2 - 4w_3$
   s.t. $2w_1 + 2w_2 - 3w_3 \ge 1$
   $-3w_1 - 2w_2 \ge 2$
   $4w_1 + w_3 \ge -1$
   and $w_1, w_2, w_3 \ge 0$

4. Min. $Z_D = 8w_1 + 10w_2 + 15w_3$
   s.t. $2w_1 + 3w_3 \ge 3$
   $3w_1 + 2w_2 + 2w_3 \ge 5$
   $5w_2 + 4w_3 \ge 4$
   and $w_1, w_2, w_3 \ge 0$

5. Max. $Z_D = 2w_1 - 3w_2 - 5w_3$
   s.t. $2w_1 - 3w_2 - w_3 \le 2$
   $3w_1 - w_2 - 4w_3 \le 2$
   $5w_1 - 7w_2 - 6w_3 \le 4$
   and $w_1, w_2, w_3 \ge 0$

6. Min. $Z_D = 6w_1 + 4w_2$
   s.t. $3w_1 + 3w_2 \ge 1$
   $2w_1 + w_2 \ge 3$
   and $w_1 \ge 0$ unrestricted in sign

7. Min. $Z_D = 2w_1 + 6w_2$
   s.t. $3w_1 + 2w_2 \ge 6$
   $7w_1 + w_2 \ge 4$
   $8w_1 + 3w_2 \ge 1$
   $5w_1 + 2w_2 \ge 7$
   $w_1 + 9w_2 = 5$
   and w_1, w_2 unrestricted in sign.

8. Min. $Z_D = -7w_1 + 12w_2 + 10w_3$
   s.t. $-3w_1 + 2w_2 - 4w_3 \le 1$
   $-w_1 + 4w_2 - 3w_3 \ge 3$
   $2w_1 - 8w_3 = 2$
   and $w_1, w \ge 0$,
   w_3 unrestricted in. sign.

§ 6·7. Fundamental Properties of Dual Problems.

In this section we shall describe a number of fundamental theorems (properties of dual problem).

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Theorem 1. *If* x *is any feasible solution of the primal, Max* $Z_P = \mathbf{c}\,\mathbf{x}$ *s.t.* $A\mathbf{x} \le \mathbf{b}$, $\mathbf{x} \ge 0$ *and* w *is any feasible solution to its dual problem Min.* $Z_D = \mathbf{b}'\mathbf{w}$, *s.t.* $A'\mathbf{w} \ge \mathbf{c}'$, $\mathbf{w} \ge 0$, *then* $\mathbf{c}\,\mathbf{x} \le \mathbf{b}'\mathbf{w}$ *i.e.* $Z_P \le Z_D$. [Meerut 85, 96; Raj. 85]

Proof. Let $\mathbf{x} = [x_1, x_2, \ldots, x_n]$ and $\mathbf{w} = [w_1, w_2, \ldots, w_m]$ be any feasible solution to the primal (1) and the dual problem (2), respectively.

$$\left.\begin{array}{ll}\textbf{Primal} & \text{Max. } Z_P = \mathbf{c}\,\mathbf{x} \\ & \text{s.t. } A\mathbf{x} \le \mathbf{b} \\ & \mathbf{x} \ge 0\end{array}\right] \quad \ldots(1)$$

$$\left.\begin{array}{ll}\textbf{Dual.} & \text{Min. } Z_D = \mathbf{b}'\mathbf{w} \\ & \text{s.t. } A'\mathbf{w} \ge \mathbf{c}' \\ \textbf{and} & \mathbf{w} \ge 0\end{array}\right] \quad \ldots(2)$$

Since w (m component column vector) is the feasible solution of the dual and the constraints of the primal are

$$A\mathbf{x} \le \mathbf{b}.$$

$\therefore$ If we multiply both sides of the above inequality by w' (m-component row vector), the sign of the inequality will again hold.

$$\therefore \quad \mathbf{w}'(A\mathbf{x}) \le \mathbf{w}'\mathbf{b}$$

or $$(A'\mathbf{w})'.\mathbf{x} \le (\mathbf{b}'\mathbf{w})' \quad \ldots(3)$$

as $(AB)' = B'A'$.

Again x (n-component column vector) is the feasible solution of the primal and the constraints of the dual are

$$A'\mathbf{w} \ge \mathbf{c}'.$$

If we multiply both sides of the above inequality by x' (n-component row vector), the sign of the inequality will again hold.

$$\therefore \quad \mathbf{x}'(A'\mathbf{w}) \ge \mathbf{x}'\mathbf{c}'$$

or $$\mathbf{x}'(\mathbf{w}'A)' \ge (\mathbf{c}\,\mathbf{x})'$$

or $$[(\mathbf{w}'A)\,\mathbf{x}]' \ge (\mathbf{c}\,\mathbf{x})'$$

or $$(\mathbf{w}'A)\,\mathbf{x} \ge \mathbf{c}\,\mathbf{x}$$

or $$(A'\mathbf{w})'\,\mathbf{x} \ge \mathbf{c}\,\mathbf{x} \quad \ldots(4)$$

From (3) and (4), we have

$$\mathbf{c}\,\mathbf{x} \le (A'\mathbf{w})'\,\mathbf{x} \le (\mathbf{b}'\mathbf{w})'.$$

[Since $(\mathbf{b}'\mathbf{w})' = \mathbf{b}'\mathbf{w}$, as b'w is a scalar value, not a matrix]

or $$\mathbf{c}\,\mathbf{x} \le \mathbf{b}'\mathbf{w}$$

or $$Z_P \le Z_D.$$

Theorem 2. *If* $\hat{\mathbf{x}}$ *is a feasible solution to the primal*

Max. $Z_P = \mathbf{c}\,\mathbf{x}$, *s.t.* $A\mathbf{x} \le \mathbf{b}$, $\mathbf{x} \ge 0$ and $\hat{\mathbf{w}}$

is a feasible solution to its dual problem Min. $Z_D = \mathbf{b}'\mathbf{w}$ *s.t.* $A'\mathbf{w} \ge \mathbf{c}'$, $\mathbf{w} \ge 0$, *such that* $\mathbf{c}\hat{\mathbf{x}} = \mathbf{b}'\hat{\mathbf{w}}$ *then* $\hat{\mathbf{x}}$ *and* $\hat{\mathbf{w}}$ *are the optimal solutions of the primal and the dual problems respectively.* [Meerut 96 (BP)]

Or Alternative Statement. *The necessary and sufficient condition for any L.P.P. and its dual to have optimal solution is that both have feasible solution.*

Proof. If x is any feasible solution, to the **primal**

$$\text{Max. } Z_P = \mathbf{c}\,\mathbf{x}, \text{ s.t. } A\mathbf{x} \le \mathbf{b}, \mathbf{x} \ge 0 \quad \ldots(1)$$

and $\hat{\mathbf{w}}$ is a feasible solution to the dual

$$\text{Min. } Z_D = \mathbf{b}'\mathbf{w}, \text{ s.t. } A'\mathbf{w} \ge \mathbf{c}', \ \mathbf{w} \ge 0 \quad \ldots(2)$$

then from theorem 1, we have

$$\mathbf{c}\,\mathbf{x} \le \mathbf{b}'\hat{\mathbf{w}}$$

or $$\mathbf{c}\,\mathbf{x} \le \mathbf{c}\hat{\mathbf{x}} \qquad \text{Since } \mathbf{c}\hat{\mathbf{x}} = \mathbf{b}'\hat{\mathbf{w}}$$

i.e. the value of the objective function of the primal at any other feasible solution is less than that at $\hat{\mathbf{x}}$. From which we conclude that $\hat{\mathbf{x}}$ is the optimal solution for the primal (maximizing problem).

Again for any feasible solution w to the dual (2) and for given feasible solution $\hat{\mathbf{x}}$ of the primal (1), from theorem 1, we have

$$\mathbf{c}\hat{\mathbf{x}} \le \mathbf{b}'\mathbf{w}$$

or $$\mathbf{b}'\hat{\mathbf{w}} \le \mathbf{b}'\mathbf{w} \qquad \text{Since } \mathbf{c}\hat{\mathbf{x}} = \mathbf{b}'\hat{\mathbf{w}}$$

i.e. the value of the objective function of the dual at any other feasible solution is greater than at $\hat{\mathbf{w}}$. From which we conclude that $\hat{\mathbf{w}}$ is the optimal solution for the dual problem (minimizing problem).

Hence the theorem.

Theorem 3. Fundamental Duality Theorem.

If either the primal or the dual problem has a finite optimal solution, then the other problem also has a finite optimal solution and the values of the two objective functions are equal.

Or

If a finite optimal feasible solution exists for the primal, there exists a finite optimal feasible solution for the dual and conversely. The values of the two objective functions are equal.

[Meerut 85, 88; Raj. T.D.C. 86, 87]

Proof. Consider the given primal and dual problems as follows.

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Primal. $\left.\begin{aligned} \text{Max. } Z_P &= \mathbf{c}\,\mathbf{x} \\ \text{s.t. } A\mathbf{x} &\geq \mathbf{b} \\ \mathbf{x} &\geq \mathbf{0} \end{aligned}\right]$ ...(1)

Dual. $\left.\begin{aligned} \text{Min. } Z_D &= \mathbf{b}'\mathbf{w} \\ \text{s.t. } A'\mathbf{w} &\leq \mathbf{c}' \\ \text{and } \mathbf{w} &\geq \mathbf{0} \end{aligned}\right]$ ...(2)

we shall prove the theorem by actually constructing an optimal solution to the dual from a given optimal solution of the primal. Here it is sufficient to prove that if primal (1) has an optimal solution, than so does its dual (2). Since the dual of the dual (2) is the primal (1) itself therefore the fact that the primal (1) has an optimal solution of the dual (2) does follows immediately.

Suppose that the primal (1) has a finite optimal feasible solution $\mathbf{x}_B$. Since $\mathbf{x}_B$ is an optimal feasible solution to (1), it can be solved by simplex method. To solve (1) by simplex method we first introduce slack variables to each of the constraints and change the inequalities in equations. Now (1) can be written as

$$\left.\begin{aligned} \text{Max. } Z_P &= \mathbf{c}\,\mathbf{x}_P \\ \text{s.t. } A\mathbf{x} + I\mathbf{x}_S &= \mathbf{b} \\ \text{and } \mathbf{x} \geq 0, \mathbf{x}_S &\geq 0 \end{aligned}\right] \quad ...(3)$$

where $\mathbf{x}_S = (x_{1}s, x_{2}s, ..., x_{m}s)$ are slack variables. Corresponding to the feasible solution $\mathbf{x}_B$ let B be the basis matrix and let $C_B = (c_{B_1}, c_{B_2}, ..., c_{B_m})$ be m component row vector containing the prices of the basic variables. It is important to note that $\mathbf{b} \geq 0$.

Since $\mathbf{x}_B$ is an optimal solution to the primal,

$\therefore \quad -Z_j + c_j \leq 0$ for all j.

Since $Z_j = C_B Y_j = C_B B^{-1} \alpha_j$

$\therefore \quad -C_B B^{-1} \alpha_j + c_j \leq 0$ for all α_j including surplus variables also

or $\quad C_B B^{-1} \alpha_j \geq c_j$ ...(4)

or $\quad C_B B^{-1} (\alpha_1, \alpha_2, ..., \alpha_n) \geq (c_1, c_2, ..., c_n)$

for vectors of $A = (\alpha_1, \alpha_2, ..., \alpha_n)$

or $\quad C_B B^{-1} A \geq \mathbf{c}$ ...(5)

If we write $(\hat{\mathbf{w}})' = C_B . B^{-1}$

where $(\hat{\mathbf{w}})' = (w_1, w_2, ... w_m) = [w_1, w_2, ..., w_m]'$.

$\therefore$ From (5), we have

$(\hat{\mathbf{w}})' A \geq \mathbf{c}$

or $\quad [(\hat{\mathbf{w}})' A]' \geq \mathbf{c}'$

(Taking transpose on both sides)

or $\quad A' [(\hat{\mathbf{w}})']' \geq \mathbf{c}'$

or $\quad A' \hat{\mathbf{w}} \geq \mathbf{c}'$

which proves that $\hat{\mathbf{w}}$ satisfies the constraints of the dual (2). Therefore $\hat{\mathbf{w}}$ is the solution of the dual (2).

Now we consider the equation (4) with α_j corresponding to slack variables. Therefore in this case $c_j = 0$.

$\therefore$ From (4), we have

$$C_B B^{-1} (e_j) \geq 0$$
$$j = 1, 2, ... m$$

or $\quad (w_1, w_2, ..., w_m) e_j \geq 0$

or $\quad w_j \geq 0$

i.e. $\quad \hat{\mathbf{w}} = [w_1, w_2, ..., w_m]$ is the feasible solution of the dual (2).

Now we prceed to prove that $\hat{\mathbf{w}}$ is an optimal solution to the dual (2).

Now $\quad Z_D = \mathbf{b}'\hat{\mathbf{w}} = [(\hat{\mathbf{w}})' \mathbf{b}]'$

$= (\hat{\mathbf{w}})' \mathbf{b} = (C_B B^{-1}) \mathbf{b} = C_B (B^{-1} \mathbf{b})$

$= C_B \mathbf{x}_B = \text{Max } Z_P.$

Hence $\hat{\mathbf{w}}$ and $\mathbf{x}_B$ are feasible solutions of the dual (2) and the primal (1) respectively such that

$$Z_P = Z_D.$$

Hence by theorem 2, $\hat{\mathbf{w}}$ is the optimal solution of the dual problem (2). Which proves the theorem.

Theorem 4. *If the primal problem has an unbounded solution, then the dual has either no solution or an unbounded solution.* **[Meerut 85]**

Proof. We shall prove this theorem by contradiction. We assume that when the primal has an unbounded solution, the dual has a finite optimal solution. But in theorem § 6·6 we have proved that the dual of the dual is the primal and in theorem 3, § 6·7 it has been proved that if the primal has a finite optimal solution

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s.t. $\sum_{i=1}^{m} a_{ji} w_j \geq c_i$ $\forall i = 1, 2, ...$

and $w_j \geq 0, \quad \forall j \neq k.$

Obviously $w_k = w_k' - w_k''$ is unrestricted in sign since $w_k >$ or $<$ according as $w_k' >$ or $< w_k''$. Hence the theorem.

Theorem 6. *If any variable of the primal is unrestricted in sign, the corresponding constraint in the dual will be a strict equality.* [Meerut]

Proof. Let the kth variable x_k of the primal be unrestricted in sign. Therefore, replacing x_k by $x_k' - x_k''$ where x_k' and x_k'' are non-negative variables, the primal can be written as

Max. $Z_P = c_1 x_1 + c_2 x_2 + ... + c_k (x_k' - x_k'') + c_{k+1} x_{k+1} + ... + c_n x_n$

s.t.

$$a_{11} x_1 + a_{12} x_2 + ... + a_{1k} (x'_k - x_k'') + a_{1,k+1} x_{k+1} + ... + a_{1n} x_n \leq b_1$$
$$a_{21} x_1 + a_{22} x_2 + ... + a_{2k} (x_k' - x_k'') + a_{2,k+1} x_{k+1} + ... + a_{2n} x_n \leq b_2$$
$$... \quad ... \quad ... \quad ... \quad ... \quad ...$$
$$a_{m1} x_1 + a_{m2} x_2 + ... + a_{mk} (x_k' - x_k'') + a_{m,k+1} x_{k+1} + ... + a_{mn} x_n \leq b_m$$

and $x_1, ..., x_{k-1}, x_k', x_k'', x_{k+1}, ..., x_n \geq 0.$

The dual to the above problem is given by

Min. $Z_D = b_1 w_1 + b_2 w_2 + ... + b_m w_m$

and

$$a_{11} w_1 + a_{21} w_2 + ... + a_{m1} w_m \geq c_1$$
$$a_{12} w_1 + a_{22} w_2 + ... + a_{m2} w_m \geq c_2$$
$$... \quad ... \quad ... \quad ... \quad ...$$
$$... \quad ... \quad ... \quad ... \quad ...$$
$$\left\{ \begin{array}{l} a_{1k} w_1 + a_{2k} w_2 + ... + a_{mk} w_m \geq c_k \\ -a_{1k} w_1 - a_{2k} w_2 - ... - a_{mk} w_m \geq -c_k \end{array} \right\}$$
$$... \quad ... \quad ... \quad ... \quad ...$$
$$... \quad ... \quad ... \quad ... \quad ...$$
$$a_{1n} w_1 + a_{2n} w_2 + ... + a_{mn} w_m \geq c_n$$

and $w_1, w_2, ..., w_m \geq 0.$

The two constraints under bracket { } in the above dual are equivalent to an equality

$$a_{1k} w_1 + a_{2k} w_2 + ... + a_{mk} w_m = c_k$$

i.e. the kth constraint in the dual is an equality.

Hence the theorem.

§ 6·8. Complementary Slackness Theorem.

For the optimal feasible solutions of the primal and dual system.

(a) *Whatever inequality oocurs in the ith relation of either system if the corresponding slack or surplus variable is positive, then the ith variable of its dual is zero.*

(b) *If the jth variable is positve in either system, the jth relation of its dual holds as a strict equality (i.e. the corresponding slack or surplus variable w_{m+1} venishes).*

Proof. The primal and the dual problems in explicit form after introducing the non-negative slack variables $x_{n+1}, x_{n+2}, ..., x_{n+m}$ in the primal constraints and introducing the non-negative surplus variables $w_{m+1}, w_{m+2}, ..., w_{m+n}$ in the dual constraints, can be written as

Primal Problem.

Max. $Z_P = c_1 x_1 + c_2 x_2 + ... + c_n x_n$ ... (1)

s.t.
$$\left. \begin{array}{lr} a_{11} x_1 + a_{12} x_2 + ... + a_{1n} x_n + x_{n+1} & = b_1 \\ a_{21} x_1 + a_{22} x_2 + ... + a_{2n} x_n \quad + x_{n+2} & = b_2 \\ ... \quad ... \quad ... \quad ... \quad ... & ... \\ ... \quad ... \quad ... \quad ... \quad ... & ... \\ a_{m1} x_1 + a_{m2} x_2 + ... + a_{mn} x_n \quad + x_{n+m} & = b_m \end{array} \right] \quad ... (2)$$

and $x_1, x_2, ..., x_{n+m} \geq 0.$

Dual Problem.

Min. $Z_D = b_1 w_1 + b_2 w_2 + ... + b_m w_m$ ... (3)

s.t.
$$\left. \begin{array}{lr} a_{11} w_1 + a_{21} w_2 + ... + a_{m1} w_m - w_{m+1} & = c_1 \\ a_{12} w_1 + a_{22} w_2 + ... + a_{m2} w_m \quad - w_{m+2} & = c_2 \\ ... \quad ... \quad ... \quad ... & ... \\ ... \quad ... \quad ... \quad ... & ... \\ a_{1n} w_1 + a_{2n} w_2 + ... + a_{mn} w_m \quad - w_{m+n} & = c_n \end{array} \right] \quad ... (4)$$

and $w_1, w_2, ..., w_m, w_{m+1}, ..., w_{m+n} \geq 0.$

Now, multiplying the equations of (2) by $w_1, w_2, ..., w_m$ respectively and then adding, we have

$$x_1 \sum_{i=1}^{m} a_{i1} w_i + x_2 \sum_{i=1}^{m} a_{i2} w_i + ... + x_n \sum_{i=1}^{m} a_{in} w_i + x_{n+1} w_1 + x_{n+2} w_2 + ... + x_{n+m} w_m = \sum_{i=1}^{m} b_i w_i \quad ... (5)$$

Subtracting (5) from (1), we have

$$(c_1 - \sum_{i=1}^{m} a_{i1} w_i) x_1 + (c_2 - \sum_{i=1}^{m} a_{i2} w_i) x_2 + ... + (c_n - \sum_{i=1}^{m} a_{in} w_i) x_n$$

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$$- w_1 x_{n+1} - w_2 x_{n+2} - \dots - w_m x_{n+m} = (Z_P - \sum_{i=1}^{m} b_i w_i)$$
$$= Z_P - Z_D \quad \dots (6)$$

From (4), we have

$$- w_{m+j} = c_j - (a_{1j} w_1 + a_{2j} w_2 + \dots + a_{mj} w_m) \quad j = 1, 2, \dots, n$$

or $$- w_{m+j} = c_j - \sum_{i=1}^{m} a_{ij} w_i \quad \dots (7)$$

$\forall j = 1, 2, \dots, n.$

Using (7) in (6), we have

$$- (w_{m+1} x_1 + w_{m+2} x_2 + \dots + w_{m+n} x_n)$$
$$- (w_1 x_{n+1} + w_2 x_{n+2} + \dots + w_n x_{n+m}) = Z_P - Z_D \quad \dots (8)$$

Now if $\mathbf{x}^* = (x_1^*, x_2^*, \dots, x_n^*)$ and $\mathbf{w}^* = (w_1^*, w_2^*, \dots, w_m^*)$ be the optimal solutions to the primal and the dual problems respectively, then by Duality theorem

$$Z^*_P = Z^*_D.$$

Thus for the optimal solutions $\mathbf{x}^*$ and $\mathbf{w}^*$ of primal and the dual problems the corresponding slack and surplus variables

$x^*_{n+i} \geq 0, i = 1, 2, \dots m$ and $w^*_{m+j} \geq 0, j + 1, 2, \dots, n$.

From equation (8), we have

$$(w^*_{m+1} x_1^* + w^*_{m+2} x_2^* + \dots w^*_{m+n} x_n^*)$$
$$+ (w_1^* x^*_{n+1} + w_2^* x^*_{n+2} + \dots + w_n^* x^*_{n+m}) = 0 \quad \dots (9)$$

Since all the variables in (9) are non-negative and so their product terms in (9) are also non-negative. The sum of these positive products in (9) will be zero if each term is individually equal to zero.

i.e. $$w^*_{m+j} x_j^* = 0 \quad (\forall\ j = 1, 2, \dots, n) \quad \dots (10)$$

and $$w_i^* x^*_{n+i} = 0 \quad (\forall\ i = 1, 2, \dots, m) \quad \dots (11)$$

(*a*) From (11), if $x^*_{n+i} > 0$ then we must have $w_i^* = 0$, *i.e.* if the slack variable in the ith relation of the primal is positive then ith variable of the dual is zero.

Again from (10), if $w^*_{m+j} > 0$ then we must have $x_j^* = 0$, *i.e.* if the surplus variable in the jth relation of the dual is positive then jth variable of the priaml (dual of the dual) is zero.

This proves the part (a) of the theorem.

(*b*) Also from (10), if $x_j^* > 0$, then we must have $w^*_{m+i} = 0$ *i.e.* if the jth variable in the primal is zero then the jth relation in the dual is strictly an equality (as $w^*_{m+j} = 0$).

Again from (11), if $w_i^* > 0$, then $x^*_{n+j} = 0$.

i.e. if the ith variable in the dual is positive then the jth relation in the primal (dual of the dual) is strictly an equality (as $x^*_{n+i} = 0$).

This proves the part (b) of the theorem.

Alternative Statement.

A necessary and sufficient condition for any pair of feasible solutions to the primal and dual to be optimal is that

$$w_i x_{n+i} = 0, i = 1, 2, \dots, m$$

where x_{n+i} is the slack variable in the primal and

$$x_j w_{m+j} = 0, j = 1, 2, \dots, n,$$

where w_{m+j} is the surplus variable for the dual.

§ 6·9. Correspondence between primal and dual.

With the help of the theorems developed in § 6·7 and § 6·8, we note the following correspondence rule between the primal and the dual problems.

Primal	Dual
1. Objective function max. Z_P.	Objective function Min. Z_D.
2. Requirement vector.	Price vector.
3. Coefficient matrix.	Transpose of the coefficient matrix.
4. Constraints with $\leq$ sign.	Constraints with $\geq$ sign.
5. Relation.	Variable.
6. i-th inequality.	i-th variable $w_i \geq 0$.
7. i-th constraints an equality.	i-th variable w_i unrestricted in sign.
8. Variable.	Relation.
9. i-th variable $x_i > 0$.	i-th relation a strict inequality.
10. i-th variable x_i unrestricted in sign.	i-th constraint a strict equality.
11. i-th slack variable positive.	i-th variable zero.
12. i-th variable zero.	i-th surplus variable positive.
13. Finite optimal solution.	Finite optimal solution with equal optimal value of objective function.
14. Unbounded solution.	No solution or an unbounded solution.

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§ 6·10. To Read the solution to the Dual from the final Simplex Table of the primal and vice-versa.

It is important to note that the final simplex table of the primal problem also contains the optimal solution of the dual and vice-versa. For this we observe the following rules :

1. *The optimal value of the primal objective function is equal to the optimal value of the dual objective function.*

   $$i.e. \, Max. \, Z_P = Min \, Z_D.$$

2. *The Δ_j's ($\Delta_j = c_j - Z_j$) with sign changed for the slack (or surplus) variables in the final simplex table for the primal are the values of the corresponding optimal dual variables in the final simplex table for the dual.*

   Proof. In the final simplex table (for primal) corresponding to the slack and surplus variables

   $$-\Delta_j = -(c_j - Z_j) = -(0 - Z_j)$$
   $$= C_B Y_j = C_B B^{-1} a_j$$

   where B is the optimal basic C_B, the corresponding price vector.

   For i-th slack variable $a_j = e_j$ unit matrix with 1 at the i-th place.

   $$\therefore \quad -\Delta_j = C_B B^{-1} e_j = (w_1, w_2, \ldots, w_m) e_j = w_j$$

   where w_j is i-th dual variable.

3. If either problem has unbounded solution, then the other will have no feasible solution.

   It is always advantageous to apply the simplex method to the problem having lesser number of constraints. Therefore, here we shall solve the problem (primal of dual) with lesser number of constraints by simplex method and then read the solution of the other from the final simplex table by the rules described above.

Illustrative Examples

Ex. 5. *Apply the simplex method to solve the following L. P. P.*

$$\left.\begin{array}{ll} \text{Max.} & Z = 30x_1 + 23x_2 + 29x_3 \\ \text{s.t.} & 6x_1 + 5x_2 + 3x_3 \le 26 \\ & 4x_1 + 2x_2 + 5x_3 \le 7 \\ \text{and} & x_1, x_2, x_3 \ge 0. \end{array}\right\}$$

Also read the solution to the dual of the above problem from the final table.

Sol. Introducing the slack variables x_4, x_5 the equations obtanied are

$$6x_1 + 5x_2 + 3x_3 + x_4 = 26$$
$$4x_1 + 2x_2 + 5x_3 + x_5 = 7$$

Taking $x_1 = 0, x_2 = 0, x_3 = 0$, we have $x_4 = 26, x_5 = 7$ which is the starting B.F.S. Proceeding as usual we get the following simplex table.

Table 6·1

B	C_B	c_j / X_B	30 / Y_1	23 / Y_2	29 / Y_3	0 / Y_4	0 / Y_5	Mini-Ratio X_B/Y_1
Y_4	0	26	6	5	3	1	0	26/6
Y_5	0	7	$\boxed{4}$	2	5	0	1	7/4 (Mini) →
$Z = C_B X_B = 0$		Δ_j	30 ↑	23	29	0	0 ↓	X_B/Y_2
Y_4	0	31/2	0	2	−9/2	1	−3/2	31/4
Y_1	30	7/4	1	$\boxed{1/2}$	5/4	0	1/4	7/2(Mini) →
$Z = 105/2$		Δ_j	0	8 ↑	−17/2	0	−15/2	
Y_4	0	17/2	−4	0	−19/2	1	−5/2	
Y_2	23	7/2	2	1	5/2	0	1/2	
$Z = 161/2$		Δ_j	−16	0	−57/2	0	−23/2	

$\therefore$ Optimal solution is $x_1 = 0, x_2 = 7/2, x_3 = 0$.

Max. $Z = 161/2$. **Ans.**

To write the dual of the problem.

The given problem can be written as

Max. $Z = (30, 23, 29)\,[x_1, x_2, x_3]$

$$\text{s.t.} \begin{bmatrix} 6 & 5 & 3 \\ 4 & 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \le \begin{bmatrix} 26 \\ 7 \end{bmatrix}$$

and $x_1, x_2, x_3 \ge 0$.

$\therefore$ the dual of the given problem is given by

Mini $Z^* = (26, 7)\,[w_1, w_2]$

$$\text{s.t.} \begin{bmatrix} 6 & 4 \\ 5 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \geq \begin{bmatrix} 30 \\ 23 \\ 29 \end{bmatrix}$$

where $w_1, w_2 \geq 0$

or

$$\left.\begin{aligned} &\text{Mini } Z^* = 26w_1 + 7w_2 \\ &\text{s.t. } 6w_1 + 4w_2 \geq 30 \\ &\quad 5w_1 + 2w_2 \geq 23 \\ &\quad 3w_1 + 5w_2 \geq 29 \end{aligned}\right\} \quad \ldots(2)$$

where $w_1, w_2 \geq 0$.

Solution of the dual (2) of the given problem (1).

The dual problem (2) can be written as

Max. $Z^*_1 = -26w_1 - 7w_2$

$$\begin{aligned} \text{s.t.} \quad 6w_1 + 4w_2 - w_3 \qquad + w_6 \qquad &= 30 \\ 5w_1 + 2w_2 \qquad - w_4 \qquad + w_7 \qquad &= 23 \\ 3w_1 + 5w_2 \qquad - w_5 \qquad + w_8 &= 29 \end{aligned}$$

and $w_1, w_2, \ldots, w_8 \geq 0.$

where w_3, w_4, w_5 are surplus variables and w_6, w_7, w_8 are the artificial variables.

Now proceeding as usual by simplex method (in two phases by big M method) the final simplex table obtained is as follows.

Final simplex table for dual (2).

Table 6·2

B	C_B	c_j	-26	-7	0	0	0	Mini-Ratio
		W_B	W_1	W_2	W_3	W_4	W_5	
W_5	0	57/2	19/2	0	0	$-5/2$	1	
W_3	0	16	4	0	1	-2	0	
W_2	-7	23/2	5/2	1	0	$-1/2$	0	
$Z_1^* = C_B W_B = -161/2$		Δ_j	$-17/2$	0	0	$-7/2$	0	

Solution of the dual problem is

$$w_1 = 0, w_2 = 23/2 \text{ and Mini } Z^* = 161/2.$$

To read the solution to the dual of the given problem (1) from the final table 6·1 page 201 of the primal problem.

The Δ_j's with the sign changed for the slack variables in table give the solution for the dual of the primal.

In the last of table 6·1, the Δ_j's for the slack variables are 0 and $-23/2$, therefore, the solution of the dual of the primal (1) is $w_1 = 0, w_2 = 23/2$, and Mini $Z^* = \text{Max.}Z = 161/2$. **Ans.**

Ex. 6. *Using the dual, solve the following L.P. P.*

$$\text{Max. } Z_P = 3x_1 - 2x_2$$

$$\begin{aligned} \textit{s.t.} \quad x_1 &\leq 4 \\ x_2 &\leq 6 \\ x_1 + x_2 &\leq 5 \\ -x_2 &\leq -1 \end{aligned}$$

and $x_1, x_2 \geq 0.$

Sol. The dual of the given problem is given by

$$\text{Mini } Z_P = 4w_1 + 6w_2 + 5w_3 - w_4$$

$$\begin{aligned} \text{s.t.} \quad w_1 + 0.w_2 + 1.w_3 + 0.w_4 &\geq 3 \\ 0.w_1 + 1.w_2 + 1.w_3 - 1.w_4 &\geq -2 \end{aligned}$$

and $w_1, w_2, w_3, w_4 \geq 0.$

Changing the minimizing objective function to maximization function and the requirements to positive requirements, the dual problem can be written as

$$\text{Max.} \quad Z'_D = -\text{Min. } Z_D = -4w_1 - 6w_2 - 5w_3 + w_4$$

$$\begin{aligned} \text{s.t.} \quad w_1 + 0.w_2 + 1.w_3 + 0.w_4 &\geq 3 \\ 0.w_1 - 1.w_2 - 1.w_3 + 1.w_4 &\leq 2 \end{aligned}$$

and $w_1, w_2, w_3, w_4 \geq 0.$

Introducing the surplus variable w_5 in the first constraint and the slack variable w_6 in the second constraint, the constraints of the dual problem reduce to the following equalities.

$$\begin{aligned} w_1 + 0.w_2 + 1.w_3 + 0.w_4 - 1.w_5 \qquad &= 3 \\ 0.w_1 - 1.w_2 - 1.w_3 + 1.w_4 \qquad + w_6 &= 2 \end{aligned}$$

$$w_1, w_2, \ldots, w_6 \geq 0.$$

Since identity matrix I_2 is present in the coefficient matrix, so there is no need to introduce artificial variable in the first constraint.

Taking $w_2 = 0 = w_3 = w_4 = w_5$, we have $w_1 = 3, w_6 = 2$ which is the starting basic feasible solution of the dual. Proceeding as usual we get following simplex table.

Table 6·3

B	C_B	c_j / W_B	-4 / W_1	-6 / W_2	-5 / W_3	1 / W_4	0 / W_5	0 / W_6	Mini-Ratio W_B/W_4
W_1	-4	3	1	0	1	0	-1	0	–
W_6	0	2	0	-1	-1	$\boxed{1}$	0	1	2/1 (Mini) $\rightarrow$
$Z'_D = -12$		Δ_j	0	-6	-1	1 $\uparrow$	-4	0	
W_1	-4	3	1	0	1	0	-1	0	
W_4	1	2	0	-1	-1	1	0	1	
$Z'_D = -10$		Δ_j	0	-5	0	0	-4	-1	

Since all $\Delta_j \le 0$, so this solution of the dual problem is optimal.

i.e. Optimal solution of the dual problem is

$$w_1 = 3, w_2 = 0, w_3 = 0, w_4 = 2.$$

$$\text{Min. } Z_D = -\text{ Max. } Z'_D = 10.$$

Solution of the original problem (Primal).

From the final simplex table 6·3 for the dual, the solution of the primal is given by

$$x_1 = -\Delta_5 = 4, x_2 = -\Delta_6 = 1$$

and $\qquad \text{Max. } Z_P = \text{Min. } Z_D = 10$

Ex. 7. *Formulate the following L. P. P. into dual problem and hence solve it :*

$$\textit{Min} \quad Z_P = 3x_1 - 2x_2 + 4x_3$$

s.t.

$$3x_1 + 5x_2 + 4x_3 \ge 7$$
$$6x_1 + x_2 + 3x_3 \ge 4$$
$$7x_1 - 2x_2 - x_3 \le 10$$
$$x_1 - 2x_2 + 5x_3 \ge 3$$
$$4x_1 + 7x_2 - 2x_3 \ge 2$$

and $\qquad x_1, x_2, x_3 \ge 0.$

Sol. In the standard primal form the given L.P.P. can be written as follows :

$$\text{Min. } Z_P = 3x_1 - 2x_2 + 4x_3$$

s.t.

$$3x_1 + 5x_2 + 4x_3 \ge 7$$

$$6x_1 + x_2 + 3x_3 \ge 4$$
$$-7x_1 + 2x_2 + x_3 \ge -10$$
$$x_1 - 2x_2 + 5x_3 \ge 3$$
$$4x_1 + 7x_2 - 2x_3 \ge 2$$

and $\qquad x_1, x_2, x_3 \ge 0.$

The dual of the above L.P.P. can be given by

$$\text{Max. } Z_D = 7w_1 + 4w_2 - 10w_3 + 3w_4 + 2w_5$$

s.t.

$$3w_1 + 6w_2 - 7w_3 + w_4 + 4w_5 \le 3$$
$$5w_1 + w_2 + 2w_3 - 2w_4 + 7w_5 \le -2$$
$$4w_1 + 3w_2 + w_3 + 5w_4 - 2w_5 \le 4$$

and $\qquad w_1, w_2, \ldots, w_5 \ge 0.$

For the solution of this dual problem by simplex method first we multiply the second constraints by -1 so that its requirement become positive and in doing so the inequality is also reversed.

i.e., the second constraint reduce to

$$-5w_1 - w_2 - 2w_3 + 2w_4 - 7w_5 \ge 2.$$

Using slack variables w_6, w_8 in the first and third constraints and the surplus variable w_7 and artificial variable w_9 in the second constraints, and assigning a large negative price M to the arificial variable, the dual problem reduce to the following form.

$$\text{Max. } Z_D = 7w_1 + 4w_2 - 10w_3 + 3w_4 + 2w_5 + 0.w_6 + 0.w_7 + 0.w_8 - Mw_9$$

s.t.

$$3w_1 + 6w_2 - 7w_3 + w_4 + 4w_5 + w_6 = 3$$
$$-5w_1 - w_2 - 2w_3 + 2w_4 - 7w_5 - w_7 + w_9 = 2$$
$$4w_1 + 3w_2 + w_3 + 5w_4 - 2w_5 + w_8 = 4$$

and $\qquad w_1, w_2, \ldots, w_9 \ge 0.$

Taking $w_1 = 0 = w_2 = w_3 = w_4 = w_5 = w_7$, we have $w_6 = 3, w_8 = 4, w_9 = 2$, which is the starting B.F.S. All computational work is shown in the table 6·4 given on page 201.

Since all $\Delta_j < 0$, so this solution is optimal $x_1 = 0 = x_2 = x_3$, $x_4 = 4/5, x_5 = 0, x_6 = 11/5, x_7 = 0, x_8 = 0, x_9 = 2/5$. But the artificial vector, appear in the basis at the positive level which indicates that the dual problem has no feasible solution.

Hence the given problem (primal) also has no solution.

(See § 3·10 Case III).

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Table 6·4

B	C_B	c_j	7	4	-10	3	2	0	0	0	$-M$	Mini-Ratio W_B/W_4
		W_B	W_1	W_2	W_3	W_4	W_5	W_6	W_7	W_8	A_1	
W_6	0	3	3	6	-7	1	4	1	0	0	0	3/1
A_1	M	2	-5	-1	-2	2	-7	0	-1	0	1	2/2
W_8	6	4	4	3	1	$\boxed{5}$	-2	0	0	1	0	4/5 (Min) →
$Z_D = -2M$		Δ_j	$7-5M$	$4-M$	$-10m-2m$	$3+2M$ ↑	$2-7M$	0	$-M$	0 ↓	0	
W_6	0	11/5	11/5	27/5	$-36/5$	0	22/5	1	0	$-1/5$	0	
A_1	$-M$	2/5	$-33/5$	$-11/5$	$-12/5$	0	$-31/5$	0	-1	$-2/5$	1	
W_4	3	4/5	4/5	3/5	1/5	1	$-2/5$	0	0	1/5	0	
$Z_D = \frac{12}{5} - \frac{2}{5}M$		Δj	$\frac{23-33M}{5}$	$\frac{11-11M}{5}$	$\frac{-53-12M}{5}$	0	$\frac{16-31M}{5}$	0	$-M$	$\frac{-3-2M}{5}$	0	

Ex. 8. *Solve the following L. P. P. by using its dual.*

Max. $Z = 5x_1 - 2x_2 + 3x_3$

s.t $2x_1 + 2x_2 - x_3 \geq 2$

$3x_1 - 4x_2 \leq 3$

$x_2 + 2x_3 \leq 5$

and $x_1, x_2, x_3 \geq 0$

Sol. In the standard primal form the problem is written as follows :

Max. $Z = 5x_1 - 2x_2 + 3x_3$

s.t. $-2x_1 - 2x_2 + x_3 \leq -2$

$3x_1 - 4x_2 \leq 3$

$x_2 + 3x_3 \leq 5$

and $x_1, x_2, x_3 \geq 0.$

The dual of the above problem is given by

Min. $Z_D = -2w_1 + 3w_2 + 5w_3$

s.t. $-2w_1 + 3w_2 + 0.w_3 \geq 5$

$-2w_1 - 4.w_2 + 1.w_3 \geq -2$

$w_1 + 0.w_2 + 3.w_3 \geq 3$

and $w_1, w_2, w_3 \geq 0.$

To solve the above dual problem by simplex method we convert it to a maximization problem and the requirement in second constraint is made positive by multiplying both sides of this constraint by -1. Thus the problem become

Max. $Z'_D = -\text{Min. } Z_D = 2w_1 - 3w_2 - 5w_3$

s.t. $-2w_1 + 3w_2 + 0.w_3 \geq 5$

$2w_1 + 4w_2 - w_3 \leq 2$

$w_1 + 0.w_2 + 3.w_3 \geq 3$

and $w_1, w_2, w_3 \geq 0$

Introducing surplus variables w_4, w_6 and artificial variables w_7, w_8 in the first and third constraint respectively and slack variable w_5 in the second constraint, and assigning a large negative price M to the artificial variables, the above dual problem reduces to

Max. $Z'_D = 2w_1 - 3w_2 - 5w_3 + 0.w_4 + 0.w_5 + 0.w_6 - Mw_7 - Mw_8$

s.t. $-2w_1 + 3w_2 + 0.w_3 - w_4 + w_7 = 5$

$2w_1 + 4w_2 - w_3 + w_5 = 2$

$w_1 + 0.w_2 + 3w_3 - w_6 + w_8 = 3$

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and $\qquad w_1, w_2, \ldots, w_8 \geq 0.$

Taking $w_1 = 0 = w_2 = w_3 = w_4 = w_6$, we get

$w_5 = 2, w_7 = 5, w_8 = 3$, which is the starting B. F. S.

Table 6·5

B	C_B	c_j / W_B	2 / W_1	−3 / W_2	−5 / W_3	0 / W_4	0 / W_5	0 / W_6	−M / A_1	−M / A_2	Mini-Ratio W_B/W_2
A_1	$-M$	5	−2	3	0	−1	0	0	1	0	5/3
W_5	0	2	2	$\boxed{4}$	−1	0	1	0	0	0	$\frac{2}{4}$(Mini) →
A_2	$-M$	3	1	0	3	0	0	−1	0	1	–
$Z'_D = -8M$		Δ_j	$2-M$	$-3+3M$ ↑	$-5+3M$	$-M$	0 ↓	$-M$	0	0	W_B/Y_3
A_1	$-M$	7/2	−7/2	0	3/4	−1	−3/4	0	1	0	14/3
W_2	−3	1/2	1/2	1	−1/4	0	1/4	0	0	0	–
A_2	$-M$	3	1	0	$\boxed{3}$	0	0	−1	0	1	$\frac{3}{3}$(Mini) →
$Z'_D = -\frac{13}{2}M - \frac{3}{2}$		Δ_j	$\frac{7-5M}{2}$	0	$\frac{15M-23}{4}$ ↑	$-M$	$\frac{3-3M}{4}$	$-M$	0	0 ↓	W_B/W_6
A_1	$-M$	11/4	−15/4	0	0	−1	−3/4	$\boxed{1/4}$	1		11(Mini) →
W_2	−3	3/4	7/12	1	0	0	1/4	−1/12	0		–
W_3	−5	1	1/3	0	1	0	0	−1/3	0		–
$Z'_D = -\frac{11}{4}M - \frac{29}{4}$		Δ_j	$\frac{65-45M}{12}$	0	0	$-M$	$\frac{3-3M}{4}$	$\frac{3M-23}{12}$ ↑	0 ↓		
W_6	0	11	−15	0	0	−4	−3	1			
W_2	−3	5/3	−2/3	1	0	−1/3	0	0			
W_3	−5	14/3	−14/3	0	1	−4/3	−1	0			
$Z'_D = -85/3$		Δ_j	−70/3	0	0	−23/3	−5	0			

Since all $\Delta_j < 0$: So this solution to the problem given in table 6·5 is an optiaml solution.

Since surplus and slack variables are w_4, w_5, w_6.

$\therefore$ Solution of the primal is given by

$$x_1 = -\Delta_4 = 23/3, x_2 = -\Delta_5 = 5, x_3 = -\Delta_6 = 0$$

and Max. Z = Min. $\quad Z_D = -$ Max. $Z'_D = 85/3$. **Ans.**

Exercise (6·2)

Use duality to solve the following L.P. problems.

1. Max. $Z = 3x_1 + x_2$
   s.t. $x_1 + x_2 \leq 1$
   $2x_1 + 3x_2 \geq 2$
   and $x_1, x_2 \geq 0.$

2. Max. $Z = x_1 - x_2$
   s.t. $2x_1 + x_2 \geq 2$
   $-x_1 - x_2 \geq 1$
   and $x_1, x_2 \geq 0.$

3. Min. $Z = 8x_1 - 2x_2 - 4x_3$
   s.t. $x_1 - 4x_2 - 2x_3 \geq 2$
   $x_1 + x_2 - 3x_3 \geq -1$
   $-3x_1 - x_2 + x_3 \geq 1,$
   $x_1, x_2, x_3 \geq 0.$

4. Max. $Z = 4x_1 + 3x_2$
   s.t. $x_1 \leq 6$
   $x_2 \leq 8$
   $x_1 + x_2 \leq 7$
   $3x_1 + x_2 \leq 15$
   $-x_2 \leq 1$
   and $x_1, x_2 \geq 0.$ **[Raj. 85, 88]**

5. Min. $Z = 10x_1 + 6x_2 + 2x_3$
   s.t. $-x_1 + x_2 + x_3 \geq 1$
   $3x_1 + x_2 - x_3 \geq 2$
   and $x_1, x_2, x_3 \geq 0.$

6. Max. $Z = 3x_1 + 2x_2$
   s.t. $2x_1 + x_2 \leq 5$
   $x_1 + x_2 \leq 3$
   $x_1, x_2 \geq 0.$

7. Max. $Z = 2x_1 + x_2,$
   s.t. $x_1 + 2x_2 \leq 10, \quad x_1 + x_2 \leq 6, \quad x_1 - x_2 \leq 2$
   $x_1 - 2x_2 \leq 1, \quad x_1, x_2, x_3 \geq 0.$ **[Meerut 95 (P)]**

8. Consider problem A.
   Minimize $Z = x_1 - 10x_2$
   s.t. $x_1 - \cdot 5x_2 \geq 0, \quad x_1 - 5x_2 \geq -5, \quad x_1, x_2 \geq 0$
   and problem B.
   Maximize $Z = -5x_2$
   s.t. $x_1 + x_2 \leq 1, -\cdot 5x_1 - 5x_2 \leq -10, x_1, x_2 \geq 0.$
   Explain how the solutions of problems A and B are related ?

9. Find the dual of the following problem and hence or otherwise solve it.
   Min. $Z = 6x + 5y + 2z$
   s.t. $x + 3y + 2z \geq 5, \quad 2x + 2y + z \geq 2,$
   $4x - 2y + 3z \geq -1 \quad x, y, z \geq 0.$

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10. Apply the principle of duality to solve the following linear programming problem :

Min. $Z = 2x_1 + 2x_2$,

s.t. $2x_1 + 4x_2 \geq 1, x_1 + 2x_2 \geq 1,$

$2x_1 + x_2 \geq 1, \quad x_1, x_2 \geq 0.$ [Meerut]

11. Use duality theory to solve the following linear programming problem

Min. $Z = 4x_1 + 3x_2 + 6x_3$

s.t. $x_1 + x_3 \geq 2, x_2 + x_3 \geq 5, x_1, x_2, x_3 \geq 0.$

12. Find the dual of the problem :

Max. $Z = 2x_1 - x_2$,

s.t. $x_1 + x_2 \leq 10, \quad -2x_1 + x_2 \leq 2,$

$4x_1 + 3x_2 \geq 12 \quad x_1, x_2 \geq 0.$

Solve the primal problem by simplex method and deduce from it the solution to the dual problem.

13. Find the dual of the problem

Max. $Z = -x_1 + 2x_2 - x_3$

s.t. $3x_1 + x_2 - x_3 \leq 10, \quad -x_1 + 4x_2 + x_3 \geq 6$

$x_2 + x_3 \leq 4, \quad x_1, x_2, x_3 \geq 0.$

Solve the primal problem by simplex method and deduce the solution of the dual problem from the optimal tableau of the primal

14. Solve the dual of the following problem by simplex method.

Max. $Z = 2x_1 + 3x_2 + 5x_3$

s.t. $x_1 + x_2 + x_3 \leq 7, \quad x_1 + 2x_2 + 2x_3 \leq 13,$

$3x_1 - x_2 + x_3 \leq 5, \quad x_1, x_2, x_3 \geq 0.$

15. Consider the problem

Max. $Z = 2x_1 + 3x_2$

s.t. $2x_1 + 2x_2 \leq 10, \quad 2x_1 + x_2 \leq 6,$

$x_1 + 2x_2 \leq 6, \quad x_1, x_2 \geq 0.$

(a) Write the complete dual problem from the canonical form of the above primal.

(b) Solve the primal problem and then find the solution to the dual.

16. Solve the following L.P. problem by simplex method.

Min. $Z = \frac{15}{2}x_1 - 3x_2$

s.t. $3x_1 - x_2 - x_3 \geq 3$

$x_1 - x_2 + x_3 \geq 2$

and $x_1, x_2, x_3 \geq 0$

Write the dual of the above problem. What should be the maximum value of the objective function of the dual ?

17. By means of the duality theory, solve and illustrate geometrically the following L.P.P.

Max. $Z = 3x_1 + 2x_2$

s.t. $x_1 + x_2 \geq 1, \; x_1 + x_2 \leq 7$

$x_1 + 2x_2 \geq 10, \; x_2 \leq 3, \; x_1, x_2 \geq 0.$

18. Consider the problem,

Max. $Z = 8x_1 + 6x_2$.

s.t $x_1 - x_2 \leq 3/5, \; x_1 - x_2 \geq 2, \; x_1, x_2 \geq 0.$

Show that both the primal and the dual problem have no feasible solution.

Answers

1. $x_1 = 1, x_2 = 0$, Min. $Z = 3$. 2. No feasible solution.
3. Unbounced solution or no solution.
4. $x_1 = 4, x_2 = 3$, Max. $Z = 25$. 5. $x_1 = 1/4, x_2 = 5/4, x_3 = 0, Z = 10$.
6. $x_1 = 2, x_2 = 1, Z = 8$. 8. No. solution
9. $x_1 = 0 = x_2, x_3 = 5/2, Z = 5$. 10. $x_1 = 1/3 = x_2, Z = 4/3$.
11. $x_1 = 0, x_2 = 3, x_3 = 2, Z = 21$.
12. $x_1 = 10, x_2 = 0$, Max. $Z = 20$;
Dual $x_1 = 2, \quad x_2 = 0 = x_3$, Mini $Z = 20$.
13. $x_1 = 0 = x_3, x_2 = 4$, Max. $Z = 8$;
Dual $x_1 = 0 = x_2 \; x_3 = 2$, Mini. $Z = 8$.
15. (i) Mini. $Z = 10x_1 + 6x_2 + 6x_3$
s.t. $2x_1 + 2x_2 + x_3 \geq 2, 2x_1 + x_2 + 2x_3 \geq 3, x_1, x_2, x_3 \geq 0.$
(ii) $x_1 = 2, x_2 = 2$, Max. $Z = 10$; Dual $x_1 = 0, x_2 = 1/3, x_3 = 4/3$, Mini. $Z = 10$.
16. $x_1 = 5/4, x_2 = 0, x_3 = 3/4, Z = 75/8$. 16. $x_1 = 4, x_2 = 3, Z = 18$.

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§ 6·11. Dual Simplex Algorithm*.

For a L.P.P. (maximization problem) the optimality criterion of the simplex method, $c_j - Z_j = c_j - C_B B^{-1} \alpha_j \le 0$, for all j where B is the basis, depends only on α_j and c_j and is independent of the requirement vector **b**. Thus every basic solutuion with all $c_j - Z_j \le 0$ will not be feasible, but any basic feasible solution with all $c_i - Z_i \le 0$ will certainly be an optimal solution. The dual simplex algorithm uses the above remarks. It starts with a basic optimal solution (not necessarily feasible) of the primal, and decrease the number of negative variables iteratively (step by step) maintaining the optimality criterion at each iteration. The dual **simplex algorithm** was discovered by **C. E. Lemke**, a student of Charnes while applying the simplex method to the dual of a L.P.P.

§ 6·12. Derivation of the Dual Simplex Algorithm :

Let the primal problem in standard form be written as

$$\left.\begin{array}{ll} \text{Max.} & Z_P = \mathbf{c\,x} \\ \text{s.t.} & A\mathbf{x} = \mathbf{b} \end{array}\right\} \quad \ldots(1)$$

and $\mathbf{x} \ge 0$.

The dual of the above problem is given by

$$\left.\begin{array}{ll} \text{Min.} & Z_D = \mathbf{b'w} \\ \text{s.t.} & A'\mathbf{w} \ge \mathbf{c'} \end{array}\right\} \quad \ldots(2)$$

The w_i of **w** are unrestricted in sign.

After adding the surplus variables to the problem (2), it can be written, in the form used in the application of the simplex method, as follows

$$\begin{array}{ll} \text{Min.} & Z_D = \mathbf{b'\,w} \\ \text{s.t.} & A'\mathbf{w} - I\,\mathbf{w}_s = \mathbf{c'} \end{array} \quad \ldots(3)$$

and $\mathbf{w}_s \ge 0$

where $\mathbf{w}_s = [\,w_{1s}, w_{2s}, \ldots, w_{ns}\,]$ is the column vector of surplus variables.

Let **x** be any solution (not necessarily feasible) to $A\mathbf{x} = \mathbf{b}$ and let **w** be any solution to the dual. Then from (3), we have

$$\mathbf{x}'A'\mathbf{w} - \mathbf{x}'\mathbf{w}_s = \mathbf{x}'\mathbf{c}' \quad \ldots(4)$$

If $\mathbf{x}'.\mathbf{w}_s = 0$ ...(5)

then from (4), we have

$$\mathbf{x}'A'\mathbf{w} = \mathbf{x}'\mathbf{c}'$$

$$(A\mathbf{x})'\mathbf{w} = (\mathbf{c\,x})'$$

*All numerical technique for solving a problem are called algorithm. In other words, an algorithm is just a prescription or specific routine for solving a given problem.

$\mathbf{b'w} = (\mathbf{c\,x})'$.

Then $Z_D = \mathbf{b'w} = (\mathbf{c\,x})' = \mathbf{c\,x} = Z_P$.

Then for any solution to the primal and any solution to the dual, the objective functions for both the problems are equal if $\mathbf{x}'\mathbf{w}_s = 0$.

Now applying the simplex method to (3), we find the constraint matrix $(A', -I)$ is of order $n \times (m+n)$. As usual we assume that the matrix A is of order $m \times n$ and $n > m$. Now order of A' is $n \times m$. If $\overline{B}$ is the basis matrix for the dual problem (3) then the order of $\overline{B}$ will obviously be equal to $n \times n$. But A' contain only m columns, hence there must always be at least $(n-m)$ surplus vectors in the basis $\overline{B}$. We assume that the first m columns of $\overline{B}$ are all the m columns of A' and the remaining $n-m$ columns of $\overline{B}$ are the surplus vectors of the form $-e_j$. Taking the surplus vectors in $\overline{B}$ to be $e_{m+1}, e_{m+2}, \ldots, e_n$, the basis matrix $\overline{B}$ can be written as

$$\overline{B} = (B'_1, B'_2, \ldots, B'_m, -e_{m+1}, \ldots, -e_n)$$

where $B_i\,;\, i = 1, 2, \ldots, m$ are the rows of A

i.e. $B'_i\,;\, i = 1, 2, \ldots, m$ are the columns of A'.

Since the first m columns of $\overline{B}$ are A',

∴ We can write

$$\overline{B} = (A', -e_{m+1}, -e_{m+2}, \ldots, -e_n)$$

But $A' = (B'_1, B'_2, \ldots, B'_m)$

$$= [B_1, B_2, \ldots, B_m]' = (\alpha_1, \alpha_2, \ldots, \alpha_m)' = [\alpha'_1, \alpha'_2, \ldots, \alpha'_m]$$

where $\alpha_1, \alpha_2, \ldots, \alpha_m$ may be any m columns of $A = (\alpha_1, \alpha_2, \ldots, \alpha_n)$.

(Note that α_1 is m component column vector while α'_1 is m component row vector).

Thus we can write

$$\overline{B} = \begin{bmatrix} \alpha'_1 & 0 & 0 & \cdots & 0 \\ \alpha'_2 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ \alpha'_m & 0 & 0 & & 0 \\ \alpha'_{m+1} & -1 & 0 & & 0 \\ \alpha'_{m+2} & 0 & -1 & & \vdots \\ \vdots & \vdots & \vdots & & \\ \alpha'_n & 0 & 0 & & -1 \end{bmatrix} \quad \ldots(6)$$

Here we have rearranged the rows of A' (columns of A) so that the surplus vector would be as shown in (6). The matrix $\overline{B}$ should be as such that it yields a basic feasible solution to the dual *i.e.* it provide a solution to the dual with all surplus variables ≥ 0 while w_i may be unrestricted.

Since $\overline{B}$ is the basis of the dual problem, therefore the vectors of $\overline{B}$ must be linearly independent. Also as $r(\overline{B}) = n > m$, the first m rows of $\overline{B}$ are linearly independent.

$$\therefore \quad B' = [\alpha'_1, \alpha'_2, \ldots, \alpha'_m]$$

in a set of m linearly independent vectors.

$$\therefore \quad B = [\alpha_1, \alpha_2, \ldots, \alpha_m] \qquad \ldots(7)$$

But $r(A) = r(A, \mathrm{b}) = m$, therefore, B is the basis for the primal problem (1). Thus in terms of B (basis of the primal), $\overline{B}$ can be written as

$$\overline{B} = \begin{bmatrix} B' & 0 \\ R & -I_{n-m} \end{bmatrix} \qquad \ldots(8)$$

where $R = [\alpha'_{m+1}, \alpha'_{m+2}, \ldots, \alpha'_n]$.

It is obvious that if the primal basis B and the dual basis $\overline{B}$ are given by (7) and (6) respectively, then when α_j is in B, $-e_j$ is not in $\overline{B}$ and conversely. Thus when $w_{sj} \neq 0$ then $x_j = 0$ and when $x_j \neq 0$ then $x_{sj} = 0$. Which implies that if B and $\overline{B}$ are selected as above, then

$$\mathrm{x}'\mathrm{w}_s = 0 \qquad \text{which is (5).}$$

Hence $Z_P = Z_D$ for the corresponding solutions to the primal and the dual problems. Also when we have a basic feasible solution to the dual with $B_1, B_2, \ldots, B_m$ in $\overline{B}$ then we can immediately find a basic solution to the primal by the method discussed above.

After getting the B.F.S. to the dual in the manner discussed above, we wish to know whether it is optimal or whether it can be improved.

Condition for optimality of x_B. (Solution of the primal corresponding to the basis B).

It is clear that when $\overline{B}$ is of the form (6), the prices of the variables in the dual basis will be b_i for the first m columns and 0 for the remaining $(n-m)$ columns. If we denote the price vector corresponding to the basis $\overline{B}$ by $\mathrm{b}\overline{B}$ then

$$\mathrm{b}\overline{B} = (\mathrm{b}', 0).$$

It is obvious that vectors for the dual problem not in the dual basis are surplus vectors. Now for the solution w (of the dual problem) to be optimal the condition $\overline{(Z_i - c_i)}^* \leq 0$ must be satisfied for the dual problem (minimization problem).

Now corresponding to the $(n-m)$ surplus vectors except for the first m surplus vector $\overline{(Z_i - c_i)} = 0$ (since these $n-m$ vectors are is $\overline{B}$. But the prices of the surplus vector are zero.

$\therefore$ The condition of optimality of the solution w_B reduces to

$\overline{Z}_i^{**} \leq 0$.

Thus the feasible basic solution to the dual will be optimal if

$$Z_i = \mathrm{b}_B (\overline{B})^{-1} \cdot (-e_i) \leq 0 \qquad \ldots(9)$$

for the surplus vectors not in the dual basis

i.e. for $i = 1, 2, \ldots, m$.

But for $\overline{B}$ given by (8) we know that

$$(\overline{B})^{-1} = \begin{bmatrix} (B')^{-1} & 0 \\ R(B')^{-1} & -I_{n-m} \end{bmatrix}$$

$\therefore$ from (9), we have

$$Z_i = (\mathrm{b}', 0) \cdot \begin{bmatrix} (B')^{-1} & 0 \\ R(B')^{-1} & -I_{n-m} \end{bmatrix} \cdot (-e_i) \leq 0$$

or $\qquad (\mathrm{b}'(B')^{-1}, 0)(-e_i) \leq 0$

(row vector)

or $\qquad (\mathrm{x}'_B, 0) \cdot (-e_i) \leq 0$

or $\qquad -x_{Bi} \leq 0$

or $\qquad x_{Bi} \geq 0$ for all $i = 2, \ldots, m$.

Thus if all $x_{Bi} \geq 0$, we have an optimal feasible solution to the dual and x_B will be the optimal solution to the primal corresponding to B.

If at least one $\overline{Z}_i > 0$

i.e. $\overline{(Z_i - c_i)} > 0$ for at least one suplus variable then the solution to the dual problem correspoding to $\overline{B}$ is not optimal and need further improvement.

* $\overline{(Z_i - c_i)}$ is $Z_i - c_i$ for the dual problem. ** $\overline{Z}_i$ is Z_i for the dual problem

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To find the improved solution of the dual problem.

If the solution of the dual problem corresponding to the basis $\overline{B}$ is not optimal, then it can be improved entering (introducing) a vector in $\overline{B}$ in place of some vector (departing vector) of $\overline{B}$ in the manner discussed below.

(a) Selection of the introducing vector.

Using the simplex criterion, for the improvement of the dual solution, we choose the vector to enter the dual basis for which

$$\overline{Z}_r = \text{Max}\,(\overline{Z}_i),\ \overline{Z}_i > 0. \qquad \ldots(10)$$

Thus if $-e_r$ satisfy (10), then $-e_r$ is the vector which must enter the dual basis (in order to improve Z_D). From this it follows that the primal variable corresponding to x_{Br} should be driven to zero, otherwise $x'\,w_s \neq 0$. Hence α_r should be removed from the primal basis where

$$x_{Br} = \text{Min}\,(x_{Bi}),\ x_{Bi} < 0.$$

Thus the vector leaving the primal basis is selected corresponding to the basic variable having the most negative value.

(b) Selection of the vector to be removed from the dual basis.

First we compute the basic feasible solution $w_{\overline{B}}$ to the dual.

We have

$$w_{\overline{B}} = (\overline{B})^{-1}\,c' = \begin{bmatrix} (B')^{-1} & 0 \\ R\,(B')^{-1} & -I_{n-m} \end{bmatrix} \begin{bmatrix} c'_B \\ c'_R \end{bmatrix}$$

Where c_B contains the prices in the primal basis B and c_R contains the prices not in the basis B.

or $$w_{\overline{B}} = \begin{bmatrix} (B')^{-1}\,c'_B \\ R\,(B')^{-1}\,c'_B - c'_R \end{bmatrix} = \begin{bmatrix} (c_B \cdot B^{-1})' \\ \{(c_B \cdot B^{-1})\,R'\}' - c'_R \end{bmatrix} \qquad \ldots(11)$$

Now since the first m components of w_B are w.

$\therefore$ For these first m components

$$w = (B')^{-1}\,c'_B\,(c_B\,B^{-1})'$$

or $$w' = c_B\,B^{-1}. \qquad \ldots(12)$$

from (11) the last $(n-m)$ components of w_B are

$\{(c_B \cdot B^{-1} \cdot R'\}' - c'_R$ or $c_B \cdot B^{-1}\,R' - c_R$ (by taking transpose)

i.e. last $(n-m)$ components of $w_{\overline{B}}$ are given by

$$w_{sj} = c_B \cdot B^{-1}\,\alpha_j - c_j = Z_j - c_j$$

for $j = m+1, \ldots, n$.

Thus the surplus variable w_{sj} in the dual has the value $Z_j - c_j$ for the primal variable x_j not in the primal basis.

Since $Z_j - c_j = 0$ for all the vectors in the primal basis and since $-e_j$ is not in $\overline{B}$ and if α_j in B, then we can write $w_{sj} = Z_j - c_j$ for the surplus variables not in the dual basis.

Hence we have,

$$w_{sj} = Z_j - c_j \qquad (j = 1, 2, \ldots, n) \qquad \ldots(13)$$

Now $$Y_r = (\overline{B})^{-1} \cdot (-e_r) = \begin{bmatrix} (B')^{-1} & 0 \\ R\,(B')^{-1} & -I_{n-m} \end{bmatrix} \cdot \begin{bmatrix} -e_r^{\,*} \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} (-B')^{-1}\,e_r^{\,*} \\ -R\,(B')^{-1} \cdot e_r^{\,*} \end{bmatrix}$$

Where $e_r^{\,*}$ is vector containing the first m-components of e_r. Since $-e_r$ corresponding to a vector which is to be introduced in the dual $r < m$, hence the last m-components of any $-e_r$ not in $\overline{B}$ are zero.

As usual the surplus vector to be removed from the basis $\overline{B}$ is determined from

$$\frac{w_{sk}}{y_{kr}} = \underset{j}{\text{Min}}\left\{\frac{w_{sj}}{y_{jr}},\ y_{jr} > 0\right\} \qquad \ldots(14)$$

$$j = m+1, \ldots, n.$$

It must be noted here that we never want to remove $B_1, B_2, \ldots, B_m$, because if we do so, we shall not be able to determine a corresponding basis for the primal problem. Also we can ignore the first m components of w_B because they are unrestricted in sign.

We have

$$R\,(B')^{-1} = (B^{-1}\,R')' = [B^{-1}\,(\alpha_{m+1}, \alpha_{m+2}, \ldots, \alpha_r)]'$$

$$= [B^{-1}\,\alpha_{m+1}, B^{-1}\,\alpha_{m+2}, \ldots, B^{-1}\,\alpha_n]^{-1}$$

$$= [Y_{m+1}, Y_{m+2}, \ldots, Y_n]'$$

Thus $R\,(B')^{-1}$ is a matrix of Y'_j, $j = m+1, \ldots, n$, *i.e.* for vectors not in the primal basis.

Also $-Y'_j \cdot e_r^{\,*} = -y_{rj}$, i.e. the last $n-m$ components of Y_r are the $-y_{rj}$.

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Hence in terms of the primal, the criterion (14) *for the vector to be removed from the dual basis, becomes*

$$\frac{Z_k - c_k}{-y_{rk}} = \text{Min.} \frac{Z_j - c_j}{-y_{rj}}, \; y_{rj} > 0$$

or

$$\frac{Z_k - c_k}{y_{rk}} = \text{Max.} \frac{Z_j - c_j}{y_{rj}}, \; y_{rj} < 0$$

or

$$\frac{\Delta_k}{y_{rk}} = \frac{c_k - Z_k}{y_{rk}} = \underset{j}{\text{Mini}} \left[\frac{\Delta_j}{y_{rj}}, \; y_{rj} < 0 \right]$$

If $y_{rj} \geq 0$, for all the vectors not in the primal basis, the dual problem has an unbounded solution and the primal problem will have no solution.

§ 6·13. Initial solution for Dual Simplex Algorithm.

It is difficult to find, a starting basic solution to the primal (maximization problem) with all $c_j - Z_j \leq 0$ which implies that it can be difficult to find a basic feasible solution to the dual. If any how we can determine a basic feasible solution of the dual, we may not find $B_1, B_2, \ldots, B_m$ in the basis of the dual and hence more work and time will be needed to obtain a basic solution to the primal with all $c_j - Z_j \leq 0$.

But there is one special case where we can easily get a basic solution to the primal with all $c_j - Z_j \leq 0$. If the primal problem (maximization problem) is a perfect inequality with $\geq$ sign *i.e.* every constraint in the primal needs a surplus variable. Also assume that all $c_j \leq 0$. For such problem if the initial basis matrix

$$B = -I, \text{ then } x_B = -I\mathbf{b} = -\mathbf{b}$$

and

$$\Delta_j = c_j - Z_j = c_j - c_B(-I)(-\alpha_j) = c_j \leq 0.$$

$$\text{as } c_j \leq 0 \text{ and } c_B = 0.$$

Hence we get a basic solution to the primal with all $\Delta_j = c_j - Z_j \leq 0$ therefore the simplex algorithm can be applied.

§ 6·14. Advantage of Dual Simplex Algorithm.

The advantage of the dual simplex algorithm over the other method is that here we do not require any artificial variables. Hence a great deal of work or a lot of labour is saved whenever this method is applicable.

§ 6·15. Computational Procedure of the Dual Simplex Algorithm

Step 1 : To unit the given L.P.P. is standard primal form.

(i) If the problem is of minimization. Convert it into the maximization problem.

(ii) Write all the constraints in the form of inequalities involving $\leq$ sign.

Note that by doing so some b's may change to negative values.

Step 2. To find initial basic solution

Introduce the slack variables in the constraints to reduce them to equalities.

Then find the initial basic solution with all $c_j - Z_j \leq 0$ which is possible only if all $c_j \leq 0$.

In this situation take all given variables equal to zero, and calculate the values of the slack variables. This solution will be the **starting basic solution.** Let $x_B = (x_{B1}, x_{B2}, \ldots, x_{Bm})$ be the initial basic solution correspondin to basic matrix. $\beta = (\beta_1, \beta_2, \ldots, \beta_m)$.

If at this stage all the basic variables are non-negative then this solution is **optimal feasible solution.** of the problem.

If at least one basic variable is negative then proceed to the next step.

Step 3. Construction of the starting Simplex table.

Construct the simplex table as usual in simplex method.

Step 4. To find the vector incoming (entering) and leaving (outgoing) the basis.

To determine the outgoing (vector) (β_r). β_r *i.e.*, r-th column in the basis is taken as the outgoing vector if

$$x_{Br} = \min. \{x_{Bi}, x_{Bi} < 0\}.$$

To determine the incoming vector (α_k)

α_k is taken as the entering (incoming) vector for that value of k, for which

$$\frac{\Delta_k}{y_{rk}} = \underset{j}{\text{Min.}} \left\{ \frac{\Delta_j}{y_{rj}}, \; y_{rj} < 0 \right\}$$

If all $y_{rj} \geq 0$, then the problem has no F.S.

Step 5. Test of optimality.

If entering the vector α_k in place of βr in the basis all basic variables reduces to non-negative values, then this solution is optimal F.S. But if at least one basic variable is negative, then this solution is not optimal F.S., in that case repeat steps 4 and 5, iteratively till an optimal F.S. is obtained.

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Illustrative Examples

Ex. 9. *Solve the following L.P.P. by the simplex dual algorithm,*

Min. $Z = 3x_1 + x_2$

s.t. $x_1 + x_2 \geq 1$

$2x_1 + 3x_2 \geq 2$

and $x_1, x_2 \geq 0.$

Sol. For the clear understanding of the procedure we shall solve this example step by step.

Step 1. First we write the given L.P.P. in the standard primal form.

Max. $Z_P = -Z = -3x_1 - x_2$

s.t. $-x_1 - x_2 \leq -1$

$-2x_1 - 3x_2 \leq -2$

and $x_1, x_2 \geq 0.$

Since objective function is that of maximization and all $c_j < 0$,

$\therefore$ It is possible to find an infisible but optimal basic solution and hence we can solve this problem by dual simplex algorithm.

Step 2. Introducing the slack variables x_3 and x_4 the constraints reduce to the following equalities

$$-x_1 - x_2 + x_3 = -1$$

$$-2x_1 - 3x_2 + x_4 = -2$$

$\therefore$ the starting basic solution to the primal is

$$x_1 = 0, x_2 = 0, x_3 = -1, x_4 = -2.$$

The starting simplex table is as follows.

Table 6·6

B	C_B	c_j	-3	-1	0	0	
		$X_B (= x_B)$	Y_1	Y_2	$Y_3 (\beta_1)$	$Y_4 (\beta_2)$	
Y_3	0	-1	-1	-1	1	0	
Y_4	0	-2	-2	$\boxed{-3}$	0	1	→
$Z_P = C_B X_B = 0$		Δ_j	-3	-1	0	0	
				↑		↓	

$\Delta_1 = c_1 - Z_1 = c_1 - C_B Y_1 = -3, \ \Delta_2 = c_2 - Z_2 = c_2 - C_B Y_2 = -1.$

Thus the starting solution $x_1 = 0, x_2 = 0, x_3 = -1, x_4 = -1$ is a basic solution which is infeasible but optimal.

$\therefore$ To find the improved solution we shall determine the leaving vector and entering vector to the basis.

To determine the leaving vector (βr).

Since $x_{Br} = \text{Min}\,(x_{Bi}, x_{Bi} < 0).$

$= \text{Min}\,(x_{B1}, x_{B2}) = \text{Min}\,(-1, -2) = -2 = x_{B2}$

$\therefore \quad r = 2$ *i.e.* $\beta_2 (= Y_4)$ is leaving vector.

To determine the entering vector (α_k).

$$\frac{\Delta_k}{y_{rk}} = \frac{\Delta_k}{y_{2k}} = \underset{j}{\text{Min}} \left\{ \frac{\Delta_j}{y_{2j}}, y_{2j} < 0 \right\}$$

$$= \text{Min} \left\{ \frac{\Delta_1}{y_{21}}, \frac{\Delta_2}{y_{22}} \right\} = \text{Min} \left\{ \frac{-3}{-2}, \frac{-1}{-3} \right\} = \frac{1}{3} = \frac{\Delta_2}{y_{22}}$$

$\therefore k = 2$ i.e. $\alpha_2 (= Y_2)$ is the entering vector.

$\therefore$ Key element $= y_{22} = -3.$

Proceeding as usual the second simplex table is as follows.

Table 6·7

B	C_B	c_j	-3	-1	0	0	
		$X_B (= x_B)$	Y_1	$Y_2 (\beta_2)$	$Y_3 (\beta_1)$	Y_4	
Y_3	0	$-1/3$	$-1/3$	0	1	$\boxed{-1/3}$	→
Y_2	-1	$2/3$	$2/3$	1	0	$-1/3$	
$Z_P = C_B X_B = -2/3$		Δ_j	$-7/3$	0	0	$-1/3$	
					↓	↑	

The solution given in table 6·7 is $x_1 = 0, x_2 = 2/3, x_3 = -1/3,$ $x_4 = 0$ which is infeasible but optimal hence it can be improved further for which we again find the leaving and entering vectors.

To determine the leaving vector (β_r).

Since $x_{Br} = \text{Min}\,\{x_{Bi}, x_{Bi} < 0\} = \text{Min}\,\{x_{Br}\} = -1/3 = x_{B1}$

$\therefore \quad r = 1,$ *i.e.* $\beta_1 (= Y_3)$ is the leaving vector.

To determine the entering vector (α_k)

$$\frac{\Delta_k}{y_{rk}} = \frac{\Delta_k}{y_{1k}} = \underset{j}{\text{Min}} \left\{ \frac{\Delta_j}{y_{1j}} \cdot y_{1j} < 0 \right\} = \text{Min} \left\{ \frac{\Delta_1}{y_{11}} \cdot \frac{\Delta_4}{y_{14}} \right\}$$

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$$= \text{Min}\left\{\frac{-7/3}{-1/3}, \frac{-1/3}{-1/3}\right\} = \text{Min}\{7, 1\} = 1 = \frac{\Delta_4}{y_{14}}$$

$\therefore$ $k = 4$ *i.e.* α_4 $(= Y_4)$ is the entering vector.

$\therefore$ Key element $= y_{14} = -1/3$.

Proceeding as usual the **third simplex table** is as follows.

Table 6·8

B	C_B	c_j	-3	-1	0	0
		$X_B\ (= x_B)$	Y_1	$Y_2\ (\beta_2)$	Y_3	$Y_4\ (\beta_1)$
Y_4	0	1	1	0	-3	1
Y_2	-1	1	1	1	-1	0
$Z_P = -1$		Δ_j	-2	0	-1	0

The solution given in table 6·8 is $x_1 = 0, x_2 = 1, x_3 = 0, x_4 = 1$ which is feasible (since all $x_j \geq 0$) and also optimal as all $\Delta_j \leq 0$.

$\therefore$ Optimal solution of the given L.P.P. is

$x_1 = 0, x_2 = 1$, Min $Z = -\text{Max}\, Z_p = 1$.

Ex. 10. *Solve the following L.P.P. by the simplex algorithm*

Min. $Z = 3x_1 + 2x_2 + x_3 + 4x_4$

s.t. $2x_1 + 4x_2 + 5x_3 + x_4 \geq 10$

$3x_1 - x_2 + 7x_3 - 2x_4 \geq 2$

$5x_1 + 2x_2 + x_3 + 6x_4 \geq 15$

and $x_1, x_2, x_3, x_4 \geq 0$. **[Meerut 85, 86, 95]**

Sol.

Step 1. The given L.P.P. in the standard primal form is

Max. $Z_P = -3x_1 - 2x_2 - x_3 - 4x_4$

s.t. $-2x_1 - 4x_2 - 5x_3 - x_4 \leq -10$

$-3x_1 + x_2 - 7x_3 + 2x_4 \leq -2$

$-5x_1 - 2x_2 - x_3 - 6x_4 \leq -15$

and $x_1, x_2, x_3, x_4 \geq 0$.

Since objective function is of maximization and all $c_j < 0$, we can solve this L.P.P. by dual simplex algorithm.

Step 2. Introducing the slack variables x_5, x_6 and x_7 the constraints of the above problem reduce to the following equalities.

$$-2x_1 - 4x_2 - 5x_3 - x_4 + x_5 = -10$$

$$-3x_1 + x_2 - 7x_3 + 2x_4 + x_6 = -2$$

$$-5x_1 - 2x_2 - x_3 - 6x_4 + x_7 = -15.$$

$\therefore$ the starting basic solution to the primal is

$x_1 = 0 = x_2 = x_3 = x_4, x_5 = -10, x_6 = -2, x_7 = -15$

which is infeasible.

The **starting simplex table** is as follows.

Table 6·9

B	C_B	c_j	-3	-2	-1	-4	0	0	0
		$X_B\ (= x_B)$	Y_1	Y_2	Y_3	Y_4	$Y_5\ (\beta_1)$	$Y_6\ (\beta_2)$	$Y_7\ (\beta_3)$
Y_5	0	-10	-2	-4	-5	-1	1	0	0
Y_6	0	-2	-3	1	-7	2	0	1	0
Y_7	0	-15	$\boxed{-5}$	-2	-1	-6	0	0	1 $\rightarrow$
$Z_P = C_B X_B = 0$		Δ_j	-3	-2	-1	-4	0	0	0
			$\uparrow$						$\downarrow$

$\Delta_1 = c_1 - Z_1 = c_1 - C_B Y_1 = -3, \quad \Delta_2 = -2, \Delta_3 = -1, \quad \Delta_4 = 4,$

$\Delta_5 = 0 = \Delta_6 = \Delta_7$.

Thus the starting basic solution is infeasible but optimal.

This solution can be improved further.

To determine the leaving vector (β_r).

Since $x_{Br} = \text{Min}\,(x_{Bi}, x_{Bi} < 0)$

$= \text{Min}(-10, -2, -15) = -15 = x_{B3}$

$\therefore$ $r = 3$ *i.e.* β_3 $(= Y_7)$ is the leaving vector.

To determine the entering vector (α_k).

$$\frac{\Delta_k}{y_{rk}} = \frac{\Delta_k}{y_{3k}} = \underset{j}{\text{Min}}\left\{\frac{\Delta_j}{y_{3j}}, y_{3j} < 0\right\}$$

$$= \text{Min}\left\{\frac{\Delta_1}{y_{31}}, \frac{\Delta_2}{y_{32}}, \frac{\Delta_3}{y_{33}}, \frac{\Delta_4}{y_{34}}\right\}$$

$$= \text{Min}\left\{\frac{-3}{-5}, \frac{-2}{-2}, \frac{-1}{-1}, \frac{-4}{-6}\right\} = \frac{3}{5} = \frac{\Delta_1}{y_{31}}$$

$\therefore$ $k = 1$ *i.e.* α_1 $(= Y_1)$ is the entering vector.

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$\therefore$ Key element $= y_{31} = -5$.

Proceeding as usual the second simplex table is as follows.

Table 6·10

B	C_B	c_j	−3	−2	−1	−4	0	0	0
		X_B (x_B)	$Y_1\ (\beta_3)$	Y_2	Y_3	Y_4	$Y_5\ (\beta_1)$	$Y_6\ (\beta_2)$	Y_7
Y_5	0	−4	0	−16/5	$\boxed{-23/5}$	7/5	1	0	−2/5 →
Y_6	0	7	0	11/5	−32/5	28/5	0	1	−3/5
Y_1	−3	3	1	2/5	1/5	6/5	0	0	−1/5
$Z_P = C_B X_B = -9$		Δ_j	0	−4/5	−2/5	−2/5	0	0	−3/5
					↑		↓		

The solution given in the table 6·10 is $x_1 = 3$, $x_2 = 0 = x_3 = x_4$, $x_5 = -4, x_6 = 7$ and $x_7 = 0$ which is infeasible and optimal.

$\therefore$ It can be improved further.

To determine the leaving vector (β_r)

Since $x_{Br} = \text{Min}(x_{Bi}, x_{Bi} < 0) = \text{Min}(x_{B1}) = \text{Min}(-4) = x_{B1}$.

$\therefore r = 1$ *i.e.* $\beta_1\ (Y_5)$ is the leaving vector.

To determine the entering vector (α_k).

$$\frac{\Delta_k}{y_{rk}} = \frac{\Delta_k}{y_{1k}} = \underset{j}{\text{Min}}\left\{\frac{\Delta_j}{y_{1j}}, y_{1j} < 0\right\} = \text{Min}\left\{\frac{\Delta_2}{y_{12}}, \frac{\Delta_3}{y_{13}}, \frac{\Delta_7}{y_{17}}\right\}$$

$$= \text{Min}\left\{\frac{-4/5}{-16/5}, \frac{-2/5}{-23/5}, \frac{-3/5}{-2/5}\right\}$$

$$= \text{Min}\left\{\frac{1}{5}, \frac{2}{23}, \frac{3}{2}\right\} = \frac{2}{23} = \frac{\Delta_3}{y_{13}}$$

$\therefore k = 3$, $\quad \therefore \alpha_3\ (Y_3)$ is the entering vector.

$\therefore$ Key element $= y_{13} = -23/5$.

Proceeding as usual the third simplex table is as follows.

Table 6·11

B	C_B	c_j	−3	−2	−1	−4	0	0	0
		X_B (x_B)	Y_1 (β_3)	Y_2	Y_3 (β_1)	Y_4	Y_5	Y_6 (β_2)	Y_7
Y_3	−1	20/23	0	16/23	1	−7/23	−5/23	0	2/23
Y_6	0	289/23	0	153/23	0	84/23	−32/23	1	−1/23
Y_1	−3	65/23	1	6/23	0	29/23	1/23	0	−5/23
$Z_P = C_B X_B = -215/23$		Δ_j	0	−12/23	0	−12/23	−2/23	0	−13/23

The solution given in table 6·11 is

$x_1 = 65/23, x_2 = 0, x_3 = 20/23, x_4 = 0 = x_5, x_6 = 289/23, x_7 = 0$

which is feasible and optimal (since all $\Delta_j \leq 0$)

Hence the optimal feasible solution of the given L.P.P. is

$x_1 = 65/23, x_2 = 0, x_3 = 20/23, x_4 = 0$

and $\quad$ Min. $Z = -\text{Max. } Z_P = 215/23$.

Ex. 11. *Use dual simplex method to solve the following L.P.P.*

Mini $\quad Z = 6x_1 + 7x_2 + 3x_3 + 5x_4$

subject to, $\quad 5x_1 + 6x_2 - 3x_3 + 4x_4 \geq 12$

$x_2 + 5x_3 - 6x_4 \geq 10$

$2x_1 + 5x_2 + x_3 + x_4 \geq 8$

and $\quad x_1, x_2, x_3, x_4 \geq 0$

[Meerut 90, 95 (BP), 98 (Old)]

Sol. Step 1. The given L.P.P. in the standard primal form is

Max. $\quad Z_P = -6x_1 - 7x_2 - 3x_3 - 5x_4$

s.t. $\quad -5x_1 - 6x_2 + 3x_3 - 4x_4 \leq -12$

$-x_2 - 5x_3 + 6x_4 \leq -10$

$-2x_1 - 5x_2 - x_3 - x_4 \leq -8$

and $\quad x_1, x_2, x_3, x_4 \geq 0,$

since objective function is of maximization and all $c_j < 0$, we can solve this L.P.P. by dual simplex algorithm.

Step 2. Introducing the slack variables x_5, x_6 and x_7 the constraints of the above problem reduce to the following equalities

$-5x_1 - 6x_2 + 3x_3 - 4x_4 + x_5 = -12$

$$-x_2 - 5x_3 + 6x_4 + x_6 = -10$$
$$-2x_1 - 5x_2 - x_3 - x_4 + x_7 = -8$$

$\therefore$ the starting basic solution to the primal is

$$x_1 = 0 = x_2 = x_3 = x_4, x_5 = -12, x_6 = -10, x_7 = -8$$

which is feasible.

The starting simplex table is as follows.

Table 6·12

B	C_B	c_j	−6	−7	−3	−5	0	0	0
		X_B (= x_B)	Y_1	Y_2	Y_3	Y_4	Y_5 β_1	Y_6 β_2	Y_7 β_3
Y_5	0	−12	−5	$\boxed{-6}$	3	−4	1	0	0 →
Y_6	0	−10	0	−1	−5	6	0	1	0
Y_7	0	−8	−2	−5	−1	−1	0	0	1
$Z_P = 0$		Δ_j	−6	−7 ↑	−3	−5	0 ↓	0	0

$$\Delta_1 = c_1 - Z_1 = c_1 - C_B Y_1 = -6, \Delta_2 = -7, \Delta_3 = -3, \Delta_4 = -5,$$
$$\Delta_5 = 0 = \Delta_6 = \Delta_7$$

Thus the starting basic solution is infeasible but optimal.

To determine the leaving vector (β_r).

Since $x_{Br} = \text{Mini}\,(x_{Bi}, x_{Bi} < 0) = \text{Mini}(-12, -10, -8)$
$= -12 = x_{B1}$

$\therefore r = 1$, *i.e.*, $\beta_1 (= Y_5)$ is the leaving vector.

To determine the entering vector (α_k)

$$\frac{\Delta_k}{y_{rk}} = \frac{\Delta_k}{y_{1k}} = \underset{j}{\text{Mini}}\left\{\frac{\Delta_j}{y_{1j}}, y_{1j} < 0\right\} = \text{Mini}\left\{\frac{\Delta_1}{y_{11}}, \frac{\Delta_2}{y_{12}}, \frac{\Delta_4}{y_{14}}\right\}$$
$$= \text{Mini}\left\{\frac{-6}{-5}, \frac{-7}{-6}, \frac{-5}{-4}\right\} = \frac{7}{6} = \frac{\Delta_2}{y_{12}}.$$

$\therefore k = 2$, *i.e.* $\alpha_2 (= Y_2)$ is the entering vector.

$\therefore$ key element $= y_{12} = -6$.

Proceeding as usual the second simplex table is as follows.

Table 6·13

B	C_B	c_j	−6	−7	−3	−5	0	0	0
		X_B (= x_B)	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
Y_2	−7	2	5/6	1	−1/2	2/3	−1/6	0	0
Y_6	0	−8	5/6	0	$\boxed{-11/2}$	20/3	−1/6	1	0 →
Y_7	0	2	13/6	0	−7/2	7/3	−5/6	0	1
$Z_P = -14$		Δ_j	−1/6	0	−13/2 ↑	−1/3	−7/6	0 ↓	0

The solution given in table 6·13 is

$$x_1 = 0 = x_3 = x_4 = x_5, x_2 = 2, x_6 = -8, x_7 = 2,$$

which is infeasible and optimal.

$\therefore$ It can be improved further.

To determine the leaving vector (β_r).

Since $x_{Br} = \text{Mini}(x_{Bi}, x_{Bi} < 0) = \text{Mini}(x_{B2})$
$= \text{Mini}(-8) = -8 = x_{B2}$

$\therefore r = 2$, *i.e.* $\beta_2 (= Y_6)$ is the leaving vector.

To determine the entering vector (α_k).

$$\frac{\Delta_k}{y_{rk}} = \frac{\Delta_k}{y_{2k}} = \underset{j}{\text{Mini}}\left\{\frac{\Delta_j}{y_{2j}}, y_{2j} < 0\right\} = \text{Mini}\left\{\frac{\Delta_3}{y_{23}}, \frac{\Delta_5}{y_{25}}\right\}$$
$$= \text{Mini}\left\{\frac{-13/2}{-11/2}, \frac{-7/6}{-1/6}\right\} = \text{Mini}\left\{\frac{13}{11}, \frac{7}{1}\right\} = \frac{13}{11} = \frac{\Delta_3}{y_{23}}$$

$\therefore k = 3$ *i.e.* $\alpha_3 (= Y_3)$ is the entering vector.

$\therefore$ Key element $= y_{23} = -11/2$.

Proceeding as usual the third simplex table is as follows.

Table 6·14

B	C_B	c_j	−6	−7	−3	−5	0	0	0
		X_B (= x_B)	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
Y_2	−7	30/11	25/33	1	0	2/33	−5/33	−1/11	0
Y_3	−3	16/11	−5/33	0	1	−40/33	1/33	−2/11	0
Y_7	0	8/11	18/11	0	0	−21/11	−8/11	−7/11	1
$Z_P = -258/11$		Δ_j	−38/33	0	0	−271/33	−32/33	−13/11	0

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The solution given in table 6·14 is

$x_1 = 0 = x_4 = x_5 = x_6, x_2 = 30/11, x_3 = 16/11$ and $x_7 = 78/11$

which is feasible and optimal.

Hence the optimal feasible solution of the given L.P.P. is

$x_1 = 0, x_2 = 30/11, x_3 = 16/11, x_4 = 0$

and Min. $Z = -\text{Max.}\ Z_p = 258/11$.

§ 6·16. Primal Dual Algorithm. [Meerut ...]

When artificial variables are introduced (necessarily) in L.P.P. then we use either two phase method or Charne's M-method for the removal of the artificial vectors from the basis i.e. for driving the artificial variables to zero. In the process of driving the artificial variables to zero, we are hardly concerned with the optimality of the problem because the criterion used to select the vector to enter the basis is only concerned with driving the artificial variable to zero. Thus at the end of phase I or when all the artificial variables are zero in Carnes M-method, the resulting feasible solution may not be any where near optimal. Hence we require a large number of iterations (improvements) to get the optimal solution in phase II. It would be very much beneficial if both the objectives of optimality and of driving the artificial variables to zero could be attained in phase I, this will reduce considerably the total number of iterations.

The **primal-dual algorithm** is developed for this purpose.

We have seen that the dual simplex algorithm can be used in special circumstances to remove the necessity of introducing artificial variables. Thus dual simplex algorithm cannot be used in general because it is not always easy to find a basic solution with all $c_j - Z_j \leq 0$.

In primal dual algorithm we introduce the artificial variables and hence require phase I. But here the computational procedure is such that at the end of phase I we found an optimal as well as feasible solution of the given L.P.P.

In this method we work simultaneously on the primal and dual problems. This method is symmetrically followed as follows:

1. We start with a solution to the dual problem *i.e.* first of all we determine an initial dual solution.
2. Then we proceed to find an infeasible solution to the primal such that the theorem of complementary slackness (§ 6·5) remains satisfied. For this we proceed as follows:

(a) We associate with the initial dual solution a restricted primal problem which is deduced from the original primal by zeroing certain variables to satisfy the theorem of complementary slackness and by replacing the original linear form by the sum of artificial variables.

(b) Then we maximize the sum of the artificial variables by using simplex method.

(c) In case the optimal solution thus obtained is not the primal solution of the original primal then we determine a new dual solution and apply steps (a) and (b).

We proceed in this way again and again till we obtain an optimal solution of the original primal.

Thus when a feasible solution to the primal is found it is also optimal.

§ 6·17. To determine the Initial Dual solution.

Let the given L. P. P. be as follows.

Primal Problem.

$$\left.\begin{array}{ll} \text{Max.} & Z_P = c_1x_1 + c_2x_2 + \dots + c_nx_n = \mathbf{c}\,\mathbf{x} \\ \text{s.t.} & A\mathbf{x} = \mathbf{b} \\ \text{and} & \mathbf{x} \geq \mathbf{0} \end{array}\right] \quad \dots(1)$$

Dual Problem. The dual of the above primal is given by

$$\left.\begin{array}{ll} \text{Max.} & Z_D = b_1w_1 + b_2w_2 + \dots + b_mw_m = \mathbf{b}'\mathbf{w} \\ \text{s.t.} & A'\mathbf{w} \geq \mathbf{c}'. \end{array}\right] \quad \dots(2)$$

In the discussion of the dual simplex algorithm we have seen that it is not easy to find an immmediate solution to the dual (2). It was Beale who suggested a very simple modification to the primal (1), which provide us an immediate solution of the dual. The procedure is as follows.

Introducing an additional constraint

$$x_0 + \mathbf{x} = x_0 + x_1 + x_2 + \dots + x_n = b_0$$

in the given primal (1), we obtain the following new primal called the **modified primal problem.**

$$\text{Max. } Z_P = 0 \cdot x_0 + c_1x_1 + c_2x_2 + \dots + c_nx_n = (0, \mathbf{c}) \cdot [x_0, \mathbf{x}]$$

$$\left.\begin{array}{ll} \text{s.t.} & x_0 + \mathbf{x} = b_0 \\ & A\mathbf{x} = \mathbf{b} \\ \text{and} & [x_0, \mathbf{x}] \geq \mathbf{0} \end{array}\right] \quad \dots(3)$$

Here it is assumed that the new constant b_0 has arbitrarily large value.

(3) can also be written as Max. $Z_P = \mathbf{c}\,\mathbf{x}$

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$$\text{s.t.} \begin{bmatrix} 1 & 1 \\ 0 & A \end{bmatrix} \begin{bmatrix} x_0 \\ \mathbf{x} \end{bmatrix} = \begin{bmatrix} b_0 \\ \mathbf{b} \end{bmatrix}, [x_0, \mathbf{x}] \geq 0 \qquad \ldots(4)$$

The dual of the modified primal (3) or (4), called the **modified dual is as follows.**

$$\text{Min. } Z_D = b_0 w_0 + \mathbf{b'w} = b_0 w_0 + b_1 w_1 + \ldots + b_m w_m$$

$$\text{s.t.} \begin{bmatrix} 1 & 0 \\ 1 & A' \end{bmatrix} \begin{bmatrix} w_0 \\ \mathbf{w} \end{bmatrix} \geq \begin{bmatrix} 0 \\ \mathbf{c'} \end{bmatrix}$$

which can also be written as

$$\left.\begin{array}{l} 1.w_0 + 0.w_1 + 0.w_2 + \ldots\ldots + 0.w_m \quad \geq 0 \\ 1.w_0 + a_{11} w_1 + a_{21} w_2 + \ldots + a_{m1} w_m \geq c_1 \\ \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ 1.w_0 + a_{1n} w_1 + a_{2n} w_2 + \ldots + a_{mn} w_m \geq c_n \end{array}\right] \qquad \ldots(5)$$

where $w_1, w_2, \ldots, w_m$ are unrestricted in sign and by virtue of first constraint in (5),

$$w_0 \geq 0 \text{ i.e. } w_0 \text{ is non-negative.}$$

In case $w_0 = 0$ then (5) is converted to the dual (2) of primal (1). We can immediately obtain a solution of this dual (5) which is

$$\left.\begin{array}{ll} w_0 = \text{Max.} \{0, c_j\} & \mathbf{w} = 0 \\ j = 1, \ldots, n. & \text{i.e. } w_1 = 0 = w_2 = \ldots = w_m. \end{array}\right] \qquad \ldots(6)$$

To solve (5) we add surplus variables w_{sj}, $j = 0, 1, 2, \ldots, n$ to the constraints of (5), and then the resulting problem is :

$$\text{Min. } Z_D = b_0 w_0 + b_1 w_1 + \ldots + b_m w_m + 0.w_{s0} + 0_{s1} + \ldots + 0w_{sn}$$

$$\left.\begin{array}{ll} \text{s.t.} & 1.w_0 + 0.w_1 + 0.w_2 + \ldots + 0.w_m - w_{s0} = 0 = c_0 \\ & 1.w_0 + a_{11} w_1 + a_{21} w_2 + \ldots + a_{m1} w_m - w_{s1} = c_1 \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ & \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\ & 1.w_0 + a_{1n} w_1 + a_{2n} w_2 + \ldots + a_{mn} w_n - w_{sn} = c_n \end{array}\right] \qquad \ldots(7)$$

and $\quad w_{s0}, w_{sj} \geq 0$

i.e. $\quad w_{s0}, w_{s1}, \ldots, w_{sn} \geq 0$

For the solution (6), from (7) we have

$$\left.\begin{array}{l} w_{s0} = w_0 - c_0 \\ w_{s1} = w_0 - c_1 \\ \ldots \quad \ldots \quad \ldots \\ \ldots \quad \ldots \quad \ldots \\ w_{sn} = w_0 - c_n \end{array}\right] \qquad \ldots(8)$$

The solution (6) is the initial dual solution with which we start in primal dual method. Since $w_0 = \text{Max}\{0, c_j\}$,

$$j = 1, \ldots, n,$$

it follows that at least one of $w_{s0}, w_{s1}, \ldots, w_{sn}$ is zero.

From (5), § 6·12, we know that if $\mathbf{x'} \cdot \mathbf{w}_s = 0$ or equivalently for dual problem (4) and (7), if

$$\left.\begin{array}{l} \begin{bmatrix} x_0 \\ \mathbf{x'} \end{bmatrix}' \begin{bmatrix} w_{s0} \\ \mathbf{w}_s \end{bmatrix} = 0, \\ \text{or} \qquad x_0 \cdot w_{s0} + \mathbf{x'} \cdot \mathbf{w}_s = 0 \end{array}\right\} \qquad \ldots(9)$$

then $Z_p = Z_D$ where $\mathbf{x}$ is any solution (not necessarily feasible) to the primal.

If along with the condition (9), $[x_0, \mathbf{x}]$ is a feasible solution to the modified primal (3) or (4) then $[x_0, \mathbf{x}]$ is an optimal feasible solution to the modified primal and $[w_0, \mathbf{w}]$ is an optimal solution to the modified dual.

If in addition $w_0 = 0$, then $Z_P = \mathbf{cx} = Z_D = \mathbf{b'w}$ and from (9) $\mathbf{x'}.\mathbf{w}_s = 0$. Consequently $\mathbf{x}$ is an optimal solution to the given primal and $\mathbf{w}$ is an optimal solution to its dual. If we get a solution to the modified dual with $w_0 \neq 0$ and a feasible solution to the modified primal such that (9) is satisfied, then it follows that $x_0 = 0$, and then

$$Z_P = Z_D = b_0 w_0 + \mathbf{b'w}.$$

§ 6·18. To determine the Restricted Primal (or Extended Primal Problem).

To solve the modified primal problem (3) or (4) of § 6·17 we add artificial variables to each of the constraints and assign price -1 to each artificial variables and zero prices corresponding to every legitimate variable. Thus we get following new primal, called the **restricted primal or extended primal.**

$$\text{Max. } Z^* = -1.x_{a0} - 1.x_{a1} - 1.x_{a2} \ldots - 1.x_{am} = -1.x_{a0} - 1.\mathbf{x}_a$$

$$\text{s.t} \begin{bmatrix} 1 & 1 \\ 0 & A \end{bmatrix} \begin{bmatrix} x_0 \\ \mathbf{x} \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & I_m \end{bmatrix} \begin{bmatrix} x_{a0} \\ \mathbf{x}_a \end{bmatrix} = \begin{bmatrix} b_0 \\ \mathbf{b} \end{bmatrix}$$

and

$$[x_0, \mathbf{x}] \geq 0, \quad [x_{a0}, \mathbf{x}_a] \geq 0.$$

The constraints can be written as

$$x_0 + x_1 + x_2 + \ldots + x_n + x_{a0} = b$$

$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \qquad + x_{a1} \qquad = 1$

$a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n \qquad + x_{a2} \qquad = 1$

....

....

$a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n \qquad + x_{am} = 1,$

and $\quad x_0, x_1, \ldots, x_n \geq 0, x_{a0}, x_{a1}, \ldots, x_{am} \geq 0$

Now we shall apply simplex method on the above restricted primal and go on improving Z^* by entering some vector in place of some leaving vector in each iteration till we obtain $Z^* = 0$.

Which follows that

$$x_{a0} = 0 = x_{a1} = \ldots = x_{am}.$$

The criterion for the selection of the entering and leaving vector will be different from that in the regular simplex method.

Here we shall use the following notations :

1. $(m+1)$ component legitimate activity vectors will be denoted by $Y_j^{(1)}, j = 0, 1, \ldots, n$.
2. A basis matrix for the constraints will be denoted by B_1.
3. The vector containing the prices c_j^* of the variables in the basis will be denoted by C_{B1}^*.
4. $\Delta_j^* = c_j^* - Z_j^* = c_j^* - C_{B1}^* \cdot B_1^{-1} \alpha_j^{(1)} = c_j^* - C^*_{B1} \cdot Y_j^{(1)}$.

§ 6·19. To find the Entering and Leaving vectors.

Entering vector. In § 6·17, we obtained the solution of the modified dual as follows

$$w_0 = \text{Max}\,\{0, c_j\}, \qquad w_1 = 0 = w_2 = \ldots = w_m$$
$$j = 1, 2, \ldots, n.$$

Which implies that at least one component w_{sj} will be zero. *The vector $Y_j^{(1)}$ for which the corresponding $w_{sj} = 0$ is taken as the vector entering the restricted primal basis.* In this way all x_j for which w_{sj} are strictly positive remain zero.

Leaving vector. The vector leaving the basis corresponding to the entering vector $Y_j^{(1)}$ is selected one by one in the same manner as in Phase 1 of ordinary simplex method (see Ch. 4).

§ 6·20. Method to obtain New Dual Solution $(\hat{w}_{sj})$ and the Improved value of $\hat{Z}_D$ and $\hat{Z}^*$.

(i) The *new dual solution* $\hat{w}_{sj}$ is obtained by using the formula

$$\hat{w}_{sj} = w_{sj} - \theta\,\Delta_j^*$$

where
$$\theta = \text{Mini}\left\{\frac{w_{sj}}{\Delta_j^*}, \Delta_j^* > 0\right\} \qquad \ldots(1)$$

Here also at least one of $\hat{w}_{sj}$ will be zero and hence the corresponding $Y_j^{(1)}$ will be selected to enter the basis at the next iteration.

(ii) The improved value of Z^* will be obtained by the formula

$$Z^* = c_{B1}^* x_B^{(1)} \qquad \ldots(2)$$

(iii) The value of $\hat{Z}_D$, will be obtained by the formula

$$\hat{Z}_D = Z_D + \theta Z^*$$

where θ and Z^* are given by (1) and (2) respectively.

§ 6·21. Test of Optimality.

The primal dual method will terminate (come to end) with one of the following results :

(i) When $Z^* = 0$, we get an optimal solution to the primal and $Z_p = Z_D$, provided $w_{s0} = 0 = w_0$.

(ii) When $Z^* = 0$ but $w_{s0} \neq 0$, the primal problem has an unbounded solution.

(iii) When $Z^* < 0, \Delta_j^* \leq 0$ for all j (at any interation), the primal have no solution.

§ 6·22. Computational Procedure of Primal-Dual Algorithm.

The primal-dual algorithm is applied in the following order :

1. First we construct the restricted primal problem (see § 6·18).
2. Then we determine the initial dual solution by
$$w_0 = \text{Max}\,\{0, c_j\}, c_j > 0, \quad w_1 = 0 = w_2 = \ldots = w_m$$
$$j = 1, \ldots, n.$$
and $\quad w_{s0} = w_0 - c_0, w_{sj} = w_0 - c_j, j = 1, 2, \ldots, n.$
3. Then we select the entering vector $Y_j^{(1)}$ corresponding to that j for which $w_{sj} = 0$.
4. Then we select the leaving vector in the similar manner as in Phase I of regular simplex method.
5. For the vector which are not in the primal basis we compute Δ_j^* by the formula $\Delta_j^* = c_j^* - C_{B1} Y_j^{*(1)}$.

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Also we compute $Z^* = C_{B1}{}^* x_B{}^{(1)}$

and $Z_w = b_0 w_0 + b_1 w_1 + \dots b_m w_m$.

6. Then we construct the starting simplex table as follows.

B_1	$C_{B1}{}^*$	c_j^* / $x_B{}^{(1)}$	$c_0^*\ c_1^*\ c_2^* \dots c_n^* \dots c_{n+m}^*$ / $Y_0^{(1)}\ Y_1^{(1)}\ Y_2^{(1)} \dots Y_n^{(1)} \dots y_{n+m}^{(1)}$	Mini $\left\{\frac{x_B^{(1)}}{Y_k^{(I)}}\right\}$
$Y_{n+1}^{(1)}$	-1	b_0		
$Y_{n+2}{}^{(1)}$	-1			
$\vdots$	$\vdots$			
$Y_{n+m}{}^{(1)}$	-1			
$Z^* = C_{B1} \cdot x_B{}^{(1)}$		x_j	... ...	
		Δ_j^*	... ...	
$Z_w = \sum_{j=0}^{m} b_j w_j$		w_{sj}		

7. If $Z^* = 0$ with all $\Delta_j^* \leq 0$, the solution under test is optimal.
If the solution is not optimal, we proceed to the next iteration.

8. The solution of the new dual is obtained by using the formula

$$\widehat{w}_{sj} = w_{sj} - \theta_0 \Delta_j^*$$

where $\theta_0 = \text{Mini}\left\{\frac{w_{sj}}{\Delta_j^*}, \Delta_j^* > 0\right\}$.

9. We again find the entering vector $Y_j^{(1)}$ for which $\widehat{w}_{sj} = 0$ and repeat steps 5 to 8 again till we get the optimal solution.

10. The optimality of solution of the primal is decided as shown in § 6·21.

Illustrative Examples

Ex. 12. *Solve the following L.P.P. by primal dual method.*

Max. $Z = x_1 + 6x_2$

s.t. $x_1 + x_2 \geq 2$

$x_1 + 3x_2 \leq 3,$

and $x_1, x_2 \geq 0.$

Sol. 1. Adding the slack and surplus variables x_3, x_4 introducing the additional constraint and introducing the artificial variables x_5, x_6, x_7 we obtain the following **restricted primal.**

$$\text{Max. } Z^* = 0.x_0 + 0.x_1 + 0.x_2 + 0.x_3 + 0.x_4 - 1.x_5 - 1.x_6 - 1.x_7.$$

$$\begin{aligned} \text{s.t. } x_0 + x_1 + x_2 + x_3 + x_4 + x_5 &= b_0 \\ x_1 + x_2 - x_3 + x_6 &= 2 \\ x_1 + 3x_2 + x_4 + x_7 &= 3 \end{aligned}$$

2. To find the initial dual solution.

For initial solution to the dual, $w_0 = \text{Max}\{0, c_j\}, c_j > 0$

$j = 1, \dots, 4.$

$= \text{Max}\{0, 1, 6\} = 6$

$w_0 = 6, w_1 = w_2 = w_3 = w_4 = 0.$

$\therefore\ w_{s0} = w_0 - c_0 = 6 - 0 = 6, w_{s1} = w_0 - c_1 = 6 - 1 = 5$

$w_{s2} = w_0 - c_2 = 6 - 6 = 0, w_{s3} = w_0 - c_3 = 6 - 0 = 6.$

$w_{s4} = w_0 - c_4 = 6 - 0 = 6.$

Also $Z_w = \sum_{j=0}^{4} b_j w_j = b_0 w_0 + b_1 w_1 + b_2 w_2 + b_3 w_3 + b_4 w_4$

$= 6b_0 + 0 = 6b_0.$

The starting simplex table is as follows.

Table 6·15

B_1	C_{B1}^*	c^*_j / $X_B^{(1)}$	0 / $Y_0^{(1)}$	0 / $Y_1^{(1)}$	0 / $Y_2^{(1)}$	0 / $Y_3^{(1)}$	0 / $Y_4^{(1)}$	-1 / $Y_5^{(1)}$ (β_1)	-1 / $Y_6^{(1)}$ (β_2)	-1 / $Y_7^{(1)}$ (β_3)	Mini Ratio x_B/Y_2
$Y_5^{(1)}$	-1	b_0	1	1	1	1	1	1	0	0	$b_0/1$
$Y_6^{(1)}$	-1	2	0	1	1	-1	0	0	1	0	2/1
$Y_7^{(1)}$	-1	3	0	1	$\boxed{3}$	0	1	0	0	1	3/3 → Min
$Z^* = C_{B1}{}^* X_B{}^{(1)}$		x_j	0	0	0	0	0	b_0	2	3	
$= -b_0 - 5$		Δ_j^*	1	3	5	0	2	0	0	0	
$Z_w = 6b_0$		w_{sj}	6	5	0	6	6	0	0	0	

↑ entering vector ↓ out going

$\Delta_0^* = c_0^* - C^*_{B1} Y_0^{(1)} = 0 - (-1, -1, -1)(1, 0, 0) = 1$

$\Delta_1^* = c_1^* - C^*_{B1} Y_1^{(1)} = 0 - (-1, -1, -1)(1, 1, 1) = 3$

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$$\Delta_2^* = c_2^* - C^*_{B1} Y_2^{(1)} = 0 - (-1, -1, -1)(1, 1, 3) = 5$$
$$\Delta_3^* = c_3^* - C^*_{B1} Y_3^{(1)} = 0 - (-1, -1, -1)(1, -1, 0) = 0$$
$$\Delta_4^* = c_4^* - C^*_{B1} Y_4^{(1)} = 0 - (-1, -1, -1)(1, 0, 1) = 2$$

Since $\Delta_j^* > 0$, so the solution is not optimal.

Therefore we proceed to improve the solution.

To find entering and outgoing vectors.

Since $w_{s2} = 0 \therefore Y_2^{(1)}$ is taken as the entering vector. By mini. ratio rule the outgoing (leaving) vector is taken as $Y_7^{(1)}$ (β_3).

$\therefore$key element $= y_{32} = 3$.

Proceeding as usual and leaving the column $Y_7^{(1)}$ (outgoing artificial vector), we get the following table.

Table 6·16

B_1	C^*_{B1}	c^*_j	0	0	0	0	0	-1	-1	Mini Ratio
		$X_B^{(1)}$	$Y_0^{(1)}$	$Y_1^{(1)}$	$Y_2^{(1)}$ (β_3)	$Y_3^{(1)}$	$Y_4^{(1)}$	$Y_5^{(1)}$ (β_1)	$Y_6^{(1)}$ (β_2)	X_B/Y_1
$Y_5^{(1)}$	-1	b_0-1	1	2/3	0	1	2/3	1	0	$(b_0-1)/\frac{2}{3}$
$Y_6^{(1)}$	-1	1	0	$\boxed{2/3}$	0	-1	$-1/3$	0	1	$1/\frac{2}{3}\rightarrow$
$Y_2^{(1)}$	0	1	0	1/3	1	0	1/3	0	0	(Mini) $1/\frac{1}{3}$
$Z^* = C^*_{B1} \cdot X_B^{(1)}$		x_j	0	0	1	0	0	b_0-1	1	
$= -b_0$		Δ_j^*	1	4/3	0	0	1/3	0	0	
$\hat{Z}_w^* = Z_w + \theta_0 Z^*$ $= 6b_0 + 15/4 \cdot (-b_0)$		w_{sj}	6	5	0	6	6	0	0	
$= 9b_0/4$		$\hat{w}_{sj}$	9/4	0	0	6	19/4	0	0	
				↑ Entering vector					↓ Out going	

$$\Delta_0^* = c_0^* - C^*_{B1} Y_0^{(1)} = 0 - (-1, -1, 0).(1, 0, 0) = 1$$
$$\Delta_1^* = c_1^* - C^*_{B1} Y_1^{(1)} = 0 - (-1, -1, 0)(2/3, 2/3, 1/3) = 4/3$$

$$\Delta_3^* = c_3^* - C^*_{B1} Y_3^{(1)} = 0 - (-1, -1, 0)(1, -1, 0) = 0$$
$$\Delta_4^* = c_4^* - C^*_{B1} Y_4^{(1)} = 0 - (-1, -1, 0).(2/3, -1/3, 1/3) = 1/3$$

Since $\Delta_j^* > 0$, so solution is not optimal.

$\therefore$ We compute the solution $\hat{w}_{sj}$ of the new dual

$$\theta_0 = \text{Mini}\left\{\frac{w_{sj}}{\Delta_j^*}, \Delta_j^* > 0\right\} = \text{Mini}\left\{\frac{w_{s0}}{\Delta_0^*}, \frac{w_{s1}}{\Delta_1^*}, \frac{w_{s4}}{\Delta_4^*}\right\}$$
$$= \text{Mini}\left\{\frac{6}{1}, \frac{5}{4/3}, \frac{6}{1/3}\right\} = \frac{15}{4} = \frac{w_{s1}}{\Delta_1^*}$$

$\therefore$ the new dual solution is

$$\hat{w}_{s0} = w_{s0} - \theta_0 . \Delta_0^* = 6 - \frac{15}{4} \cdot 1 = \frac{9}{4}$$
$$\hat{w}_{s1} = w_{s1} - \theta_0 \Delta_1^* = 5 - \frac{15}{4} \cdot \frac{4}{3} = 0$$
$$\hat{w}_{s2} = w_{s2} - \theta_0 . \Delta_2^* = 0 - \frac{15}{4} \cdot 0 = 0$$
$$\hat{w}_{s3} = w_{s3} - \theta_0 . \Delta_3^* = 6 - \frac{15}{4} \cdot 0 = 6$$
$$\hat{w}_{s4} = w_{s4} - \theta_0 . \Delta_4^* = 6 - \frac{15}{4} \cdot \frac{1}{3} = \frac{19}{4}$$

To find the improved solution we select the vector $Y_1^{(1)}$ as the **entering vector**, since $\hat{w}_{s1} = 0$, [$Y_2^{(1)}$ is already in the basic so it cannot be selected, even $\hat{w}_{s2} = 0$].

By minimum ratio rule we find the vector $Y_6^{(1)}$ (β_2) as the **outgoing vector.**

$\therefore$ key element $y_{22} = 2/3$.

Proceeding as usual and leaving the column $Y_6^{(1)}$ (outgoing artificial vector), we get the following table. 6·17. (See on page 238).

$$\Delta_0^* = c_0^* - C^*_{B1} Y_0^{(1)} = 0 - (-1, 0, 0).(1, 0, 0) = 1$$
$$\Delta_3^* = c_3^* - C^*_{B1} Y_3^{(1)} = 0 - (-1, 0, 0).(2, -3/2, 1/2) = 2$$
$$\Delta_4^* = c_4^* - C^*_{B1} Y_4^{(1)} = 0 - (-1, 0, 0).(1, -1/2, 1/2) = 1$$

Since $\Delta_j^* > 0$, so the solution is not optimal.

$\therefore$ we compute the solution $\hat{w}_{sj}$ of the new dual

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Table 6·17

B_1	C^*_{B1}	c_j^*	0	0	0	0	0	−1	
		$X_B^{(1)}$	$Y_0^{(1)}$	$Y_1^{(1)}$ (β_2)	$Y_2^{(1)}$ (β_3)	$Y_3^{(1)}$	$Y_4^{(1)}$	$Y_5^{(1)}$ (β_1)	Mini Ratio X_B/Y_0
$Y_5^{(1)}$	−1	b_0-2	$\boxed{1}$	0	0	2	1	1	$(b_0-2)/1 \rightarrow$
$Y_1^{(1)}$	0	3/2	0	1	0	−3/2	−1/2	0	(Mini)
$Y_2^{(1)}$	0	1/2	0	0	1	1/2	1/2	0	
$Z^* = C^*_B \cdot x_B^{(1)}$		x_j	0	3/2	1/2	0	0	−1	
$= -b_0 + 2$		Δ_j^*	1	0	0	2	1	0	
$= \hat{Z}_w + \theta_0 Z^*$ $= 9b_0/4 + 9/4.$		w_{sj}	9/4	0	0	6	19/4	0	
$(b_0 + 2) = 9/2$		$\hat{w}_{sj}$	0	0	0	3/2	5/2	0	
			↑ entering vector					↓	

$$\theta_0 = \text{Mini}\left\{\frac{\hat{w}_{sj}}{\Delta_j^*}, \Delta_j^* > 0\right\} = \text{Mini}\left\{\frac{\hat{w}_{s0}}{\Delta_0^*}, \frac{\hat{w}_{s3}}{\Delta_3^*}, \frac{\hat{w}_{s4}}{\Delta_4^*}\right\}$$

$$= \text{Mini}\left\{\frac{9/4}{1}, \frac{6}{2}, \frac{19/4}{1}\right\} = \frac{9}{4} = \frac{\hat{w}_{s0}}{\Delta_0^*}$$

$\therefore$ the new dual solution is

$$\hat{w}'_{s0} = \hat{w}_{s0} - \theta_0 . \Delta_0^* = 9/4 - 9/4.1 = 0$$
$$\hat{w}'_{s1} = \hat{w}_{s1} - \theta_0 . \Delta_1^* = 0 - 9/4.0 = 0$$
$$\hat{w}'_{s2} = \hat{w}_{s2} - \theta_0 . \Delta_2^* = 0 - 9/4.0 = 0$$
$$\hat{w}'_{s3} = \hat{w}_{s3} - \theta_0 . \Delta_3^* = 6 - 9/4.2 = 3/2$$
$$\hat{w}'_{s4} = \hat{w}_{s4} - \theta_0 . \Delta_4^* = 19/4 - 9/4.1 = 5/2.$$

To find the improved solution we select the vector $Y_0^{(1)}$ as the entering vector, since $w'_{s0} = 0$, $[Y_1^{(1)}, Y_2^{(1)}$, are already in basis].

By minimum ratio rule we find the vector $Y_5^{(1)}(\beta_1)$ as the outgoing vector.

$\therefore$ key element $= y_{11} = 1$.

Proceeding as usual and leaving the column $Y_5^{(1)}$ (outgoing artificial vector), we get the following table.

Table 6·18

B_1	C_{B1}^*	c_j^*	0	0	0	0	0
		$X_B^{(1)}$	$Y_0^{(1)}$	$Y_1^{(1)}$	$Y_2^{(1)}$	$Y_3^{(1)}$	$Y_4^{(1)}$
$Y_0^{(1)}$	0	b_0-2	1	0	0	2	1
$Y_1^{(1)}$	0	3/2	0	1	0	−3/2	−1/2
$Y_2^{(1)}$	0	1/2	0	0	1	1/2	1/2
$Z^* = C^*_{B1} . x_B^{(1)}$		x_j	$b_0 - 2$	3/2	1/2	0	0
$= 0$		Δ_j^*	0	0	0	0	0
$Z''_w = Z'_w + \theta_0 . Z^*$ $= 9/2 + \theta_0 . 0$ $= 9/2$		w'_{sj}	0	0	0	3/2	5/2

$$\Delta_3^* = c_3^* - C^*_{B1} . Y_3^{(1)} = 0 - (0,0,0).(2, -3/2, 1/2) = 0$$
$$\Delta_4^* = c_4^* - C^*_{B1} . Y_4^{(1)} = 0 - (0,0,0).(1, -1/2, 1/2) = 0.$$

Here all $\Delta_j^* \le 0$ $\therefore$ The solution under test is optimal. $Z^* = 0$. Since $w_0 = 0 = \hat{w}_{s0}$. $\therefore$ The sol. of the given primal problem is also optimal. Thus the optimal sol. of the given primal is $x_1 = 3/2$, $x_2 = 1/2$, $Z = \hat{Z}_w'' = 9/2$. **Ans.**

Exercise on Chapter VI

1. Define the dual of a L.P.P. State the fundamental properties of duality with reference of a L.P.P. and prove any one of them.
2. Write a short note on duality theory. **[Meerut 86]**
3. Define the dual of a L.P.P. and prove that if the primal has an unbounded solution, the dual has no solution and vice versa.
4. Prove that if either the primal or the dual problem has a finite optimum solution, then the other problem has a finite optimum solution and the extrema of the linear functions are equal, *i.e.* Max Z = Mini Z_D. And hence show that if either problem has an unbounded optimum solution then the other problem has no feasible solution.
5. Write a short note on Primal Dual Algorithm. **[Meerut (Spe.), 86, 90]**

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6. Write short note on "Reading the solution to the dual from the final simplex table of the primal."
7. What is the essential difference betwen regular simplex method and dual simplex method. [Meerut 90]
8. Explain the Primal Dual Algorithm, pointing out its advantages over other algorithms used for similar purpose.
9. Solve the following L. P. P. by the dual simplex algorithm.

Mini $Z = x_1 + 2x_2$
s.t. $2x_1 + x_2 \ge 4$
$x_1 + 7x_2 \ge 7$
and $x_1, x_2 \ge 0$
[Meerut 94]

10. Mini $Z = x_1 + 2x_2 + 3x_3$
s.t. $2x_1 - x_2 + x_3 \ge 4$
$x_1 + x_2 + 2x_3 \le 8$
$x_2 - x_3 \ge 2$
and $x_1, x_2, x_3 \ge 0$
[Meerut 93]

11. Max. $Z = -2x_1 - 2x_2 - 4x_3$
s.t. $2x_1 + 3x_2 + 5x_3 \ge 2$
$3x_1 + x_2 + 7x_3 \le 3$
$x_1 + 4x_2 + 6x_3 \le 5$
and $x_1, x_2, x_3 \ge 0$

12. Mini $Z = 2x_1 + x_2$
s.t. $3x_1 + x_2 \ge 3$
$4x_1 + 3x_2 \ge 6$
$x_1 + 2x_2 \ge 3$
and $x_1, x_2 \ge 0$

13. Max. $Z = -2x_1 - x_3$
s.t. $x_1 - 2x_2 + 4x_3 \ge 8, x_1 + x_2 - x_3 \ge 5$ and $x_1, x_2 x_3 \ge 0$.

Solve the following problems by primal dual method.

14. Max. $Z = x_1 + 2x_2$
s.t. $3x_1 + 2x_2 \ge 6$
$x_1 + 6x_2 \ge 3$
and $x_1, x_2 \ge 0$

15. Max. $Z = 7x_1 + 2x_2 + x_3 + 4x_4 + 6x_5$
s.t. $3x_1 + 5x_2 - x_3 + 2x_4 + 4x_5 = 27$
$x_1 + 2x_2 + 3x_3 - 7x_4 + 6x_5 \ge 2$
$9x_1 - 4x_2 + 2x_3 + 5x_4 - 2x_5 = 16$
and $x_j \ge 0, j = 1, 2, \ldots 5$.

16. Can you apply Dual Simplex Algorithm to the following L.P.P.
Max. $Z = 5x_1 + 3x_2$
s.t. $3x_1 + 5x_2 \le 15,\ 5x_1 + 2x_2 \le 10, x_1,\ x_2 \ge 0$.
17. A diet conscious house wife wishes to ensure certain minimum intake of Vitamins A, B and C for the family. The minimum daily (quantity) needs of the vitamins A, B and C for the family are respectively 30, 20 and 16 units. For the supply of these minimum vitamin requirements, the housewife relies on two fresh foods. The first one provides 7, 5, 2 units of the three vitamins per gram respectively and the second one provides 2, 4, 8 units of the same three vitamins per gram of the food stuff respectively. The first food stuff costs Rs. 3 per gram and the second Rs. 2 per gram. The

problem is how many grams of each foodstuff should the housewife buy everyday to keep her food bill as low as possible ?
(i) Formulate the underlying L. P. problem.
(ii) Write the "Dual" problem.
(iii) Solve the "Dual" problem by using simplex method.
(iv) Solve the primal problem graphically.
(v) Interpret the dual problem and its solution.

18. A company makes three products X, Y, Z out of three materials P_1, P_2 and P_3. The three products use units of the three materials according to the following table :

	P_1	P_2	P_3	Profit/unit
X	1	2	3	3
Y	2	1	1	4
Z	3	2	1	5
Amount available	10	12	15	

Determine the product mix which will maximize the total profit. Solve the primal problem and write the dual and give its economic interpretation.

Answers

9. $x_1 = 21/13, x_2 = 10/13, Z = 41/13$.
10. $x_1 = 3, x_2 = 2, x_3 = 0, Z = 7$.
11. $x_1 = 0, x_2 = 2/3, x_3 = 0, Z = -4/3$.
12. $x_1 = 3/5, x_2 = 6/5, Z = 12/5$.
13. $x_1 = 0, x_2 = 14, x_3 = 9, Z = -9$.
14. $x_1 = 15/8, x_2 = 3/16, Z = 9/4$.
17. (i) Mini $Z = 3x_1 + 2x_2$, s.t. $7x_1 + 2x_2 \ge 30, 5x_1 + 4x_2 \ge 20,$
$2x_1 + 8x_2 \ge 16, x_1, x_2, x_3 \ge 0$.
(ii) Max. $Z_D = 30w_1 + 20w_2 + 16w_3$, s.t. $7w_1 + 5w_2 + 2w_3 \le 3,$
$2w_1 + 4w_2 + 8w_3 \le 2, w_1, w_2, w_3 \ge 0$.
(iii) $w_1 = 5/13, w_2 = 0, w_3 = 2/13, Z_D = 14$.
(iv) $x_1 = 4,\ x_2 = 1,\ Z = 14$.

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10

Assignment Problem

§ 10·1. Introduction

It is a special type of linear programming problem in which the objective is to find the optimum allocation of a number of tasks (jobs) to an equal number of facilities (persons). Here we make the assumption that each person can perform each job but with varying degree of efficiency. For example, a departmental head may have four persons available for assignment and four jobs to fill. Then his interest is to find the best assignment which will be in the best interest of the department.

Although simplex method is powerful enough to solve all the L.P. problems, but the above type of the problems may be solved by special procedures which are described in the following sections.

§ 10·2. Assignment Problem

The assignment problem can be stated in the form of $n \times n$ matrix $[c_{ij}]$ called cost or effectiveness matrix, where c_{ij} is the cost of assigning i-th facility (person) to the j-th job.

Effectiveness matrix

Jobs

$$\text{Persons}\quad \begin{array}{c|ccccccc|} & 1 & 2 & 3 & \cdots & j & \cdots & n \\ \hline 1 & c_{11} & c_{12} & c_{13} & \cdots & c_{1j} & \cdots & c_{1n} \\ 2 & c_{21} & c_{22} & c_{23} & \cdots & c_{2j} & \cdots & c_{2n} \\ 3 & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \vdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ i & c_{i1} & c_{i2} & c_{i3} & \cdots & c_{ij} & \cdots & c_{in} \\ \vdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ n & c_{n1} & c_{n2} & c_{n3} & \cdots & c_{nj} & \cdots & c_{nn} \\ \hline \end{array}$$

Mathematical formulation of assignment problem

[Raj. 85; Meerut 96 (BP)]

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Mathematically an assignment problem can be stated as follows :

Minimize the total cost

$$Z=\sum_{i=1}^{n}\sum_{j=1}^{n} c_{ij}\,.x_{ij}$$

where $x_{ij}=\begin{cases}1, \text{ if } i\text{th person is assigned to the } j\text{th job}\\ 0, \text{ if } i\text{th person is not assigned to the } j\text{th job}\end{cases}$

Subject to the conditions

(i) $\sum_{i=1}^{n} x_{ij}=1, \quad j=1,2,\ldots,n$

which means that only one job is done by the i-th person, $i=1,2,\ldots,n$.

(ii) $\sum_{j=1}^{n} x_{ij}=1, \quad i=1,2,\ldots,n$

which means that only one person should be assigned to the j-th job, $j=1,2,\ldots,n$.

§ 10·3. Important theorems

Now we shall prove the following two important theorems on which the procedure of solution of an assignment problem is based.

Theorem 1. (Reduction theorem). If, in an assignment problem, a constant is added or subtracted to every element of a row (or column) of the cost matrix $[c_{ij}]$, then an assignment which minimizes the total cost for one matrix, also minimize the total cost for the other matrix.

[Meerut 94, 96; Bhopal (Stat.) 86; Agra 84]

Or Mathematically the theorem may be stated as follows :

If $x_{ij}=X_{ij}$, **minimizes** $Z=\sum_{i=1}^{n}\sum_{j=1}^{n} c_{ij}x_{ij}$

s.t. $\sum_{i=1}^{n} x_{ij}=1=\sum_{j=1}^{n} x_{ij}$ **and** $x_{ij}\geq 0$

then $x_{ij}=X_{ij}$ **also minimizes** $Z'=\sum_{i=1}^{n}\sum_{j=1}^{n} c'_{ij}\,.x_{ij}$

where $c'_{ij}=c_{ij}\pm a_i\pm b_j$.

a_i, b_j **are constants,** $i=1,2,\ldots,n;\ j=1,2,\ldots,n$.

Proof : We have

$$Z'=\sum_{i=1}^{n}\sum_{j=1}^{n} c'_{ij}x_{ij}$$

$$=\sum_{i=1}^{n}\sum_{j=1}^{n}(c_{ij}\pm a_i\pm b_j)\,x_{ij}$$

$$=\sum_{i=1}^{n}\sum_{j=1}^{n} c_{ij}x_{ij}\pm\sum_{i=1}^{n}\sum_{j=1}^{n} a_i x_{ij}\pm\sum_{i=1}^{n}\sum_{j=1}^{n} b_j x_{ij}$$

$$=Z\pm\sum_{i=1}^{n} a_i\,.\sum_{j=1}^{n} x_{ij}\pm\sum_{j=1}^{n} b_j\sum_{i=1}^{n} x_{ij}$$

$$=Z\pm\sum_{i=1}^{n} a_i\,.1\pm\sum_{j=1}^{n} b_j\,.1$$

$$=Z\pm\sum_{i=1}^{n} a_i\pm\sum_{j=1}^{n} b_j$$

Since $\sum_{i=1}^{n} a_i$, $\sum_{j=1}^{n} b_j$ are independent of x_{ij}, if follows that Z' is minimized when Z is minimized.

Hence $x_{ij}=X_{ij}$ which minimizes Z also minimize Z'.

Theorem 2. If all $c_{ij}\geq 0$ and there exists a solution $x_{ij}=X_{ij}$.

s.t. $$\sum_{i=1}^{n}\sum_{j=1}^{n} c_{ij}x_{ij}=0$$

then this solution is an optimal solution (*i.e.*, this solution minimizes Z).

Proof : Since all $c_{ij}\geq 0$.

$\therefore \quad Z=\sum_{i=1}^{n}\sum_{j=1}^{n} c_{ij}x_{ij}$ cannot be negative.

Thus its minimum value is 0, when $x_{ij}=X_{ij}$.

Hence the solution $x_{ij}=X_{ij}$ for which $\sum_{i=1}^{n}\sum_{j=1}^{n} c_{ij}=0$ is an optimal solution.

§ 10·4. Hungarian Method for solving an assignment problem (Assignment algorithm)

From the above two theorems we get a powerful method termed as **"assignment algorithm"** for solving an assignment problem. The procedure for the solution is as follows :

Step 1. Subtract the minimum element of each row in the cost matrix $[c_{ij}]$ from every element of the corresponding row.

Step 2. Subtract the minimum element of each column in the reduced matrix obtained in the step 1 from every element of the corresponding column.

Step 3. (*a*) Starting with row 1 of the matrix obtained in step 2, examine rows successively until a row with exactly one zero element is found. Mark (☐) at this zero, as an assignment will be made there. Mark (×) at all others zeros in the column (in which we mark ☐) to show that they cannot be used to make other assignments. Proceed in this way until the last row is examined.

(*b*) After examining all the rows completely, proceed similarly examining the columns. Examine columns starting with column 1 until a column containing exactly one unmarked zero is found. Mark (☐) at this zero and cross mark (×) at all zeros of the row in which ☐ is marked. Proceed in this way until the last column is examined.

(*c*) Continue these operations (*a*) and (*b*) successively until we reach to any of the two situations.

(i) all the zeroes are marked ☐ or crossed.

or (ii) the remaining unmarked zeroes lie at least two in each row and column.

In case (i), we have a maximal assignment (assignment as much as we can) and in case (ii) still we have some zeroes to be treated for which we use the trial and error method to avoid the use of highly complicated algorithm.

Now there are two possibilites :

(i) If it has as assignment in every row and every column (*i.e.*, total number of marked ☐ zeroes is exactly *n*), then the complete optimal assignment is obtained. (See example 1).

(ii) If it does not contain assignment in every row and every column (*i.e.*, the total number of marked ☐ zero is less than *n*), then one has to modify the cost (effectiveness) matrix by adding or subtracting to create some more zeroes in it. For this proceed to the next step 4.

Step 4. When the matrix obtained in step 3 does not contain assignment in every row and every column then we draw the minimum number of horizontal and vertical lines necessary to cover all zeros at least once. For this the following procedure is adopted.

(i) Mark (√) all rows in which assignment have not been made.

(ii) Mark (√) column which have zeros in marked rows.

(iii) Mark (√) rows (not already marked) which have assignments in marked columns.

(iv) Repeat step (ii) and (iii) until the chain of marking ends.

(v) Draw minimum number of lines through unmarked rows and through marked columns to cover all the zeros.

This procedure will yield the minimum number of lines (equal to the number of assignments in the maximal assignment obtained in step 3) that will pass through all zeros.

Step 5. Select the smallest of the elements that do not have a line through them, subtract it from all the element that do not have a line through them, add it to every element that lies at the intersection of two lines and leave the remaining elements of the martix unchanged.

Step 6. At the end of step 5, number of zeros are increased (never decrease) in the matrix than that in step 3.

Now re-apply the step 3 to the modified matrix obtained in step 5, to obtain the desired solution. For the clear understanding of the procedure see Ex. 4.

Illustrative Examples

I

Ex. 1. *Solve the following minimal assignment problem.*

Job ↓ \ Man →	1	2	3	4
I	12	30	21	15
II	18	33	9	31
III	44	25	24	21
IV	23	30	28	14

Sol. For the clear understanding, this example is solved step by step systematically.

Step 1. Subtracting the smallest element of each row from every element of the corresponding row, we get the following matrix :

	1	2	3	4
I	0	18	9	3
II	9	24	0	22
III	23	4	3	0
IV	9	16	14	0

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Step 2. Subtracting the smallest element of each column from every element of the corresponding column, we get the following matrix.

Table 10·1

	1	2	3	4
I	0	14	9	3
II	9	20	0	22
III	23	0	3	0
IV	9	12	14	0

Step 3. Now we test whether it is possible to make an assignment using the zeros by the method described in step 3 page 332.

Starting with row 1, we mark □ (*i.e.* make assignment) in the rwo containing only one zero and cross (×) the zeros in the corresponding column in which □ lies. Thus we get the following table 10·2.

Table 10·2

	1	2	3	4
I	$\boxed{0}$	14	9	3
II	9	20	$\boxed{0}$	22
III	23	0	3	$\cancel{0}$
IV	9	12	14	$\boxed{0}$

Again starting with column 1, we mark □ (*i.e.* make assignment) in column containing only one unmarked or uncrossed zero in table 10·2 and cross out the zeros in corresponding row in which this assignment □ is marked. Thus we get the table 10·3.

Table 10·3

	1	2	3	4
I	$\boxed{0}$	14	9	3
II	9	20	$\boxed{0}$	22
III	23	$\boxed{0}$	3	$\cancel{0}$
IV	9	12	14	$\boxed{0}$

Since in the table 10·3, every row and every column have one assignment, so we have the complete optimal zero assignment.

Job.	I	II	III	IV
Man.	1	3	2	4

which is the optimal assignment.

II **Ex. 2.** *Solve the minimal assignment problem whose effectiveness matrix is*

	I	*II*	*III*	*IV*
A	2	3	4	5
B	4	5	6	7
C	7	8	9	8
D	3	5	8	4

[Meerut 95 (P)]

Sol.

Step 1. Subtracting the smallest element of each row from every element of the corresponding row, we get the following matrix (Table 10·4).

Table 10·4

	I	*II*	*III*	*IV*
A	0	1	2	3
B	0	1	2	3
C	0	1	2	1
D	0	2	5	1

Step 2. Substracting the smallest element of each column of the matrix (Table 10·4) from the corresponding column, we get the following matrix

Table 10·5

	I	*II*	*III*	*IV*
A	0	0	0	2
B	0	0	0	2
C	0	0	0	0
D	0	1	3	0

Step 3. Now we test whether it is possible to make an assignment using the zeroes of table 10·5 by the method described in step 3 page 332.

Since none of the rows or columns contain exactly one zero, therefore, the trial and error method is followed. Now we start searching two zeros. Starting with the row 1 we find row 4 which contain two zeros. We make the assignment at the first zero as shown in table 10·6, and cross out all the other zeros of the first column in which we have made the assignment □ Now starting with

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column 1 we find column 4 which contain two zeros and make assignment □ at the zero of row 3 (we can not make assignment at the zero of row 4 which already contain an assignment) and cross out all other zeros of this row. Now again starting with row 1 we find row containing only one zero but we cannot make an assignment in this row in which we have already given an assignment. The same is true when we check the columns for one zero. Again we start with row 1 searching two zeros and find the row (row 1) containing two unmarked zeros. We make an assignment at any one of these zeros and cross out the other zeros of the corresponding coloumn in which assignment is made (see table 10·6 and 10·7). If we made an assignment in first row second column and crossed zeros of second column then the second row contain only one zero in third column where we can make an assignment (see table 10·6). Similarly if we make an assignment in first row, third column and crossed zeros of third column then the second row contain only one zero in second column where we can make an assignment (see table 10·7).

Table 10·6

	I	II	III	IV
A	$\cancel{0}$	$\boxed{0}$	$\cancel{0}$	2
B	$\cancel{0}$	$\cancel{0}$	$\boxed{0}$	2
C	$\cancel{0}$	$\cancel{0}$	$\cancel{0}$	0
D	$\boxed{0}$	1	3	$\cancel{0}$

Table 10·7

	I	II	III	IV
A	$\cancel{0}$	$\cancel{0}$	$\boxed{0}$	2
B	$\cancel{0}$	$\boxed{0}$	$\cancel{0}$	2
C	$\cancel{0}$	$\cancel{0}$	$\cancel{0}$	$\boxed{0}$
D	$\boxed{0}$	1	3	$\cancel{0}$

Thus we get the following two assignments.

$$A \to II,\ B \to III,\ C \to IV,\ D \to I$$

and
$$A \to III,\ B \to II,\ C \to IV,\ D \to I.$$

Note : Other assignments may exist. Students must try to find them.

III **Ex. 3.** *Solve the assignment problem represented by the following matrix.*

	I	II	III	IV	V	VI
A	9	22	58	11	19	27
B	43	78	72	50	63	48
C	41	28	91	37	45	33
D	74	42	27	49	39	32
E	36	11	57	22	25	18
F	3	56	53	31	17	28

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Sol.

Step 1. Subtracting the smallest element of each row from every element of the corresponding row, we get the following reduced matrix (table 10·8).

Table 10·8

	I	*II*	*III*	*IV*	*V*	*VI*
A	0	13	49	2	10	18
B	0	35	29	7	20	5
C	13	0	63	9	17	5
D	47	15	0	22	12	5
E	25	0	46	11	14	7
F	0	53	50	28	14	25

Step 2. Subtracting the smallest element of each column from every element of the corresponding column, we get the following reduced matrix (table 10·9).

Table 10·9

	I	*II*	*III*	*IV*	*V*	*VI*
A	0	13	49	0	0	13
B	0	35	29	5	10	0
C	13	0	63	7	7	0
D	47	15	0	20	2	0
E	25	0	46	9	4	2
F	0	53	50	26	4	20

Step 3. Now we give the zero assignments in our usual manner and get the following matrix (table 10·10).

Since row 3 and column 5 have no assignments so we proceed to the next step.

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Table 10·10

⊠	13	49	[0]	⊠	13
⊠	35	29	5	10	[0]
13	⊠	63	7	7	⊠
47	15	[0]	20	2	⊠
25	[0]	46	9	4	2
[0]	53	50	26	4	20

Step 4. In this step we draw minimum number of lines to cover all zeros at least once. For this we proceed as follows. (see table 10·11).

Table 10·11

	L_1	L_2				L_3	
L_4	⊠	13	49	[0]	⊠	13	
	⊠	35	29	5	10	[0]	✓ (5)
	13	⊠	63	7	7	⊠	✓ (1)
L_5	47	15	[0]	20	2	⊠	
	25	[0]	46	9	4	2	✓ (4)
	[0]	53	50	26	4	20	
	✓ (6)	✓ (2)				✓ (3)	

(i) We mark (√) row 3 in which there is no assignment.

(ii) Then we mark (√) column 2 and 6 which have zeros in marked row 3.

(iii) Then we mark (√) rows 5 and 2 which have assignments in the marked column 2 and 6.

(iv) Then we mark column 1 (not already marked) which has zero in the marked row 2.

(v) Then we mark row 6 which has assignment in the marked column 1.

(vi) Now we draw lines through all marked columns 1, 2, 6. Then we draw lines through unmarked row 1 and 4 having zeros through which there is no line. Thus we get five lines (minimum number) to cover all the zeros.

Step 5. Now the smallest of the elements that do not have a line through them is 4. Subtracting this element 4 from all the elements that do not have a line through them and adding to every element that lies at the intersection of two lines and leaving the remaining elements unchanged, we get the following matrix (table 10·12).

Step 6. Again repeating the step 3 we make the zero assignments and get the following matrix (table 10·12).

Table 10·12

	I	*II*	*III*	*IV*	*V*	*VI*
A	4	17	49	[0]	⊠	17
B	[0]	35	25	1	6	⊠
C	13	⊠	59	3	3	[0]
D	51	19	[0]	20	2	4
E	25	[0]	42	5	⊠	2
F	⊠	53	46	22	[0]	20

Thus the optimal assignment is

$$A \to IV,\ B \to I,\ C \to VI,\ D \to III,\ E \to II,\ F \to V,$$

and minimum cost, $Z = 11 + 43 + 33 + 27 + 11 + 17 = 142$

(from the given cost matrix)

Another optimal solution of the assignment problem is as follows :

$A \to IV,\ B \to VI,\ C \to II,$
$D \to III,\ E \to V,\ F \to I.$

In this case minimum cost

$$Z = 11 + 48 + 28 + 27 + 25 + 3 = 142$$

(from the given cost matrix) which is equal to the minimum cost for the first solution.

Table 10·13

	I	*II*	*III*	*IV*	*V*	*VI*
A	4	17	49	[0]	⊠	17
B	⊠	35	25	1	6	[0]
C	13	[0]	59	3	3	⊠
D	51	19	[0]	20	2	4
E	25	⊠	42	5	[0]	2
F	[0]	53	46	22	⊠	20

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Ex. 4. *A company has four territories open, and four salesman available, for assignment. The territories are not equally rich in their sales potential; it is estimated that a typical salesman operating in each territory would bring in the following annual sales :*

Territories	*I*	*II*	*III*	*IV*
Annual sales (Rs.)	60,000	50,000	40,000	30,000

The four salesman are also considered to differ in ability; it is estimated that, working under the some conditions their yearly sales would be proportionally as follows :

Salesman :	*A*	*B*	*C*	*D*
Proportion :	7	5	5	4

If the criterion is maximum expected total sales, the intuitive answer is to assign the best salesman to the richest territory the next best salesman to the second richest, and so on. Verify this answer by the assignment technique.

Sol. To construct the effectiveness matrix.

The sum of proportions of sales of four salesman

$$= 7 + 5 + 5 + 4 = 21.$$

Taking the salesmen in the four territories, sales are as follows:

for A, $\frac{7}{21} \times 6$, $\frac{7}{21} \times 5$, $\frac{7}{21} \times 4$, $\frac{7}{21} \times 3$

for B, $\frac{5}{21} \times 6$, $\frac{5}{21} \times 5$, $\frac{5}{21} \times 4$, $\frac{5}{21} \times 3$

for C, $\frac{5}{21} \times 6$, $\frac{5}{21} \times 5$, $\frac{5}{21} \times 4$, $\frac{5}{21} \times 3$

for D, $\frac{4}{21} \times 6$, $\frac{4}{21} \times 5$, $\frac{4}{21} \times 4$, $\frac{4}{21} \times 3$.

To avoid fractions, we consider the sales in 21 years which are as follows

for A,	42,	35,	28,	21
for B,	30,	25,	20,	15
for C,	30,	25,	20,	15
for D,	20,	20,	16,	12

Thus the problem is to determine the assignment of (salesmen to territories) which make the total sales maximum, and effectiveness matrix is given by

Table 10·14

	I	*II*	*III*	*IV*
A	42	35	28	21
B	30	25	20	15
C	30	25	20	15
D	24	20	16	12

To convert the maximization problem to a minimization problem.

This is a maximization problem. By multiplying each element of this matrix (table 10·14) by −1, it is converted to minimization problem. Thus the resulting matrix is as follows.

Table 10·15

	I	*II*	*III*	*IV*
A	−42	−35	−28	−21
B	−30	−25	−20	−15
C	−30	−25	−25	−15
D	−24	−20	−16	−12

Solution of the minimization problem

Step 1 and 2. Applying step 1 and 2, we obtain following matrix (table 10·16).

Table 10·16

L_1				
0	3	6	9	
0	1	2	3	
0	1	2	3	
0	0	0	0	L_2

Step 3. Since only two lines (less than 4, the number of rows or columns

in matrix) cover all the zeros, the optimal assignment is not possible at this stage. So we proceed to the next step.

Step 4. Subtracting 1 minimum of all uncovered elements by the lines, from all the elements through which no line passes, adding to the elements which lie at the intersection of two lines and leaving other elements as they are, we obtain the following matrix (table 10·17).

Table 10·17

L_1	L_2			
$\boxed{0}$	2	5	8	4 ✓
$\cancel{0}$	$\boxed{0}$	1	2	5 ✓
$\cancel{0}$	$\cancel{0}$	1	2	1 ✓
1	$\cancel{0}$	$\boxed{0}$	$\cancel{0}$	L_3
✓ 2	✓ 3			

Step 5. Giving the zero assignments in matrix (table 10·17) we see that there is no assignment in row 3 and column 4. Therefore, we again proceed to draw minimum number of lines to cover all zeros.

Step 6. Proceeding in our usual manner we find that here are three lines (minimum number) to cover all zeros (see table 10·17).

Step 7. Now subtracting 1, the minimum of all uncovered elements by the lines, from all the elements through which no line passes, adding the elements which lie at the intersection of two lines and leaving other elements as they are, we get the following matrix (table 10·18).

Table 10·18

0	2	4	7
0	0	0	1
0	0	0	1
2	1	0	0

Step 8. Giving assignments in our usual manner we get the following two assignments. (table 10·19 and 10·20).

Table 10·19

	I	II	III	IV
A	$\boxed{0}$	2	4	7
B	$\cancel{0}$	$\boxed{0}$	$\cancel{0}$	1
C	$\cancel{0}$	$\cancel{0}$	$\boxed{0}$	1
D	2	1	$\cancel{0}$	$\boxed{0}$

Table 10·20

	I	II	III	IV
A	$\boxed{0}$	2	4	7
B	$\cancel{0}$	$\cancel{0}$	$\boxed{0}$	1
C	$\cancel{0}$	$\boxed{0}$	$\cancel{0}$	1
D	2	1	$\cancel{0}$	$\boxed{0}$

Thus we get the following two optimal solutions (assignments)

(i) $A \to I$, $B \to II$ $C \to III$, $C \to IV$

and (ii) $A \to I$, $B \to III$, $C \to II$, $D \to IV$.

From both the solutions it is obvious that the best salesman A is assigned to the richest teritory I and the worst salesman D to the poorest territory D. The salesman B and C are equal in efficiency, so either of them may be assigned to territories II and III.

This assignment verify the given problem. **Proved.**

Ex. 5. *An airline that operates seven days a week has timetable shown below. Crews must have a minimum layover of 5 hours between flights. Obtain the pairing of flights that minimizes layover time away from home. For any given pairing the crew will be based at the city that results in the smaller layover.*

Delhi-Jaipur Flight No.	Depart	Arrive	Jaipur-Delhi Flight No.	Depart	Arrive
1	7.00 A.M.	8.00 A.M.	101	8.00 A.M.	9.15 A.M.
2	8.00 A.M.	9.00 A.M.	102	8.30 A.M.	9.45 A.M.
3	1.30 P.M.	2.30 P.M.	103	12.00 Noon	1.15 P.M.
4	6.30 P.M.	7.30 P.M.	104	2.30 P.M.	6.45 P.M.

For each pair also mention the town where the crew should be based.

Sol.

Step 1. First we construct the table for layover times between flights when crew is based at Delhi. (so that they start from and come back to Delhi with minimum stay at jaipur)

Since the crew must have a minimum layover of 5 hours between flights, the layover time between flights 1 and 101 will

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be 24 hours. Also the layover times between flights 1 and 102, flights. 1 and 103, flights 1 and 104 are 24·5 hours, 28 hours, 9·5 hours. Similarly the layover times between other pair of flights may be calculated which are shown in table 10·21.

Table 10·21
Layover times when crew based at Delhi

Flights → ↓	101	102	103	104
1	24	24.5	28	9.5
2	23	23.5	27	8.5
3	17.5	18	21.5	27
4	12.5	13	16.5	22

Since the plane arrives at Delhi at 9·15 A.M. by flight No. 101 and will depart to Jaipur at 7·00 A.M. by flight No. 1 after 21·75 hours. Therefore, layover times between pair of flights No. 101 and 1 is 21·75 hours. Similarly, the layover times between other pairs of fights may be calculated.

The layover times between the pair of flights when the crew is based at Jaipur (so that they start from and come back to Jaipur with minimum stay at Delhi) are shown in table 10·22 or 10·23.

Table 10·22
Layover times when crew based at Jaipur

Flights → ↓	1	2	3	4
101	21.75	22.75	28.25	9.25
102	21.25	22.75	27.75	8.75
103	17.75	18.75	24.25	5.25
104	12.25	13.25	18.75	23.75

Table 10·23

Flights → ↓	101	102	103	104
1	21.75	21.25	17.75	12.25
2	22.75	22.25	18.75	13.25
3	28.25	27.75	24.25	18.75
4	9.25	8.75	5.25	23.75

Step 2. To avoid the fractions we consider either the layovers times in terms of quarter hour as one unit of time or the layovers times for four weeks. Thus multiplying the matrices (table 10·21 and 10·23) by 4, the modified matrices are as follows. (table 10·24 and 10·25).

Table 10·24 (Crew based at Delhi)

Flights → ↓	101	102	103	104
1	96	98	112	38
2	92	94	108	34
3	70	72	86	108
4	50	52	66	88

Table 10·25 (Crew based at Jaipur)

Flights → ↓	101	102	103	104
1	87	85	71	49
2	91	89	75	53
3	113	111	97	75
4	37	35	21	95

Step 3. Now we combine the tables 10·24 and 10·25, choosing the base which gives a lesser time for each pairing. The layover times

marked with '*' denote the crue based at Jaipur. Otherwise, the crew is based at Delhi. Thus we get the following table :

Table 10·26 (Minimum layover times table)

87*	85*	71*	38
91*	86*	75*	34
70	72	86	75*
37*	35*	21*	88

Step 4. Subtracting the smallest element of each row from every element of the corresponding row and then subtracting the smallest element of each column from every element of the corresponding column, we get the following matrix (table 10·27)

Table 10·27

			L3	
49*	45*	33*	$\boxed{0}$	3
57*	53*	41*	$\cancel{0}$	1
$\boxed{0}$	$\cancel{0}$	16	5*	L1
16*	12*	$\boxed{0}$*	67	L2
			2	

Step 5. Giving the zero assignment we find the there is no assignment in row 2 and column 2, so we draw minimum number of lines to cover all the zeros as shown in table 10·27.

Step 6. Now subtracting the smallest uncovered element 33 from all uncovered elements and adding it to the elements which lies at the intersection of lines and leaving other elements as usual the matrix obtained is as follows : (table 10·28).

Table 10·28

		L2	L3	
16*	12*	0*	0	1
24*	20*	8*	$\boxed{0}$	5
$\boxed{0}$	$\cancel{0}$	16	38*	L1
16*	12*	$\boxed{0}$*	100	4
		2	3	

Step 7. Giving the zero assignments we can find that there is no assignments in row 1 and column 2, so we again draw minimum number of lines to cover all the zeros as shown in table 10·28.

Step 8. Proceeding again as in Step 6, the final matrix obtained is as follows (table 10·29).

Table 10·29

	101	102	103	104
1	4*	0*	0*	0
2	12*	8*	8*	0
3	0	0	28	50*
4	4*	0*	0*	100

Step 9. Giving the zero assignments, we get the following tables :

Table 10·30

	101	102	103	104
1	4*	$\boxed{0}$*	$\cancel{0}$*	$\cancel{0}$
2	12*	8*	8*	$\boxed{0}$
3	$\boxed{0}$	$\cancel{0}$	28	50*
4	4*	$\cancel{0}$*	$\boxed{0}$*	100

Table 10·31

	101	102	103	104
1	4*	$\cancel{0}$*	$\boxed{0}$*	$\cancel{0}$
2	12*	8*	8*	$\boxed{0}$
3	$\boxed{0}$	$\cancel{0}$	28	50*
4	4*	$\boxed{0}$*	$\cancel{0}$*	100

Two optimal assignments (tables 10·30 and 10·31) :

(i) 1 → 102 (Crew at Jaipur), 2 → 104 (Crew at Delhi)
3 → 101 (Crew at Delhi), 4 → 103 (Crew at Jaipur)

(ii) 1 → 103 (Crew at Jaipur) 2 → 104 (Crew at Delhi)
3 → 101 (Crew at Delhi), 4 → 102 (Crew at Jaipur),

In both the cases minimum layover times is 210 hours for four weeks *i.e.*, 52 hours 30 minutes per week.

§ 10·5. Unbalanced Assignment Problem

An assignment problem is called an **Unbalanced assignment problem** whenever the number of tasks (jobs) is not equal to the number of facilities (persons). Thus the cost matrix of an unbalanced assignment problem is not a square matrix. For the solution of such problem we add the dummy rows or columns to the given matrix to make it a square matrix. The costs in these dummy rows or columns are taken to be 0. Now the problem reduces to the balanced assignment problem and can be solved by assignment algorithm § 10·4.

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Illustrative Example

Ex. 6. *A company is faced with the problem of assigning six different machines to five different jobs. The costs are estimated as follows (hundred of rupees).*

Job

Machine	1	2	3	4	5
1	2.5	5	1	6	1
2	2	5	1.5	7	3
3	3	6.5	2	8	3
4	3.5	7	2	9	4.5
5	4	7	3	9	6
6	6	9	5	10	6

Solve the problem assuming that the objective is to minimise total cost.

Sol. Since the given matrix is not a square matrix, we add one fictitious job 6 (sixth column) to make it a square matrix. Thus the resulting matrix obtained is as follows :

	1	2	3	4	5	6
1	2.5	5	1	6	1	0
2	2	5	1.5	7	3	0
3	3	6.5	2	8	3	0
4	3.5	7	2	9	4.5	0
5	4	7	3	9	6	0
6	6	9	5	10	6	0

Proceeding as usual the solution to the given problem is

(i) $1 \to 4, 2 \to 1, 3 \to 5, 4 \to 3, 5 \to 2.$

(ii) $1 \to 4, 2 \to 2, 3 \to 5, 4 \to 3, 5 \to 1.$

Minimum Cost = Rs. 2000.

Exercise on Chapter X

1. Define Assignment problem.
2. Write a short note on "Assignment" "problem".
3. Give an algorithm to solve an assignment problem.
4. There are five jobs to be assigned, on each to five machines and the associated cost matrix is as follows :

Job, Machine →

↓	1	2	3	4	5
A	11	17	8	16	20
B	9	7	12	6	15
C	13	16	15	12	16
D	21	24	17	28	26
E	14	10	12	11	15

Ans. $A \to 1$, $B \to 4$, $C \to 5$, $D \to 3$, $E \to 2$, $Z^* = 60$

Solve the following minimal assignment problem.

5. One car is available at each of the stations 1, 2, 3, 4, 5, 6; and one car is required at each of the stations 7, 8, 9, 10, 11, 12. The distances between the various stations are given in the matrix below. How could the cars be despatched so as to minimize the total mileage travelled.

To

From	7	8	9	10	11	12
1	42	72	39	52	25	51
2	22	29	49	65	81	50
3	27	39	60	51	32	32
4	45	50	48	52	37	43
5	29	40	39	26	30	33
6	82	40	40	60	51	30

Ans. $1 \to 11$, $2 \to 8$, $3 \to 7$, $4 \to 9$, $5 \to 10$, $6 \to 12$. Minimum mileage $Z^* = 185$.

Solve the following minimal assigment problems.

6. Job

Problem	1	2	3	4	5
A	8	4	2	6	1
B	0	9	5	5	4
C	3	8	9	2	6
D	4	3	1	0	3
E	9	5	8	9	5

Ans. $A \to 5$, $B \to 1$, $C \to 4$, $D \to 3$, $E \to 2$, Mini Cost = 9

[Rohilkhand 86]

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7.

Job → Machine ↓	I	II	III	IV	V
A	5	11	10	12	4
B	2	4	6	3	5
C	3	12	5	14	6
D	6	14	4	11	7
E	7	9	8	12	5

Ans. $A \to V$, $B \to IV$, $C \to I$, $D \to III$, $E \to II$, $Z^* = 23$

8.

Man → Job ↓	1	2	3	4	5
I	12	8	7	15	14
II	7	9	17	14	10
III	9	6	12	6	7
IV	7	6	14	6	10
V	9	6	12	10	6

Ans. $I \to 3$ or $I \to 3$, $II \to 1$ or $II \to 1$, $III \to 2$ or $III \to 4$, $IV \to 4$ or $IV \to 2$, $V \to 5$ or $V \to 5$

9. A national truck-rental service has a surplus of one truck in each of the cities 1, 2, 3, 4, 5, 6 and a deficit of one truck in each of the cities 7, 8, 9, 10, 11 and 12. The distances (in kms) between the cities with a surplus and the cities with a deficit are displaced below :

From \ To	7	8	9	10	11	12
1	31	62	29	42	15	41
2	12	19	39	55	71	40
3	17	29	50	41	22	22
4	35	40	38	42	27	33
5	19	30	29	16	20	23
6	72	30	30	50	41	20

How should the trucks be dispersed so as to minimize the total distance travelled?

Ans. $1 \to 11, 2 \to 8, 3 \to 7, 4 \to 9, 5 \to 10, 6 \to 12$, Mini: $Z = 125$.

10. Solve the following assignment problem.

	1	2	3	4
A	10	12	19	11
B	5	10	7	8
C	12	14	13	11
D	8	15	11	9

Ans. $A \to 2$, $B \to 3$, $C \to 4$, $D \to 1$, $Z^* = 38$

11. A marketing manager has 5 salesman and 5 sales-districts. Considering the capabilities of the salesman and the nature of districts, the marketing manager estimates that sales per month (in hundred rupees) for each salesman in each district would be as follows :

Job	A	B	C	D	E
1	32	38	40	28	40
2	40	24	28	21	36
3	41	27	33	30	37
4	22	38	41	36	36
5	29	33	40	35	39

Ans. $1 \to B$, $2 \to A$, $3 \to E$, $4 \to C$, $5 \to D$, Max. $Z \to 191$.

Find the assigment of salesman to dictrcts that will result in maximum sales..

12. Find the minimum cost solution for the 5×5 assignment problem whose coefficients are as given below.

	I	II	III	IV	V
1	−2	−4	−8	−6	−1
2	0	−9	−5	−5	−4
3	−3	−8	−9	−2	−6
4	−4	−3	−1	0	−3
5	−9	−5	−8	−9	−5

Ans. $1 \to III$ or $1 \to IV$, $2 \to II$ or $3 \to II$, $3 \to V$ or $2 \to III$, $4 \to I$ or $4 \to V$, $5 \to IV$ or $5 \to I$, Cost = 36

13. Five men are available to do five different jobs. From past records, the time (in hours) that each man takes to do each job is known and is given in the following table.

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		Job					
		I	II	III	IV	V	Ans.
	1	2	9	2	7	1	1 → III, III, III
	2	6	8	7	6	1	2 → V, V, V
Man	3	4	6	5	3	1	3 → I, IV, IV
	4	4	2	7	3	1	4 → IV, II, I
	5	5	3	9	5	1	5 → II, I, II

Find the assignment of men to jobs that will minimize the total time taken.

14. A department head has four subordinates, and four tasks to be performed. The subordinates differ in efficiency, and the tasks in their intrinsic difficulty. His estimate of the times each man would take to perform each task is given in the effectiveness matrix below. How should the tasks be allocated, one to a man, so as to minimize the total men hours?

		Subordinates				
		I	II	III	IV	Ans.
	A	8	26	17	11	A → I
Task	B	13	28	4	26	B → III
	C	38	19	18	15	C → II
	D	19	26	24	10	D → IV

[Meerut O.R. 85]

15. Six wagons are available at six stations A, B, C, D, E and F. These are required at stations I, II, III, IV, V and VI. The mileage between various stations is given by the following table :

	I	II	III	IV	V	VI
A	20	23	18	10	16	20
B	50	20	17	16	15	11
C	60	30	40	55	8	7
D	6	7	10	20	100	9
E	18	19	28	17	60	70
F	9	10	20	30	40	55

How should the wagons be transported in order to minimize the total mileage covered.

[**Ans.** $A \to IV, B \to VI, C \to V, D \to III, E \to I, F \to II$

Total milage 67]

16. Four engineers are available to design four projects. Engineer 2 is not competent to design the project B. Given the following time estimates needed to each engineer to design a given project, find how should the engineers be assigned to projects so as to minimize the total design time of four projects.

		Project			
		A	B	C	D
	1	12	10	10	8
Engineer,	2	14	Not suitable	15	11
	3	6	10	16	4
	4	8	10	9	7

[**Ans.** $1 \to B, 2 \to D, 3 \to A, 4 \to C$, Total time = 36 hours.]

17. A car hire company has one car at each of five depots a, b, c, d and e. A customer required a car in each town namely A, B, C, D and E. Distances (in kms) between depots (origins) and towns (destinations) are given in the following distance matrix.

	a	b	c	d	e
A	160	130	175	190	200
B	135	120	130	160	175
C	140	110	155	170	185
D	50	50	80	80	110
E	54	34	70	80	105

How should cars be assigned to customers so as to minimize the distance travelled?

[**Ans.** $A \to a, B \to c, C \to b, D \to a, E \to d$,

Total distance = 570 kms]

18. An airline that operates seven days a week has a time table shown below. Crews must have minimum layover of 6 hours between flights. Obtain the pairing of flights that minimizes layover time away from home. For any given pairing the crew will be based at the city that results in the smaller layover.

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	Delhi-Calcutta			Calcutta-Delhi	
Flight No.	Depart	Arrival	Flight No.	Depart	Arrival
1	7.00 A.M. –	9.00 A.M.	101	9.00 A.M.–	11.00 A.M.
2	9.00 A.M. –	11.00 A.M.	102	10.00 A.M.–	12.00 Noon
3	1.30 P.M. –	3.30 P.M.	103	3.30 P.M. –	5.30 P.M.
4	7.30 P.M. –	9.30 P.M.	104	8.00 P.M. –	10.00 P.M.

For each pair also mention the town where the crew should be based ? [I.C.W.A. Final June 80]

19. Solve the following assignment problem.

	Man				
	I	*II*	*III*	*IV*	*V*
A	1	3	2	3	6
B	2	4	3	1	5
C	5	6	3	4	6
D	3	1	4	2	2
E	1	5	6	5	4

[Meerut O.R. 87]

[Ans. $A \to I, B \to IV, C - III, D \to II, E \to V$]

20 Use Hugariam method to solve the following cost minimizing assignment problem.

	Jobs					
	J_1	J_2	J_3	J_4	J_5	Ans.
P_1	8	9	7	6	3	$P_1 \to J_3, J_3$
P_2	9	10	11	8	4	$P_2 \to J_5, J_4$
P_3	11	9	10	9	3	$P_3 \to J_2, J_5$
P_4	7	8	11	10	5	$P_4 \to J_1, J_2$
P_5	9	10	11	8	4	$P_5 \to J_4, J_1$

Mini cost = Rs. 35.

[Meerut 96]

21. Four salesmen are to be assigned to four districts. Estimates of the sales revenue in hundred of Rs. for sale are as under

Salesmen/Districts	*A*	*B*	*C*	*D*	Ans.
1	320	350	400	280	$1 \to C$
2	400	250	300	220	$2 \to A$
3	420	270	340	300	$3 \to D$
4	250	390	410	350	$4 \to B$

Give the assignment pattern that maximises the sales revenue [Meerut 95]

22. The owner of a small machine shop has four mechanists available to assign to Jobs for the day. Five jobs are affered with the expected profit in rupees for each mechanist on each job being as follows :

	Jobs					
	A	*B*	*C*	*D*	*E*	Ans.
1	6.20	7.80	5.00	10.10	8.20	$1 \to D$
2	7.10	8.40	6.10	7.30	5.90	$2 \to B$
3	8.70	9.20	11.10	7.10	8.10	$3 \to C$
4	4.80	6.40	8.70	7.70	8.00	$4 \to E$

Job *A* not offered

Find the assignment of mechanists to jobs that will result in a maximum profit. [Meerut 95 (BP)]

23. Solve the following unbalanced assignment problem of minimizing total time for doing all the jobs.

Operator	1	2	3	4	5
1	6	2	5	2	6
2	2	5	8	7	7
3	7	8	6	9	8
4	6	2	3	4	5
5	9	3	8	9	0
6	4	7	5	6	8

[C.A. Final May 85]

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[Ans. $1 \to 4, 2 \to 1, 4 \to 2, 5 \to 5, 6 \to 3$ Total time = 11 units]

24. A company is faced with the problem of assigning 4 machines to 6 different jobs (one machine to one job only). The profits are estimated as follows :

Job \ Machine	A	B	C	D
1	3	6	2	6
2	7	1	4	4
3	3	8	5	8
4	6	4	3	7
5	5	2	4	3
6	5	7	6	4

Solve the problem to minimize the total profits.

[Meerut O.R. 89]

[**Hint :** Change the problem to minimization assignment problem by multiplying each element by -1, then change the unbalanced assignment problem to balanced one by adding two more columns with zero costs.]

[Ans. Job 2 to A, 3 to B, 4 to D, 6 to C. Max. profit = 28].

25. Solve the following minima assignment problem :

Men/Jobs	1	2	3	4	5	6	Ans.
1	2	9	2	7	9	1	$1 \to 3$ or 3
2	6	8	7	6	14	1	$2 \to 6$ or 6
3	4	6	5	3	8	1	$3 \to 1$ or 4
4	4	2	7	3	10	1	$4 \to 4$ or 2
5	5	3	9	5	12	1	$5 \to 2$ or 1

Job 5 is not done.

[Meerut 98 (Old)]

26. Use the Humgarian method to find which of the two jobs should be left undone when each of the 4 persons will do only one job in the following cost minimizing assignment problem :

Person \ Job	J_1	J_2	J_3	J_4	J_5	J_6	Ans.
P_1	10	9	11	12	8	5	$P_1 \to J_5$
P_2	12	10	9	11	9	4	$P_2 \to J_6$
P_3	8	11	10	7	12	6	$P_3 \to J_4$
P_4	10	7	8	10	10	5	$P_4 \to J_2$

Jobs J_1 and J_3 left undone.

[Meerut 96 (BP)]

27. There are 3 persons: P_1, P_2 and P_3 and 5 jobs $J_1, J_2, \ldots, J_5$. Each person can do only one job and a job is to be done by one person only. Using Humgarian method, find which 2 jobs should be left undone in the following cost minimizing assignment problem.

	J_1	J_2	J_3	J_4	J_5	Ans.
P_1	7	8	6	5	9	$P_1 \to J_4$
P_2	9	6	7	6	10	$P_2 \to J_2$
P_3	8	7	9	5	6	$P_3 \to J_5$

Jobs J_1 and J_3 left undone

[Meerut 97, 97 (BP), 98 (P)]

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11

Transportation Problem

§ 11·1. Introduction

The transportation problem is a particular class of linear programming problem and can be regarded as a generalization of the assignment problem.

§ 11·2. Transportation Problem

The transportation problem can be described as follows.

Suppose that the factories F_i $(i = 1, 2, 3, \ldots, m)$, called the **origins** or **sources** produce the non-negative quantities a_i $(i = 1, 2, \ldots, m)$ of a product and the non-negative quantities b_j $(j = 1, 2, \ldots, n)$ of the same product are required at other n places, called the **destinations**, such that the total quantity produced is equal to the total quantity required

i.e.
$$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j \quad \ldots(1)$$

Also suppose that c_{ij} is the cost of transportation of a unit from the i-th source to the j-th destination. Then the problem is to determine x_{ij}, the quantity transported from the i-th source to the j-th destination, in such a way that the total transportation cost $\sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij}$ is minimized.

The transportation problem as described above can be represented in a tabular form as follows :

Destination → Sources ↓	W_1	W_2	…	W_j	…	W_n	Capacities of the sources
F_1	c_{11}	c_{12}	…	c_{1j}	…	c_{1n}	a_1
F_2	c_{21}	c_{22}	…	c_{2j}	…	c_{2n}	a_2
⋮	…	…	…	…	…	…	⋮
⋮	…	…	…	…	…	…	⋮
F_i	c_{i1}	c_{i2}	…	c_{ij}	…	c_{in}	a_i
⋮	…	…	…	…	…	…	⋮
⋮	…	…	…	…	…	…	⋮
F_m	c_{m1}	c_{m2}	…	c_{mj}	…	c_{mn}	a_m
Requirements →	b_1	b_2	…	b_j	…	b_n	$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j$

The calculations are made directly on the transportation array given below which gives the current trial solution.

Destination → Sources ↓	W_1	W_2	…	W_i	…	W_n	Capacities of the sources
F_1	x_{11}	x_{12}	…	x_{1j}	…	x_{1n}	a_1
F_2	x_{21}	x_{22}	…	x_{2j}	…	x_{2n}	a_2
⋮	…	…	…	…	…	…	⋮
⋮	…	…	…	…	…	…	⋮
F_i	x_{i1}	x_{i2}	…	x_{ij}	…	x_{in}	a_i
⋮	…	…	…	…	…	…	⋮
⋮	…	…	…	…	…	…	⋮
F_m	x_{m1}	x_{m2}	…	x_{mj}	…	x_{mn}	a_m
Requirements →	b_1	b_2	…	b_j	…	b_n	$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j$

The above two tables can be combined together by writing the costs c_{ij} within the bracket ( ), as follows :

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Destinations → Sources ↓	W_1	W_2	...	W_j	...	W_n	Capacities of the sources
F_1	$x_{11}(c_{11})$	$x_{12}(c_{12})$	...	$x_{1j}(c_{1j})$	...	$x_{1n}(c_{1n})$	a_1
F_2	$x_{21}(c_{21})$	$x_{22}(c_{22})$	...	$x_{2j}(c_{2j})$	...	$x_{2n}(c_{2n})$	a_2
$\vdots$	...	...	...	...	...	...	$\vdots$
$\vdots$	...	...	...	...	...	...	$\vdots$
F_i	$x_{i1}(c_{i1})$	$x_{i2}(c_{i2})$	...	$x_{ij}(c_{ij})$	...	$x_{in}(c_{in})$	a_i
$\vdots$	...	...	...	...	...	...	$\vdots$
$\vdots$	...	...	...	...	...	...	$\vdots$
F_m	$x_{m1}(c_{m1})$	$x_{m2}(c_{m2})$	...	$x_{mj}(c_{mj})$	...	$x_{mn}(c_{mn})$	a_m
Requirements	b_1	b_2	...	b_j	...	b_n	$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j$

Mathematical formulation of a transportation problem [Raj. 87]

Mathematically the transportation problem can be stated as follows.

Find x_{ij} $(i = 1, 2, ..., m,\ j = 1, 2, ..., n)$ for which total transportation cost

$$Z = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij} \qquad ...(2)$$

is minimized, subject to the restrictions

$$\left.\begin{aligned} &\sum_{j=1}^{n} x_{ij} = a_i,\ i = 1, 2, 3, ..., m \\ &\sum_{i=1}^{m} x_{ij} = b_j,\ j = 1, 2, 3, ..., n \\ &\sum_{i=1}^{m} a_i = \sum_{j=1}^{m} b_j \end{aligned}\right\} \qquad ...(3)$$

and $x_{ij} \geq 0$, for all $i = 1, 2, ..., m$; $j = 1, 2, ..., n$.

§ 11·3. Difference between a transportation and an Assignment problem [Meerut 89 Raj. 85]

An assignment problem is a special case of the transportation problem in which $m = n$, all the a_i and b_j are unity, and each x_{ij} is limited to one of the two values 0 and 1.

In these circumstances, exactly n of x_{ij} can be non-zero, one in each row of the array and one in each column showing that only one source (person) can be assigned to each destination (job).

§ 11·4. Few important definitions

Now we shall define few terms that are used in the transportation problem.

1. A feasible solution (A.F.S.) A **feasible solution** to a transportation problem is a set of non-negative individual allocations $(x_{ij} \geq 0)$ which satisfies the row and column sum restrictions [(*i.e.* restrictions (3), § 11·3)].

2. Basic feasible solution (B.F.S.) : A feasible solution of a m by n transportation problem is said to be a **basic feasible solution** if the total number of positive allocations x_{ij} is exactly equal to $m + n - 1$ *i.e.*, one less than the sum of the number of rows and columns.

3. Optimal solution : A feasible solution (not necessarily basic) is said to be **optimal** if it minimizes the total trapsportation cost.

4. Non-degenerate basic feasible solution : A feasible solution of m by n transportation problem is said to be **non-degenerate basic feasible solution** if [Agra 84]

(i) *Total number of positive allocations is exactly equal to* $(m + n - 1)$.

and (ii) *These allocations are in independent positions.*

In other words, if *A.F.S. involves exactly* $(m + n - 1)$ *independent individual positive allocations, then it is known as* **non-degenerate B.F.S.**, otherwise, it is said to be **degenerate B.F.S.**

Here by independent positions of the allocations we mean that it is always impossible to form any closed circuit (loop) by joining these allocations by horizontal and vertical lines only. See tables 11·1 and 11·2 below in which the positions of allocations are indicated by•.

Table 11·1
Independent positions

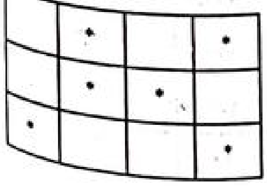
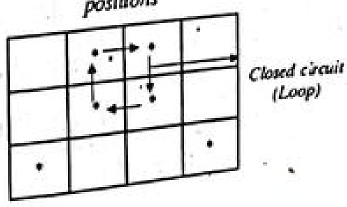


Table 11·2
Non-Independent positions



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§ 11·5. Solution of a transportation problem

The solution (optimal) of a transportation problem consists of the following two steps.

Step 1. To find an initial basic feasible solution.

Step 2. To obtain an optimal solution by making successive improvements to initial basic feasible solution (obtained in step 1) until no further decrease is the transportation cost is possible.

§ 11·6. To find an Initial Feasible Solution

There are several methods for finding the initial feasible solution of the given transportation problem. Here we describe the following three simple methods.

Method 1 : North-west Corner Rule

[Meerut 97 (BP), 98 (P); C.A. Final (May) 87]

By this rule we allocate a set of allocations in the cells so that the row totals and column totals will be, as indicated before each, as follows :

(i) Start with the cell (1, 1) at the north-west corner *i.e.*, the top most left corner and allocate it maximum possible amount.

(ii) Then move to the right hand cell (1, 2) if there is still any available quantity left, otherwise move to the down cell (2, 1) and allocate it maximum possible amount.

(iii) Repeat the step (ii) again and continue until all the available quantity is exhausted.

The method is well explained by taking the following numerical example :

		To W_1	W_2	W_3	Supply
From	F_1	(2)	(7)	(4)	5
	F_2	(3)	(3)	(1)	8
	F_3	(5)	(4)	(7)	7
	F_4	(1)	(6)	(2)	14
Demand		7	9	18	34

Table 11·3

		To W_1	W_2	W_3	
From	F_1	5 (2)			$5 = a_1$
	F_2	2 (3)	6 (3)		$8 = a_2$
	F_3		3 (4)	4 (7)	$7 = a_3$
	F_4			14 (2)	$14 = a_4$
		$7 = b_1$	$9 = b_2$	$18 = b_3$	

(i) We start with the top most left corner and allocate it maximum possible amount 5. (Since Mini. of $a_1 = 5$ and $b_1 = 7$ is 5).

(ii) Since there is no amount left available at source 1 so we move downwards to the cell (2, 1), in place of moving to right, and allocate it maximum possible amount. Since the column 1 still need the amount 2 and the amount 8 is available in row 2 so we allocate the maximum amount 2 to this cell (2, 1). Now the allocation for column 1 is complete, so we move to the right of this cell. Since the amount 6 is still available in row 2 and amount 9 is needed in column 2, so we allocate the maximum amount 6 in this cell (2, 2). Thus allocation for row 2 is complete, so we move downwards to the cell (3, 2). Since the amount 3 is still needed in column 2 and amount 7 is available in row 3, so we allocate the maximum amount 3 in this cell (3, 2). Thus allocation for column 2 is complete, so we move to the right of this cell. Since the amount 4 is still available in row 3 and amount 18 is needed in column 3, so we allocate the maximum amount 4 to the cell (3, 3). Thus the requirement of the row 3 is complete so we move to the downward cell (4, 3), Since amount 14 is still available in row 3 and an equal amount 14 is needed in row 4, so we allocate this amount 14 to this cell (4, 3). This completes the allocations and the resulting feasible solution (allocations) are shown in table 11·3.

In the end it may be checked that the sum of rows and columns are as needed.

On multiplying each individual allocation by its corresponding unit in ( ), and adding, the total transportation cost to this F.S. is

$$= 5 \times 2 + 2 \times 3 + 6 \times 3 + 3 \times 4 + 4 \times 7 + 14 \times 2 = \text{Rs. } 102$$

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In this north-west corner rule we move to the right or down, so no loop (circuit) can be formulated here by drawing horizontal and vertical lines to the allocations. Also at each step (allocation) at least one row or column is discarted from further consideration and at the last allocation both rows and columns are discarded. So we cannot get more than $(m + n - 1)$ individual positive allocations by this rule. **Thus we always get a non-degenerate basic feasible solution by this north-west corner rule.**

Method 2 : Lowest Cost Entry Method

(or Method of matrix minima)

In this method we write the costs and the requirement matrix. The cost are written within brackets ( ). Now we examine the cost matrix carefully and choose the cell with lowest cost and allocate there as much as possible. If such cell of lowest cost is not unique, we can select any one of these cells. Again we examine the cost matrix and select the cell with the lowest cost (the cell in which allocation has been made is not considered) and allocate there as much as possible. We continue this process until all the available quantity is exhaused.

The method is well explained by taking the same numerical example as in method I.

First we write the cost and requirement matrix as follows (table 11·4).

Table 11·4

		To			
		W_1	W_2	W_3	
	F_1	(2)	2 (7)	3 (4)	5
From	F_2	(3)	(3)	8 (1)	8
	F_3	(5)	7 (4)	(7)	7
	F_4	7 (1)	(6)	7 (2)	14
		7	9	18	

Examine the cost matrix (table 11·4) we find that there is lowest cost 1 in cell (2, 3) and in (4, 1). We choose any one of these say the cell (2, 3) and allocate the maximum amount 8 to this cell. Leaving this cell we find that there is lowest cost 1 in cell (4, 1) where we allocate the maximum amount 7. Continuing in this way we get the required feasible solution shown in table 11·4.

The total transportation cost to this F.S. is

$= 2 \times 7 + 3 \times 4 + 8 \times 1 + 7 \times 4 + 7 \times 1 + 7 \times 2 = \text{Rs.} 83.$

This cost is less than the cost associated with the F.S. obtained by *north west corner rule*.

The initial feasible solution obtained by the method usually gives a lower transportation cost than that obtained in *north west corner rule*.

Method 3 : Unit Cost-Penalty Method

(Vogels Approximation Method)

[Rohilkhand 85; C.A. Final May 87]

In this method we write the differences of the smallest and second smallest costs in each column, below the corresponding column and write the similar differences of each row to the right of the corresponding row. These individual differences can be thought of a **penalty** for making allocations in the second lowest cost cell instead of lowest cost cell in each row or column. Now we select the row or column for which the penalty is the largest and allocate the maximum possible amount to the cell with lowest cost in that particular row or column. If there are more than one largest penalty rows or columns, then we may select any one of them. Then we cross (or leave) that row or column in which the requirement (or demand) has been satisfied and construct the reduced matrix. We continue this process on the reduced matrices till all the allocations have been made.

The method is well explained by taking the same numerical example as in method I.

First we write the cost and requirement matrix as follows (table 11·5) :

Table 11·5

	W_1	W_2	W_3	Available	Penalties	
F_1	5 (2)	(7)	(4)	5	(2)	←
F_2	(3)	(3)	(1)	8	(2)	
F_3	(5)	(4)	(7)	7	(1)	
F_4	(1)	(6)	(2)	14	(1)	
Demand	7	9	18			
Penalties	(1)	(1)	(1)			

Since the maximum penalty (2) is associated with row 1 and row 2, so we may select any one of these. Suppose we select row 1 then we allocate the maximum possible amount 5 to the cell (1, 1) with lowest cost and cross this row 1. The first reduced matrix after leaving row with remaining demand and availables is as follows.

(Note that the amount 5 has been allocated to column 1, so the amount still needed to column 1 is 2).

Table 11·6

	W_1	W_2	W_3	Available	Penalties
F_2	(3)	(3)	8 (1)	8	(2) ←
F_3	(5)	(4)	(7)	7	(1)
F_4	(1)	(6)	(2)	14	(1)
Demand	2	9	18		
Penalties	(2)	(1)	(1)		

Since the maximum penalty (2) is associated with row 1 (table 11·6), so the maximum possible amount 8 is allocated to the cell (1, 3) with lowest cost in this row.

The second reduced matrix after leaving row 1 of matrix in table 11·6 with remaining demands and availables is as follows (table 11·7).

Table 11·7

	W_1	W_2	W_3	Available	Penalties
F_3	(5)	(4)	(7)	7	(1)
F_4	(1)	(6)	10 (2)	14	(1)
Demand	2	9	10		
Penalties	(4)	(2)	(5) ↑		

Since the maximum penalty (5) is associated with column 3 of matrix in table 11·7, so the maximum possible amount 10 is allocated to the cell (2, 3) with lowest cost in this column.

The third reduced matrix after leaving column 3 of the matrix in table 11·7 with remaining demands and availables is as follows (table 11·8).

Table 11·8

	W_1	W_2	Available	Penalties
F_3	(5)	7 (4)	7	(1)
F_4	2 (1)	2 (6)	4	(5) ←
Demand	2	9		
Penalties	(4)	(2)		

Since the maximum penalty (5) is associated with row 2 (table 11·8), so the maximum possible amount 2 is allocated to the cell (2, 1) with lowest cost in this row. The remaining amount 2 in row second is allocated to the cell (2, 2) and then we allocate the remaining amount 7 to the cell (1, 2) with minimum cost after crossing (or leaving) the row 2.

Thus we get the required feasible solution as shown in table 11·9.

Table 11·9

	W_1	W_2	W_3	Available
F_1	5 (2)	(7)	(4)	5
F_2	(3)	(3)	8 (1)	8
F_3	(5)	7 (4)	(7)	7
F_4	2 (1)	2 (6)	10 (2)	14
Demand	7	9	18	

The total transporation cost to this F.S.

$= 5 \times 2 + 8 \times 1 + 7 \times 4 + 2 \times 1 + 2 \times 6 + 10 \times 2 =$ Rs. 80.

which is less than the cost associated with the feasible solutions obtained by the previous methods 1 and 2.

It is important to note that the Vogel's method which take more time as compared with the other methods 1 and 2, gives the better initial F.S. Since the initial F.S. obtained by Vogel's method is much more close to the optimal solution so the time in reaching to the optimal solution from this initial F.S. will be saved.

Note : The students are advised to find the initial B.F. solution by the *Vogel's method* which will save their valuable time to reach the optimal solution.

§11·7. Optimal Test

After getting the initial F.S. of a transportation problem, we test this solution for optimality *i.e.*, we check whether the solution thus obtained minimizes the transportation cost or not. It is important to note that the *optimality test is applicable to a F.S. consisting of $(m + n - 1)$ allocations in independent positions.*

To get the optimal solution we start with an empty cell (*i.e.*, the cell in which there is no allocation) and allocate +1 unit

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to this cell. In order to maintain the row and column sum unchanged we make necessary adjustment in the solution. The net change in total cost resulting from this adjustments, is called the evaluation for the empty cell. If the cell evaluation is positive, the adjustments would increase the total cost; if negative, it would decrease the total cost. Since there are $mn - (m + n - 1) = (m - 1) \cdot (n - 1)$ empty cells, therefore there are $(m - 1)(n - 1)$ such cell evaluation.

If all the cell evaluations are positive or zero, then we cannot decrease the total transportation cost and hence the solution under test is the required optimal solution. The problem of computing cell evaluations for all unoccupied cells individually is very complicated. To avoid this complication, first we shall prove a theorem § 11·8 and then give a optimality test routine for calculating all of them simultaneously.

§ 11·8. Theorem

If we have a B.F.S. consisting of $(m + n - 1)$ independent positive allocations and a set of arbitrary numbers u_i and v_j, $i = 1, 2, \ldots, m$; $j = 1, 2, \ldots, n$; such that

$$c_{rs} = u_r + v_s$$

for all occupied cells (r, s), then the evaluation d_{ij} corresponding to each empty cell (i, j) is given by

$$d_{ij} = c_{ij} - (u_i + v_j).$$

Proof : Mathematically a transportation problem is to determine $x_{ij} \geq 0$, which minimize

$$Z = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} \cdot x_{ij} \qquad \ldots(1)$$

subject to the restrictions

$$\sum_{j=1}^{n} x_{ij} = a_i \quad \text{or} \quad 0 = a_i - \sum_{j=1}^{n} x_{ij} \qquad \ldots(2)$$

for $i = 1, 2, \ldots, m$

$$\sum_{i=1}^{m} x_{ij} = b_j \quad \text{or} \quad 0 = b_j - \sum_{i=1}^{m} x_{ij} \qquad \ldots(3)$$

for $j = 1, 2, \ldots, n$.

Multiplying (2) by u_i $(i = 1, 2, \ldots, m)$, (3) by v_j $(j = 1, 2, \ldots, n)$ and adding to (1), we have

$$Z = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij} + \sum_{i=1}^{m} u_i \left(a_i - \sum_{j=1}^{n} x_{ij}\right) + \sum_{j=1}^{n} v_j \left(b_j - \sum_{i=1}^{m} x_{ij}\right)$$

or
$$Z = \sum_{i=1}^{m} \sum_{j=1}^{n} [c_{ij} - (u_i + v_j)] x_{ij} + \sum_{i=1}^{m} u_i a_i + \sum_{j=1}^{n} v_j b_j \qquad \ldots(4)$$

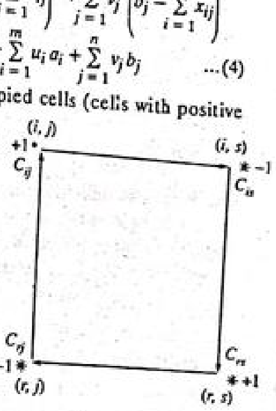
But it is given that for all occupied cells (cells with positive allocations)

$$c_{rs} = u_r + v_s \qquad \ldots(5)$$

$\therefore$ In the objective function (4), all the terms of positive allocations vanish as their cofficients are zero.

For this F.S. the value of the objective function (4) reduce to

$$Z = \sum_{i=1}^{m} u_i a_i + \sum_{j=1}^{n} v_j b_j \qquad \ldots(6)$$



(Fig. 11.1)

To prove the required result, let us first determine the cell evaluation for the empty cell (i,j) when we allocate +1 unit to this empty cell, the number of positive allocations become $(m + n)$, hence they become independent in position and so a closed loop can be formed. Let the closed loop formed by joining this cell (i,j) to the occupied cells be as shown in the fig. 11·1.

Here cell $(i\,s)$, (r,s) and (r,j) are occupied cells.

$\therefore c_{is} = u_i + v_s$, $c_{rs} = u_r + v_s$, $c_{rj} = u_r + v_j$.

When we allocate +1 unit to the empty cell (i,j) then to maintain the row and column sums unchanged we shall decrease the individual allocation at the cell (i,s) by 1, increase that at the cell (r,s) by 1, and decrease that at cell (r,j) by 1. Thus the allocations at the occupied cells (i,s) (r,s), (r,j) are changed.

Thus the cell evaluation d_{ij} corresponding to this empty cell (i,j) is given by

d_{ij} = Difference in the total cost for the new solution and the original one

$$= c_{ij} - c_{is} + c_{rs} - c_{rj}$$
$$= c_{ij} - (u_i + v_s) + (u_r + v_s) - (u_r + v_j)$$

or
$$d_{ij} = c_{ij} - (u_i + v_j)$$

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This gives the evaluation d_{ij} connecting empty cell (i,j) to the occupied cells by a square (or rectangle) shaped loop. By similar reasoning, we may generalize it to the case of a loop of arbitrary shape connecting empty cell (i,j) to the occupied cells.

Thus we conclude that the evaluation d_{ij} corresponding to each empty cell (i,j) is given by

$$d_{ij} = c_{ij} - (u_i + v_j).$$

This completes the proof of the theorem.

§ 11·9. Computational procedure of optimal test

After giving the initial B.F.S. of a transportation problem, we test this solution for optimality as follows.

1. For a B.F.S. we determine a set of $(m+n)$ numbers

$$u_i, \quad i = 1, 2, \ldots, m$$
$$v_j, \quad j = 1, 2, \ldots, n$$

such that, for each occupied cell (r,s)

$$c_{rs} = u_r + v_s$$

For this we assign an arbitrary value to one of the u_i's or v_j's then the rest $(m+n-1)$ of them can easily be solved algebraically from the relation $c_{rs} = u_r + v_s$, for occupied cells. Generally we choose that u_i or $v_j = 0$ for which the corresponding row of column have the maximum number of individual allocations.

2. Then we calculate the cell evaluations d_{ij} for each unoccupied cell (i,j) by using the formula

$$d_{ij} = c_{ij} - (u_i + v_j)$$

3. Then we examine the matrix of cell evaluations and conclude that
   (i) if all $d_{ij} > 0$, then the solution under test is optimal and unique.
   (ii) if all $d_{ij} \geq 0$, with at least one $d_{ij} = 0$, then the solution under test is optimal and alternative optimal solution exists.
   (iii) if at least one $d_{ij} < 0$, in step 3, then the solution is not optimal.

   In the last case we proceed to the next step 4.

4. If at the least one $d_{ij} < 0$, in step 3, then we form a new B.F.S. In the next B.F.S. we give maximum allocation to the cell for which d_{ij} is minimum and negative, by making an occupied cell empty.

5. Then we repeat the steps 1 to 3 to test the optimality to this new B.F.S.

§ 11·10. Transportation Algorithm or Modi Method (Modified Distribution) **[Rohilkhand 86]**

For the solution of a minimization transportation problem proceed systematically as follows. This iterative procedure of determining an optimum solution of a minimization transportation problem is known as **Modi Method.**

Step 1. Construct a transportation table entering the capacities $a_1, a_2, \ldots, a_m$ of the sources and the requirements $b_1, b_2, \ldots, b_n$. Enter the various costs c_{ij} at the upper left corners of all the cells. Find an initial B.F. solution (allocations in independent positions) of the problem by any of the methods given in § 11·6 enter the allocations at the centres of the cells.

Step 2. Find the set of numbers u_i $(i = 1, 2, \ldots, m)$ and v_j $(j = 1, 2, \ldots, n)$ by the method given in step 2 of § 11·9 such that for each occupied cell (r,s), $c_{rs} = u_r + v_r$.

Step 3. Find the cell evaluations $u_i + v_j$ for each unoccupied cell (i,j) and enter at the upper right corner of the corresponding cell (i,j).

Step 4. Find the cell evaluations $d_{ij} = c_{ij} - (u_i + v_j)$ for each unoccupied cell (i,j) and enter at the lower right corners of the corresponding cells.

Step 5. Examine the cell evaluations d_{ij} for unoccupied cells and conclude that
   (i) if all $d_{ij} > 0$, then the solution under test is optimal and unique.
   (ii) if all $d_{ij} \geq 0$, with at least one $d_{ij} = 0$, then the solution under test is optimal and an alternative optimal solution exists.
   (iii) if at least one $d_{ij} < 0$, then the solution is not optimal. In the last case proceed to the next step 6.

Step 6. Form a new B.F.S. by giving maximum allocation to the cell for which d_{ij} is minimum and negative, by making an occupied cell empty.

Step 7. Then repeat the step (2) to (5) to test the optimality to this new B.F. solution.

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Continue improving the B.F.S. iteratively using the step (2) to (6) till an optimal solution is attained.

Thus in the table after making all the entries the occupied cells having allocations and the unoccupied cells will be as follows.

(c_{ij}) Allocation

(c_{ij}) $(u_i + v_j)$ $d_{ij} = c_{ij} - (u_i + v_j)$

Illustrative Examples

Ex. 1. *A company has four plants P_1, P_2, P_3, P_4 from which it supplies to three markets M_1, M_2, M_3. Determine the optimal transportation plan from the following data giving the plant to market shifting costs, quantities available at each plant and quantities required at each market.*

Market ↓	Plant P_1	P_2	P_3	P_4	Required at Market
M_1	19	14	23	11	11
M_2	15	16	12	21	13
M_3	30	25	16	39	19
Available at plant	6	10	12	15	43

Sol.

Step 1. By Vogel's method an initial B.F.S. of the given problem is given by the following table.

Table 11·10

				a_i
(19)	(14)	(23)	(11) 11	11
(15) 6	(16) 3	(12)	(21) 4	13
(30)	(25) 7	(16) 12	(39)	19
b_j: 6	10	12	15	

Total transportation cost $= 6 \times 15 + 3 \times 16 + 7 \times 25 + 12 \times 16 + 11 \times 11 + 4 \times 21$
$=$ Rs. 710.

Step 2. Here we determine a set of u_i and v_j s.t. for each occupied cell (r, s), $c_{rs} = u_r + v_s$.

For this we choose $u_2 = 0$ (since row 2 contains maximum number of allocations).

Since $c_{21} = 15 = u_2 + v_1$, $c_{22} = 16 = u_2 + v_2$, $c_{24} = 21 = u_2 + v_4$

$\therefore$ $v_1 = 15$, $v_2 = 16$, $v_4 = 21$

Also $c_{14} = 11 = u_1 + v_4$, $c_{32} = 25 = u_3 + v_2$, $c_{33} = 16 = u_3 + v_3$

$\therefore$ $u_1 = 11 - v_4 = -10$, $u_3 = 25 - v_2 = 9$, $v_3 = 16 - u_3 = 7$.

Step 3. Now we find the cell evaluations $u_i + v_j$ for each unoccupied cell (i, j) and enter at the upper right corner of the corresponding unoccupied cell.

Step 4. Then we find the cell evaluations $d_{ij} = c_{ij} - (u_i + v_j)$ (*i.e.*, the difference of the upper right corner entry from the upper left corner entry) for each unoccupied cell (i, j) and enter at the lower right corner of the corresponding unoccupied cell.

Thus we get the following table.

Table 11·11

				u_i
(19) (5) (14)	(14) (6) (8)	(23) (−3) (26)	(11) 11	−10 (u_1)
(15) 6	(16) 3	(12) (7) (5)	(21) 4	0 (u_2)
(30) (24) (6)	(25) 7	(26) 12	(39) (30) (9)	9 (u_3)
v_j: 15 (v_1)	16 (v_2)	7 (v_3)	21 (v_4)	

Step 5. Since all $d_{ij} > 0$ (see table 11·11). Hence the B.F.S. shown by table 11·11 is an optimal solution and the optimal transportation cost $=$ Rs. 710.

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Continue improving the B.F.S. iteratively using the step (2) to (6) till an optimal solution is attained.

Thus in the table after making all the entries the occupied cells having allocations and the unoccupied cells will be as follows.

(c_{ij})
Allocation

(c_{ij})	$(u_i + v_j)$
	$d_{ij} = c_{ij} - (u_i + v_j)$

Illustrative Examples

Ex. 1. *A company has four plants P_1, P_2, P_3, P_4 from which it supplies to three markets M_1, M_2, M_3. Determine the optimal transportation plan from the following data giving the plant to market shifting costs, quantities available at each plant and quantities required at each market.*

Market ↓	Plant P_1	P_2	P_3	P_4	Required at Market
M_1	19	14	23	11	11
M_2	15	16	12	21	13
M_3	30	25	16	39	19
Available at plant	6	10	12	15	43

Sol.

Step 1. By Vogel's method an initial B.F.S. of the given problem is given by the following table.

Table 11·10

				a_i
(19)	(14)	(23)	(11) 11	11
(15) 6	(16) 3	(12)	(21) 4	13
(30)	(25) 7	(16) 12	(39)	19
b_j: 6	10	12	15	

Total transportation cost $= 6 \times 15 + 3 \times 16 + 7 \times 25 + 12 \times 16 + 11 \times 11 + 4 \times 21$
$= \text{Rs. } 710.$

Step 2. Here we determine a set of u_i and v_j s.t. for each occupied cell (r, s), $c_{rs} = u_r + v_s$.

For this we choose $u_2 = 0$ (since row 2 contains maximum number of allocations).

Since $c_{21} = 15 = u_2 + v_1$, $c_{22} = 16 = u_2 + v_2$, $c_{24} = 21 = u_2 + v_4$

$\therefore \quad v_1 = 15,\ v_2 = 16,\ v_4 = 21$

Also $c_{14} = 11 = u_1 + v_4$, $c_{32} = 25 = u_3 + v_2$, $c_{33} = 16 = u_3 + v_3$

$\therefore \quad u_1 = 11 - v_4 = -10,\ u_3 = 25 - v_2 = 9, \quad v_3 = 16 - u_3 = 7.$

Step 3. Now we find the cell evaluations $u_i + v_j$ for each unoccupied cell (i, j) and enter at the upper right corner of the corresponding unoccupied cell.

Step 4. Then we find the cell evaluations $d_{ij} = c_{ij} - (u_i + v_j)$ (*i.e.*, the difference of the upper right corner entry from the upper left corner entry) for each unoccupied cell (i, j) and enter at the lower right corner of the corresponding unoccupied cell.

Thus we get the following table.

Table 11·11

				u_i
(19) (5) (14)	(14) (6) (8)	(23) (−3) (26)	(11) 11	−10(u_1)
(15) 6	(16) 3	(12) (7) (5)	(21) 4	0 (u_2)
(30) (24) (6)	(25) 7	(26) 12	(39) (30) (9)	9 (u_3)
v_j: 15 (v_1)	16 (v_2)	7 (v_3)	21 (v_4)	

Step 5. Since all $d_{ij} > 0$ (see table 11·11). Hence the B.F.S. shown by table 11·11 is an optimal solution and the optimal transportation cost = Rs. 710.

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Thus the solution of the given transportation problem is
From source 1 transport 11 units to destination 4.
From soruce 2 transport 6, 3 and 4 units to destinations 1, 2 and 4 respectively
and From source 3 transport 7 and 12 units to destinations 2 and 3 respectively.

Ex. 2. *Solve the following transportation problem*

		To 1	2	3	Supply
From	1	2	7	4	5
	2	3	3	1	8
	3	5	4	7	7
	4	1	6	2	14
Demand		7	9	18	34

Sol.

Step 1. The initial B.F.S. of the above problem (by Vogel's method) is given by table 11·12.
(For initial B.F.S. see § 11·6, method 3 on page 365).
Total transportation cost

$$= 5 \times 2 + 2 \times 1 + 7 \times 4 + 2 \times 6 + 8 \times 1 + 10 \times 2$$
$$= \text{Rs. } 80$$

Table 11·12

			a_i
(2) 5	(7)	(4)	5
(3)	(3)	(1) 8	8
(5)	(4) 7	(7)	7
(1) 2	(6) 2	(2) 10	14
b_j: 7	9	18	

Step 2. Now we determine a set of u_i and v_j s.t. for each occupied cell (r, s), $c_{rs} = u_r + v_s$.

For this we choose $u_4 = 0$ (since row 4 contains maximum number of allocations).

Since $c_{41} = 1 = u_4 + v_1$, $c_{42} = 6 = u_4 + v_2$, $c_{43} = 2 = u_4 + v_3$

$\therefore$ $v_1 = 1 - u_4 = 1$, $v_2 = 6 - u_4 = 6$, $v_3 = 2 - u_4 = 2$.

Also $c_{11} = 2 = u_1 + v_1$, $c_{23} = 1 = u_2 + v_3$, $c_{32} = 4 = u_3 + v_2$.

$\therefore$ $u_1 = 2 - v_1 = 1$, $u_2 = 1 - v_3 = -1$, $u_3 = 4 - v_2 = -2$.

Step 3. Then we find the cell evaluations $u_i + v_j$ for each unoccupied cell (i, j) and enter at the upper right corner of the corresponding unoccupied cell.

Step 4. Then we find the cell evaluations $d_{ij} = c_{ij} - (u_i + v_j)$ (*i.e.*, the difference of the upper right corner entry from the upper left corner entry) for each unoccupied cell (i, j) and enter at the lower right corner of the corresponding unoccupied cell.

Thus we get the following table.

Table 11·13

			u_i
(2) 5	(7) (7) (0)	(4) (3) (1)	1 (u_1)
(3) (0) (3)	(3) (5) (−2*)	(1) 8	−1 (u_2)
(5) (−1) (6)	(4) 7	(7) (0) (7)	−2 (u_3)
(1) 2	(6) 2	(2) 10	0 (u_4)
v_j: 1 (v_1)	6 (v_2)	2 (v_3)	

Step 5. Since cell evaluation $d_{22} = -2 < 0$, so the solution under test is not optimal.

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Step 6. Since minimum d_{ij} is $d_{22} = -2$ (negative), so we give maximum allocation to this cell from an occupied cell and make the necessary changes in other allocation as shown in table 11·14.

Table 11·14

5		
	+2	−2 8
	7	
2	−2 2	+2 10

Step 7. The new B.F.S. (allocations in independent positions) thus obtained is shown in table 11·15 For this B.F.S. Total transportation cost

$$= 5 \times 2 + 2 \times 1 + 2 \times 3 + 7 \times 4 + 6 \times 1 + 12 \times 2$$
$$= \text{Rs. } 76,$$

which is less than that for the initial B.F.S.

Table 11·15

			a_i
(2) 5	(7)	(4)	5
(3)	(3) 2	(1) 6	8
(5)	(4) 7	(7)	7
(1) 2	(6)	(2) 12	14
b_j 7	9	18	

Step 8. Proceeding as in step 2, 3 and 4 we get the following table.

Table 11·16

			u_i
(2) 5	(7) (5) 　 (2)	(4) (3) 　 (1)	1 (u_1)
(3) (0) 　 (3)	(3) 2	(1) 6	−1 (u_2)
(5) (1) 　 (4)	(4) 7	(7) (2) 　 (5)	0 (u_3)
(1) 2	(6) (4) 　 (2)	(2) 12	0 (u_4)
b_j 1 (v_1)	4 (v_2)	2 (v_3)	

Since all $d_{ij} > 0$, hence the B.F.S. shown by table 11·16 is an optimal solution which is also unique.

Thus the solution of the given transportation problem is

From source 1 transport 5 units to destination 1,
From source 2 transport 2 and 6 units to destination 2 and 3 respectively,
From source 3 transport 7 units to destination 2,
and From source 4 transport 2 and 12 units to destinations 1 and 3 respectively.

And the total transportation cost (optimal) = Rs.76.

Ex. 3. *Solve the following transportation problem.*

	S_1	S_2	S_3	S_4	a_i
0_1	1	2	1	4	30
0_2	3	3	2	1	50
0_3	4	2	5	9	20
b_j	20	40	30	10	100

[Meerut 88]

Sol. By 'Lowest Cost Entry' method, we get the following B.F.S. of the problem.

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Table 11·17

	S_1	S_2	S_3	S_4	a_i
0_1	(1) 20	(2)	(2) 10	(4)	30
0_2	(3)	(3) 20	(2) 20	(1) 10	50
0_3	(4)	(2) 20	(5)	(9)	20
b_j	20	40	30	10	

and Total transportation cost

$$= 20 \times 1 + 10 \times 1 + 20 \times 3 + 20 \times 2 + 10 \times 1 + 20 \times 2$$
$$= \text{Rs.}180$$

To test solution for optimality

Finding the set of u_i $(i = 1, 2, 3)$, v_j $(j = 1, 2, 3, 4)$ such that $c_{rs} = u_r + v_s$ for occupied cells and entering $u_i + v_j$ and d_{ij} in the unoccupied cells, the table giving the optimal solution is as follows

Table 11·18

					u_i
	(1) 20	(2) (2) (0)	(1) 10	(4) (0) (4)	$-1\ (u_1)$
	(3) (2) (1)	(3) 20	(2) 20	(1) 10	$0\ (u_2)$
	(4) (1) (3)	(2) 20	(5) (1) (4)	(9) (0) (9)	$-1\ (u_3)$
v_j	2 (v_1)	3 (v_2)	2 (v_3)	1 (v_4)	

Since all $d_{ij} \geq 0$, the solution given by the above table is optimal. But an alternative optimal solution will also exist.

Thus the solution of the given problem is

From 0_1 transport 20 and 10 units to destinations S_1 and S_3 respectively

From 0_2 transport 20, 20 and 10 units to destinations S_2, S_3 and S_4 respectively

From 0_3, transport 20 units to destination S_2.

and the total transportation cost = Rs. 180.

Ex. 4. *Solve the following problem.*

		To 1	2	3	4	Supply
From	1	21	16	25	13	11
	2	17	18	14	23	13
	3	32	27	18	41	19
	Demand	6	10	12	15	43

Sol. The final table giving optimal solution is as follows.

Table 11·19

		To 1	2	3	4	u_i
From	1	(21) (7) (14)	(16) (8) (8)	(25) (−1) (26)	(13) 11	$-10\ (u_1)$
	2	(17) 6	(18) 3	(14) (9) (5)	(23) 4	$0\ (u_2)$
	3	(32) (26) (6)	(27) 7	(18) 12	(41) (32) (9)	$9\ (u_3)$
	v_j	17 (v_1)	18 (v_2)	9 (v_3)	23 (v_4)	

Hence the solution of the given problem is

From source 1 transport 11 units to destination 4

From source 2 transport 6, 3 & 4 units to destination 1, 2 & 4 respectively

From source 3 transport 7 and 12 units to destination 2 & 3 respectively

Optimal transportation cost

$$= 6 \times 17 + 3 \times 18 + 7 \times 27 + 12 \times 18 + 11 \times 13 + 4 \times 3$$
$$= \text{Rs. } 796.$$

§11·11. Degeneracy in Transportation Problems

[Meerut 88; Rohilkhand 86]

In a transportation problem, degeneracy occurs whenever the number of independent individual allocations is less than

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$(m + n - 1)$. Degeneracy may occur either at the initial stage or at an intermediate stage at some subsequent iteration.

In such cases, to resolve degeneracy, we allocate an extremely small amount (close to zero) to one or more empty cells of the matrix (generally lowest cost cells if possible), so that the total number of occupied (allocated) cells becomes $(m + n - 1)$ at independent positions. We denote this small amount by Δ (delta) or ε (epsilon) satisfying the following conditions.

1. $0 < \Delta x_{ij}$ for all $x_{ij} > 0$.
2. $\Delta + 0 = \Delta = 0 + \Delta$.
3. $x_{ij} \pm \Delta = x_{ij}$, for all $x_{ij} > 0$.
4. If there are more than one Δ's introduced in the solution then
   (i) if Δ, Δ' are in the same row, $\Delta < \Delta'$ when Δ is to the left to Δ'.

and (ii) if Δ, Δ' are in the same column, $\Delta < \Delta'$ when Δ is above Δ'.

It is clear that after introducing Δ satisfying the above conditions, the original solution of the problem is not changed. It is merely a technique to apply the optimality test and is omitted ultimately.

For the clear understanding of the method. See Ex. 5,

Problems of Degeneracy

Ex. 5. *Given below the unit cost array with supplies a_i; $i = 1, 2, 3$ and demand b_j; $j = 1, 2, 3, 4$.*

Sink

		1	2	3	4	a_i
	1	8	10	7	6	50
Source	2	12	9	4	7	40
	3	9	11	10	8	30
	b_j	25	32	40	23	120

Find the optimal solution to the above Hitchock problem.

Sol. By "Lowest Cost Entry Method" the initial B.F.S. of the given problem is given by the following :

Table 11·20

	1	2	3	4	a_i
1	(8) 25	(10) 2	(7)	(6) 23	50
2	(12)	(9)	(4) 40	(7) Δ	40
3	(9)	(11) 30	(10)	(8)	30
b_j	25	32	40	23	

Since the total number of allocations is 5 which is one less than $m + n - 1 = 6$. Hence this solution is a *degenerate solution*.

Now to resolve this degeneracy we allocate a very small amount Δ to the cell (2, 4) getting 6 allocation at independent positions.

To test the solution for optimality

Now to test the solution for optimality we construct the following table 11·21.

Table 11·21

	1	2	3	4	u_i
1	(8) 25	(10) 2	(7) (3) (4)	(6) 23	0 (u_1)
2	(12) (9) (3)	(9) (11) (−2)	(4) 40	(7) Δ	1 (u_2)
3	(9) (9) (0)	(11) 30	(10) (4) (6)	(8) (7) (1)	1 (u_3)
(v_j)	8 (v_1)	10 (v_2)	3 (v_3)	6 (v_4)	

Since $d_{22} = -2 < 0$, so the solution under test is not optimal. Now we shall allocate to this cell (2, 2) as much as possible. Thus we take Δ from cell (2, 4) to cell (2, 2) and form the new table 11·22 to check the optimality of the solution.

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Table 11·22

	1	2	3	4	u_i
1	(8) 25	(10) 2	(7) (5) (2)	(6) 23	0
2	(12) (7) (5)	(9) Δ	(4) 40	(7) (5) (2)	−1
3	(9) (9) (0)	(11) 30	(10) (6) (4)	(8) (7) (1)	1
v_j	8	10	5	6	

Since all $d_{ij} \geq 0$, so the solution under test is optimal.

Also $d_{31} = 0$, implies that an alternative optimal solution exist.

The optimal solution is as given in table 11·21.

and the optimal cost $= 25 \times 8 + 2 \times 10 + 30 \times 11 + 40 \times 4 + 23 \times 6$
$= \text{Rs.}848.$

§ 11·12. Unbalanced Transportation Problems.

A transportation problem is said to be an **unbalanced transportation problem,** if the sum of all available amounts (quantities) is not equal to the sum of all requirements, *i.e.*, if $\sum_{i=1}^{m} a_i \neq \sum_{j=1}^{n} b_j$.

An unbalanced transportation problem is converted into a balanced transportation problem, by introducing a fictitious source or destination which will provide the surplus supply or demand. The costs of transporting an unit from the fictitious source (or to the fictitious destination) is taken to be zero. After converting the unbalanced transportation problem to balanced transportation problem, by the introduction of fictitious source or destination, it is solved by the earlier methods.

For the clear understanding of the method. See Ex. 6.

Problem on Unbalanced transportation problem

Ex. 6. *Determine the optimal transportation plan from the following table giving the plant to market shipping costs, and*

quantities required at each market and available at each plant :

Plant	Market W_1	W_2	W_3	W_4	Availability
F_1	11	20	7	8	50
F_2	21	16	10	12	40
F_3	8	12	18	9	70
Requirement	30	25	35	40	

Sol. Here total requirement of the market
$= 30 + 25 + 35 + 40 = 130$

and total availability at the plants $= 50 + 40 + 70 = 160$

Since the total availability at three plants is 30 more than the total requirements in four markets W_1, W_2, W_3, W_4, therefore, this transportation problem is unbalanced and so we convert this problem to a balanced one by introducing a fictitious market W_5 with requirements 30 such that the cost of transportation from plants to this market W_5 are zero.

∴ The balanced transportation problem is given by the table 11·23.

Table 11·23

Plant	Market W_1	W_2	W_3	W_4	W_5	Availability
F_1	11	20	7	8	0	50
F_2	21	16	10	12	0	40
F_3	8	12	18	9	0	70
Requirement	30	25	35	40	30	160 (Total)

By 'Vogel's Method', we get the following initial B.F. solution of the problem.

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Table 11·24

Plant	W_1	W_2	W_3	W_4	W_5	a_i
F_1	(11)	(20)	(7) 25	(8) 25	(0)	50
F_2	(21)	(16)	(10) 10	(12)	(0) 30	40
F_3	(8) 30	(12) 25	(18)	(9) 15	(0)	70
b_j	30	25	35	40	30	160

The solution given in table 11·24 is an optimal solution (students are advised to check).

Thus the optimal solution is

Transport from plant F_1 to Market W_3, 25 units.

Transport from plant F_1 to Market W_4, 25 units.

Transport from plant F_2 to Market W_3, 10 units.

Transport from plant F_3 to Market W_1, 30 units.

Transport from plant F_3 to Market W_2, 25 units.

Transport from plant F_3 to Market W_4, 15 units.

Total transportation cost

$$= 25 \times 7 + 25 \times 8 + 10 \times 10 + 30 \times 8 + 25 \times 12 + 15 \times 9$$
$$= \text{Rs.}1150.$$

It is important to note that 30 units are despatched from plant F_2 to market (Fictitious) W_5. In other words, we can say that 30 units are left undispatched at the plant F_2.

Exercise on Chapter XI

1. Write a short note on "Transportation problem. [Meerut O.R. 87]
2. Explain the difference between a transportation problem and an assignment problem.
3. Describe methods to obtain an initial feasible solution for a transportation problem.
4. Give a brief outline of the procedure for solving a transportation problem.
5. State the transportation problem in general terms and explain, in this connection, degeneracy. Also explain how it is tested whether a B.F.S. of a transportation problem is optimal or not.
6. Write short note on degeneracy in T.P. [Meerut O.R. 88]

Solve the following transportation problems.

7.

		To 1	2	3	Available	Ans.
From	I	2	7	4	5	I to 1 units 5
	II	3	3	1	8	II to 2 units 2, to 3 units 6
	III	5	4	7	7	III to 2 units 7
	IV	1	6	2	14	IV to 1 units 2, to 3 units 12
Required		7	9	18	34	

[Optimal T.C. = Rs. 76·00]

[Meerut O.R. 87]

8. Obtain an initial B.F.S. to the following Transportation Problem using *VAM*.

		Stores S_1	S_2	S_3	S_4	Availabilit	Ans.
Ware house	A	5	1	3	3	34	A to S_2 units 25, to S_3 units 9
	B	3	3	5	4	15	B to S_1 units 15
	C	6	4	4	3	12	C to S_3 units 8, to S_4 units 4
	D	4	1	4	2	19	D to S_1 units 6, to S_4 units 13
		21	25	17	17	80	and others.

9. Determine an initial B.F.S. to the following transportation problem by using the N-W-corner rule.

		Destination A_1	B_1	C_1	D_1	E_1	Supply	Ans.
Origin	A	2	11	10	3	7	4	A to A_1 units 3, to B_1 units 1
	B	1	4	7	2	1	8	B to B_1 units 2, to C_1 units 4, to D_1 units 2
	C	3	9	4	8	12	9	C to D_1 units 3, to E_1 units 6
		3	3	4	5	6	21	

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10.

Ware house / Factory ↓	W_1	W_2	W_3	W_4	Factory capacity
F_1	19	30	50	10	7
F_2	70	30	40	60	9
F_3	40	8	70	20	18
Ware-house Requirement	5	8	7	14	34

Ans. F_1 to W_1 units 5, to W_4 units 2; F_2 to W_2 units 2, to W_3 units 7; F_3 to W_2 units 6, to W_4 units 12. Total cost = Rs. 743·00

[Meerut 95 (P), Agra 86]

11.

	To	W_1	W_2	W_3	W_4	W_5	Available
	F_1	3	4	6	8	9	20
From	F_2	2	10	1	5	8	30
	F_3	7	11	20	40	3	15
	F_4	2	1	9	14	16	13
	Required	40	6	8	18	6	78

Ans. F_1 to W_1 units 20; F_2 to W_1 units 4, to W_3 units 8, to W_4 units 18; F_3 to W_1 units 9, to W_5 units 6; F_4 to W_1 units 7, to W_2 units 6

12.

	D_1	D_2	D_3	D_4	D_5	D_6	a_i
0_1	9	12	9	6	9	10	5
0_2	7	3	7	7	5	5	6
0_3	6	5	9	11	3	11	2
0_4	6	8	11	2	2	10	9
b_j	4	4	6	2	4	2	22

Ans. 0_1 to D_3 units 5; 0_2 to D_2 units 4, to D_6 units 2; 0_3 to D_1 units 1, to D_3 units 1; 0_4 to D_1 units 3, to D_4 units 2, to D_5 units 4. Total T. cost = Rs. 112·00

[Meerut 95]

Plant	Market A	B	C	D	Available
X	10	22	10	20	8
Y	15	20	12	8	13
Z	20	12	10	15	11
Required	5	11	8	8	32

Ans. X to A units 5, to C units 3; Y to C units 5, to D units 8; Z to B units 11. Total T. cost = Rs. 336·00

Plant	Market A	B	C	D	Available
X	14	9	18	6	11
Y	10	11	7	16	3
Z	25	20	11	34	19
Required	6	10	12	15	43

Ans. X to D units 11; Y to A units 6, to B units 3, to D units 4; Z to B units 7, to C units 12. Total T. cost = Rs. 495·00

Plant	Warehouse A	B	C	D	Available at plant
X	30	20	50	20	75
Y	20	10	20	40	120
Z	40	30	40	30	105
Required at warehouse	65	60	80	95	300

Ans. X to D units 75; Y to B units 40, to C units 80; Z to A units 65, to B units 20, to D units 20. Total T. cost = Rs. 7300·00

Plant	Market A	B	C	Available at plant
X	11	21	16	14
Y	7	17	13	26
Z	11	23	21	36
Required at Market	18	28	25	

Ans. X to C units 14; Y to B units 15, to C units 11; Z to A units 18, to B units 13. 5 units are left at plant Z. Total T. cost = Rs. 1119·00

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17. A manufacture wants to ship 8 loads of his product as shown below. The matrix gives the mileage from origin 0 to the destination.

$D \rightarrow$ / $\downarrow 0$	A	B	C	Available
X	50	30	220	1
Y	90	45	170	3
Z	250	200	50	4
Reqd.	4	2	2	8

Shipping costs are Rs. 10 per load mile. What shipping schedule should be used ?

[**Ans.** X to A units 1, Y to A units 3, Z to B units 2 and to C units 2; cost Rs. 8200]

18. A textile firm has three factories F_1, F_2, F_3 and four warehouse W_1, W_2, W_3, W_4. The transportation costs, factory capacities and warehouse requirements are given in the following table :

	W_1	W_2	W_3	W_4	Capacity
F_1	15	24	11	12	5,000
F_2	25	20	14	16	4,000
F_3	12	16	22	13	7,000
Req.	3,000	2,500	3,500	4,000	

Find the least cost transportation schedule.

[**Ans.** F_1 to W_3 units 2500, to W_4 units 2500; F_2 to W_3 units 1000; F_3 to W_1 units 3000, to W_2 units 2500, to W_4 units 1500. Total cost Rs. 1,67,000.]

Solve the following transportation problems.

19.

From \ To	D_1	D_2	D_3	Supply
0_1	2	4	1	40
0_2	6	3	2	50
0_3	4	5	6	20
0_4	3	2	1	30
0_5	5	2	5	10
Demand	50	60	40	150

Ans. 0_1 to D_1 units 30, to D_3 units 10; 0_2 to D_2 units 50; 0_3 to D_1 units 20; 0_4 to D_3 units 30; 0_5 to D_2 units 10; Optimal T.C. = Rs. 350·00

	W_1	W_2	W_3	W_4	Capacity
F_1	95	105	80	15	12
F_2	115	180	40	30	7
F_3	155	180	95	70	5
Requirements	5	4	4	11	

Ans. F_1 to W_1 units 5, to W_2 units 4, to W_4 units 3; F_2 to W_3 units 4, to W_4 units 3; F_3 to W_4 units 5; Optimal T.C. = Rs. 1540

	D_1	D_2	D_3	D_4	a_i
0_1	1	2	1	4	30
0_2	4	2	5	9	50
0_3	20	40	30	10	20
b_j	20	40	30	10	100

Ans. 0_1 to D_3 units 30; 0_2 to D_1 units 10, to D_2 units 40; 0_3 to D_1 units 10, to D_4 units 10; Optimal T.C. = Rs. 450

	D_1	D_2	D_3	D_4	a_i
0_1	6	7	3	4	5
0_2	7	9	1	2	7
0_3	6	5	16	7	8
0_4	18	9	10	2	10
b_j	10	5	10	5	30

Ans. 0_1 to D_1 units 2, to D_3 units 3; 0_2 to D_3 units 7; 0_3 to D_1 units 8; 0_4 to D_2 units 5, to D_4 units 5; Optimal T.C. = Rs. 131.

Factory \ Store	A	B	C	a_i
I	10	9	8	8
II	10	7	10	7
III	11	9	7	9
IV	12	14	10	4
b_j	10	10	8	

Ans. I to A units 6, to B units 2; II to B units 7; III to B units 1, to C units 8; IV to A units 4; Optimal T.C. = Rs. 240.

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24. Obtain an optimum B.F.S. to the following degenerate T.P.

	To			Available
From	7	3	4	2
	2	1	3	3
	3	4	6	5
Demand	4	1	5	10

Ans. 1 to 3, 2 units; 2 to 2, 1 unit; 2 to 3, 2 units; 3 to 1, 4 units; 3 to 3, 1 unit; Total T. Cost = Rs.33.

25. Consider the transportation problem shown in the following table :

		Demand Points 1	2	3	4	Supply
Source	1	2	3	11	7	6
	2	1	0	6	1	1
	3	5	8	15	9	10
Demand		7	5	3	2	

Ans. 1 to 2 units 5; 1 to 3 units 1; 2 to 3 units 1; 3 to 1 units 7; 3 to 3 units 1; 3 to 4 units 2; Mini. Dis. 100 miles.

The matrix shows the cost of transportation from source to demand points. Find the optimal solution. [Meerut 89]

26. Consider the following unbalanced T.P.

		To 1	2	3	
From	1	5	1	7	10
	2	6	4	6	80
	3	3	2	5	15
Demand		75	20	50	

Since there is not enough supply, some of the demands at these destination may not be satisfied. Suppose there are penalty costs for every unsatisfied demand unit which are given by 5, 3 and 2 for destination 1, 2 and 3 respectively. Find the optimal solution.

27. Solve the following cost minimizing T.P.

		Destinations D_1	D_2	D_3	D_4	Availability
Sources	S_1	5	3	6	5	15
	S_2	10	7	12	4	11
	S_3	7	5	8	4	13
Demand		8	12	13	6	

Ans. S_1 to D_1 units 8, to D_2 units 7; S_2 to D_2 units 5, to D_4 units 6; S_3 to D_3 units 13; T.C. = 234

[Meerut 97, 97 (BP), 98 (P)]

28. Solbe the following cost minimizing T.P.

		Destinations D_1	D_2	D_3	D_4	D_5	Availability
Sources	S_1	5	5	6	4	2	9
	S_2	6	9	7	8	5	13
	S_3	5	6	4	6	3	9
Demand		3	7	8	5	8	

Ans. S_1 to D_2 units 4, to D_4 units 5; S_2 to D_1 units 3, to D_2 units 2, to D_5 units 8; S_3 to D_2 units 1, to D_3 units 8. T.C. = 154

[Meerut 96 (BP)]

29. Solve the following T.P.

		Destinations D_1	D_2	D_3	D_4	D_5	Availability
Sources	S_1	5	7	6	8	7	13
	S_2	7	6	8	9	8	6
	S_3	6	4	7	7	9	8
Requirement		6	7	3	5	6	

Ans. S_1 to D_1 units 6, to D_3 units 3, to D_5 units 4; S_2 to D_4 units 4, to D_5 units 4; S_3 to D_2 units 7, to D_4 units 1; T.C. = 154

[Meerut 96]